

THÈSE PRÉSENTÉE
POUR OBTENIR LE GRADE DE
DOCTEUR
DE L'UNIVERSITÉ DE BORDEAUX

ÉCOLE DOCTORALE SCIENCES PHYSIQUES & DE L'INGÉNIEUR
AUTOMATIQUE, PRODUCTIQUE, SIGNAL ET IMAGE, INGÉNIERIE COGNITIVE

Par **Bastien HUBERT**

Cartographie et Navigation Autonome sur Champs Physiques
par Filtrage Particulaire à Processus Gaussiens

Autonomous Navigation and Mapping on Physical Fields
by Gaussian Process-based Particle Filtering

Sous la direction de : **Audrey GIREMUS**
la co-direction de : **Karim DAHIA**
et l'encadrement de : **Nicolas MERLINGE**

Thèse soutenue le 8 décembre 2026

Membres du jury :

Manon KOK	Professeure associée	TU Delft	Rapporteure
Carl MILNER	Professeur associé	ENAC	Rapporteur
Vincent FRÉMONT	Professeur	Centrale Nantes	Examinateur
Pierre MINVIELLE	Ingénieur de recherche	CEA	Examinateur
Audrey GIREMUS	Professeure	Université de Bordeaux	Directrice
Karim DAHIA	Ingénieur de Recherche	ONERA	Co-directeur
Nicolas MERLINGE	Ingénieur de Recherche	ONERA	Encadrant

“ *Throw yourself into the unknown
With pace and a fury defiant
Clothe yourself in beauty untold
And see life as means to a triumph* ”

Gang of Youths, Achilles Come Down

REMERCIEMENTS

	ABSTRACTS
--	-----------

RÉSUMÉ ÉTENDU EN FRANÇAIS

CONTENTS

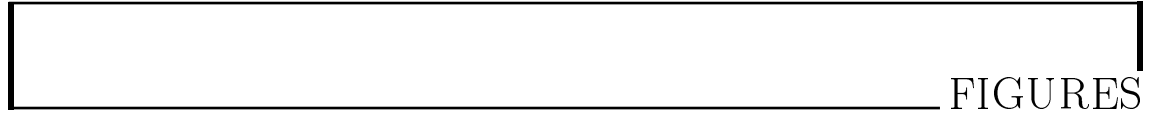
Remerciements	iii
Abstracts	v
Résumé étendu en français	vii
Contents	xii
Figures	xiv
Tables	xv
Algorithms	xvii
Statements	xix
Nomenclature	xxi
1 Introduction	1
1.1 Motivation and Context	1
1.2 Objective and Contributions	3
1.3 Outline	3
1.4 Publications	4
I Mathematical Foundations	5
2 Bayesian Estimation	7
2.1 Introduction	8
2.2 State Estimation	9
2.2.1 Posterior Expectation Estimator	9
2.2.2 Posterior Median Estimator	11
2.2.3 Maximum <i>a Posteriori</i> Estimator	12
2.3 Analytical Density Estimation	13
2.3.1 Optimal Bayesian Filter	13
2.3.2 Kalman Filter	15

2.3.3	Extended Kalman Filter	17
2.3.4	Other Suboptimal Kalman Filters	18
2.4	Monte Carlo Methods	20
2.4.1	Monte Carlo Integration	20
2.4.2	Importance Sampling	22
2.4.3	Sequential Importance Sampling	24
2.4.4	Weight Degeneracy	27
2.4.5	Sampling Importance Resampling	29
2.5	Advanced Particle Filters	31
2.5.1	Generic Particle Filter	32
2.5.2	Regularised Particle Filter	33
2.5.3	Auxiliary Particle Filter	37
2.5.4	Rao-Blackwellised Particle Filter	39
3	Gaussian Processes	45
3.1	Introduction	46
3.2	Definition and Properties	46
3.2.1	Basic Properties	47
3.2.2	Common Kernels	49
3.3	Gaussian Process Regression	52
3.3.1	Simple Kriging	54
3.3.2	Ordinary Kriging	57
3.3.3	Universal Kriging	58
3.4	Reduced-Rank Kernel Approximation	60
3.4.1	Mercer's Theorem	60
3.4.2	Efficient Kriging Estimation	61
3.4.3	Karhunen-Loève Decomposition	63
3.5	Reproducing Kernel Hilbert Spaces	64
3.5.1	Reproducing Property and Definition	64
3.5.2	Link with Gaussian Processes	66
3.5.3	Kernel Mean Embedding	67
3.5.4	Kernel Kalman Filter	67
II	Navigation with Gaussian Processes	69
4	Kriging-Based Navigation	71
4.1	Introduction	72
4.2	Kriging the Observation Model	73
4.2.1	Choosing the Kriging Method	73
4.2.2	Hyperparameter Estimation	78
4.2.3	Simulation Scenario	80
4.2.4	Numerical Results	83
4.3	Adaptive Kriging	88
4.3.1	Random Subsampling	89
4.3.2	Adaptive Subsampling Window	92
4.3.3	Numerical Results	95
4.4	Non-Spherical Subsampling Windows	100
4.4.1	Nearest Neighbour Selection	101

4.4.2	Clustering	102
4.4.3	Clustered Adaptive Kriging	106
4.4.4	Numerical Results	109
4.5	Subsampling Error Bounds	109
4.5.1	Sources of Error	111
4.5.2	Subsampling Error Metrics	115
4.5.3	Importance of the Gram Matrix	116
4.5.4	Block Inversion and Residual Quantities	117
4.5.5	Relations Between the Errors	119
4.5.6	Simple Kriging and Trend Contributions	120
4.5.7	Nested Subsampling	121
4.5.8	Entrywise Bounds	122
5	Simultaneous Localisation and Mapping	125
5.1	Introduction	126
5.2	The RBPF Approach	126
5.2.1	A Finite-Dimensional Map	127
5.2.2	A Conditionally Linear Model	128
5.3	The RKHS Approach	130
5.3.1	Embedding the Filter	130
5.3.2	Random Fourier Features	131
5.3.3	Recursive Least Squares	132
5.3.4	The AKKF-SLAM Algorithm	132
5.3.5	Numerical Results	133
5.4	The Learning Bias	135
6	Conclusion	137
6.1	Limitations	137
6.2	Perspectives	138
III	Appendices	139
A	Useful Matrix Identities	141
A.1	Gaussian Conditioning Formula	142
A.2	Schur Inversion Formula	143
A.3	Woodbury Inversion Formula	144
B	Deferred Proofs	147
B.1	Proof of Proposition 2.3	148
B.2	Proof of Proposition 2.10	149
B.3	Proof of Proposition 3.2	150
B.4	Proof of Proposition 3.3	151
B.5	Proof of Proposition 3.8	152
B.6	Proof of Proposition 3.9	153
B.7	Proof of Proposition 3.10	154
B.8	Proof of Proposition 3.11	156
B.9	Proof of Proposition 3.12	157
B.10	Proof of Proposition 3.14	159

B.11 Proof of Proposition 3.15	160
B.12 Proof of Proposition 3.16	161
B.13 Proof of Proposition 3.18	163
B.14 Proof of Proposition 4.3	163
B.15 Proof of Lemma 4.2	164
B.16 Proof of Proposition 4.7 and Corollary 4.2	165
B.17 Proof of Lemma 4.3	165

Bibliography	174
---------------------	------------



FIGURES

2.1	Schematic representation of the Kalman filter prediction/correction cycle. Each ellipsoid represents a confidence region associated with its corresponding Gaussian density.	16
2.2	An example CDF of the particle weights. The width of each step is equal to the corresponding weight, so heavier particles are more likely to be selected.	30
2.3	Particle impoverishment after resampling. Discarding non-resampled particles progressively shrinks the support of the particle approximation, degrading the filter's ability to explore the state-space.	32
2.4	An example Monte Carlo approximation of the PDF (2.110) on the left and its kernel regularisation (2.111) on the right, for a given particle set. The latter approximation bridges the gaps between particles and spans a larger region of the state space.	34
3.1	Radial profile of Matérn kernels with increasing smoothness, together with their squared exponential limit. Larger values of ν produce smoother kernels, with flatter behaviours near the origin and faster-decaying tails.	51
3.2	Ordinary kriging (red mesh) of an unknown field (blue surface) from noisy samples (black crosses). The confidence band (red envelope) is proportional to the square root of the interpolation covariance, and widens away from the samples.	53
4.1	Simple kriging under a correct mean function, on the left, and under a wrong one, on the right. Since R_{SK} does not depend on m in (3.35), both shaded areas have the same width along the y -axis.	74
4.2	Simultaneous kriging of the training data and mean estimation by ordinary kriging. The constant mean is reconstructed here at $\hat{m}_{\text{OK}} = 10.56$	74
4.3	Universal kriging over an affine trend basis, on the left, and over a degree five monomial basis, on the right. The trend coefficients are estimated here at $\hat{m}_{\text{UK}} = [9.78, 0.15]^\top$ and $\hat{m}_{\text{UK}} = [4.95, 15.85, -10.82, 2.40, -0.20, 0.01]^\top$ respectively.	74

4.4	The magnetic field anomaly used throughout this chapter. The black line represents the overflow trajectory, and the red crosses correspond to the training points on a 7.5 km-resolution grid. The black dot signals the beginning of the trajectory.	81
4.5	Reconstruction ARMSE of the field over the trajectory corridor for ordinary kriging and bilinear interpolation as a function of the training grid resolution. The domain E considered is a rectangular corridor around the true trajectory, with a 28 km margin. It covers 74% of the map.	84
4.6	Estimation RMSE of the navigation algorithms using the known map, bilinear interpolation and static kriging. Each panel corresponds to a different grid resolution, and the true map result is repeated for comparison.	85
4.7	Estimation NEES of the navigation algorithms using the known map, bilinear interpolation and static kriging. The shaded area delimits the quantiles of a χ^2 law with d_x degrees of freedom, within which the NEES of a consistent filter falls with probability 0.99.	86
4.8	Average median RMSE and NEES at the end of the flight using the known map, bilinear interpolation and static kriging as a function of grid resolution.	87
4.9	Percentage of converged runs for the known map, bilinear interpolation and static kriging against the grid resolution for the two metrics. The true map does not depend on the resolution, and has a convergence rate of 98% under both criteria.	87
4.10	Comparative RMSE of the navigation algorithms using static kriging or random subsampling across the six grid resolutions. The performance of the filter with access to the true map is given as a reference.	90
4.11	Comparative RMSE of the navigation algorithms using static kriging or random subsampling across the six grid resolutions. The shaded area corresponds to that of the previous NEES figure.	91
4.12	Convergence rates for the navigation algorithms using static kriging and random subsampling as a function of grid resolution.	92
4.13	RMSE curves for the AK-RPF compared with that of an RPF using static kriging across the six grid resolutions.	97
4.14	NEES curves for the AK-RPF compared with that of an RPF using static kriging across the six grid resolutions.	98
4.15	Convergence rates of the AK-RPF and the RPF using static kriging using the threshold and the Mahalanobis criteria.	99
4.16	Evolution of the average number of training points retained by the AK-RPF during the mission for each grid resolution. A logarithmic scale is used before the first turn at $k = 33$ to capture the large variations in $\tilde{N}_{AK,k}^r$, while a linear scale is used afterwards, where these variations are smaller.	99
4.17		111
4.18		112
4.19		113
4.20		114

TABLES

4.1	An example kriging scenario. The SK and UK estimators are given either mean function and trend bases to illustrate their reliance on prior knowledge, as shown in their respective figures.	73
4.2	ARMSE of the five estimators over the three subsets E_{in} , E_{gap} , and E_{out} of the prediction domain.	76
4.3	Average confidence interval half-width and average share of the covariance caused by the estimation of the mean for the five estimators over each region. The share is computed as the pointwise ratio of $R_{\hat{m}}$ over $R_K = R_{\hat{m}} + R_{\text{SK}}$	76
4.4	Computational cost of the five estimators, split into the setup and the query costs. The simple kriging row is the cost of reconstructing the field, paid by every estimator, and the following ones are what estimating the mean adds to it. Since their cost is independent of the mean they were given, the two simple estimators share the same row.	77
4.5	Simulation parameters for the navigation scenario. For improved readability, covariance matrices are represented by their diagonal coefficients only.	82
4.6	Memory and computational costs of the navigation algorithm using the known map, bilinear interpolation, or static kriging. Since the cost of kriging depends on the number of training data, one row is assigned per grid resolution.	88
4.7	Total computational cost, in flops, of using static kriging or random subsampling throughout the mission, for each grid resolution.	92
4.8		100
4.9		114
5.1	The SLAM scenario. The observation function is scalar and non-injective, so that the likelihood is multimodal at every time step. . . .	134
5.2	Transient and asymptotic trajectory errors of the four filters. The three rightmost columns are particle filters navigating on a known map degraded at level γ	134

ALGORITHMS

2.1	Optimal Bayesian Filter	14
2.2	Kalman Filter	16
2.3	Extended Kalman Filter	18
2.4	Sequential Importance Sampling	26
2.5	Sampling Importance Resampling	29
2.6	Kitagawa Resampling	31
2.7	Generic Particle Filter	33
2.8	Regularised Particle Filter	37
2.9	Auxiliary Particle Filter	39
2.10	Rao-Blackwellised Particle Filter	42
3.1	Adaptive Kernel Kalman Filter	68
4.1	Hyperparameter estimation by evidence maximisation	80
4.2	Adaptive Kriging Regularised Particle Filter	96
4.3	Nearest Neighbour Adaptive Kriging RPF	102
4.4	Independent Clustering Adaptive Kriging RPF	108
4.5	Merged Clustering Adaptive Kriging RPF	110
5.1	Rao-Blackwellised SLAM	129
5.2	AKKF-SLAM	133

STATEMENTS

Definitions

2.1	Symmetric positive definite matrix	15
2.2	Empirical measure	21
2.3	Weighted empirical measure	22
2.4	Self-normalised measure	23
2.5	Regularisation kernel	34
2.6	Regularised measure	34
3.1	Gaussian process	47
3.2	Symmetric positive definite kernel	48
3.3	Radial kernel	49
3.4	Matérn kernel	50
3.5	Self-adjoint positive semi-definite operator	60
3.6	Reproducing kernel Hilbert space	65
3.7	Feature map	66
4.1		115
4.2		115
4.3		116
4.4		117
4.5		117

Propositions

2.1	MMSE estimator	10
2.2	MMAE estimator, simplified assumptions	11
2.3	Gaussian closure under affine assumptions	15
2.4	Monte Carlo estimator	21
2.5	Weighted MC estimator	22
2.6	Self-normalised estimator	23
2.7	Weight degeneracy	27
2.8	Optimal proposal density	28

2.9	Regularised estimator	35
2.10	Noise decorrelation	40
3.1	Finite-dimensional marginals	47
3.2	Definite kernel/matrix equivalence	49
3.3	Properties of Matérn kernels	50
3.4	Separable kernel	51
3.5	Simple kriging estimator	54
3.6	Exact interpolation in the noiseless case	55
3.7	Gaussian conditioning interpretation	56
3.8	Ordinary kriging estimator	57
3.9	Universal kriging estimator	58
3.10	Conditioning under a vague trend prior	60
3.11	Mercer operator	61
3.12	Mercer's theorem	61
3.13	Reduced-rank approximation	61
3.14	Reduced-rank simple kriging	63
3.15	Karhunen-Loève decomposition	63
3.16	Karhunen-Loève converse	64
3.17	Reproducing kernel	65
3.18	Native RKHS	65
3.19	Regularised least-squares in RKHS	66
4.1	Maximum likelihood hyperparameters	78
4.2		116
4.3		116
4.4		118
4.5		118
4.6		118
4.7		118
4.8		119
4.9		119
4.10		120
4.11		120
4.12		121
4.13		123
4.14		123
A.1	Gaussian conditioning [117]	142
A.2	Schur inversion formula [118]	143
A.3	Inversion identity	144
A.4	Push-through identity [94]	144
A.5	Generalised Woodbury formula [119]	145

NOMENCLATURE

Abbreviations

AK	Adaptive Kriging
AKKF	Adaptive Kernel Kalman Filter
APF	Auxiliary Particle Filter
ARMSE	Averaged Root Mean Squared Error
a.s.	almost surely
BLUE	Best Linear Unbiased Estimator
CAK	Clustered Adaptive Kriging
CDF	Cumulative Distribution Function
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
EKF	Extended Kalman Filter
flop	Floating-Point Operations
GP	Gaussian Process
HCA	Hierarchical Clustering Analysis
ICAK	Independently Clustered Adaptive Kriging
IEKF	Iterative Extended Kalman Filter
i.i.d.	independent and identically distributed
KKF	Kernel Kalman Filter
KL	Karhunen-Loève
KME	Kernel Mean Embedding
MC	Monte Carlo
MCAK	Merged Clustered Adaptive Kriging
MAP	Maximum <i>A Posteriori</i>
MISE	Mean Integrated Squared Error
ML	Maximum Likelihood
MMAE	Minimum Mean Absolute Error
MMSE	Minimum Mean Squared Error
NNAK	Nearest-Neighbour Adaptive Kriging
OK	Ordinary Kriging
PDF	Posterior Probability Density Function
PF	Particle Filter
RBPF	Rao-Blackwellised Particle Filter
RFF	Random Fourier Features

RKHS	Reproducing Kernel Hilbert Space
RLS	Recursive Least Squares
RMSE	Root Mean Squared Error
RPF	Regularised Particle Filter
RS	Random Subsampling
SE	Squared Exponential
SIR	Sampling Importance Resampling
SIS	Sequential Importance Sampling
SK	Simple Kriging
SLAM	Simultaneous Localisation and Mapping
SPD	Symmetric Positive Definite
SPSD	Symmetric/Self-adjoint Positive Semi-Definite
UK	Universal Kriging

Symbols

$\triangleq$	Equality by definition
$\propto$	Proportionality
$\sim$	"is distributed according to"
$\overset{i.i.d.}{\sim}$	Independently and identically distributed according to
$\xrightarrow{a.s.}$	Almost sure convergence
$\xrightarrow{\mathcal{L}}$	Convergence in law
$\cdot \succeq 0$	Symmetric positive semi-definite
$\cdot \succ 0$	Symmetric positive definite
$\langle \cdot \cdot \rangle, \langle \cdot \cdot \rangle_{\mathcal{E}}$	Inner product, subscript: of the space $\mathcal{E}$
$\langle \cdot, \cdot \rangle$	Measure/function duality pairing, $\langle \mu, \varphi \rangle = \int \varphi d\mu$
$\ \cdot \ , \ \cdot \ _2, \ \cdot \ _F$	Generic, Euclidean and Frobenius norms
$\cdot^{\top}$	Transpose of a matrix
$\cdot^{-1}$	Inverse of a matrix
$*$	Convolution operator
$\circ$	Function composition operator
$\otimes$	Kronecker product
$\mathbb{1}$	Indicator function
$[\cdot, \cdot]$	Discrete interval
$\cdot_{m n}$	"at time m conditioned on the observations from 1 to n "
$\alpha, \hat{\alpha}_k$	Map coefficients and their estimate
$\alpha_{\text{reg}}, \alpha_{\text{cov}}$	Regularisation and window dilation coefficients
β	Forgetting factor
Γ	Gamma function
δ_x	Dirac measure at x
$\varepsilon, \varepsilon_i$	Query and training noises
θ	Kernel hyperparameters
$\lambda_k^{(i)}$	Auxiliary weights
$\lambda_{\min}, \lambda_{\max}, \lambda_i$	Smallest, largest and i -th eigenvalues of a matrix
$\lambda_x, \lambda_z, \lambda_{\text{RLS}}$	Regularisation coefficients
μ	Target probability measure, of density p
$\mu_k^{(i)}$	Auxiliary particle

μ_X	Kernel mean embedding of a random variable X
ν	Proposal (importance) measure, of density q
π_k	Projection of a trajectory onto its last component
Π_{pos}	Projection matrix onto the position subspace
ρ	Distance on a space of estimators
$\Sigma_{k k-1}, \Sigma_{k k}$	Covariances of the embedded weights
φ	Mapping of the state
$\hat{\psi}_{\text{MMSE}}, \hat{\psi}_{\text{MMAE}}, \hat{\psi}_{\text{MAP}}, \hat{\psi}_{\text{ML}}$	State estimators
ψ_i, Ψ, Λ	Mercer eigenfunctions and eigenvalue matrix
Φ	Feature map of an RKHS
ω_j, b_j	Frequencies and phases of the basis functions
$\text{argmin}, \text{argmax}$	Argument of the minimum, of the maximum
$b_k^{(e)}, b_k^{(o)}$	Evolution and observation bias vectors
C_μ, C_ν	Normalising constants of μ and ν
cdf	Cumulative distribution function
d_x, d_z, d_φ	Dimensions of the state, observation, and mapping
d_w, d_n	Dimensions of the evolution and observation noises
d_{pos}	Dimension of the position subspace
$\det$	Determinant of a matrix
$\mathbb{E}[\cdot], \mathbb{E}_p[\cdot]$	Expectation, subscript: under the law p
$\exp$	Exponential function
$f_k, f_k^l, f_k^n, \bar{f}_k^l$	Evolution functions
$F_k, F_k^l, F_k^n, \bar{F}_k^l$	Evolution matrices
g, g_l	Trend functions
$\mathcal{GP}(m, k)$	Gaussian process
G, G_θ	Gram matrix
h_k	Observation function
H_k	Observation matrix
I, I_n	Identity matrix, subscript: of size n
$\mathbf{1}_n$	Vector of ones of size n
J_f	Jacobian matrix of a function f
$k(\cdot, \cdot), k_{\text{SE}}, k_\nu, k_{\text{sep}}, k_{\text{MS}}, k_B, k_{r_{\text{KL}}}$	Covariance kernels
$K, K_h, K_{\text{Epa}}, K_{\text{Gauss}}$	Regularisation kernels
K_ν	Modified Bessel function of the second kind
K_k	Kalman gain
$\mathcal{K}, \mathcal{K}_{r_{\text{KL}}}$	Mercer operator and its reduced-rank truncation
$\mathcal{L}^0(E, F), \mathcal{L}^p(E, F)$	Sets of measurable and p -integrable functions from E to F
m	Mean function of a Gaussian process
$\hat{m}_{\text{OK}}, \hat{m}_{\text{UK}}$	Estimated ordinary and universal-kriging means
$\mathcal{M}_{m,n}(\mathbb{R})$	Set of real $m \times n$ matrices
n_k	Observation noise
$N, \tilde{N}$	Number of particles, of training points
$\tilde{N}^r, \tilde{N}_{\min}$	Number of retained training points and its floor
$\hat{N}_{\text{eff}}, N_{\text{eff}}$	Estimated and real effective sample size
N_{th}	Resampling threshold
$\mathbb{N}, \mathbb{N}^*$	Sets of natural and strictly positive natural integers
$\mathcal{N}(m, P)$	Gaussian distribution
$o(\cdot)$	Landau symbol for asymptotic negligibility

$\mathcal{O}(\cdot)$	Landau symbol for asymptotic boundedness
p	Target probability density
p_0	Probability density of the initial state
$p_k^{(e)}, p_k^{(o)}$	Probability densities of the evolution and observation noises
$p_{A B}(a b)$	Conditional density of A given B
$P_0, P_{k k-1}, P_{k k}$	State covariance matrices
P_k^α	Covariance of the map coefficients
$\mathbb{P}$	Probability of an event
q	Proposal (importance) density
$Q_k, Q_k^n, Q_k^l, Q_k^{nl}, \overline{Q}_k^l$	Evolution noise covariances
$r_{\text{UK}}, r_{\text{KL}}, r_{\text{RFF}}$	Number of trend functions, KL rank and number of features
$r_{\text{cov},k}, r_{\text{min},k}$	Covariance and minimum-count window radii
rank	Rank of a kernel or operator
$R, R_k, \tilde{R}$	Observation and training noise covariances
$R_{\text{SK}}, R_{\text{OK}}, R_{\text{UK}}, R_{\text{K}}, R_{\hat{m}}$	Kriging error covariances
$\mathbb{R}, \mathbb{R}_+, \mathbb{R}_+^*$	Sets of real, positive real, and strictly positive real numbers
$S^N(\mu), S^N(\mu, \nu), \tilde{S}^N, S_{K_h}^N$	Monte Carlo measures
S_k	Innovation covariance
S_ν, p_ν	Spectral density of k_ν and its normalisation
$\mathcal{S}_d^+, \mathcal{S}_d^{++}$	Sets of SPSD and SPD matrices
supp	Support of a measure or function
tr	Trace of a matrix
$\mathcal{U}$	Input set of a stochastic process
$\mathcal{U}([a, b])$	Uniform distribution
$\mathbb{V}[\cdot], \mathbb{V}_\mu[\cdot]$	Variance, subscript: under the measure μ
$w_k, w_k^l, w_k^n, \overline{w}_k^l$	Evolution noises
$w_k^{(i)}, \tilde{w}_k^{(i)}$	Normalised and unnormalised particle weights
$x_k, x_k^n, x_k^l, \mathbf{x}_{0:k}$	System states and trajectory
$\mathbf{x}, \mathbf{x}', \tilde{\mathbf{x}}, \tilde{\mathbf{x}}_k^r$	Input tuples, training inputs and their retained subset
$x_k^{(i)}$	Particle
$x_{k k-1}, x_{k k}, \hat{x}_k$	Predicted, corrected and estimated state
y_k	Innovation vector
$z_k, \mathbf{z}_{1:k}$	Observation vector and sequence
$z, \tilde{\mathbf{z}}$	Query and training observations
$\hat{z}, \hat{z}_{\text{SK}}, \hat{z}_{\text{OK}}, \hat{z}_{\text{UK}}, \hat{z}_{\text{K}}$	Kriging estimators

CHAPTER 1

INTRODUCTION

1.1 Motivation and Context

In order for an unmanned vehicle to move autonomously from one point to another, the integration of several disciplines is required, ranging from motor control to trajectory planning. This work focuses on autonomous navigation, which corresponds to the estimation of the vehicle's state, including its position, angular orientation and velocity, using only the information available to the vehicle at any given time.

One of the simplest ways to do this is through the use of inertial measurement units or wheel encoders, when applicable, to estimate the odometry of the vehicle over short periods of time. While this solution is acceptable for short missions, the inertial measurements are imperfect due to bias or scale factors. Since inertial navigation integrates these measurements once or twice, the resulting solution drifts over time, and cannot be used for longer missions. In order to keep the inertial solution bounded, data fusion algorithms can use an additional source of information to periodically correct the estimated state of the vehicle.

In open environments, this is usually achieved by using the Global Navigation Satellite System (GNSS). In many situations however, such as underwater, underground or indoors, GNSS data may be unavailable. Moreover, signal interference, jamming or spoofing may prevent reliable GNSS-based navigation algorithms.

In GNSS-denied scenarios, navigation can instead be aided by the physical fields through which the vehicle travels, such as sea floor depth, terrain or anomalies in the ambient gravitational or magnetic field. Given a map of such a field, the values measured by an embedded sensor along the trajectory can be compared with the map to estimate the likely position of the vehicle.

Physical field-aided navigation falls under the scope of Bayesian estimation methods, the most famous of which is the Kalman filter. In practice, however, two main issues prevent the use of this algorithm. First, the observed physical field may be strongly non-linear and non-injective, so that a single observation may remain compatible with several, possibly distant, hypotheses about the vehicle state. This can result in highly multimodal state distributions that cannot be adequately represented by a Kalman filter. Second, the initial state of the vehicle may be poorly known at the beginning of the mission, which may prevent localised approximations

from converging towards the true state.

An alternative to Kalman filtering are particle filters, which use Monte Carlo approximations to represent the probability density of the vehicle state throughout the mission. While these methods are robust to multimodal densities and high initial uncertainty, they suffer from two problems, known as weight degeneracy and particle impoverishment, potentially limiting their ability to operate reliably over long missions. More advanced particle filters, used throughout this work, are thus required.

Although particle filters provide a reliable solution to the navigation problem, they require an embedded map of the environment. When no such map is available, the observation model is itself unknown and must be inferred during the mission. In this work, we examine two scenarios depending on the amount of prior information available about the environment.

The first one arises when the environment is poorly known, and the vehicle only has access to a degraded map consisting of sparse and noisy measurements of the field.

TODO: Ancienne intro + à finir

A diffuse initial belief and a multimodal likelihood together rule out the linear-Gaussian estimators on which classical navigation is built. In the linear-Gaussian case, the conditional state density remains Gaussian and the optimal estimate is delivered in closed form by the Kalman filter; under strong non-linearities, this density is no longer Gaussian, and the Kalman filter and its linearised variants fail to represent it. Particle filters address this regime by approximating the conditional density with a weighted set of samples, at the cost of two further difficulties: the samples must be numerous enough to cover the state space, and the filter is prone to the degeneracy of its weights and to the impoverishment of its particle set over long missions.

Reconstructing the observation function from the available field data, rather than relying on a fixed embedded map, is the approach pursued in this thesis, and Gaussian-process regression – kriging – is the natural tool for the task. This reconstruction, however, carries its own difficulties. Each evaluation of a kriged field is costly, which is at odds with the real-time budget of an embedded filter that must query the field for every particle at every step.

Challenge 1 (Computational cost of kriging). *Kriging-based reconstructions of the environment may be computationally expensive, making its direct use impractical under real-time constraints.*

Kriging also rests, in its standard form, on a stationarity assumption that physical fields seldom satisfy over large areas, so that data gathered far from the vehicle may bias the local reconstruction rather than improve it.

Challenge 2 (Non-stationary environment). *The observed physical field is not necessarily stationary, potentially causing data from distant regions to introduce errors or bias into local reconstructions.*

In the most demanding scenario, finally, no prior data are available at all: the field must be discovered and mapped during the mission itself, jointly with the localisation of the vehicle on it.

Challenge 3 (Unknown environment). *In the most challenging scenarios, no prior data are available from the environment, making kriging impossible.*

1.2 Objective and Contributions

The objective of this thesis is to navigate autonomously on physical fields by reconstructing the observation function with Gaussian processes and by embedding this reconstruction within the Bayesian navigation framework. The Gaussian-process model turns the raw, possibly sparse and degraded field data into a smooth observation function together with a quantification of its own uncertainty, which the particle filter exploits to correct the inertial drift. The two probabilistic languages – that of the state and that of the map – are thereby carried within a single estimator. Around this idea, the contributions of the thesis are the following.

- A particle-filtering scheme in which the observation function is reconstructed by kriging rather than read from a fixed map, allowing navigation on a field known only through sparse and noisy data, and naturally propagating the uncertainty of the reconstruction into the state estimate.
- An adaptive kriging scheme that restores the tractability of the reconstruction under real-time constraints and accommodates the non-stationarity of the field, by kriging only from the data relevant to the current region; the error incurred by this subsampling is bounded theoretically.
- A treatment of the map-free case, in which localisation and mapping are performed jointly – through both a Rao-Blackwellised particle filter and a reproducing-kernel-Hilbert-space reconstruction from a dictionary of basis functions – together with the identification of the learning bias that this joint estimation introduces.

1.3 Outline

The manuscript is organised in two parts. The first, *Mathematical Foundations*, sets up the two pillars of the approach. Chapter 2 recalls the Bayesian formulation of state estimation and the filters that solve it, from the Kalman filter in the linear-Gaussian case to the Monte Carlo and particle filters – including their regularised and Rao-Blackwellised variants – that address the non-linear, multimodal regime of navigation. Chapter 3 develops the Gaussian-process model later used to reconstruct the observation function: kriging and its universal form, the reduced-rank approximations that make it affordable, and the reproducing-kernel-Hilbert-space viewpoint that underpins the map-free case.

The second part, *Navigating on Gaussian Processes*, brings the two pillars together. Chapter 4 embeds the kriged observation function within the particle filter and introduces adaptive kriging, which reconstructs the field locally from a relevant subset of the data, with a theoretical bound on the resulting subsampling error. Chapter 5 addresses the case of an unknown field, where the map is built online jointly with the navigation, through a Rao-Blackwellised and a reproducing-kernel-Hilbert-space approach, and analyses the learning bias inherent to this joint estimation. A concluding chapter summarises the contributions and opens onto the fusion of several physical fields.

1.4 Publications

Part I

Mathematical Foundations

CHAPTER 2

BAYESIAN ESTIMATION

Contents

2.1	Introduction	8
2.2	State Estimation	9
2.2.1	Posterior Expectation Estimator	9
2.2.2	Posterior Median Estimator	11
2.2.3	Maximum <i>a Posteriori</i> Estimator	12
2.3	Analytical Density Estimation	13
2.3.1	Optimal Bayesian Filter	13
2.3.2	Kalman Filter	15
2.3.3	Extended Kalman Filter	17
2.3.4	Other Suboptimal Kalman Filters	18
2.4	Monte Carlo Methods	20
2.4.1	Monte Carlo Integration	20
2.4.2	Importance Sampling	22
2.4.3	Sequential Importance Sampling	24
2.4.4	Weight Degeneracy	27
2.4.5	Sampling Importance Resampling	29
2.5	Advanced Particle Filters	31
2.5.1	Generic Particle Filter	32
2.5.2	Regularised Particle Filter	33
2.5.3	Auxiliary Particle Filter	37
2.5.4	Rao-Blackwellised Particle Filter	39

2.1 Introduction

The goal of this chapter is to formally introduce the tools necessary to solve the general filtering problem. More precisely, given a dynamical system whose state evolves according to a known stochastic model and is partially observed through noisy measurements, we seek to sequentially estimate the system state at each time step in an optimal sense. We first define what optimality means in a rigorous setting, then derive exact and approximate solutions to this problem. We also discuss the limitations of such methods.

Formally, for $d_x, d_z, d_\varphi \in \mathbb{N}^*$, we consider a hidden Markov chain $(x_k)_{k \geq 0}$ taking values in $\mathbb{R}^{d_x}$ and a series of observations $(z_k)_{k \geq 1}$ taking values in $\mathbb{R}^{d_z}$, each associated with the unknown state x_k . The quantity of interest is a mapping of the state under a known function $\varphi : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_\varphi}$, which allows us to target any derived quantity of the state (*e.g.* position, velocity, or a subset of coordinates). When solving a navigation problem, the state and observation processes are linked by the state-space model [1]:

$$\begin{cases} x_k = f_k(x_{k-1}, w_k) \\ z_k = h_k(x_k, n_k) \end{cases} \quad (2.1)$$

where the processes $w_k \in \mathbb{R}^{d_w}$ and $n_k \in \mathbb{R}^{d_n}$ denote the evolution and observation noises respectively, and are assumed to be white noises (*i.e.* temporally independent, centered noises of finite variance), mutually independent and independent of x_0 .

The known data of this problem are:

$$\begin{cases} p_0 & \text{the probability density of the initial state } x_0 \\ f_k & \text{the evolution function} \\ p_k^{(e)} & \text{the probability density of the evolution noise } w_k \\ z_k & \text{the current noisy observation} \\ h_k & \text{the observation function} \\ p_k^{(o)} & \text{the probability density of the observation noise } n_k \end{cases} \quad (2.2)$$

Remark 2.1 (Control input). *In many navigation applications, a control input u_k is also available, and the evolution function takes the form $f_k(x_{k-1}, u_k, w_k)$. For the sake of clarity, we omit u_k from the general formulation as it can always be absorbed into f_k or treated as a known deterministic input.*

Depending on how observations are used relative to the time index of the state to estimate, three problems arise [2]:

- **Filtering:** estimate $\varphi(x_k)$ given $\mathbf{z}_{1:k} \triangleq (z_1, \dots, z_k)$. It is the most common case in real-time navigation.
- **Smoothing:** estimate $\varphi(x_l)$ for $l < k$ given $\mathbf{z}_{1:k}$. It corresponds to refining past estimates using future observations.
- **Prediction:** estimate $\varphi(x_l)$ for $l > k$ given $\mathbf{z}_{1:k}$. This amounts to anticipating future states from current observations.

This thesis focuses on the filtering problem. The rest of this chapter is organised as follows:

- Section 2.2 rigorously defines what it means to optimally estimate $\varphi(x_k)$ from the observations $\mathbf{z}_{1:k}$, and shows that all classical optimality criteria reduce to computing the posterior probability density function (PDF) $p(x_k|\mathbf{z}_{1:k})$.
- Section 2.3 derives analytical filtering algorithms to propagate this density over time, and discusses their limitations.
- Section 2.4 then presents asymptotically optimal numerical methods based on Monte Carlo (MC) sampling that alleviate such flaws.
- Finally, section 2.5 presents advanced particle filtering methods that improve upon standard MC techniques.

2.2 State Estimation

The first issue to tackle is to define precisely the meaning of "best estimator" of $\varphi(x_k)$ given $\mathbf{z}_{1:k}$ [3]. Formally, given a functional metric space $(\mathcal{F}, \rho)$, it amounts to finding a measurable function $\psi \in \mathcal{F}$ that minimises the expected loss:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\rho(\psi(\mathbf{z}_{1:k}), \varphi(x_k))) \quad (2.3)$$

Depending on the considered space and distance, different optimal estimators arise. We present three classical ones below. For the sake of simplicity, we write Z for a generic observation vector and X for the underlying state, as opposed to the time-indexed processes x_k and z_k . At a fixed time step k , (2.3) thus becomes:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\rho(\psi(Z), \varphi(X))) \quad (2.4)$$

For the remainder of this document, we let, for $p \in \mathbb{N}^*$, E a measure space and F a normed vector space:

$$\begin{aligned} \mathcal{L}^0(E, F) &\triangleq \{f : E \rightarrow F \mid f \text{ measurable}\} \\ \mathcal{L}^p(E, F) &\triangleq \left\{ f \in \mathcal{L}^0(E, F) \mid \int_E \|f\|^p < \infty \right\} \end{aligned} \quad (2.5)$$

where almost everywhere equal functions with respect to the Lebesgue measure on E are identified.

2.2.1 Posterior Expectation Estimator

Let:

$$\mathcal{F} \triangleq \{\psi \in \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi}) \mid \mathbb{E}[\|\psi(Z)\|^2] < \infty\} \quad (2.6)$$

To simplify notations, we identify each estimator $\psi \in \mathcal{F}$ with the random variable $\psi(Z)$ it produces. Under this identification, $\mathcal{F}$ is equal to $\mathcal{L}^2(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi})$, and can be equipped with the inner product:

$$(A, B) \in \mathcal{F}^2 \mapsto \langle A|B \rangle \triangleq \mathbb{E}[A(Z)^\top B(Z)] \quad (2.7)$$

where $\mathbb{E}[X]$ denotes the expectation of the random variable X .

Let $\|\cdot\|$ be the norm associated with $\langle \cdot | \cdot \rangle$. When $\varphi(X)$ is square-integrable, that is $\mathbb{E}[\|\varphi(X)\|^2] < \infty$, the minimisation problem (2.4) is equivalent to:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\mathbb{E} [\|\psi(Z) - \varphi(X)\|^2]) \quad (2.8)$$

An estimator solving (2.8) is called a minimum mean squared error (MMSE) estimator.

Proposition 2.1 (MMSE estimator). *The unique solution to (2.8) is the posterior expectation:*

$$\hat{\psi}_{\text{MMSE}}(z) \triangleq \mathbb{E}[\varphi(X)|Z = z] \quad (2.9)$$

Proof. Let $\hat{\psi}(z) \triangleq \mathbb{E}[\varphi(X)|Z = z]$ and $\psi \in \mathcal{F}$. It comes that:

$$\begin{aligned} \rho(\psi(Z), \varphi(X))^2 &= \langle \psi(Z) - \varphi(X) | \psi(Z) - \varphi(X) \rangle \\ &= \langle \psi(Z) - \hat{\psi}(Z) + \hat{\psi}(Z) - \varphi(X) | \psi(Z) - \hat{\psi}(Z) + \hat{\psi}(Z) - \varphi(X) \rangle \\ &= \langle \psi(Z) - \hat{\psi}(Z) | \psi(Z) - \hat{\psi}(Z) \rangle + \langle \hat{\psi}(Z) - \varphi(X) | \hat{\psi}(Z) - \varphi(X) \rangle \\ &\quad + 2\langle \psi(Z) - \hat{\psi}(Z) | \hat{\psi}(Z) - \varphi(X) \rangle \end{aligned} \quad (2.10)$$

The cross term vanishes by the tower property of conditional expectation:

$$\begin{aligned} \langle \psi(Z) - \hat{\psi}(Z) | \hat{\psi}(Z) - \varphi(X) \rangle &= \mathbb{E} \left[(\psi(Z) - \hat{\psi}(Z))^\top (\hat{\psi}(Z) - \varphi(X)) \right] \\ &= \iint (\psi(z) - \hat{\psi}(z))^\top (\hat{\psi}(z) - \varphi(x)) p(dx, dz) \\ &= \iint (\psi(z) - \hat{\psi}(z))^\top (\hat{\psi}(z) - \varphi(x)) p(dx|z) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \left(\int (\hat{\psi}(z) - \varphi(x)) p(dx|z) \right) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top (\hat{\psi}(z) - \mathbb{E}[\varphi(X)|Z = z]) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \times 0 p(dz) \\ &= 0 \end{aligned} \quad (2.11)$$

by definition of $\hat{\psi}(z)$.

Therefore:

$$\begin{aligned} \rho(\psi(Z), \varphi(X))^2 &= \rho(\psi(Z), \hat{\psi}(Z))^2 + \rho(\hat{\psi}(Z), \varphi(X))^2 \\ &\geq \rho(\hat{\psi}(Z), \varphi(X))^2 \end{aligned} \quad (2.12)$$

with equality if and only if $\|\psi(Z) - \hat{\psi}(Z)\|^2 = 0$, i.e. $\psi = \hat{\psi}$ almost surely (a.s.). $\square$

Remark 2.2 (Geometric interpretation). *The MMSE estimator is the orthogonal projection of $\varphi(X)$ onto $\mathcal{F}$, viewed as the set of square-integrable $\mathbb{R}^{d_\varphi}$ -valued random variables that are measurable with respect to Z . The disappearance of the cross term in the proof above is precisely the orthogonality condition characterising this projection.*

Remark 2.3 (MMSE assumptions). *Square-integrability of $\varphi(X)$ is only required for the minimisation (2.8) to be well-posed, not the estimator itself. The posterior expectation $\mathbb{E}[\varphi(X)|Z = z]$ remains well-defined as soon as $\varphi(X)$ is integrable, that is $\mathbb{E}[|\varphi(X)|] < \infty$, but it only minimises the mean squared error when $\varphi(X)$ is square-integrable.*

2.2.2 Posterior Median Estimator

Because the MMSE estimator minimises a quadratic loss, it is highly sensitive to extreme values and may perform poorly for heavy-tailed distributions. To mitigate this sensitivity to outliers, one can use an absolute loss instead. In dimension $d_\varphi = 1$, let:

$$\mathcal{F} \triangleq \{\psi \in \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}) \mid \mathbb{E}[|\psi(Z)|] < \infty\} \quad (2.13)$$

As previously, $\mathcal{F}$ is identified as a subspace of $\mathcal{L}^1(\mathbb{R})$, and is equipped with the following non-Euclidean distance:

$$(A, B) \in \mathcal{F}^2 \mapsto \rho(A, B) \triangleq \mathbb{E}[|A(Z) - B(Z)|] \quad (2.14)$$

The minimisation problem (2.4) thus becomes:

$$\underset{\psi \in \mathcal{F}}{\operatorname{argmin}} (\mathbb{E}[|\psi(Z) - \varphi(X)|]) \quad (2.15)$$

While the terminology is less standard than MMSE, an estimator solving (2.15) is often called a minimum mean absolute error (MMAE) estimator.

Proposition 2.2 (MMAE estimator, simplified assumptions). *Assume that, for every z , the conditional distribution of $\varphi(X)$ given $Z = z$ is absolutely continuous with a strictly positive density. Then, the unique solution to (2.15) is the posterior median of $\varphi(X)$ given $Z = z$, i.e. the value $\hat{\psi}_{\text{MMAE}}(z) \in \mathbb{R}$ that satisfies:*

$$\mathbb{P}(\varphi(X) \geq \hat{\psi}_{\text{MMAE}}(z) | Z = z) = \mathbb{P}(\varphi(X) < \hat{\psi}_{\text{MMAE}}(z) | Z = z) = \frac{1}{2} \quad (2.16)$$

Proof. By the law of total expectation, for any $\psi \in \mathcal{F}$, we have:

$$\mathbb{E}[|\psi(Z) - \varphi(X)|] = \mathbb{E} \left[\int |\psi(z) - \varphi(x)| p(dx | Z = z) \right] \quad (2.17)$$

Since the outer expectation is over non-negative quantities, this is minimised by minimising the inner integral pointwise for each z , i.e. by solving:

$$\underset{u \in \mathbb{R}}{\operatorname{argmin}} \int |\varphi(x) - u| p(dx | Z = z) \quad (2.18)$$

As $u \mapsto |\varphi(x) - u|$ is convex, so is the integral, hence any stationary point is a global minimum. Since the conditional law is absolutely continuous, the integral is differentiable in u . By differentiating this expression with respect to $u \in \mathbb{R}$, it

follows that:

$$\begin{aligned}
& \frac{d}{du} \left(\int |\varphi(x) - u| p(dx|Z = z) \right) = 0 \\
\iff & \frac{d}{du} \left(\int ((\varphi(x) - u) \mathbb{1}_{\varphi(x) \geq u}(x) - (\varphi(x) - u) \mathbb{1}_{\varphi(x) < u}(x)) p(dx|Z = z) \right) = 0 \\
\iff & \int (-\mathbb{1}_{\varphi(x) \geq u}(x) + \mathbb{1}_{\varphi(x) < u}(x)) p(dx|Z = z) = 0 \\
\iff & \mathbb{P}(\varphi(X) \geq u|Z = z) = \mathbb{P}(\varphi(X) < u|Z = z)
\end{aligned} \tag{2.19}$$

Since the total probability is one, this is equivalent to $u = \hat{\psi}_{\text{MMAE}}(z)$ as defined in (2.16), with the strict positivity of the density ensuring uniqueness. $\square$

Remark 2.4 (Median assumptions). *The absolute continuity and positivity assumptions are only required for the clean statement above, not for the existence of a solution. Since the inner integral is convex, continuous and coercive in u , a minimum always exists for any integrable conditional distribution. In full generality, the set of solutions is the closed interval of medians:*

$$\left\{ u \in \mathbb{R} \mid \mathbb{P}(\varphi(X) \leq u|Z = z) \geq \frac{1}{2} \text{ and } \mathbb{P}(\varphi(X) \geq u|Z = z) \geq \frac{1}{2} \right\} \tag{2.20}$$

The absence of atoms then guarantees that this interval is characterised by the equality (2.16), while the strict positivity of the density ensures that it reduces to a single point.

Remark 2.5 (Generalised median). *The extension to higher dimensions $d_\varphi > 1$ is non-trivial. The geometric median, which minimises the mean absolute error, is the most common generalisation. In the remainder of this thesis, we focus on the MMSE and MAP estimators (presented below), which extend naturally to any dimension.*

2.2.3 Maximum *a Posteriori* Estimator

As they produce a single point estimate that may lie in a low-probability region, both the MMSE and the MMAE estimators may not be suited for multimodal posteriors. In such cases, a natural alternative is to select a maximum of the posterior density.

Let $\mathcal{F} \triangleq \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi})$, with no integrability constraint. The maximum *a posteriori* (MAP) estimator is defined as the global mode of the posterior density of $\varphi(X)$:

$$\hat{\psi}_{\text{MAP}} \triangleq \operatorname{argmax}_{\psi \in \mathcal{F}} (\mathbb{E} [p_{\varphi(X)|Z}(\psi(Z)|Z)]) \tag{2.21}$$

As the objective is separable over z , the maximisation can be carried out point-wise, yielding for each z :

$$\hat{\psi}_{\text{MAP}}(z) = \operatorname{argmax}_{u \in \mathbb{R}^{d_\varphi}} (p_{\varphi(X)|Z}(u|z)) \tag{2.22}$$

Using Bayes' theorem, this is equivalent to:

$$\hat{\psi}_{\text{MAP}}(z) = \operatorname{argmax}_{u \in \mathbb{R}^{d_\varphi}} (p_{Z|\varphi(X)}(z|u) p_{\varphi(X)}(u)) \tag{2.23}$$

since the normalising constant $p_Z(z)$ is independent of u .

Although the MAP estimator avoids low-density regions, it still reduces the posterior to a single point. In particular, in multimodal situations, only one mode is retained, while the remaining ones are discarded.

Remark 2.6 (MAP optimisation problem). *The MAP estimator can be seen as the solution of an optimisation problem of the form (2.4), associated with the threshold loss:*

$$\forall \varepsilon > 0, (A, B) \in \mathcal{F}^2 \mapsto \rho_\varepsilon(A, B) \triangleq \mathbb{1}_{\|A(Z) - B(Z)\| > \varepsilon} \quad (2.24)$$

in the limit $\varepsilon \rightarrow 0^+$:

$$\hat{\psi}_{\text{MAP}} = \lim_{\varepsilon \rightarrow 0^+} \operatorname{argmin}_{\psi \in \mathcal{F}} (\rho_\varepsilon(\psi(Z), \varphi(X))) \quad (2.25)$$

Maximum Likelihood Estimator

In the special case where no prior distribution on $\varphi(X)$ is assumed, (2.23) reduces to the maximum likelihood (ML) estimator:

$$\hat{\psi}_{\text{ML}}(z) \triangleq \operatorname{argmax}_{u \in \mathbb{R}^{d_\varphi}} (p_{Z|\varphi(X)}(z|u)) \quad (2.26)$$

As it does not take any prior information on $\varphi(X)$ into account, this estimator is not a Bayesian estimator and may be poorly suited when the available data is scarce. While it coincides with the MAP estimator under a uniform prior, it will not be considered in the rest of this document.

2.3 Analytical Density Estimation

As established in section 2.2, all three estimators ultimately rely on the PDF $p(x_k | \mathbf{z}_{1:k})$, from which an estimate of $\varphi(x_k)$ can be derived. Point estimators provide a concise and directly interpretable representation of the state, which is often desirable for our applications. However, reducing the posterior density to a single point estimate may discard valuable information, particularly when it is heavy-tailed or multimodal.

The goal of the following section is thus to present algorithms that propagate the PDF itself at each discrete time step k in order to solve the inference problem presented by (2.1).

2.3.1 Optimal Bayesian Filter

In the Bayesian filtering framework, the PDF can be propagated recursively through two steps. However, each step relies on a specific conditional independence hypotheses as presented by Assumptions 2.1 and 2.2.

Assumption 2.1 (Markov property). *The state sequence $(x_k)_{k \geq 0}$ is a Markov chain. As such, it verifies:*

$$p(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}) = p(x_k | x_{k-1}) \quad (2.27)$$

with $\mathbf{x}_{0:k} \triangleq (x_0, \dots, x_k)$.

Assumption 2.2 (Conditional observation independence). *The current observation z_k only depends on the current state x_k :*

$$p(z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) = p(z_k | x_k) \quad (2.28)$$

Both assumptions follow from the state-space model (2.1) and, in particular, from the whiteness of w_k and n_k and their independence from x_0 . Building on them, the optimal estimation algorithm computes the PDF in two steps: prediction and correction.

Prediction. The prior density at time k can be obtained by marginalising the previous PDF:

$$\begin{aligned} p(x_k | \mathbf{z}_{1:k-1}) &= \int p(x_k, x_{k-1} | \mathbf{z}_{1:k-1}) dx_{k-1} \\ &= \int p(x_k | x_{k-1}, \mathbf{z}_{1:k-1}) p(x_{k-1} | \mathbf{z}_{1:k-1}) dx_{k-1} \end{aligned} \quad (2.29)$$

Assumption 2.1 simplifies (2.29) into the Chapman-Kolmogorov equation [4]:

$$p(x_k | \mathbf{z}_{1:k-1}) = \int p(x_k | x_{k-1}) p(x_{k-1} | \mathbf{z}_{1:k-1}) dx_{k-1} \quad (2.30)$$

where $p(x_k | x_{k-1})$ is the evolution, or transition, density induced by the state equation $x_k = f_k(x_{k-1}, w_k)$ and the density $p_k^{(e)}$ of the evolution noise w_k .

Correction. Upon receiving the new observation z_k , Bayes' theorem gives:

$$p(x_k | \mathbf{z}_{1:k}) = \frac{p(z_k | x_k, \mathbf{z}_{1:k-1}) p(x_k | \mathbf{z}_{1:k-1})}{p(z_k | \mathbf{z}_{1:k-1})} \quad (2.31)$$

Applying Assumption 2.2 to (2.31) yields:

$$p(x_k | \mathbf{z}_{1:k}) = \frac{p(z_k | x_k) p(x_k | \mathbf{z}_{1:k-1})}{\int p(z_k | x_k) p(x_k | \mathbf{z}_{1:k-1}) dx_k} \quad (2.32)$$

where the likelihood $p(z_k | x_k)$ depends on the observation model $z_k = h_k(x_k, n_k)$ and the density $p_k^{(o)}$ of the observation noise n_k .

These two equations allow for a recursive algorithm to compute the PDF of the system at each time step. This algorithm, usually referred to as the optimal Bayesian filter [5], is given by Algorithm 2.1.

Algorithm 2.1 Optimal Bayesian Filter

- | | |
|--|--------------|
| Compute $p(x_k \mathbf{z}_{1:k-1})$ using (2.30) | ▷ Prediction |
| Compute $p(x_k \mathbf{z}_{1:k})$ using (2.32) | ▷ Correction |
-

While theoretically optimal, this algorithm has no closed-form solution in general. The integrands in both (2.30) and (2.32) are analytically intractable for arbitrary f_k , h_k .

2.3.2 Kalman Filter

While the optimal estimation algorithm cannot be used in the form given by Algorithm 2.1, a closed-form solution can be obtained by introducing additional hypotheses on (2.1), known as the Kalman assumptions [6]. In this section, we assume that:

$$\begin{cases} x_0 & \sim \mathcal{N}(\bar{x}_0, P_0) \\ f_k & : x_{k-1} \mapsto F_k x_{k-1} + b_k^{(e)} \\ w_k & \sim \mathcal{N}(0, Q_k), \text{ independent of } x_0 \\ z_k & \text{ is known} \\ h_k & : x_k \mapsto H_k x_k + b_k^{(o)} \\ n_k & \sim \mathcal{N}(0, R_k), \text{ independent of } x_0 \text{ and } w_k \end{cases} \quad (2.33)$$

where $F_k \in \mathcal{M}_{d_x, d_x}(\mathbb{R})$ and $H_k \in \mathcal{M}_{d_z, d_x}(\mathbb{R})$ are known matrices, and $b_k^{(e)} \in \mathbb{R}^{d_x}$ and $b_k^{(o)} \in \mathbb{R}^{d_z}$ are known bias vectors. $P_0 \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$ is the initial state covariance, and $Q_k \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$ and $R_k \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ are the evolution and observation noise covariance matrices. Here, $\mathcal{S}^{++}$ denotes the set of symmetric positive definite (SPD) matrices [7], as introduced by Definition 2.1.

Definition 2.1 (Symmetric positive definite matrix). *Let $n \in \mathbb{N}^*$ and $M \in \mathcal{M}_{n,n}(\mathbb{R})$. M is symmetric positive semi-definite (SPSD), denoted $M \succeq 0$, if it is:*

$$\begin{cases} M^\top = M & \text{symmetric} \\ \forall a \in \mathbb{R}^n, a^\top M a \geq 0 & \text{positive semi-definite} \end{cases} \quad (2.34)$$

If the inequality is strict for every non-zero vector, then M is SPD. We write $\mathcal{S}_n^+(\mathbb{R})$ and $\mathcal{S}_n^{++}(\mathbb{R})$ respectively for the sets of PSD and SPD matrices of size n .

In particular, the state-space model (2.1) becomes:

$$\begin{cases} x_k = F_k x_{k-1} + b_k^{(e)} + w_k \\ z_k = H_k x_k + b_k^{(o)} + n_k \end{cases} \quad (2.35)$$

Finally, $x_0, (w_k)_{k \geq 0}$ and $(n_k)_{k \geq 0}$ are assumed to be mutually independent. Under these assumptions, the true state x_k follows a Gaussian distribution at each time step as an affine transformation of Gaussian random variables.

Proposition 2.3 (Gaussian closure under affine assumptions). *For $m, n \in \mathbb{N}$, let the index $m|n$ denote "at time m conditioned on the observations from 1 to n ". Under the hypotheses of (2.33), if $p(x_{k-1} | \mathbf{z}_{1:k-1}) = \mathcal{N}(\hat{x}_{k-1|k-1}, P_{k-1|k-1})$, then:*

$$\begin{cases} p(x_k | \mathbf{z}_{1:k-1}) = \mathcal{N}(\hat{x}_{k|k-1}, P_{k|k-1}) \\ p(x_k | \mathbf{z}_{1:k}) = \mathcal{N}(\hat{x}_{k|k}, P_{k|k}) \end{cases} \quad (2.36)$$

where the predicted and corrected states $\hat{x}_{k|k-1}$ and $\hat{x}_{k|k}$, and covariances $P_{k|k-1} \succ 0$ and $P_{k|k} \succ 0$ are given by Algorithm 2.2.

Proof. See section B.1. □

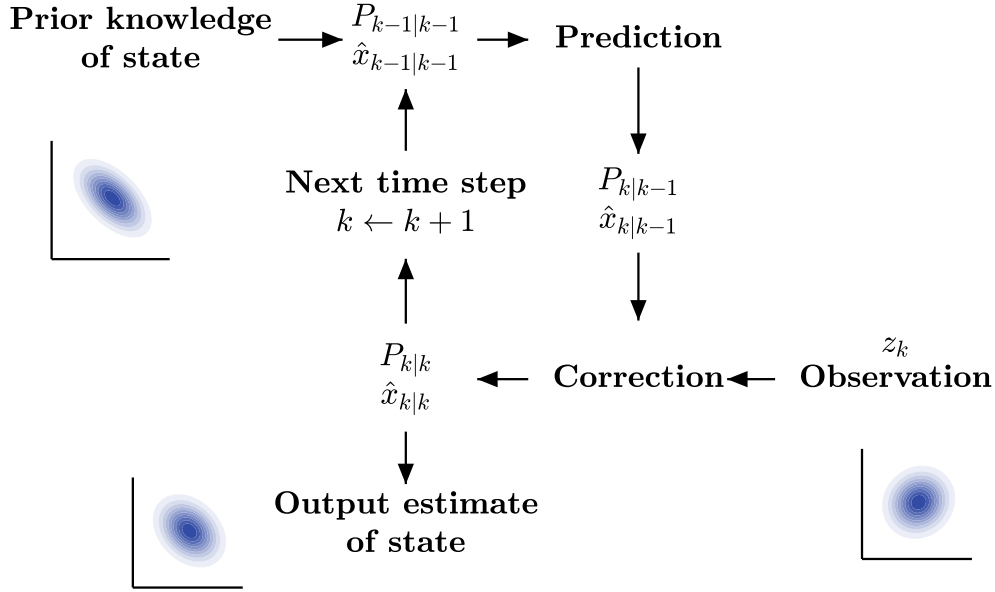


Figure 2.1: Schematic representation of the Kalman filter prediction/correction cycle. Each ellipsoid represents a confidence region associated with its corresponding Gaussian density.

Algorithm 2.2 Kalman Filter

$\hat{x}_{k k-1} = F_k \hat{x}_{k-1 k-1} + b_k^{(e)}$	▷ State prediction
$P_{k k-1} = F_k P_{k-1 k-1} F_k^\top + Q_k$	▷ Covariance prediction
$y_k = z_k - \left(H_k \hat{x}_{k k-1} + b_k^{(o)} \right)$	▷ Innovation
$S_k = H_k P_{k k-1} H_k^\top + R_k$	▷ Innovation covariance
$K_k = P_{k k-1} H_k^\top S_k^{-1}$	▷ Kalman gain
$\hat{x}_{k k} = \hat{x}_{k k-1} + K_k y_k$	▷ State correction
$P_{k k} = (I - K_k H_k) P_{k k-1}$	▷ Covariance correction

The resulting algorithm, known as the Kalman filter [8], is depicted by Figure 2.1 and presented in Algorithm 2.2.

Remark 2.7 (Kalman state estimation). *Under the Kalman assumptions, the PDF is Gaussian, and so every Bayesian estimator presented in section 2.2 coincides with $\hat{x}_{k|k}$:*

$$\hat{\psi}_{\text{MMSE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\text{MAE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\text{MAP}}(\mathbf{z}_{1:k}) = \hat{x}_{k|k} \quad (2.37)$$

However, since the above equations use a predictive prior, the ML estimator does not generally correspond to the corrected state.

Remark 2.8 (Precomputation of gains). *Under the assumptions of (2.33), the matrices $P_{k|k-1}$, K_k and $P_{k|k}$ depend only on the system matrices F_k , H_k , Q_k , R_k and the initial covariance P_0 , but not on the observations z_k . They can therefore be computed before any measurement is received.*

Joseph's Formula

In finite-precision arithmetic, the standard update $P_{k|k} = (I - K_k H_k) P_{k|k-1}$ may lose symmetry or positive definiteness due to rounding errors. Joseph's formula [9, 10] gives an alternative way to compute $P_{k|k}$ while ensuring that this property remains numerically satisfied:

$$\begin{aligned}
P_{k|k} &= (I - K_k H_k) P_{k|k-1} \\
&= (I - K_k H_k) P_{k|k-1} - K_k S_k K_k^\top + K_k S_k K_k^\top \\
&= (I - K_k H_k) P_{k|k-1} - (P_{k|k-1} H_k^\top S_k^{-1}) S_k K_k^\top + K_k (H_k P_{k|k-1} H_k^\top + R_k) K_k^\top \\
&= (I - K_k H_k) P_{k|k-1} - P_{k|k-1} H_k^\top K_k^\top + K_k H_k P_{k|k-1} H_k^\top K_k^\top + K_k R_k K_k^\top \\
&= (I - K_k H_k) P_{k|k-1} - (I - K_k H_k) P_{k|k-1} H_k^\top K_k^\top + K_k R_k K_k^\top \\
&= (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^\top + K_k R_k K_k^\top
\end{aligned} \tag{2.38}$$

Under this more robust form, $P_{k|k}$ is SPSD as the sum of two SPSD matrices.

2.3.3 Extended Kalman Filter

The hypothesis that the evolution and observation functions are affine in x_k is often too restrictive in practice. The extended Kalman filter (EKF) [11] relaxes this assumption by locally linearising f_k and h_k around the current state estimate via a first-order Taylor expansion.

Let $m, n \in \mathbb{N}^*$, let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at some $a \in \mathbb{R}^n$. f admits the following first-order expansion:

$$\forall x \in \mathbb{R}^n, f(x) = f(a) + J_f(a) (x - a) + o(\|x - a\|_2) \tag{2.39}$$

where $J_f(a) \in \mathcal{M}_{m,n}(\mathbb{R})$ is the Jacobian matrix of f evaluated at a , and defined as:

$$J_f \triangleq \frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f^{(1)}}{\partial x^{(1)}} & \cdots & \cdots & \frac{\partial f^{(1)}}{\partial x^{(n)}} \\ \vdots & \ddots & \ddots & \vdots \\ \frac{\partial f^{(m)}}{\partial x^{(1)}} & \cdots & \cdots & \frac{\partial f^{(m)}}{\partial x^{(n)}} \end{bmatrix} \tag{2.40}$$

and $f^{(i)}$ and $x^{(j)}$ respectively denote the i -th and j -th component of f and x .

Assuming that f_k and h_k are differentiable along the state trajectory, we can apply the first-order expansion (2.39) to the evolution and observation model. Successive linearisations at each time step can then be used to apply the Kalman filter equations locally. The resulting algorithm is given by Algorithm 2.3.

While less restrictive than the standard Kalman filter, the EKF is suboptimal for three reasons:

- First, f_k and h_k must be differentiable along the state trajectory, which excludes discontinuous or piecewise-defined models.
- Second, the linearisation in (2.39) neglects higher-order terms of the Taylor expansion, which introduces an approximation error that grows with the degree of nonlinearity of f_k and h_k . This error increases over time as it propagates through the equations of the Kalman filter due to the sequential formulation.

Algorithm 2.3 Extended Kalman Filter

$\tilde{F}_k = J_{f_k}(\hat{x}_{k-1 k-1})$	▷ Evolution Jacobian
$\hat{x}_{k k-1} = f_k(\hat{x}_{k-1 k-1}) + b_k^{(e)}$	▷ State prediction
$P_{k k-1} = \tilde{F}_k P_{k-1 k-1} \tilde{F}_k^\top + Q_k$	▷ Covariance prediction
$\tilde{H}_k = J_{h_k}(\hat{x}_{k k-1})$	▷ Observation Jacobian
$y_k = z_k - \left(h_k(\hat{x}_{k k-1}) + b_k^{(o)} \right)$	▷ Innovation
$S_k = \tilde{H}_k P_{k k-1} \tilde{H}_k^\top + R_k$	▷ Innovation covariance
$K_k = P_{k k-1} \tilde{H}_k^\top S_k^{-1}$	▷ Kalman gain
$\hat{x}_{k k} = \hat{x}_{k k-1} + K_k y_k$	▷ State correction
$P_{k k} = (I - K_k \tilde{H}_k) P_{k k-1}$	▷ Covariance correction

- Finally, it relies on Gaussian assumptions and thus fails to handle multimodal or heavy-tailed distributions, as does its affine counterpart.

2.3.4 Other Suboptimal Kalman Filters

Iterative and Higher-Order EKF

The iterative EKF (IEKF) [12] improves upon the EKF correction step by iteratively re-linearising h_k around successive state estimates within the same measurement update, while keeping the prediction step unchanged. Starting from the initialisation $\hat{x}_{k|k}^{(0)} = \hat{x}_{k|k-1}$, each iteration produces a new $\hat{x}_{k|k}^{(i)}$ by repeating the correction step from the predicted state, with the Jacobian evaluated at the latest estimate. This is done until:

$$\left\| \hat{x}_{k|k}^{(i)} - \hat{x}_{k|k}^{(i-1)} \right\|_2 \leq \varepsilon \quad (2.41)$$

for some $\varepsilon > 0$, or a maximum number of iterations is reached. The corrected state and covariance matrix are then defined as the result of the last loop.

This procedure is equivalent to performing a Gauss-Newton descent [13] on the posterior density maximisation problem, and converges to the MAP estimator under Gaussian assumptions. It reduces the linearisation error of the correction step when h_k is strongly non-linear, at the cost of several Jacobian evaluations per time step.

The second-order EKF (SO-EKF) [14] can also be used to extend the linearisation (2.39) to second-order terms via the Hessians of f_k and h_k , which reduces the approximation error for highly nonlinear models. However, it requires computing and storing the Hessian matrices of f_k and h_k at each time step, making it computationally prohibitive in high dimensions.

Unscented Kalman Filter

Rather than linearising the model, the unscented Kalman filter (UKF) [15] propagates a small deterministic set of points through the non-linear functions themselves. Let $d \in \mathbb{N}^*$, and let X be a random vector of $\mathbb{R}^d$ with mean μ and covariance $P \succ 0$.

The $2d + 1$ sigma points associated with X , and their weights, are:

$$\begin{cases} \mathcal{X}^{(0)} \triangleq \mu, & W^{(0)} \triangleq \frac{\lambda}{d + \lambda} \\ \mathcal{X}^{(\pm i)} \triangleq \mu \pm \sqrt{d + \lambda} (A)_i, & W^{(\pm i)} \triangleq \frac{1}{2(d + \lambda)}, i \in \llbracket 1, d \rrbracket \end{cases} \quad (2.42)$$

where A is a Cholesky factor [16] of P , $(A)_i$ is its i -th column, and $\lambda > -d$ is a scaling parameter. In particular, the weights are independent of X and add up to 1.

For a function φ , the first and second moments of $\varphi(X)$ can then be approximated by the weighted moments of the images of the sigma points:

$$\begin{cases} \hat{\varphi} \triangleq \sum_{i=-d}^d W^{(i)} \varphi(\mathcal{X}^{(i)}) \\ \hat{P} \triangleq \sum_{i=-d}^d W^{(i)} (\varphi(\mathcal{X}^{(i)}) - \hat{\varphi}) (\varphi(\mathcal{X}^{(i)}) - \hat{\varphi})^\top \end{cases} \quad (2.43)$$

This procedure is known as the unscented transform [17], and is applied twice per time step by the UKF. It is first used with $\mu = \hat{x}_{k-1|k-1}$, $P = P_{k-1|k-1}$ and $\varphi = f_k$ to compute the predicted state, then with $\mu = \hat{x}_{k|k-1}$, $P = P_{k|k-1}$ and $\varphi = h_k$ to obtain the predicted observation and thus the innovation. In both cases, Q_k and R_k are added to the transformed covariance $\hat{P}$, as in Algorithm 2.3, to account for evolution and observation uncertainty.

This removes the linearisation shortcomings of the EKF, at the cost of $2d_x + 1$ evaluations of f_k and h_k per time step. A square-root formulation [18] propagates a factor of the covariance rather than the covariance itself, which preserves its symmetry and positive definiteness throughout the recursion. However, the UKF still relies on the Gaussian assumption, the posterior remaining summarised by a mean and a covariance matrix.

Ensemble Kalman Filter

The ensemble Kalman filter (EnKF) [19] avoids both the local linearisation of the EKF and the deterministic sigma-point approximation of the UKF. Instead, the state distribution is represented by a set of $N \in \mathbb{N}^*$ samples $(x_{k|k}^{(1)}, \dots, x_{k|k}^{(N)})$, which are propagated through the true evolution and observation functions.

For two sets $\mathbf{X} \triangleq (X^{(1)}, \dots, X^{(N)})$ and $\mathbf{Z} \triangleq (Z^{(1)}, \dots, Z^{(N)})$, we let:

$$\begin{cases} \hat{X} \triangleq \frac{1}{N} \sum_{i=1}^N X^{(i)}, & \hat{Z} \triangleq \frac{1}{N} \sum_{i=1}^N Z^{(i)} \\ \hat{P}_{\mathbf{XZ}} \triangleq \frac{1}{N-1} \sum_{i=1}^N (X^{(i)} - \hat{X}) (Z^{(i)} - \hat{Z})^\top \end{cases} \quad (2.44)$$

Each predicted sample $x_{k|k-1}^{(i)}$ is obtained by propagating the corresponding sample $x_{k-1|k-1}^{(i)}$ through f_k . The predicted observation $z_k^{(i)}$ is then computed using h_k .

The approximate Kalman gain is defined as:

$$\hat{K}_k \triangleq \hat{P}_{\mathbf{X}_{k|k-1}\mathbf{Z}_k} \left(\hat{P}_{\mathbf{Z}_k\mathbf{Z}_k} + R_k \right)^{-1} \quad (2.45)$$

where $\mathbf{X}_{k|k-1}$ is the predicted sample set, and $\mathbf{Z}_k$ is the predicted observation set. Finally, each corrected sample $x_{k|k}^{(i)}$ is computed from its associated predicted sample through a standard Kalman update, using $\hat{K}_k$ in place of the exact gain.

The computational cost of the EnKF is primarily determined by the value of N instead of the dimensions d_x and d_z . It is therefore particularly attractive for high-dimensional systems, such as those encountered in meteorological and oceanographic data assimilation [20].

It should be noted that, while the EnKF is more flexible than its counterparts, the correction step remains a Kalman update based on the empirical mean and covariance matrix of the sample set. As a result, it may provide a poor approximation when the PDF is strongly non-Gaussian.

2.4 Monte Carlo Methods

While Kalman-based extensions can alleviate some of the limitations of the original Kalman filter, they all enforce the PDF to be Gaussian at each time step. This assumption is often violated in practice, as navigation problems frequently involve highly non-linear and non-injective functions. In turn, these functions induce multimodal or heavy-tailed distributions, which cannot be adequately characterised by Gaussian hypotheses.

Monte Carlo (MC) methods [21, 22] offer an alternative approach that remains consistent under very mild assumptions. Their accuracy improves asymptotically with the number of samples, at the expense of an increased computational cost. This section introduces the main concepts for MC-based algorithms to solve the navigation problem (2.1).

2.4.1 Monte Carlo Integration

Even though MC methods originally stem from the computation of expectations, they approximate the entire distribution [23], from which every estimator of section 2.2 can be derived. Being itself an expectation, the MMSE estimator is simply the most immediate, and is where we begin.

Let X be a random variable taking values in $\mathbb{R}^{d_x}$ with probability density p , and let $\varphi \in \mathcal{L}^0(\mathbb{R}^{d_x}, \mathbb{R}^{d_\varphi})$ such that $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$. We wish to compute the expectation of $\varphi(X)$:

$$\mathbb{E}_p[\varphi(X)] = \int \varphi(x) p(x) dx \quad (2.46)$$

Let $\mu(dx) \triangleq p(x) dx$ be the probability measure with density p with respect to the Lebesgue measure. (2.46) can be rewritten as:

$$\langle \mu, \varphi \rangle \triangleq \int \varphi(x) \mu(dx) = \mathbb{E}_p[\varphi(X)] \quad (2.47)$$

Since μ is a continuous measure, evaluating (2.47) exactly is intractable in most practical situations. The MC method aims to approximate μ by an empirical measure induced by N independent samples, as presented by Definition 2.2 and Proposition 2.4.

Definition 2.2 (Empirical measure). *Let $N \in \mathbb{N}^*$ and $X^{(1)}, \dots, X^{(N)}$ be independent and identically distributed (i.i.d.) random variables drawn from μ . The empirical measure associated with $(X^{(i)})_{i \in \llbracket 1, N \rrbracket}$ is defined as:*

$$S^N(\mu) \triangleq \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}} \quad (2.48)$$

where δ_x denotes the Dirac measure at x .

Proposition 2.4 (Monte Carlo estimator). *The MC estimator:*

$$\langle S^N(\mu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \langle \delta_{X^{(i)}}, \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \varphi(X^{(i)}) \quad (2.49)$$

is unbiased and consistent:

$$\mathbb{E}_p [\langle S^N(\mu), \varphi \rangle] = \mathbb{E}_p [\varphi(X)] = \langle \mu, \varphi \rangle \quad (2.50)$$

$$\langle S^N(\mu), \varphi \rangle \xrightarrow{a.s.} \langle \mu, \varphi \rangle \quad (2.51)$$

If moreover $\varphi(X) \in \mathcal{L}^2(\mathbb{R}^{d_\varphi})$, its mean squared error is:

$$\mathbb{E}_p \left[|\langle S^N(\mu) - \mu, \varphi \rangle|^2 \right] = \frac{1}{N} \text{tr } \mathbb{V}_\mu[\varphi] \quad (2.52)$$

where $\mathbb{V}_\mu[\varphi] \triangleq \langle \mu, (\varphi - \langle \mu, \varphi \rangle)(\varphi - \langle \mu, \varphi \rangle)^\top \rangle$ is the covariance matrix of φ under μ , and its rescaled error converges in distribution to a centred Gaussian:

$$\sqrt{N} \langle S^N(\mu) - \mu, \varphi \rangle \xrightarrow{\mathcal{L}} \mathcal{N}(0, \mathbb{V}_\mu[\varphi]) \quad (2.53)$$

Proof. By linearity of the expectation and the transfer theorem, it comes that:

$$\begin{aligned} \mathbb{E}_p [\langle S^N(\mu), \varphi \rangle] &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_p [\varphi(X^{(i)})] \\ &= \frac{1}{N} \sum_{i=1}^N \langle \mu, \varphi \rangle \\ &= \langle \mu, \varphi \rangle \end{aligned} \quad (2.54)$$

Furthermore, by independence of the $X^{(i)}$:

$$\begin{aligned} \mathbb{E}_p \left[|\langle S^N(\mu) - \mu, \varphi \rangle|^2 \right] &= \mathbb{E}_p \left[\left| \frac{1}{N} \sum_{i=1}^N (\varphi(X^{(i)}) - \langle \mu, \varphi \rangle) \right|^2 \right] \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E}_p \left[|\varphi(X^{(i)}) - \langle \mu, \varphi \rangle|^2 \right] \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E} [|\varphi(X) - \langle \mu, \varphi \rangle|^2] \\ &= \frac{1}{N} \text{tr } \mathbb{V}_\mu[\varphi] \end{aligned} \quad (2.55)$$

Finally, the a.s. convergence and the convergence in distribution follow respectively from the strong law of large numbers and the central limit theorem, applied to the i.i.d. variables $\varphi(X^{(i)})$. $\square$

Informally, this means that:

$$S^N(\mu) = \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}} \approx \mu \quad (2.56)$$

in the sense that, for any measurable function φ such that $\varphi(X)$ is integrable:

$$\langle S^N(\mu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \varphi(X^{(i)}) \approx \langle \mu, \varphi \rangle \quad (2.57)$$

2.4.2 Importance Sampling

The standard MC method requires sampling directly from μ , which can be complex in practice. Importance sampling [24] circumvents this issue by drawing from a chosen importance, or proposal, distribution ν with density q .

Assuming that $\text{supp}(q) \subseteq \text{supp}(p)$, and for $x \in \text{supp}(q)$, it follows from the identity $p(x) = \frac{p(x)}{q(x)} q(x)$ and the transfer theorem that:

$$\mathbb{E}_p[\varphi(X)] = \int \left(\varphi(x) \frac{p(x)}{q(x)} \right) q(x) dx = \mathbb{E}_q \left[\varphi(X) \frac{p(X)}{q(X)} \right] \quad (2.58)$$

Using measure theory notations, this is equivalent to writing:

$$\langle \mu, \varphi \rangle = \left\langle \nu, \frac{d\mu}{d\nu} \varphi \right\rangle \quad (2.59)$$

The density $\frac{d\mu}{d\nu}$ is known as the Radon-Nikodym derivative. This motivates the introduction of the weighted empirical measure, given by Definition 2.3.

Definition 2.3 (Weighted empirical measure). *Let $\nu(dx) = q(x) dx$ with $\text{supp}(\mu) \subseteq \text{supp}(\nu)$, and let $X^{(1)}, \dots, X^{(N)} \stackrel{i.i.d.}{\sim} \nu$. The associated weighted empirical measure is:*

$$S^N(\mu, \nu) \triangleq \frac{d\mu}{d\nu} S^N(\nu) = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \delta_{X^{(i)}} \quad (2.60)$$

Pairing this measure with a test function φ yields the weighted MC estimator of Proposition 2.5.

Proposition 2.5 (Weighted MC estimator). *The weighted MC estimator:*

$$\langle S^N(\mu, \nu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \varphi(X^{(i)}) \quad (2.61)$$

is unbiased and consistent:

$$\mathbb{E}_p [\langle S^N(\mu, \nu), \varphi \rangle] = \mathbb{E}_p[\varphi(X)] \quad (2.62)$$

$$\langle S^N(\mu, \nu), \varphi \rangle \xrightarrow{a.s.} \mathbb{E}_p[\varphi(X)] \quad (2.63)$$

Furthermore, when $\mathbb{V}_\nu \left[\frac{d\mu}{d\nu} \varphi \right] < \infty$, its mean squared error and rescaled error satisfy:

$$\mathbb{E}_q \left[\left| \langle S^N(\mu, \nu) - \mu, \varphi \rangle \right|^2 \right] = \frac{1}{N} \text{tr} \mathbb{V}_\nu \left[\frac{d\mu}{d\nu} \varphi \right] \quad (2.64)$$

$$\sqrt{N} \langle S^N(\mu, \nu) - \mu, \varphi \rangle \xrightarrow{\mathcal{L}} \mathcal{N} \left(0, \mathbb{V}_\nu \left[\frac{d\mu}{d\nu} \varphi \right] \right) \quad (2.65)$$

Proof. This is direct using the results of Proposition 2.4. $\square$

For $\varphi \equiv 1$, the total mass:

$$\langle S^N(\mu, \nu), 1 \rangle = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \quad (2.66)$$

converges a.s. to 1, but each finite sample is not necessarily normalised. Furthermore, p and q are often known only up to normalising constants $\tilde{p} = C_\mu p$ and $\tilde{q} = C_\nu q$. To alleviate these issues, Definition 2.4 introduces the self-normalised measure [25], whose estimator remains consistent despite these unknown constants, as proven by Proposition 2.6.

Definition 2.4 (Self-normalised measure). *Let $X^{(1)}, \dots, X^{(N)} \stackrel{i.i.d.}{\sim} \nu$, with $\tilde{\mu} = C_\mu \mu$ and $\tilde{\nu} = C_\nu \nu$. The self-normalised measure is:*

$$\tilde{S}^N(\mu, \nu) \triangleq \sum_{i=1}^N w^{(i)} \delta_{X^{(i)}} \quad (2.67)$$

$$w^{(i)} \triangleq \tilde{w}^{(i)} / \sum_{j=1}^N \tilde{w}^{(j)} \quad (2.68)$$

$$\tilde{w}^{(i)} \triangleq \frac{\tilde{p}(X^{(i)})}{\tilde{q}(X^{(i)})} = \frac{d\tilde{\mu}}{d\tilde{\nu}}(X^{(i)}) \quad (2.69)$$

Proposition 2.6 (Self-normalised estimator). *The self-normalised estimator:*

$$\langle \tilde{S}^N(\mu, \nu), \varphi \rangle = \sum_{i=1}^N w^{(i)} \varphi(X^{(i)}) = \frac{\langle S^N(\tilde{\mu}, \tilde{\nu}), \varphi \rangle}{\langle S^N(\tilde{\mu}, \tilde{\nu}), 1 \rangle} \quad (2.70)$$

is biased at finite N , but consistent as soon as $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$ and $\text{supp}(\mu) \subseteq \text{supp}(\nu)$:

$$\langle \tilde{S}^N(\mu, \nu), \varphi \rangle \xrightarrow{a.s.} \langle \mu, \varphi \rangle \quad (2.71)$$

Proof. Although the bias of $\tilde{S}^N(\mu, \nu)$ admits no simple closed-form expression in general, it can be computed for $N = 1$. In this situation, the self-normalisation forces the single weight to 1, such that:

$$\langle \tilde{S}^1(\mu, \nu), \varphi \rangle = \varphi(X^{(1)}) \quad (2.72)$$

and thus:

$$\mathbb{E} \left[\langle \tilde{S}^1(\mu, \nu), \varphi \rangle \right] = \langle \nu, \varphi \rangle \neq \langle \mu, \varphi \rangle \quad (2.73)$$

in general.

Furthermore, applying the strong law of large numbers separately to both the numerator and denominator of the self-normalised estimator yields:

$$\langle S^N(\tilde{\mu}, \tilde{\nu}), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \tilde{w}^{(i)} \varphi(X^{(i)}) \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}} \varphi \right\rangle = \frac{C_\mu}{C_\nu} \langle \mu, \varphi \rangle \quad (2.74)$$

and:

$$\langle S^N(\tilde{\mu}, \tilde{\nu}), 1 \rangle = \frac{1}{N} \sum_{j=1}^N \tilde{w}^{(j)} \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}} \right\rangle = \frac{C_\mu}{C_\nu} \langle \mu, 1 \rangle = \frac{C_\mu}{C_\nu} \neq 0 \quad (2.75)$$

Combining both expressions finally gives the desired result. $\square$

2.4.3 Sequential Importance Sampling

In the filtering context, the target distribution is the joint posterior $p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k})$ over the entire state trajectory. As new observations arrive, this distribution grows in dimension, making a static importance sampling approach impractical. Sequential importance sampling (SIS) [26, 27] addresses this by updating the weights recursively rather than re-approximating the full trajectory at every time step.

Assumption 2.3 (SIS hypotheses). *In addition to Assumptions 2.1 and 2.2, we assume that the proposal density of the previous state does not depend on the current observation:*

$$q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k}) = q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1}) \quad (2.76)$$

This assumption enables the following factorisation:

$$\begin{aligned} q(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) &= q(x_k, \mathbf{x}_{0:k-1} | \mathbf{z}_{1:k}) \\ &= q(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k}) \\ &= q(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1}) \end{aligned} \quad (2.77)$$

While Assumption 2.3 is sufficient to derive the SIS algorithm, an additional assumption is often considered:

$$q(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) = q(x_k | x_{k-1}, z_k) q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1}) \quad (2.78)$$

so that a new particle $x_k^{(i)}$ can be propagated from $x_{k-1}^{(i)}$ and z_k alone.

Using Assumptions 2.1 and 2.2 at time k , the PDF factorises as:

$$\begin{aligned} p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) &= p(\mathbf{x}_{0:k} | z_k, \mathbf{z}_{1:k-1}) \\ &= \frac{p(z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k-1})}{p(z_k | \mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(x_k, \mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1})}{p(z_k | \mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k | \mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}) p(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1})}{p(z_k | \mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k | x_k) p(x_k | x_{k-1}) p(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1})}{p(z_k | \mathbf{z}_{1:k-1})} \\ &\propto p(z_k | x_k) p(x_k | x_{k-1}) p(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1}) \end{aligned} \quad (2.79)$$

Let $\mathbf{x}_{0:k}^{(i)}$ denote the sequence $(x_0^{(i)}, \dots, x_k^{(i)})$. The recursive weight update is obtained by combining (2.79) with Assumption 2.3:

$$\begin{aligned} w_k^{(i)} &\propto \frac{\tilde{p}(\mathbf{x}_{0:k}^{(i)} | \mathbf{z}_{1:k})}{\tilde{q}(\mathbf{x}_{0:k}^{(i)} | \mathbf{z}_{1:k})} \\ &\propto \frac{\tilde{p}(z_k | x_k^{(i)}) \tilde{p}(x_k^{(i)} | x_{k-1}^{(i)}) \tilde{p}(\mathbf{x}_{0:k-1}^{(i)} | \mathbf{z}_{1:k-1})}{\tilde{q}(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}) \tilde{q}(\mathbf{x}_{0:k-1}^{(i)} | \mathbf{z}_{1:k-1})} \\ &\propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k | x_k^{(i)}) \tilde{p}(x_k^{(i)} | x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})} \end{aligned} \quad (2.80)$$

with the further simplification under the hypothesis of (2.78):

$$w_k^{(i)} \propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k | x_k^{(i)}) \tilde{p}(x_k^{(i)} | x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} \quad (2.81)$$

While the current formulation allows us to approximate the density of the full trajectory $p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k})$ through the self-normalised measure of Definition 2.4, we often only want to estimate the current state x_k . This can be done naturally by letting π_k denote the projection of a $(k+1)$ -tuple onto its last element:

$$\pi_k : \begin{cases} (\mathbb{R}^{d_x})^{k+1} \rightarrow \mathbb{R}^{d_x} \\ (x_0, \dots, x_k) \mapsto x_k \end{cases} \quad (2.82)$$

and computing the self-normalised estimator for $\varphi = \pi_k$, which approximates the MMSE estimator of section 2.2:

$$\hat{x}_k \triangleq \left\langle \tilde{S}^N(\mu_k, \nu_k), \pi_k \right\rangle = \sum_{i=1}^N w_k^{(i)} x_k^{(i)} \approx \mathbb{E}_{p(\cdot | \mathbf{z}_{1:k})}[x_k] \quad (2.83)$$

where $\mu_k(dx) \triangleq \tilde{p}(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) dx$ and $\nu_k(dx) \triangleq \tilde{q}(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) dx$. The corresponding empirical covariance matrix is:

$$\hat{P}_k \triangleq \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top \quad (2.84)$$

The other estimators of section 2.2 remain accessible, though they are not as immediate:

- For a scalar state, the MMAE estimator only requires the cumulative distribution function, which $\tilde{S}^N(\mu_k, \nu_k)$ does define, and is obtained by accumulating the weights of the sorted particles until the total exceeds $\frac{1}{2}$. Its extension to a vector state is however not straightforward.

- The MAP estimator, on the other hand, calls for a density, which the particle approximation does not possess. In particular, $\hat{\psi}_{\text{MAP}}(\mathbf{z}_{1:k})$ cannot be computed using the weights, since they are importance ratios rather than density values. The heaviest particle maximises the Radon-Nikodym derivative rather than the PDF itself, and maximising $\tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$ over the particles only locates a mode among the sampled locations.

As for the optimal and Kalman filters, the SIS algorithm follows the prediction/correction loop. The prediction step corresponds to the sampling of the new particle set through the proposal density, and is often called particle propagation. Conversely, the correction step is the weight update of (2.80). The SIS algorithm is given by Algorithm 2.4.

Algorithm 2.4 Sequential Importance Sampling

$$\begin{aligned}
\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} &\sim q(\cdot | x_{k-1}^{(i)}, z_k) &> \text{Particle propagation} \\
\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} &= w_{k-1}^{(i)} \frac{\tilde{p}(z_k | x_k^{(i)}) \tilde{p}(x_k^{(i)} | x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})} &> \text{Weight update} \\
\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} &= \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)} &> \text{Normalisation}
\end{aligned}$$

Remark 2.9 (Feynman-Kac distribution). *The recurrence (2.79) can be expanded:*

$$p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) \propto p(x_0) \prod_{j=1}^k p(x_j|x_{j-1}) \times \prod_{j=1}^k p(z_j|x_j) \quad (2.85)$$

to show that the PDF is a Feynman-Kac distribution [28, 29], with the transitions $p(x_j|x_{j-1})$ acting as the mutation and the likelihoods $p(z_j|x_j)$ as the selection function. Rather than being an artifice used to cancel the unknown constants, the self-normalisation introduced in Definition 2.4 is thus, more fundamentally, the normalisation of this distribution.

Numerical correction of the covariance matrix

For numerical reasons, the estimated covariance matrix $\hat{P}_k$ may not satisfy SPD properties in practice. While an analytic reformulation existed for the Kalman filter, the numerical nature of MC methods prevents a similar solution. Formally, we want to find the closest matrix $M_+^* \succ 0$ to $\hat{P}_k$ in the sense of the Frobenius norm:

$$A \in \mathcal{M}_{d_x, d_x}(\mathbb{R}) \mapsto \|A\|_F \triangleq \sqrt{\sum_{i=1}^{d_x} \sum_{j=1}^{d_x} A_{i,j}^2} \quad (2.86)$$

If $d_x = 1$, the positive real $m_+ \in \mathbb{R}_+$ closest to some real $m \in \mathbb{R}$ is:

$$m_+ = \frac{m + |m|}{2} \quad (2.87)$$

While this only applies to scalars, a similar operation can be performed in higher dimensions by using the SPSD term of the polar decomposition. The singular values decomposition is used to simplify the computation of the polar decomposition.

Formally, the correction algorithm [30] is as follows:

- Symmetrisation of $\hat{P}_k$: $M \triangleq \frac{\hat{P}_k + \hat{P}_k^\top}{2}$
- Singular values decomposition: $M = U D V^\top$
- Polar decomposition: $M = \Omega \Sigma$ with $\Omega \triangleq U V^\top$ and $\Sigma \triangleq V D V^\top$
- Computation of M_+ : $M_+ \triangleq \frac{M + \Sigma}{2}$

With this algorithm, M_+ is ensured to be SPSD. However, we wish to achieve strict positive definiteness. To solve this issue, we recall that a symmetric matrix is SPD if and only if all its eigenvalues are strictly positive, and we introduce a small correction $\varepsilon > 0$. Let $\lambda_{\min}$ denote the smallest eigenvalue of M_+ . Let:

$$\delta \triangleq \max(0, -\lambda_{\min}) + \varepsilon \quad (2.88)$$

where the first term is introduced to compensate any numerical errors introduced during the symmetrisation and SVD steps.

Since M_+ and δI commute, they are simultaneously diagonalisable, and the eigenvalues of:

$$M_+^* \triangleq M_+ + \delta I \quad (2.89)$$

are thus:

$$\lambda_i(M_+^*) = \lambda_i(M_+) + \delta \geq \lambda_{\min} + \delta > 0 \quad (2.90)$$

For small enough values of ε , M_+^* is the desired matrix.

2.4.4 Weight Degeneracy

A fundamental limitation of SIS is that the variance of the importance weights accumulates along the trajectory [31] as shown by Proposition 2.7.

Proposition 2.7 (Weight degeneracy). *Let $\mathbf{x}_{0:k} \sim \nu_k$, and let $\tilde{w}_k$ denote the non-normalised weight associated with the entire trajectory:*

$$\tilde{w}_k \triangleq \frac{\tilde{p}(\mathbf{x}_{0:k} | \mathbf{z}_{1:k})}{\tilde{q}(\mathbf{x}_{0:k} | \mathbf{z}_{1:k})} \quad (2.91)$$

Its variance decomposes as:

$$\mathbb{V}_{\nu_k} [\tilde{w}_k] = \mathbb{E}_{\nu_k} [\mathbb{V}_{\nu_k} [\tilde{w}_k | \mathbf{x}_{0:k-1}]] + \mathbb{V}_{\nu_k} [\mathbb{E}_{\nu_k} [\tilde{w}_k | \mathbf{x}_{0:k-1}]] \quad (2.92)$$

where the first term is non-negative, and vanishes if and only if $\tilde{w}_k$ is a.s. determined by $\mathbf{x}_{0:k-1}$.

Proof. (2.92) is the law of total variance. Being the expectation of a non-negative quantity, its first term vanishes if and only if $\mathbb{V}_{\nu_k} [\tilde{w}_k | \mathbf{x}_{0:k-1}]$ is a.s. zero, that is if and only if $\tilde{w}_k$ is a.s. determined by $\mathbf{x}_{0:k-1}$. $\square$

Given $\mathbf{x}_{0:k-1}$, the only randomness remaining in (2.92) is the draw of x_k . The inner variance measures its effect, while the inner expectation integrates it out and leaves a function of $\mathbf{x}_{0:k-1}$. The first term therefore corresponds to the variance injected by x_k and the second is the variance inherited from the past.

As each step adds to the previous ones, the weight distribution typically concentrates on a small number of particles, and so (2.67) becomes a poor approximation of the PDF. While degeneracy occurs in most cases, Proposition 2.8 exhibits a proposal density that eliminates the additional variance introduced at the current step.

Proposition 2.8 (Optimal proposal density). *When the proposal density is the posterior transition:*

$$q(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k | x_{k-1}, z_k) \quad (2.93)$$

the trajectory weight is determined by $\mathbf{x}_{0:k-1}$, so that the variance introduced at step k vanishes:

$$\mathbb{V}_{\nu_k} [\tilde{w}_k | \mathbf{x}_{0:k-1}] = 0 \quad (2.94)$$

Proof. Plugging (2.93) into (2.80) yields:

$$\begin{aligned} \tilde{w}_k &\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k | x_k) \tilde{p}(x_k | x_{k-1})}{\tilde{q}(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k})} \\ &\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k | x_k) \tilde{p}(x_k | x_{k-1})}{\tilde{p}(x_k | x_{k-1}, z_k)} \\ &\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k | x_k) \tilde{p}(x_k | x_{k-1})}{\tilde{p}(z_k | x_k, x_{k-1}) \tilde{p}(x_k | x_{k-1})} \\ &\propto \tilde{w}_{k-1} \tilde{p}(z_k | x_{k-1}) \frac{\tilde{p}(z_k | x_k)}{\tilde{p}(z_k | x_k, x_{k-1})} \\ &\propto \tilde{w}_{k-1} \tilde{p}(z_k | x_{k-1}) \end{aligned} \quad (2.95)$$

Both factors being functions of $\mathbf{x}_{0:k-1}$ alone, $\tilde{w}_k$ is determined by the conditioning, and its conditional variance is therefore zero. $\square$

Using Proposition 2.8, the draw of x_k thus leaves $\tilde{w}_k$ untouched, but reaches $\tilde{w}_{k+1}$ through $\tilde{p}(z_{k+1} | x_k)$. Its contribution is deferred by one step rather than suppressed, and so this density delays the degeneracy rather than completely removing it. Furthermore, sampling from the posterior transition requires computing the marginal likelihood:

$$p(z_k | x_{k-1}^{(i)}) = \int p(z_k | x_k^{(i)}) p(x_k^{(i)} | x_{k-1}^{(i)}) dx_k^{(i)} \quad (2.96)$$

which is generally intractable.

A common alternative is to use the prior, or transition, density:

$$q(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k | x_{k-1}) \quad (2.97)$$

which is easier to compute, but ignores the current observation during propagation.

The prior density also simplifies the weight update (2.80) to:

$$w_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k | x_k^{(i)}) \quad (2.98)$$

However, because the particles are propagated without regard to z_k , their incremental weight is simply the likelihood $\tilde{p}(z_k|x_k^{(i)})$. When this likelihood is peaked, most particles land where it is negligible. Only a few particles then carry most of the weight, which causes the degeneracy to accelerate. The prior is thus a less effective proposal in highly informative observation regimes.

2.4.5 Sampling Importance Resampling

Weight degeneracy is intrinsic to SIS and worsens with time unless the proposal distribution is chosen extremely close to the optimal proposal density. The sampling importance resampling (SIR) filter [32], also known as the bootstrap filter [33], mitigates this by introducing a resampling step after each weight update. Resampling replaces the weighted particle set $\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N$ by an unweighted set $\{x_k^{*(j)}, 1/N\}_{j=1}^N$ drawn according to a chosen resampling algorithm:

$$\{x_k^{*(j)}\}_{j=1}^N = \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right) \quad (2.99)$$

It also uses the transition density as proposal (2.97), which further reduces (2.98) to:

$$w_k^{(i)} \propto \tilde{p}(z_k|x_k^{(i)}) \quad (2.100)$$

since all weights are reset to $1/N$ after resampling. This makes the SIR applicable under very weak assumptions, only requiring the ability to sample from $p(\cdot|x_{k-1})$ and compute the likelihood $p(z_k|x_k)$. Its pseudo-code is given by Algorithm 2.5.

Algorithm 2.5 Sampling Importance Resampling

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot x_{k-1}^{(i)})$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = \tilde{p}(z_k x_k^{(i)})$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation
$\{x_k^{(j)}, w_k^{(j)}, -\}_{j=1}^N \leftarrow \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right)$	▷ Resampling

Resampling Algorithms

Several resampling algorithms [34, 35] can be used in (2.99). We present three algorithms, which all select a new particle set from the particle approximation of the PDF:

$$p(x_k|z_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} \delta_{x_k^{(i)}} \quad (2.101)$$

Drawing a particle from this discrete measure amounts to drawing its index $i \in \llbracket 1, N \rrbracket$ from the categorical distribution of the weights $(w_k^{(i)})_{i \in \llbracket 1, N \rrbracket}$ through its

cumulative distribution function (CDF):

$$\begin{cases} \text{cdf}_i \triangleq \sum_{j=1}^i w_k^{(j)}, & i \in \llbracket 1, N \rrbracket \\ \text{cdf}_0 = 0 \end{cases} \quad (2.102)$$

As illustrated by Figure 2.2, a draw of $u_j \in [0, 1]$ uniquely determines the new particle $x_k^{*(j)}$ as the parent particle $x_k^{(i)}$ of index i satisfying:

$$\text{cdf}_{i-1} \leq u_j < \text{cdf}_i \quad (2.103)$$

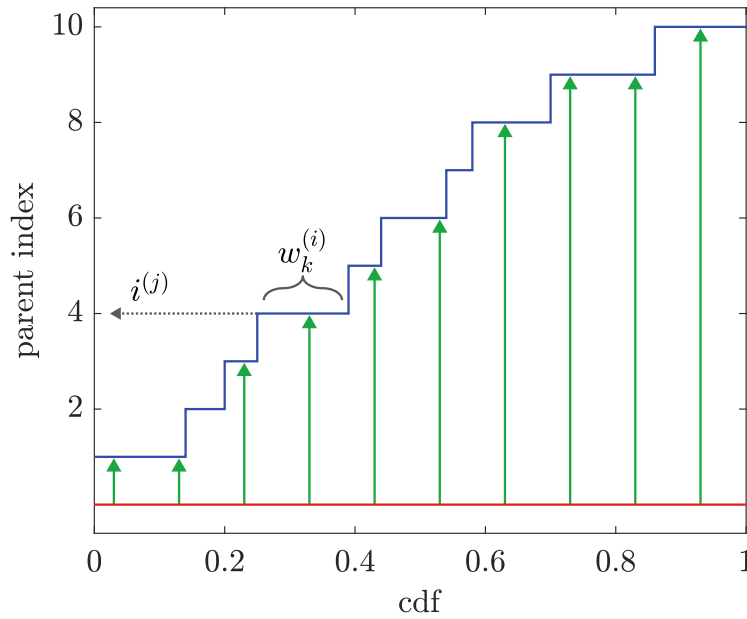


Figure 2.2: An example CDF of the particle weights. The width of each step is equal to the corresponding weight, so heavier particles are more likely to be selected.

The main difference between the three algorithms lies in how the N samples $(u_j)_{j \in \llbracket 1, N \rrbracket}$ are generated:

- **Multinomial resampling.** All the draws are independent:

$$u_j \stackrel{i.i.d.}{\sim} \mathcal{U}([0, 1]) \quad (2.104)$$

This is the most natural method, but has the highest resampling variance.

- **Systematic or Kitagawa resampling [36].** A single draw generates all samples via:

$$\begin{cases} u_1 \sim \mathcal{U}([0, 1/N]) \\ u_j = u_1 + (j-1)/N, & j \in \llbracket 2, N \rrbracket \end{cases} \quad (2.105)$$

This significantly reduces the variance of the resampling step and is generally preferred in practice.

- **Deterministic resampling.** All samples are deterministic such that, for some $m \in [0, 1]$:

$$u_j = (j - m)/N, \quad j \in \llbracket 1, N \rrbracket \quad (2.106)$$

This eliminates resampling variance entirely at the cost of a deterministic bias.

Some advanced filters, such as the auxiliary particle filter, presented in section 2.5, require keeping track of the parent index $i^{(j)}$ of each resampled particle. This value is thus recorded at the creation of each new particle. Algorithm 2.6 presents a linear implementation of the Kitagawa resampling algorithm, using the fact that both the CDF values and the Kitagawa samples are increasing to compute the parent indices sequentially.

Algorithm 2.6 Kitagawa Resampling

```

cdf0 = 0                                ▷ Computation of the CDF
∀i ∈ ⌊1, N⌋, cdfi = cdfi-1 + wk(i)
u1 ~ U([0, 1/N])                        ▷ Kitagawa samples
∀j ∈ ⌊2, N⌋, uj = u1 + (j - 1)/N
i = 1
for j = 1 : N do
    while uj ≥ cdfi do                  ▷ Computation of the parent index
        i ← i + 1
    end while
    xk*(j) = xk(i)                        ▷ Generation of the new particle
    wk*(j) = 1/N
    i(j) = i
end for

```

Remark 2.10 (Loss of independence). *Because resampling duplicates particles, the resulting set is no longer independent, and the i.i.d. results of Propositions 2.4 and 2.5 no longer apply. However, the resampling is unbiased and the propagation step draws the particles independently conditionally on the past. The Feynman-Kac formalism proves [37, 38] that consistency and asymptotic Gaussianity still hold in this scenario, at the price of a variance larger than in the independent case and which accumulates over time.*

2.5 Advanced Particle Filters

Even though the SIR applies under very mild assumptions, it has several caveats. Indeed, while resampling at every step eliminates the weight degeneracy of the SIS, it wastes computational load, and erodes particle diversity. Furthermore, the number of particles required grows quickly with the state dimension, a phenomenon known as the curse of dimensionality [39].

The SIR admits several refinements [40], each targeting a specific issue. We first present the generic particle filter, which reduces the SIR's computational cost by

only resampling when necessary. We then introduce the regularised and auxiliary particle filters to restore particle diversity and improve on the prior proposal. Finally, we derive the equations of the Rao-Blackwellised particle filter, which reduces the dimension of the explored space under linear-Gaussian substructure assumptions.

2.5.1 Generic Particle Filter

The SIR resamples at every time step, which eliminates weight degeneracy but comes with two drawbacks:

- It introduces unnecessary computational cost when weights are already well-distributed, and increases the variance of the estimated state, as mentioned by Remark 2.10.
- It also replicates the heaviest particles and discards the lightest, so the set progressively concentrates on a few distinct locations. This loss of diversity, known as particle impoverishment [41], is especially severe in low evolution noise or high-dimensional settings, where little diversity is re-introduced at the propagation step. It is illustrated by Figure 2.3.

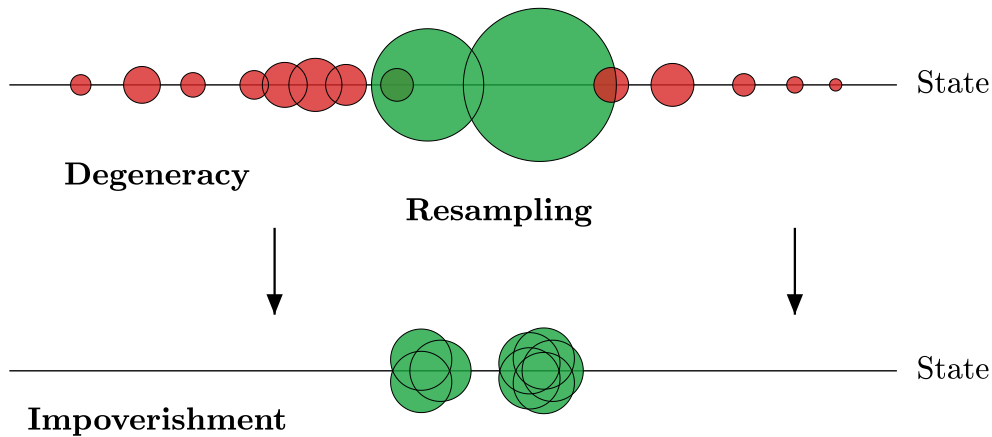


Figure 2.3: Particle impoverishment after resampling. Discarding non-resampled particles progressively shrinks the support of the particle approximation, degrading the filter’s ability to explore the state-space.

To limit both issues, the generic particle filter (PF) [42] resamples only when the weight distribution has degraded significantly. This degradation is measured by the effective sample size [31]:

$$N_{\text{eff}} \triangleq \frac{N}{1 + \mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})} \left[\frac{p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}{q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})} \right]} \quad (2.107)$$

Its denominator increases with the variance of the importance weights, whose growth over time corresponds to the weight degeneracy characterised by Proposition 2.7. When all the weights are uniform, the variance is zero, and thus $N_{\text{eff}} = N$.

Conversely, the more unevenly they are distributed, the smaller N_{eff} , with $N_{\text{eff}} \rightarrow 1$ when the mass concentrates on a single particle.

Since the true importance weights are unknown, N_{eff} is estimated by replacing $\mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})}[\cdot]$ with the empirical spread of the normalised weights [43], which gives:

$$\hat{N}_{\text{eff}} \triangleq \frac{1}{\sum_{i=1}^N \left(w_k^{(i)}\right)^2} \leq N \quad (2.108)$$

Like N_{eff} , this quantity ranges from N for uniform weights, down to 1 in the fully degenerate case. The PF then performs the resampling step only when this estimate falls below a threshold:

$$\hat{N}_{\text{eff}} < N_{\text{th}} \quad (2.109)$$

where N_{th} is a predetermined value, typically chosen proportional to N .

Unlike the SIR, which traditionally uses the transition density as its proposal distribution, the generic PF is not typically restricted to any particular choice of proposal distribution. It is presented by Algorithm 2.7.

Algorithm 2.7 Generic Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q\left(\cdot x_{k-1}^{(i)}, z_k\right)$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}\left(z_k x_k^{(i)}\right) \tilde{p}\left(x_k^{(i)} x_{k-1}^{(i)}\right)}{\tilde{q}\left(x_k^{(i)} \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}\right)}$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation
$\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N \left(w_k^{(i)}\right)^2$	▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ then	
$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^N \leftarrow \text{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^N\right)$	▷ Resampling
end if	

2.5.2 Regularised Particle Filter

Since each resampling step still collapses diversity, the PF only mitigates impoverishment, but does not prevent it. Indeed, the self-normalised measure of Definition 2.4 approximates the PDF by a mixture of Diracs:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} \delta_{x_k^{(i)}} \quad (2.110)$$

Since this measure is singular, resampling from it can only reproduce the existing particles. The regularised particle filter (RPF) [44] addresses this issue through

kernel density estimation [45, 46, 47], by replacing each Dirac with a smooth kernel to obtain the absolutely continuous measure:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} K_h \left(x_k - x_k^{(i)} \right) dx_k \quad (2.111)$$

New particles are then drawn from this continuous density rather than from the discrete set, which restores diversity. This is illustrated by Figure 2.4 and formalised by Definition 2.5 and proposition 2.9. Being absolutely continuous, this approximation also supplies the density that (2.110) lacks, and thus makes the MAP estimator of section 2.2 well-defined at the cost of a numerical maximisation.

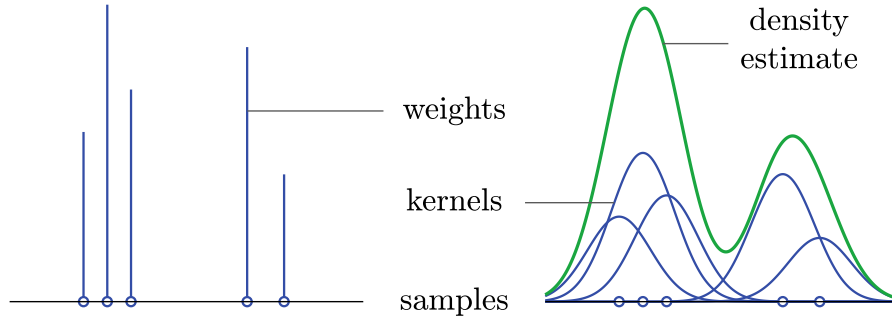


Figure 2.4: An example Monte Carlo approximation of the PDF (2.110) on the left and its kernel regularisation (2.111) on the right, for a given particle set. The latter approximation bridges the gaps between particles and spans a larger region of the state space.

Definition 2.5 (Regularisation kernel). *Let $K : \mathbb{R}^{d_x} \rightarrow \mathbb{R}$. K is a regularisation kernel if:*

$$\left\{ \begin{array}{ll} \forall x \in \mathbb{R}^{d_x}, K(x) \geq 0 & \text{positive} \\ \forall x \in \mathbb{R}^{d_x}, K(-x) = K(x) & \text{symmetric} \\ \int K(x) dx = 1 & \text{normalised} \\ \int \|x\|^2 K(x) dx < \infty & \text{of finite variance} \end{array} \right. \quad (2.112)$$

Furthermore, for K a regularisation kernel and $h > 0$, we define the rescaled kernel K_h of bandwidth h as:

$$K_h : \left\{ \begin{array}{l} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ x \mapsto \frac{1}{h^{d_x}} K \left(\frac{x}{h} \right) \end{array} \right. \quad (2.113)$$

Definition 2.6 (Regularised measure). *Let $*$ denote the convolution operator. The regularised measure $S_{K_h}^N$ of some empirical measure S^N with respect to a regularisation kernel K_h is:*

$$S_{K_h}^N(\mu) \triangleq K_h * S^N(\mu) \quad (2.114)$$

In particular, the regularised self-normalised measure verifies:

$$\tilde{S}_{K_h}^N(\mu, \nu) = \sum_{i=1}^N w^{(i)} (K_h * \delta_{X^{(i)}}) \quad (2.115)$$

Proposition 2.9 (Regularised estimator). *The regularised self-normalised estimator is:*

$$\left\langle \tilde{S}_{K_h}^N(\mu, \nu), \varphi \right\rangle = \sum_{i=1}^N w^{(i)} (K_h * \varphi) (X^{(i)}) = \left\langle \tilde{S}^N(\mu, \nu), K_h * \varphi \right\rangle \quad (2.116)$$

It is a biased and inconsistent estimator in general:

$$\left\langle \tilde{S}_{K_h}^N(\mu, \nu), \varphi \right\rangle \xrightarrow{a.s.} \langle \mu, K_h * \varphi \rangle \quad (2.117)$$

However, it becomes consistent when φ is linear. In particular:

$$\left\langle \tilde{S}_{K_h}^N(\mu_k, \nu_k), \pi_k \right\rangle \xrightarrow{a.s.} \langle \mu_k, \pi_k \rangle = \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}[x_k] \quad (2.118)$$

Proof. Let $i \in \llbracket 1, N \rrbracket$. By definition of $*$, we have:

$$(K_h * \delta_{X^{(i)}})(dx) = K_h(x - X^{(i)}) dx \quad (2.119)$$

and thus, by symmetry of K_h :

$$\begin{aligned} \langle K_h * \delta_{X^{(i)}}, \varphi \rangle &= \int \varphi(x) K_h(x - X^{(i)}) dx \\ &= \int \varphi(x) K_h(X^{(i)} - x) dx \\ &= (K_h * \varphi)(X^{(i)}) \end{aligned} \quad (2.120)$$

In turn, this result gives:

$$\left\langle \tilde{S}_{K_h}^N(\mu, \nu), \varphi \right\rangle = \sum_{i=1}^N w^{(i)} (K_h * \varphi) (X^{(i)}) = \left\langle \tilde{S}^N(\mu, \nu), K_h * \varphi \right\rangle \quad (2.121)$$

and Proposition 2.6, applied to $K_h * \varphi$, yields the a.s. limit $\langle \mu, K_h * \varphi \rangle$.

Assume that φ is linear, that is $\varphi : x \mapsto A x$ for some $A \in \mathcal{M}_{d_\varphi, d_x}(\mathbb{R})$. Then, by normalisation and symmetry of K_h , for all $x \in \mathbb{R}^{d_x}$:

$$\begin{aligned} (K_h * \varphi)(x) &= \int K_h(y) \varphi(x - y) dy \\ &= \varphi(x) \int K_h(y) dy - A \int y K_h(y) dy \\ &= \varphi(x) \end{aligned} \quad (2.122)$$

Hence $K_h * \varphi = \varphi$, and the limit reduces to $\langle \mu, \varphi \rangle$. In particular, for the linear projection π_k , it becomes $\langle \mu_k, \pi_k \rangle = \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}[x_k]$. $\square$

The choice of bandwidth h is critical to the performance of the RPF. If h is too small, K_h concentrates around the Dirac measure, and the regularisation has no practical effect on particle diversity. Conversely, the kernel density estimate over-smooths the posterior when h is too large, artificially spreading particles across regions of low probability, and degrading filter accuracy. In this section, we present two classical choices for K_h :

- **Epanechnikov kernel.** This kernel minimises the mean integrated squared error (MISE) among all regularisation kernels [48], and the associated bandwidth is optimal when the density is Gaussian with identity covariance. It is defined by:

$$\begin{cases} K_{\text{Epa}} : x \mapsto \frac{d_x + 2}{2c_{d_x}} (1 - \|x\|^2) \mathbb{1}_{\|x\| \leq 1}(x) \\ h_{\text{Epa}} = \left(\frac{8(d_x + 4)(2\sqrt{\pi})^{d_x}}{c_{d_x}} \right)^{\frac{1}{d_x+4}} N^{-\frac{1}{d_x+4}} \end{cases} \quad (2.123)$$

where:

$$c_{d_x} \triangleq \frac{\pi^{d_x/2}}{\Gamma(d_x/2 + 1)} \quad (2.124)$$

is the volume of the $\mathbb{R}^{d_x}$ unit ball, and Γ is the Gamma function.

- **Gaussian kernel.** While suboptimal with respect to the Epanechnikov kernel, it is simpler to compute [49]. Its expression is given by:

$$\begin{cases} K_{\text{Gauss}} : x \mapsto (2\pi)^{-d_x/2} \exp\left(-\frac{\|x\|^2}{2}\right) \\ h_{\text{Gauss}} = \left(\frac{4}{d_x + 2} \right)^{\frac{1}{d_x+4}} N^{-\frac{1}{d_x+4}} \end{cases} \quad (2.125)$$

It must be noted that the assumptions for the Epanechnikov, and consequently the Gaussian, regularisation are not usually met in practice, since MC methods are precisely used when the PDF is not Gaussian.

However, the two conditions do not play the same role. The Gaussian hypothesis is only used to compute the optimal bandwidth, whose expression would otherwise depend on an unknown PDF. It is thus a calibration convention rather than a modelling assumption. In particular, it tends to overestimate h when the PDF is multimodal.

On the other hand, the unit covariance is a scaling requirement, and can be enforced exactly by a whitening step before the regularisation. This amounts to replacing every particle by $A_k^{-1} x_k^{(i)}$, where A_k is a Cholesky factor of $\hat{P}_k$:

$$\hat{P}_k = A_k A_k^\top \quad (2.126)$$

which ensures that the covariance matrix of the new particle set is the identity. While seemingly computationally heavy, this reduces to using the following regularisation kernel on the standard particle set:

$$\tilde{K}_h : x \mapsto \frac{\det(A_k)^{-1}}{h^{d_x}} K\left(\frac{1}{h} A_k^{-1} x\right) \quad (2.127)$$

Moreover, the bandwidth h is multiplied by a modulation factor $\alpha_{\text{reg}} \in [0, 1]$ to avoid over-smoothing in cases where the PDF is multimodal. The pseudo-code for the RPF is given by Algorithm 2.8.

Algorithm 2.8 Regularised Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q(\cdot x_{k-1}^{(i)}, z_k)$	$\triangleright$ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}(z_k x_k^{(i)}) \tilde{p}(x_k^{(i)} x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})}$	$\triangleright$ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	$\triangleright$ Normalisation
$\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N (w_k^{(i)})^2$	$\triangleright$ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ then	
$\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$	$\triangleright$ Whitening
$\hat{P}_k = \sum_{i=1}^N w_k^{(i)} (x_k^{(i)} - \hat{x}_k) (x_k^{(i)} - \hat{x}_k)^\top$	
$A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top$	
$\{x_k^{(j)}, w_k^{(j)}, -\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{ x_k^{(i)}, w_k^{(i)} \right\}_{i=1}^N \right)$	$\triangleright$ Resampling
$\forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K$	$\triangleright$ Regularisation
$\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)}$	
end if	

2.5.3 Auxiliary Particle Filter

The auxiliary particle filter (APF) [50, 51] addresses the degeneracy problem at its root by exploiting a different weakness of the SIR. Since particles are propagated before the new observation z_k is taken into account, particles that will ultimately receive low likelihood waste computational effort. The APF thus introduces a look-ahead step by using a prediction of their future likelihood to resample promising parent particles before propagation.

Formally, the APF targets the joint distribution $p(x_k, i | \mathbf{z}_{1:k})$, where $i \in \llbracket 1, N \rrbracket$ indexes the parent particle at time $k-1$. Using Bayes' theorem and the chain rule gives:

$$p(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | x_k, i, \mathbf{z}_{1:k-1}) p(x_k | i, \mathbf{z}_{1:k-1}) p(i | \mathbf{z}_{1:k-1}) \quad (2.128)$$

Each term of this decomposition can be simplified as follows:

- Assumption 2.2 states that the current observation depends only on the current state:

$$p(z_k | x_k, i, \mathbf{z}_{1:k-1}) = p(z_k | x_k) \quad (2.129)$$

- Since the parent index uniquely determines the ancestor trajectory, conditioning on i is equivalent to conditioning on $\mathbf{x}_{0:k-1}^{(i)}$. Moreover, Assumption 2.1 establishes that $(x_k)_{k \geq 0}$ is a Markov chain, and thus:

$$p(x_k | i, \mathbf{z}_{1:k-1}) = p(x_k | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k-1}) = p(x_k | x_{k-1}^{(i)}) \quad (2.130)$$

- Finally, assuming that the particle approximation is exact gives:

$$p(i | \mathbf{z}_{1:k-1}) \propto w_{k-1}^{(i)} \quad (2.131)$$

Combining these equations reduces (2.128) to:

$$p(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | x_k) p(x_k | x_{k-1}^{(i)}) w_{k-1}^{(i)} \quad (2.132)$$

However, $p(z_k | x_k)$ cannot be evaluated before x_k is drawn. It is thus approximated by $p(z_k | \mu_k^{(i)})$, where:

$$\mu_k^{(i)} \sim p(\cdot | x_{k-1}^{(i)}) \quad (2.133)$$

is an auxiliary particle drawn from the transition density, and used to assess the consistency of the future particle $x_k^{(i)}$ with the new observation z_k . Alternatively, $\mu_k^{(i)}$ can be chosen as the mean of this density, a deterministic choice which avoids an additional draw.

In any case, the importance density is therefore:

$$q(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | \mu_k^{(i)}) p(x_k | x_{k-1}^{(i)}) w_{k-1}^{(i)} \quad (2.134)$$

Since they are drawn from the prior distribution, (2.98) ensures that the auxiliary weights $\lambda_k^{(i)}$ follow:

$$\lambda_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k | \mu_k^{(i)}) \quad (2.135)$$

In the hope that the evolution noise is sufficiently weak for the auxiliary particles to correctly represent the future particle set, the parent indices $\{i^{(j)}\}_{j=1}^N$ are resampled according to the auxiliary weights.

The new particle set is then propagated through a standard SIS algorithm using the transition density:

$$x_k^{(j)} \sim p(\cdot | x_{k-1}^{(i^{(j)})}) \quad (2.136)$$

with their weights updating as:

$$\begin{aligned} w_k^{(j)} &\propto \frac{\tilde{p}(x_k, i^{(j)} | \mathbf{z}_{1:k})}{\tilde{q}(x_k, i^{(j)} | \mathbf{z}_{1:k})} \\ &\propto \frac{\tilde{p}(z_k | x_k^{(j)}) \tilde{p}(x_k^{(j)} | x_{k-1}^{(i^{(j)})}) w_{k-1}^{(i^{(j)})}}{\tilde{p}(z_k | \mu_k^{(i^{(j)})}) \tilde{p}(x_k^{(j)} | x_{k-1}^{(i^{(j)})}) w_{k-1}^{(i^{(j)})}} \\ &\propto \frac{\tilde{p}(z_k | x_k^{(j)})}{\tilde{p}(z_k | \mu_k^{(i^{(j)})})} \end{aligned} \quad (2.137)$$

The full APF algorithm is given in algorithm 2.9.

Algorithm 2.9 Auxiliary Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, \mu_k^{(i)} \sim p(\cdot x_{k-1}^{(i)})$	▷ Auxiliary propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{\lambda}_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k \mu_k^{(i)})$	▷ Auxiliary update
$\forall i \in \llbracket 1, N \rrbracket, \lambda_k^{(i)} = \tilde{\lambda}_k^{(i)} / \sum_{j=1}^N \tilde{\lambda}_k^{(j)}$	▷ Normalisation
$\{-, -, i^{(j)}\}_{j=1}^N = \text{Resampling} \left(\left\{ \mu_k^{(i)}, \lambda_k^{(i)} \right\}_{i=1}^N \right)$	▷ Resampling
$\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \sim p(\cdot x_{k-1}^{(i^{(j)})})$	▷ Particle propagation
$\forall j \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(j)} \propto \frac{\tilde{p}(z_k x_k^{(j)})}{\tilde{p}(z_k \mu_k^{(i^{(j)})})}$	▷ Weight update
$\forall j \in \llbracket 1, N \rrbracket, w_k^{(j)} = \tilde{w}_k^{(j)} / \sum_{i=1}^N \tilde{w}_k^{(i)}$	▷ Normalisation

Remark 2.11 (Comparison between the RPF and the APF). *Both the RPF and the APF aim to alleviate the degeneracy and impoverishment problems of the standard PF, but they target different sources:*

- *The RPF restores diversity through kernel density estimation, addressing the severe impoverishment caused by near-deterministic dynamics. It is therefore suited to low evolution noise scenarios.*
- *The APF instead pre-selects promising parents from the incoming observation. It is preferred when the observation noise is small, since a peaked likelihood triggers an important single-step weight degeneracy.*

These preferences are rules of thumb rather than theoretical results, and the APF's pre-selection also degrades under strong evolution noise. Acting at different stages, both filters are ultimately complementary rather than exclusive.

2.5.4 Rao-Blackwellised Particle Filter

The marginalised particle filter [52], also known as the Rao-Blackwellised particle filter (RBPf) [53, 54], reduces the dimension of the state-space explored by the particle filter by exploiting a partial linear-Gaussian structure in the state-space model. Suppose that the state can be decomposed as:

$$x_k = [x_k^n, x_k^l] \quad (2.138)$$

such that (2.1) is linear and Gaussian in x_k^l conditionally on x_k^n . The navigation system can be rewritten as:

$$(S) : \begin{cases} x_k^n = F_k^n(x_{k-1}^n) & x_{k-1}^l + f_k^n(x_{k-1}^n) + w_k^n \\ x_k^l = F_k^l(x_{k-1}^n) & x_{k-1}^l + f_k^l(x_{k-1}^n) + w_k^l \\ z_k = H_k(x_k^n) & x_k^l + h_k(x_k^n) + n_k \end{cases} \quad (2.139)$$

with:

$$w_k = \begin{bmatrix} w_k^n \\ w_k^l \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} Q_k^n & (Q_k^{nl})^\top \\ Q_k^{nl} & Q_k^l \end{bmatrix} \right), \quad n_k \sim \mathcal{N}(0, R_k) \quad (2.140)$$

x_0^n of known distribution, and x_0^l Gaussian.

The idea behind the RBPF is to factorise the PDF as:

$$p(\mathbf{x}_{0:k}^l, \mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) = p(\mathbf{x}_{0:k}^l | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k}) p(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) \quad (2.141)$$

where $p(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k})$ is estimated via particle filtering techniques, and $p(\mathbf{x}_{0:k}^l | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k})$ is analytically solved by a Kalman filter. Here, $\mathbf{x}_{0:k}^l$ and $\mathbf{x}_{0:k}^n$ are defined similarly to $\mathbf{x}_{0:k}$.

When $Q_k^{nl} \neq 0$, the noises w_k^l and w_k^n are correlated, and the Kalman filter cannot be applied directly. Proposition 2.10 decouples them by removing the component of w_k^l correlated with w_k^n .

Proposition 2.10 (Noise decorrelation). *The decorrelated noise:*

$$\bar{w}_k^l \triangleq w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n \quad (2.142)$$

is centred, uncorrelated with w_k^n , and has covariance:

$$\bar{Q}_k^l \triangleq \mathbb{V}[\bar{w}_k^l] = Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \quad (2.143)$$

Proof. See section B.2. □

Let:

$$\bar{F}_k^l(x_{k-1}^n) \triangleq F_k^l(x_{k-1}^n) - Q_k^{nl} (Q_k^n)^{-1} F_k^n(x_{k-1}^n) \quad (2.144)$$

$$\bar{f}_k^l(x_{k-1}^n) \triangleq f_k^l(x_{k-1}^n) - Q_k^{nl} (Q_k^n)^{-1} f_k^n(x_{k-1}^n) \quad (2.145)$$

The system (2.139) can be rewritten as:

$$(\bar{S}) : \begin{cases} x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^n + f_k^n(x_{k-1}^n) + w_k^n \\ x_k^l = \bar{F}_k^l(x_{k-1}^n) x_{k-1}^l + \bar{f}_k^l(x_{k-1}^n) + Q_k^{nl} (Q_k^n)^{-1} x_k^n + \bar{w}_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases} \quad (2.146)$$

which can in turn be decomposed as:

$$\begin{aligned} (S_1) : & \begin{cases} x_{k-1}^l = x_{k-1}^l \\ x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^n + f_k^n(x_{k-1}^n) + w_k^n \end{cases} \\ (S_2) : & \begin{cases} x_k^l = \bar{F}_k^l(x_{k-1}^n) x_{k-1}^l + \bar{f}_k^l(x_{k-1}^n) + Q_k^{nl} (Q_k^n)^{-1} x_k^n + \bar{w}_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases} \end{aligned} \quad (2.147)$$

Note the similarities between each of the systems (S_1) and (S_2) , and the navigation equations under Kalman hypotheses (2.35). Indeed, (S_1) can be treated as if the propagation of the non-linear state x_k^n were an observation of x_{k-1}^l . Its absence of dynamics leads to only performing the (pre-)correction step of the Kalman filter. On the other hand, (S_2) can be handled by the usual Kalman prediction/correction.

In full generality, a standard RBPF is computed in five steps:

- **Particle propagation.** The non-linear state is propagated according to the proposal density:

$$x_k^{n(i)} \sim q\left(\cdot | x_{k-1}^{n(i)}, z_k\right) \quad (2.148)$$

- **Kalman pre-correction.** The previous linear state is corrected according to the Kalman equations applied to (S_1) :

$$\tilde{F}_k^{n(i)} = F_k^n \left(x_{k-1}^{n(i)} \right) \quad (2.149)$$

$$y_k^{n(i)} = x_k^{n(i)} - \left(\tilde{F}_k^{n(i)} \hat{x}_{k-1|k-1}^{l(i)} + f_k^n \left(x_{k-1}^{n(i)} \right) \right) \quad (2.150)$$

$$S_k^{n(i)} = \tilde{F}_k^{n(i)} P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{n(i)} \right)^\top + Q_k^n \quad (2.151)$$

$$K_k^{n(i)} = P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{n(i)} \right)^\top \left(S_k^{n(i)} \right)^{-1} \quad (2.152)$$

$$\hat{x}_{k-1|k-1}^{l(i)} \leftarrow \hat{x}_{k-1|k-1}^{l(i)} + K_k^{n(i)} y_k^{n(i)} \quad (2.153)$$

$$P_{k-1|k-1}^{l(i)} \leftarrow \left(I - K_k^{n(i)} \tilde{F}_k^{n(i)} \right) P_{k-1|k-1}^{l(i)} \quad (2.154)$$

- **Kalman prediction.** The current linear state is then predicted via the dynamics of (S_2) :

$$\tilde{F}_k^{l(i)} = \bar{F}_k^l \left(x_{k-1}^{n(i)} \right) \quad (2.155)$$

$$\hat{x}_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)} \hat{x}_{k-1|k-1}^{l(i)} + \bar{f}_k^l \left(x_{k-1}^{n(i)} \right) + Q_k^{nl} (Q_k^n)^{-1} x_k^{n(i)} \quad (2.156)$$

$$P_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)} P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{l(i)} \right)^\top + \bar{Q}_k^l \quad (2.157)$$

$$\tilde{H}_k^{(i)} = H_k \left(x_k^{n(i)} \right) \quad (2.158)$$

$$y_k^{l(i)} = z_k - \left(\tilde{H}_k^{(i)} \hat{x}_{k|k-1}^{l(i)} + h_k \left(x_k^{n(i)} \right) \right) \quad (2.159)$$

$$S_k^{l(i)} = \tilde{H}_k^{(i)} P_{k|k-1}^{l(i)} \left(\tilde{H}_k^{(i)} \right)^\top + R_k \quad (2.160)$$

- **Weight update.** The weights corresponding to each particle are updated with respect to the new observation:

$$\tilde{w}_k^{n(i)} = w_{k-1}^{n(i)} \frac{\tilde{p} \left(z_k | x_k^{n(i)} \right) \tilde{p} \left(x_k^{n(i)} | x_{k-1}^{n(i)} \right)}{\tilde{q} \left(x_k^{n(i)} | x_{k-1}^{n(i)}, z_k \right)} \quad (2.161)$$

$$w_k^{n(i)} = \tilde{w}_k^{n(i)} / \sum_{j=1}^N \tilde{w}_k^{n(j)} \quad (2.162)$$

- **Kalman correction.** The linear state is corrected through the standard Kalman correction for (S_2) :

$$K_k^{l(i)} = P_{k|k-1}^{l(i)} \left(\tilde{H}_k^{(i)} \right)^\top \left(S_k^{l(i)} \right)^{-1} \quad (2.163)$$

$$\hat{x}_{k|k}^{l(i)} = \hat{x}_{k|k-1}^{l(i)} + K_k^{l(i)} y_k^{l(i)} \quad (2.164)$$

$$P_{k|k}^{l(i)} = \left(I - K_k^{l(i)} \tilde{H}_k^{(i)} \right) P_{k|k-1}^{l(i)} \quad (2.165)$$

In the RBPF, only the non-linear state is sampled, while the conditional distribution of the linear substate is propagated analytically. The particle approximation is thus the mixture [55]:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{n(i)} \delta_{x_k^{n(i)}}(dx_k^n) \mathcal{N}(\hat{x}_{k|k}^{l(i)}, P_{k|k}^{l(i)})(dx_k^l) \quad (2.166)$$

The MMSE estimator of section 2.2 can then be approximated by:

$$\hat{x}_k = \sum_{i=1}^N w_k^{n(i)} x_k^{(i)} \quad (2.167)$$

where

$$x_k^{(i)} \triangleq \begin{bmatrix} x_k^{n(i)} \\ \hat{x}_{k|k}^{l(i)} \end{bmatrix} = \mathbb{E} \left[x_k | \mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k} \right] \quad (2.168)$$

Furthermore, since the non-linear state is deterministic when conditioned on the sampled trajectory, we have:

$$\mathbb{V} \left[x_k | \mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k} \right] = \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \quad (2.169)$$

Applying the law of total covariance finally gives the associated empirical covariance matrix:

$$\hat{P}_k = \sum_{i=1}^N w_k^{n(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top + \sum_{i=1}^N w_k^{n(i)} \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \quad (2.170)$$

where the first term is the covariance of the conditional means, and the second is the average conditional covariance.

The full RBPF algorithm is summarised in Algorithm 2.10.

Algorithm 2.10 Rao-Blackwellised Particle Filter

$\forall i \in \llbracket 1, N \rrbracket$, propagate $x_k^{n(i)}$	using (2.148)
$\forall i \in \llbracket 1, N \rrbracket$, update $\hat{x}_{k-1 k-1}^{l(i)}$ and $P_{k-1 k-1}^{l(i)}$	using (2.149) through (2.154)
$\forall i \in \llbracket 1, N \rrbracket$, predict $\hat{x}_{k k-1}^{l(i)}$ and $P_{k k-1}^{l(i)}$	using (2.155) through (2.160)
$\forall i \in \llbracket 1, N \rrbracket$, update $w_k^{n(i)}$	using (2.161) through (2.162)
$\forall i \in \llbracket 1, N \rrbracket$, compute $\hat{x}_{k k}^{l(i)}$ and $P_{k k}^{l(i)}$	using (2.163) through (2.165)

It must be noted that the version presented by Algorithm 2.10 uses a simple SIS to handle the particles, but any more advanced algorithm presented in this section can be used instead. Furthermore, the likelihood $\tilde{p}(z_k | x_k^{n(i)})$ and the transition $\tilde{p}(x_k^{n(i)} | x_{k-1}^{n(i)})$ in the weight update are white Gaussian densities, with covariances $S_k^{l(i)}$ and $S_k^{n(i)}$ respectively. This is why $S_k^{l(i)}$ is computed within the Kalman prediction step, before the weight update. However, the correction step itself must be performed after the weight update in situations where a resampling step is involved.

Remark 2.12 (Single Kalman filter). *In general, one Kalman filter is required per particle, since $\tilde{F}_k^{n(i)}$, $\tilde{F}_k^{l(i)}$ and $\tilde{H}_k^{(i)}$ all depend on $x_{k-1}^{n(i)}$ and $x_k^{n(i)}$. However, if these matrices do not depend on the non-linear state, then the gains $K_k^{n(i)}$ and $K_k^{l(i)}$, and the covariances $P_{k|k}^{l(i)}$ are identical for all particles. In this scenario, a single Kalman filter can be used for the whole particle set, which substantially reduces the computational cost of the RBPF.*

Remark 2.13 (Triangular model). *If (2.139) is such that $F_k^n = 0$, the system (S) is called triangular. In this scenario, x_k^n does not depend on x_{k-1}^l , which decouples the two filters entirely. The Kalman pre-correction step can thus be omitted, and the RBPF becomes a simple interleaving between a PF and a collection of Kalman filters.*

CHAPTER 3

GAUSSIAN PROCESSES

Contents

3.1	Introduction	46
3.2	Definition and Properties	46
3.2.1	Basic Properties	47
3.2.2	Common Kernels	49
3.3	Gaussian Process Regression	52
3.3.1	Simple Kriging	54
3.3.2	Ordinary Kriging	57
3.3.3	Universal Kriging	58
3.4	Reduced-Rank Kernel Approximation	60
3.4.1	Mercer's Theorem	60
3.4.2	Efficient Kriging Estimation	61
3.4.3	Karhunen-Loève Decomposition	63
3.5	Reproducing Kernel Hilbert Spaces	64
3.5.1	Reproducing Property and Definition	64
3.5.2	Link with Gaussian Processes	66
3.5.3	Kernel Mean Embedding	67
3.5.4	Kernel Kalman Filter	67

3.1 Introduction

The analytical and Monte Carlo methods of chapter 2 all assume the observation function h_k to be fully known. However, many navigation scenarios depend on a physical field that is either unavailable or known only through scarce and noisy measurements.

A first way to fill this gap is to reconstruct the field directly from the measurements: **TODO: j'avais oublié, mais à reprendre + présenter des cas d'application au krigeage**

- Deterministic interpolation, through nearest-neighbour selection, multilinear interpolation over a regular grid, or spline fitting;
- Linear regression on a fixed basis, such as polynomials or sinusoids;
- More expressive models, such as neural networks.

All of them return a single value at the queried location, without quantifying how much it can be trusted. Their use in the filters of chapter 2, however, requires the observation likelihood $p(z_k|x_k)$, a full distribution rather than a point prediction. The reconstruction error must then be absorbed into the observation noise, usually as a covariance held constant over the whole domain, although this error is almost zero next to a measurement and grows in sparsely sampled regions.

This chapter instead models such fields as a Gaussian process (GP), an adaptable framework that encodes both a prior expectation of the field and its spatial correlation structure. Together with its uncertainty, the resulting estimator can then serve as a surrogate observation model [56, 57] for the filters of chapter 2.

The rest of this chapter is organised as follows:

- Section 3.2 defines GPs through their mean functions and covariance kernels. It discusses the choice of kernel.
- Section 3.3 then introduces kriging as a map reconstruction tool in situations where we are given preexisting data samples.
- Section 3.4 addresses the cost of inverting large Gram matrices in kriging by approximating k with a reduced-rank kernel, therefore lowering the effective complexity of kriging methods.
- Finally, section 3.5 derives two representations of the mean function over a dictionary of functions, used to reconstruct the field when no prior knowledge is available.

3.2 Definition and Properties

Formally, a stochastic process generalises the notion of random variable to a random function, *i.e.* a family $(f(x))_{x \in \mathcal{U}}$ of random variables indexed by a set $\mathcal{U}$, which may represent time $\mathbb{R}_+$, physical space $\mathbb{R}^2$ or $\mathbb{R}^3$, or any other domain. As shown by Definition 3.1, a GP is a special case of stochastic process [58].

Definition 3.1 (Gaussian process). *For $d_z \in \mathbb{N}^*$, a stochastic process $(f(x))_{x \in \mathcal{U}}$ taking values in $\mathbb{R}^{d_z}$ is a GP if every finite linear combination of its values follows a Gaussian distribution. Formally, for all $n \in \mathbb{N}^*$, all $x_1, \dots, x_n \in \mathcal{U}$ and all $a_1, \dots, a_n \in \mathbb{R}^{d_z}$, there exist $\mu \in \mathbb{R}^{d_z}$ and $\Sigma \in \mathcal{S}_{d_z}^+(\mathbb{R})$ such that:*

$$\sum_{i=1}^n a_i^\top f(x_i) \sim \mathcal{N}(\mu, \Sigma) \quad (3.1)$$

The GP is said to be non-degenerate if $\Sigma \succ 0$ whenever the $x_1, \dots, x_n$ are pairwise distinct and the coefficients $a_1, \dots, a_n$ are not all zero.

A GP is associated with a mean function:

$$m : \begin{cases} \mathcal{U} \rightarrow \mathbb{R}^{d_z} \\ x \mapsto \mathbb{E}[f(x)] \end{cases} \quad (3.2)$$

and a covariance kernel:

$$k : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ x, x' \mapsto \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))^\top] \end{cases} \quad (3.3)$$

3.2.1 Basic Properties

As shown by Proposition 3.1, m and k entirely characterise every finite-dimensional marginal of f . By Kolmogorov's extension theorem, they also uniquely determine the law of f . We thus write $f \sim \mathcal{GP}(m, k)$ for a GP with mean function m and covariance kernel k .

Proposition 3.1 (Finite-dimensional marginals). *For $n \in \mathbb{N}^*$, let $x_1, \dots, x_n$ be pairwise distinct elements of $\mathcal{U}$. The joint distribution of $F \triangleq [f(x_1)^\top, \dots, f(x_n)^\top]^\top$ is the multivariate Gaussian distribution:*

$$\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \right) \quad (3.4)$$

where each block $k(x_i, x_j) \in \mathcal{M}_{d_z, d_z}(\mathbb{R})$.

Proof. According to Definition 3.1, every linear combination:

$$a^\top F = \sum_{i=1}^n a_i^\top f(x_i), \quad a \triangleq \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^{nd_z} \quad (3.5)$$

is Gaussian. By the Cramér-Wold theorem [59] F is thus also a Gaussian vector of $\mathbb{R}^{nd_z}$.

Such a vector is entirely determined by its mean and covariance, which we compute. In particular, we have:

$$\mathbb{E}[F] = \begin{bmatrix} \mathbb{E}[f(x_1)] \\ \vdots \\ \mathbb{E}[f(x_n)] \end{bmatrix} = \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \quad (3.6)$$

By definition of k , it comes that:

$$\begin{aligned}\mathbb{V}[F] &= \left[\mathbb{E} \left[(f(x_i) - m(x_i))(f(x_j) - m(x_j))^\top \right] \right]_{i,j=1}^n \\ &= \left[k(x_i, x_j) \right]_{i,j=1}^n\end{aligned}\tag{3.7}$$

which is exactly (3.4). $\square$

For two tuples of inputs $\mathbf{x} \triangleq (x_1, \dots, x_n) \in \mathcal{U}^n$ and $\mathbf{x}' \triangleq (x'_1, \dots, x'_p) \in \mathcal{U}^p$ of sizes $n, p \in \mathbb{N}^*$, we introduce the following vectorised notation:

$$f(\mathbf{x}) \triangleq \begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix}\tag{3.8}$$

$$m(\mathbf{x}) \triangleq \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}\tag{3.9}$$

$$k(\mathbf{x}, \mathbf{x}') \triangleq \begin{bmatrix} k(x_1, x'_1) & \cdots & k(x_1, x'_p) \\ \vdots & \ddots & \vdots \\ k(x_n, x'_1) & \cdots & k(x_n, x'_p) \end{bmatrix}\tag{3.10}$$

Using these notations, Proposition 3.1 reads compactly:

$$\forall n \in \mathbb{N}^*, \forall \mathbf{x} \in \mathcal{U}^n \text{ pairwise distinct, } f(\mathbf{x}) \sim \mathcal{N}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}))\tag{3.11}$$

A GP is thus a consistent generalisation of the multivariate Gaussian distribution to an infinite and potentially continuous index set. Both of its parameters admit a direct interpretation. The mean function m encodes prior knowledge about the average behaviour of f , while the kernel k captures the correlation structure. The diagonal blocks $k(x, x)$ represent the covariance of $f(x)$, while the off-diagonal blocks $k(x, x')$, for $x \neq x'$, describe the cross-covariance between $f(x)$ and $f(x')$.

In order for the GP to be non-degenerate, the matrix $k(\mathbf{x}, \mathbf{x})$ must be SPD in the sense of Definition 2.1 for every finite tuple $\mathbf{x}$ of pairwise distinct inputs. Definition 3.2 recasts this as a condition on k alone [60], and Proposition 3.2 proves the equivalence between the two.

Definition 3.2 (Symmetric positive definite kernel). *A kernel $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is symmetric positive semi-definite (SPSD) if it is:*

- *Symmetric:*

$$\forall x, x' \in \mathcal{U}, k(x', x) = k(x, x')^\top\tag{3.12}$$

- *Positive semi-definite:*

$$\begin{aligned}\forall n \in \mathbb{N}^*, \forall x_1, \dots, x_n \in \mathcal{U}, \forall a_1, \dots, a_n \in \mathbb{R}^{d_z}, \\ \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j \geq 0\end{aligned}\tag{3.13}$$

If the inequality is strict for every choice of pairwise distinct points $x_1, \dots, x_n$ and coefficients $a_1, \dots, a_n$ not all zero, then k is symmetric positive definite (SPD).

Proposition 3.2 (Definite kernel/matrix equivalence). *The following equivalences hold:*

- $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is a SPSP kernel if and only if, for every $n \in \mathbb{N}^*$ and every $\mathbf{x} \in \mathcal{U}^n$, $k(\mathbf{x}, \mathbf{x}) \in \mathcal{M}_{nd_z, nd_z}(\mathbb{R})$ is a SPSP matrix.
- k is a SPD kernel if and only if, for every such tuple $\mathbf{x}$ of pairwise distinct points, $k(\mathbf{x}, \mathbf{x})$ is a SPD matrix.

Proof. See section B.3. □

Remark 3.1 (Necessity of distinct points). *While pairwise distinctness does not intervene in the computation of the strict version above, it is necessary for the SPD condition to be satisfiable. Indeed, at coinciding inputs $x_1 = x_2$, the choice $a_1 = -a_2 = 1$, with the other coefficients set to zero, cancels the quadratic form for non-zero coefficients, so no kernel could be SPD.*

Remark 3.2 (Non-diagonal blocks). *While Proposition 3.2 equates the SPDness of k with that of $k(\mathbf{x}, \mathbf{x})$ at distinct points, it does not make the off-diagonal blocks $k(x, x')$ SPD themselves. For $x \neq x'$, these may be non-positive or even non-symmetric.*

Throughout the rest of this document, we assume that the covariance kernel k is SPD and that the points of every collection $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{U}^n$ are pairwise distinct. This ensures that we only work with non-degenerate GPs.

3.2.2 Common Kernels

Assume in this section that $\mathcal{U}$ is a finite-dimensional normed vector space, *i.e.* $\mathcal{U} = \mathbb{R}^{d_x}$ for some $d_x \in \mathbb{N}^*$, up to isomorphism. A natural way to build kernels on $\mathcal{U}$ is to let the correlation between the field values at two points x, x' depend only on the distance $\|x - x'\|$ between them. Kernels of this form are called radial kernels [61], and are formally defined by Definition 3.3.

Since f typically represents a physical field, observations tend to become less correlated as the distance between them increases. Definition 3.4 introduces a common family of radial kernels modelling this behaviour, called the Matérn kernels [62]. Proposition 3.3 then establishes their continuity and SPDness.

Definition 3.3 (Radial kernel). *A kernel $k : \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is radial if there exists a function $\kappa : \mathbb{R}_+ \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ and a norm $\|\cdot\|$ on $\mathbb{R}^{d_x}$ such that:*

$$\forall x, x' \in \mathbb{R}^{d_x}, k(x, x') = \kappa(\|x - x'\|) \quad (3.14)$$

To simplify notations, we identify κ with k in the remainder of this document, such that we write:

$$\forall x, x' \in \mathbb{R}^{d_x}, k(x, x') = k(\|x - x'\|) \quad (3.15)$$

Definition 3.4 (Matérn kernel). Assume that $d_z = 1$, and let $\nu, l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel of smoothness ν , correlation length l , and characteristic variance σ_f^2 is the radial kernel defined as:

$$k_\nu : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{l} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} r}{l} \right) \end{cases} \quad (3.16)$$

where Γ is the Gamma function, and K_ν is the modified Bessel function of the second kind [63].

This family extends to $\nu = +\infty$ by taking the pointwise limit, yielding the squared exponential (SE) kernel:

$$k_{\text{SE}} \triangleq k_{+\infty} : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \exp \left(-\frac{r^2}{2l^2} \right) \end{cases} \quad (3.17)$$

Proposition 3.3 (Properties of Matérn kernels). Let $\nu \in \mathbb{R}_+^* \cup \{+\infty\}$ and $l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel associated with ν , l and σ_f^2 is a continuous SPD radial kernel.

Proof. See section B.4. □

The parameter ν controls the regularity. Indeed, for $p \in \mathbb{N}$, it can be shown [64] that a GP equipped with k_ν is p times mean-square differentiable if and only if $\nu > p$. For half-integer $\nu = p + \frac{1}{2}$, Matérn kernels can be written as the product of an exponential and a polynomial function of degree p :

$$k_{p+1/2}(r) = \sigma_f^2 \exp \left(-\frac{\sqrt{2p+1}r}{l} \right) \frac{\Gamma(p+1)}{\Gamma(2p+1)} \sum_{i=0}^p \frac{(p+i)!}{i!(p-i)!} \left(\frac{2\sqrt{2p+1}r}{l} \right)^{p-i} \quad (3.18)$$

In particular, the following kernels correspond to usual choices of ν , the SE kernel being recalled here for comparison:

- **Exponential kernel.**

$$k_{1/2}(r) = \sigma_f^2 \exp \left(-\frac{r}{l} \right) \quad (3.19)$$

- **Once-differentiable Matérn kernel.**

$$k_{3/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{3}r}{l} \right) \exp \left(-\frac{\sqrt{3}r}{l} \right) \quad (3.20)$$

- **Twice-differentiable Matérn kernel.**

$$k_{5/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{l} + \frac{5r^2}{3l^2} \right) \exp \left(-\frac{\sqrt{5}r}{l} \right) \quad (3.21)$$

- **Squared exponential kernel.**

$$k_{+\infty}(r) = \sigma_f^2 \exp \left(-\frac{r^2}{2l^2} \right) \quad (3.22)$$

Figure 3.1 illustrates the radial decay of these kernels, evaluated with unit variance $\sigma_f^2 = 1$ and correlation length $l = 1$ to facilitate comparison.

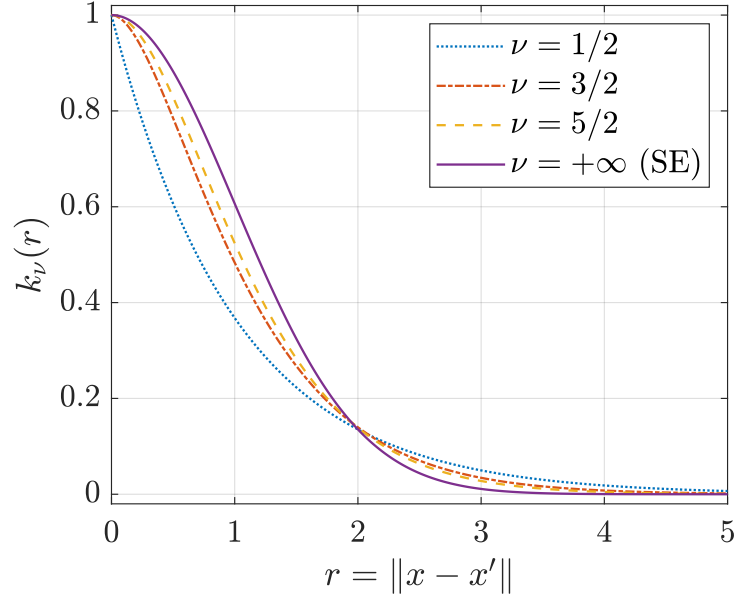


Figure 3.1: Radial profile of Matérn kernels with increasing smoothness, together with their squared exponential limit. Larger values of ν produce smoother kernels, with flatter behaviours near the origin and faster-decaying tails.

Remark 3.3 (Anisotropic kernel). *The norm used in Definition 3.3 does not necessarily need to be the canonical norm of $\mathbb{R}^{d_x}$. In particular, for $M \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$, the Mahalanobis norm:*

$$x \in \mathbb{R}^{d_x} \mapsto \|x\|_M \triangleq \sqrt{x^\top M^{-1} x} \quad (3.23)$$

may produce anisotropic radial kernels in situations where M is not proportional to the identity matrix.

The main limitation of Matérn kernels is that they are scalar, while our observation model takes values in $\mathbb{R}^{d_z}$. To alleviate this issue, Proposition 3.4 presents a method to create a vector-valued kernel from a scalar one.

Proposition 3.4 (Separable kernel). *Let $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ be a scalar SPD kernel, and let $\Sigma \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$. Then:*

$$k_{\text{sep}} : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ x, x' \mapsto \Sigma k(x, x') \end{cases} \quad (3.24)$$

is a valid SPD kernel for an $\mathbb{R}^{d_z}$ -valued GP.

Furthermore, when k is continuous, k_{sep} is also continuous.

Proof. The symmetry and potential continuity of k_{sep} are direct from that of k and Σ . The positive definiteness can be established by letting A be a Cholesky factor of the SPD matrix Σ , and considering, for every $n \in \mathbb{N}^*$ and every not all zero coefficients $a_1, \dots, a_n \in \mathbb{R}^{d_z}$:

$$\forall i \in \llbracket 1, n \rrbracket, b_i \triangleq A^\top a_i \quad (3.25)$$

Since A is invertible, these coefficients are not all zero, and thus:

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=1}^n a_i^\top k_{\text{sep}}(x_i, x_j) a_j &= \sum_{i=1}^n \sum_{j=1}^n a_i^\top A A^\top a_j k(x_i, x_j) \\
&= \sum_{i=1}^n \sum_{j=1}^n b_i^\top b_j k(x_i, x_j) \\
&= \sum_{l=1}^{d_z} \sum_{i=1}^n \sum_{j=1}^n b_i^{(l)} b_j^{(l)} k(x_i, x_j)
\end{aligned} \tag{3.26}$$

where $b_i^{(l)}$ denotes the l -th component of b_i . Each inner double sum is a quadratic form of k , hence non-negative, and at least one of them is strictly positive since at least one of the vectors $(b_1^{(l)}, \dots, b_n^{(l)})$ is non-zero. $\square$

Remark 3.4 (Multi-scale kernels). For $n \in \mathbb{N}^*$, $\Sigma_1, \dots, \Sigma_n \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ and $k^{(1)}, \dots, k^{(n)}$ Matérn kernels of correlation lengths $l_1 < \dots < l_n \in \mathbb{R}_+^*$, the multi-scale kernel is:

$$k_{\text{MS}} : \begin{cases} \mathbb{R}_+ \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ r \mapsto \sum_{i=1}^n \Sigma_i k^{(i)}(r) \end{cases} \tag{3.27}$$

Each of its terms is separable in the sense of Proposition 3.4, hence SPD, and a sum of SPD kernels with positive coefficients is itself SPD. When the correlation lengths have different orders of magnitude, such kernels can be used to capture local fluctuations and global trends simultaneously.

Every kernel considered in this section is specified by a set of hyperparameters, such as ν , l and σ_f^2 for a Matérn kernel, Σ for its separable extension, or one such set per scale for a multi-scale model. Their estimation is a standard step in GP modelling, and is independent of bayesian estimation considerations. However, the specific characteristics of our applications can be used to improve the estimation of such hyperparameters. These aspects are thus discussed in section 4.2.2, and the hyperparameters are assumed to be fully specified for the rest of this chapter.

3.3 Gaussian Process Regression

Gaussian process regression, also known as kriging [65, 66, 67], estimates the value of a field at an unobserved location using noisy measurements of the same field at known locations. Formally, let $f \sim \mathcal{GP}(m, k) : \mathcal{U} \rightarrow \mathbb{R}^{d_z}$, and $x \in \mathcal{U}$. We wish to estimate the law of:

$$z \triangleq f(x) + \varepsilon \in \mathbb{R}^{d_z}, \quad \varepsilon \sim \mathcal{N}(0, R), \quad R \in \mathcal{S}_{d_z}^+(\mathbb{R}) \tag{3.28}$$

where ε is independent of f .

While kriging usually aims to reconstruct $f(x)$ itself, the query noise ε models the observation uncertainty, so that kriging estimates the noisy observation from (2.1). The standard formulation can nonetheless be recovered by taking $R = 0$.

Without further information regarding f , we would only be able to state that z follows a Gaussian distribution of expectation $m(x)$ and covariance $k(x, x) + R$. However, we are given $\tilde{N} \in \mathbb{N}^*$ distinct training points $\tilde{\mathbf{x}} \triangleq (\tilde{x}_1, \dots, \tilde{x}_{\tilde{N}}) \in \mathcal{U}^{\tilde{N}}$ and their corresponding noisy observations:

$$\tilde{\mathbf{z}} \triangleq \begin{bmatrix} f(\tilde{x}_1) + \varepsilon_1 \\ \vdots \\ f(\tilde{x}_{\tilde{N}}) + \varepsilon_{\tilde{N}} \end{bmatrix} \in \mathbb{R}^{\tilde{N}d_z}, \quad \varepsilon_i \sim \mathcal{N}(0, \tilde{R}), \quad \tilde{R} \in \mathcal{S}_{d_z}^+(\mathbb{R}) \quad (3.29)$$

where $\varepsilon_1, \dots, \varepsilon_{\tilde{N}}$ are mutually independent and independent of f .

The goal of this section is to derive the best linear unbiased estimator (BLUE) of z with respect to $\tilde{\mathbf{z}}$, *i.e.* the estimator:

$$\hat{z} = W^\top \tilde{\mathbf{z}} \quad (3.30)$$

parametrised by a weight matrix:

$$W \triangleq \begin{bmatrix} w_1 \\ \vdots \\ w_{\tilde{N}} \end{bmatrix} \in \mathcal{M}_{\tilde{N}d_z, d_z}(\mathbb{R}) \quad (3.31)$$

such that the error vector has an expectation of 0 and a minimum covariance. This setting is illustrated by Figure 3.2 for a $\mathbb{R}^2 \rightarrow \mathbb{R}$ field.

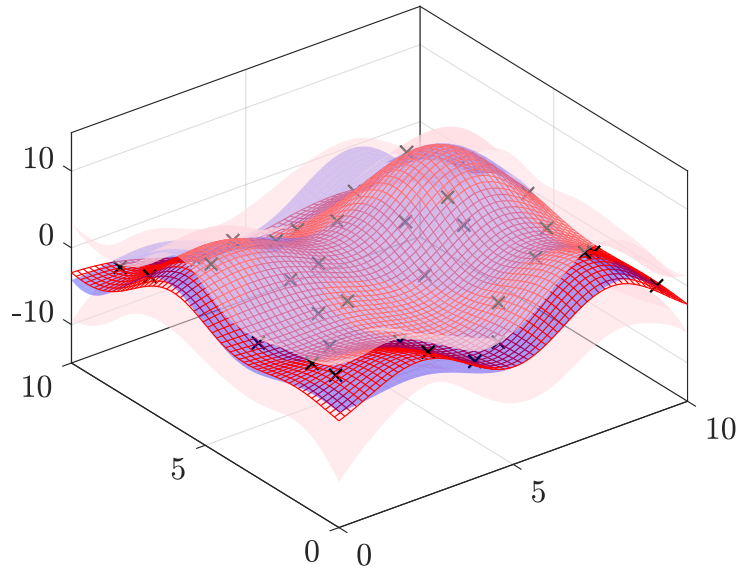


Figure 3.2: Ordinary kriging (red mesh) of an unknown field (blue surface) from noisy samples (black crosses). The confidence band (red envelope) is proportional to the square root of the interpolation covariance, and widens away from the samples.

Since $\hat{z} - z$ is a vector, its covariance is a matrix, and cannot be minimised as such. To alleviate this issue, we minimise its trace, which is exactly the mean squared error:

$$\text{tr}(\mathbb{V}[\hat{z} - z]) = \mathbb{E} [\|\hat{z} - z\|^2] \quad (3.32)$$

In the following, we present three kriging estimates, based on increasingly relaxed assumptions made on our knowledge of the underlying GP. Each method relies on the inversion of the Gram matrix:

$$\begin{aligned} G(\tilde{\mathbf{x}}) &\triangleq k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R} \\ &= \begin{bmatrix} k(\tilde{x}_1, \tilde{x}_1) + \tilde{R} & \cdots & k(\tilde{x}_1, \tilde{x}_{\tilde{N}}) \\ \vdots & \ddots & \vdots \\ k(\tilde{x}_{\tilde{N}}, \tilde{x}_1) & \cdots & k(\tilde{x}_{\tilde{N}}, \tilde{x}_{\tilde{N}}) + \tilde{R} \end{bmatrix} \end{aligned} \quad (3.33)$$

of $\tilde{\mathbf{x}}$ with respect to f , where $\otimes$ is the Kronecker matrix product. Since k is SPD and the training points are distinct, it comes that:

$$G(\tilde{\mathbf{x}}) \in \mathcal{S}_{\tilde{N}d_z}^{++}(\mathbb{R}) \quad (3.34)$$

and so $G(\tilde{\mathbf{x}})$ is invertible, even in the noiseless case $\tilde{R} = 0$.

We also assume that the query point x differs from the training points $\tilde{\mathbf{x}}$, so that the joint law of $(z, \tilde{\mathbf{z}})$ is non-degenerate. This hypothesis is not restrictive and is only needed for the proofs. Indeed, the estimators are continuous in x and thus remain valid at the training points as soon as m and k are continuous, which we assume.

Moreover, $\mathbb{V}[\hat{z} - z]$ is a covariance matrix, and therefore SPSPD whatever the weight matrix W . It is furthermore positive definite when x is not one of the training points, since $\hat{z} - z$ is a non-degenerate linear combination of z and $\tilde{\mathbf{z}}$.

3.3.1 Simple Kriging

Simple kriging (SK) is the simplest form of kriging, and assumes that the mean function m is known. Proposition 3.5 gives a closed-form expression of the SK estimator and its associated covariance matrix, while Proposition 3.7 shows that it coincides with the conditional distribution of z given $\tilde{\mathbf{z}}$.

Proposition 3.5 (Simple kriging estimator). *Using the notations introduced in section 3.3, the SK estimator of z and its error covariance are:*

$$\begin{cases} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = m(x) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\ R_{\text{SK}}(x, \tilde{\mathbf{x}}) = k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (3.35)$$

Proof. Since m is known, we may assume it to be identically equal to 0 by working with $f - m$ instead of f , without any loss of generality. In the centred case, (3.32) can be developed as follows:

$$\mathbb{V}[W^\top \tilde{\mathbf{z}} - z] = W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \quad (3.36)$$

By symmetry of $G(\tilde{\mathbf{x}})$, differentiating (3.32) with respect to W yields:

$$\begin{aligned} &\frac{\partial}{\partial W} (\text{tr} (W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R)) = 0 \\ \iff &\frac{\partial}{\partial W} (\text{tr} (W^\top G(\tilde{\mathbf{x}}) W)) - 2 \frac{\partial}{\partial W} (\text{tr} (W^\top k(\tilde{\mathbf{x}}, x))) = 0 \\ \iff &2G(\tilde{\mathbf{x}}) W - 2k(\tilde{\mathbf{x}}, x) = 0 \\ \iff &W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (3.37)$$

The SK estimator is thus:

$$\hat{z}_{\text{SK}} = W^\top \tilde{\mathbf{z}} = k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \quad (3.38)$$

and its corresponding covariance:

$$\begin{aligned} R_{\text{SK}} &= W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\ &= k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} G(\tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &\quad - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) + k(x, x) + R \\ &= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (3.39)$$

The general case is obtained by adding m back into f . $\square$

Remark 3.5 (Additive query noise). *In Proposition 3.5, the term R appears because the noisy query z is predicted rather than the field value $f(x)$. The two share the same estimator, and differ in covariance only by an additive R term. Kriging the field can thus be done by subtracting R , or equivalently by taking $R = 0$ as mentioned in the introduction to kriging.*

Remark 3.6 (Error covariance minimisation). *While the SK weight matrix $W = G^{-1}k(\tilde{\mathbf{x}}, x)$ was obtained by minimising the trace of $\mathbb{V}[\hat{z} - z]$, it minimises the whole covariance in the SPSPD sense. Indeed:*

$$\begin{aligned} \mathbb{V}[\hat{z} - z] - R_{\text{SK}} &= W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &= (W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x))^\top G(\tilde{\mathbf{x}}) (W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)) \\ &\succeq 0 \end{aligned} \quad (3.40)$$

with equality if and only if $W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)$ since $G(\tilde{\mathbf{x}}) \succ 0$.

Proposition 3.6 (Exact interpolation in the noiseless case). *Assume that $\tilde{R} = 0$, and let $i \in \llbracket 1, \tilde{N} \rrbracket$. The SK estimator interpolates the data at $\tilde{x}_i$ exactly:*

$$\begin{cases} \hat{z}_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = \tilde{z}_i \\ R_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}) = R \end{cases} \quad (3.41)$$

In particular, the error covariance reduces to the query noise R alone, and vanishes when $R = 0$. That degenerate case is the one set aside at the start of this section, and is reached by continuity in x . On the contrary, a positive training noise $\tilde{R} \succ 0$ turns kriging into a smoother rather than an interpolator [68].

Proof. Since $\tilde{R} = 0$, the Gram matrix reduces to $G(\tilde{\mathbf{x}}) = k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})$. Let $e_i \triangleq [0, \dots, 0, 1, 0, \dots, 0]^\top$ be the i -th canonical basis vector of $\mathbb{R}^{\tilde{N}}$. The matrix $e_i^\top \otimes I_{d_z}$ extracts the i -th block of a stacked matrix, such that:

$$(e_i^\top \otimes I_{d_z}) G(\tilde{\mathbf{x}}) = (e_i^\top \otimes I_{d_z}) k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) = k(\tilde{x}_i, \tilde{\mathbf{x}}) \quad (3.42)$$

The kriging weights thus reduce to this selector:

$$k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} = (e_i^\top \otimes I_{d_z}) G(\tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} = e_i^\top \otimes I_{d_z} \quad (3.43)$$

Evaluating (3.35) at $x = \tilde{x}_i$ by the continuity yields:

$$\begin{aligned}
\hat{z}_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) &= m(\tilde{x}_i) + k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\
&= m(\tilde{x}_i) + (e_i^\top \otimes I_{d_z}) (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\
&= m(\tilde{x}_i) + \tilde{z}_i - m(\tilde{x}_i) \\
&= \tilde{z}_i
\end{aligned} \tag{3.44}$$

Similarly:

$$\begin{aligned}
R_{\text{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}) &= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, \tilde{x}_i) \\
&= k(\tilde{x}_i, \tilde{x}_i) + R - (e_i^\top \otimes I_{d_z}) k(\tilde{\mathbf{x}}, \tilde{x}_i) \\
&= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{x}_i) \\
&= R
\end{aligned} \tag{3.45}$$

□

Proposition 3.7 (Gaussian conditioning interpretation). *The conditional distribution of z knowing $\tilde{\mathbf{z}}$ is Gaussian, parametrised by the SK estimator and its error covariance matrix:*

$$z \mid \tilde{\mathbf{z}} \sim \mathcal{N}(\hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m), R_{\text{SK}}(x, \tilde{\mathbf{x}})) \tag{3.46}$$

Proof. Since they are evaluations of the same GP under an additive Gaussian white noise, z and $\tilde{\mathbf{z}}$ are jointly Gaussian. Their joint distribution is:

$$\begin{bmatrix} z \\ \tilde{\mathbf{z}} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x) \\ m(\tilde{\mathbf{x}}) \end{bmatrix}, \begin{bmatrix} k(x, x) + R & k(x, \tilde{\mathbf{x}}) \\ k(\tilde{\mathbf{x}}, x) & G(\tilde{\mathbf{x}}) \end{bmatrix} \right) \tag{3.47}$$

As noted at the start of this section, this joint covariance is SPD. The Gaussian conditioning formula (see Proposition A.1) can thus be used to give the desired result. □

This result is a special case of the more general result established in Proposition 2.1. Since the unbiased hypothesis of the BLUE is automatically verified by the SK estimator, the minimisation criterion (3.32) amounts to finding the MMSE estimator of z given $\tilde{\mathbf{z}}$, which is always the conditional expectation.

Remark 3.7 (Simultaneous estimation). *The simultaneous estimation of f at several query points $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{U}^n$ can be done by directly replacing x with $\mathbf{x}$ in (3.35). In particular, the $\mathcal{O}(\tilde{N}^3)$ cost of inverting $G(\tilde{\mathbf{x}})$ can be shared across the n estimations whether they are performed successively or simultaneously.*

However, computing only the n diagonal blocks of $R_{\text{SK}}(\mathbf{x}, \tilde{\mathbf{x}})$ costs $\mathcal{O}(\tilde{N}^3 + n\tilde{N}^2)$, whereas forming the full matrix before discarding its off-diagonal blocks costs $\mathcal{O}(\tilde{N}^3 + n\tilde{N}^2 + n^2\tilde{N})$ due to the extra off-diagonal matrix products. The off-diagonal blocks describe the correlation between the estimation errors made at distinct query points, and should only be computed when that correlation is needed.

3.3.2 Ordinary Kriging

Assuming the mean function to be known is a strong assumption, and can be unrealistic in practical scenarios. While an approximated model could be used instead of the true m , this would cause the kriging estimator to be overly confident in its estimation. Ordinary kriging (OK) extends its simple counterpart by assuming that the mean is an unknown constant $m \in \mathbb{R}^{d_z}$, and estimating it alongside the weights.

The unbiased assumption of the OK estimator yields:

$$\begin{aligned}
 & \mathbb{E}[\hat{z} - z] = 0 \\
 \iff & W^\top \mathbb{E}[\tilde{\mathbf{z}}] - \mathbb{E}[z] = 0 \\
 \iff & \left(\sum_{i=1}^{\tilde{N}} w_i^\top - I_{d_z} \right) m = 0 \\
 \iff & C^\top W - I_{d_z} = 0
 \end{aligned} \tag{3.48}$$

with:

$$C \triangleq \mathbf{1}_{\tilde{N}} \otimes I_{d_z} \in \mathcal{M}_{\tilde{N}d_z, d_z}(\mathbb{R}), \quad \mathbf{1}_{\tilde{N}} \triangleq \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^{\tilde{N}} \tag{3.49}$$

The OK weights thus minimise (3.32) under the constraint (3.48). Using the Lagrange multiplier method [69], Proposition 3.8 derives the corresponding estimator and covariance matrix, and Remark 3.8 establishes the link between ordinary and simple kriging.

Proposition 3.8 (Ordinary kriging estimator). *The OK estimator of z given $\tilde{\mathbf{z}}$ and its error covariance are:*

$$\begin{cases} \hat{z}_{\text{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - C \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})) \\ R_{\text{OK}}(x, \tilde{\mathbf{x}}) = R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \tag{3.50}$$

with:

$$\begin{cases} \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = (C^\top G(\tilde{\mathbf{x}})^{-1} C)^{-1} C^\top G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) = (I_{d_z} - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} C) (C^\top G(\tilde{\mathbf{x}})^{-1} C)^{-1} \\ \quad (I_{d_z} - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} C)^\top \end{cases} \tag{3.51}$$

and $R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) \succeq 0$.

Proof. See section B.5. □

Remark 3.8 (Comparison between simple and ordinary kriging). *The formulas obtained in (3.50) closely resemble those of simple kriging. In fact, they can be rewritten as follows:*

$$\begin{cases} \hat{z}_{\text{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})) \\ R_{\text{OK}}(x, \tilde{\mathbf{x}}) = R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) + R_{\text{SK}}(x, \tilde{\mathbf{x}}) \end{cases} \tag{3.52}$$

where, by abuse of notation, $\hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ denotes either the estimated mean vector or its replication for each the training point $\tilde{\mathbf{x}}$, namely:

$$\underbrace{\begin{bmatrix} \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ \vdots \\ \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \end{bmatrix}}_{\tilde{N} \text{ times}} = C \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \quad (3.53)$$

Ordinary kriging can therefore be interpreted as simple kriging in which the mean m is concurrently estimated from the observations. The uncertainty associated with this second estimation is captured by the covariance term $R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) \succeq 0$, and is added to the SK covariance.

3.3.3 Universal Kriging

Universal kriging (UK) [70] generalises the constant mean of ordinary kriging to a linear combination of $r_{\text{UK}} \in \mathbb{N}^*$ known trend functions $g_1, \dots, g_{r_{\text{UK}}} : \mathcal{U} \rightarrow \mathbb{R}$:

$$m(x) = g(x) \beta = \sum_{i=1}^{r_{\text{UK}}} g_i(x) \beta_i \quad (3.54)$$

where $g(x)$ is the trend matrix:

$$g(x) \triangleq \begin{bmatrix} g_1(x) & \cdots & g_{r_{\text{UK}}}(x) \end{bmatrix} \otimes I_{d_z} \in \mathcal{M}_{d_z, r_{\text{UK}} d_z}(\mathbb{R}) \quad (3.55)$$

and β gathers the unknown coefficients $\beta_i \in \mathbb{R}^{d_z}$:

$$\beta \triangleq \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_{r_{\text{UK}}} \end{bmatrix} \in \mathbb{R}^{r_{\text{UK}} d_z} \quad (3.56)$$

The trend matrix at the training points stacks these evaluations:

$$g(\tilde{\mathbf{x}}) = \begin{bmatrix} g(\tilde{x}_1) \\ \vdots \\ g(\tilde{x}_{\tilde{N}}) \end{bmatrix} \in \mathcal{M}_{\tilde{N} d_z, r_{\text{UK}} d_z}(\mathbb{R}) \quad (3.57)$$

so that $m(\tilde{\mathbf{x}}) = g(\tilde{\mathbf{x}}) \beta$ lies in $\mathbb{R}^{\tilde{N} d_z}$, as does $\tilde{\mathbf{z}}$.

This matrix is assumed to have full column rank, which requires at least $\tilde{N} \geq r_{\text{UK}}$ training points, and the trend functions to be linearly independent once restricted to the training points. Proposition 3.9 generalises Proposition 3.8 accordingly.

Proposition 3.9 (Universal kriging estimator). *Assume that $g(\tilde{\mathbf{x}})$ has full column rank. The UK estimator of z given $\tilde{\mathbf{z}}$ and its error covariance are:*

$$\begin{cases} \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = g(x) \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - g(\tilde{\mathbf{x}}) \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)) \\ R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) = R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (3.58)$$

with:

$$\begin{cases} \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = (g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}))^{-1} g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) = (g(x) - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})) (g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}))^{-1} \\ \quad (g(x) - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}))^\top \end{cases} \quad (3.59)$$

and $R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) \succeq 0$.

Proof. See section B.6. $\square$

Remark 3.9 (Link with ordinary and simple kriging). *As mentioned in the beginning of this section, ordinary kriging is a special case of universal kriging, which arises when $r_{\text{UK}} = 1$ and $g_1 \equiv 1$. In this scenario, the trend matrices reduce to $g(x) = I_{d_z}$ and $g(\tilde{\mathbf{x}}) = C$. This, in turn, yields:*

$$\begin{cases} \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 1) = \hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, 1) = R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) \end{cases} \quad (3.60)$$

and thus:

$$\begin{cases} \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 1) = \hat{z}_{\text{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ R_{\text{UK}}(x, \tilde{\mathbf{x}}, 1) = R_{\text{OK}}(x, \tilde{\mathbf{x}}) \end{cases} \quad (3.61)$$

Similarly to Remark 3.8, universal kriging can be seen as the simultaneous simple kriging from the training points and mean estimation from the trend functions. Indeed, comparing (3.58) with Proposition 3.5 shows that the following relation:

$$\begin{cases} \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)) \\ R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) = R_{\hat{m}_{\text{UK}}}(x, \tilde{\mathbf{x}}, g) + R_{\text{SK}}(x, \tilde{\mathbf{x}}) \end{cases} \quad (3.62)$$

Remark 3.8 is the special case $r_{\text{UK}} = 1$, $g_1 \equiv 1$ of this reading, where the estimated mean function is constant.

Remark 3.10 (Kriging with a coarse model). *Universal kriging is particularly relevant when a coarse model of the field is available, such as a polynomial trend [71] or a simplified reference model [72]. In the simplest case, setting $r_{\text{UK}} = 1$, and choosing g_1 as this model amounts to kriging the discrepancy between the field and its approximation. Universal kriging can then be viewed as correcting the approximation, rather than reconstructing the field entirely from the observations.*

Remark 3.11 (Cost of estimating the mean). *In addition to the $\mathcal{O}(\tilde{N}^3)$ cost of inverting $G(\tilde{\mathbf{x}})$, and the $\mathcal{O}(\tilde{N}^2)$ required to query the simple kriging model, universal kriging incurs an additional cost to estimate the mean. This can be decomposed into an $\mathcal{O}(r_{\text{UK}}\tilde{N}^2 + r_{\text{UK}}^2\tilde{N} + r_{\text{UK}}^3)$ computation of $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})$, and an additional $\mathcal{O}(r_{\text{UK}}\tilde{N} + r_{\text{UK}}^2)$ per query to compute the estimated mean and covariance matrix.*

In particular, when $\tilde{N} \gg r_{\text{UK}}$, universal kriging retains the same dominant complexity as simple kriging.

Proposition 3.7 shows that simple kriging can be seen as a Gaussian conditioning, which ordinary and universal kriging do not admit as such, their estimators being BLUEs of an unknown mean rather than posterior expectations. This interpretation can nonetheless be recovered by using a Gaussian prior on the trend coefficients, as proven by Proposition 3.10 [73, 74].

Proposition 3.10 (Conditioning under a vague trend prior). *Let $B \in \mathcal{S}_{r_{\text{UK}}d_z}^{++}$, and assume that the UK coefficients follow:*

$$\beta \sim \mathcal{N}(0, B) \quad (3.63)$$

Then:

$$f \sim \mathcal{GP}(0, k_B) \quad (3.64)$$

where k_B is the augmented kernel:

$$k_B : x, x' \mapsto k(x, x') + g(x) B g(x')^\top \quad (3.65)$$

We write $\hat{z}_{\text{SK},B}$ and $R_{\text{SK},B}$ for the SK estimator and error covariance of Proposition 3.5 taken under k_B rather than k . The following limit holds:

$$z \mid \tilde{\mathbf{z}} \sim \mathcal{N}(\hat{z}_{\text{SK},B}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0), R_{\text{SK},B}(x, \tilde{\mathbf{x}})) \xrightarrow{B^{-1} \rightarrow 0} \mathcal{N}(\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g), R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)) \quad (3.66)$$

which gives an asymptotic equivalent of Proposition 3.7 for universal, and thus ordinary, kriging.

Proof. See section B.7. □

3.4 Reduced-Rank Kernel Approximation

Exact GP regression requires inverting the $\tilde{N}d_z \times \tilde{N}d_z$ Gram matrix $G(\tilde{\mathbf{x}})$, which scales as $\mathcal{O}(\tilde{N}^3)$. When the number of training points is large, this operation becomes prohibitive. Reduced-rank methods [75, 76] circumvent this issue by approximating the kernel as a linear combination of $r_{\text{KL}} \in \mathbb{N}^*$ basis functions, with $r_{\text{KL}} < \tilde{N}$. The inversion cost is reduced to $\mathcal{O}(r_{\text{KL}}^3)$, bringing down the overall kriging cost to $\mathcal{O}(r_{\text{KL}}^2 \tilde{N})$.

Mercer's theorem [77], presented in Proposition 3.12, establishes the existence of such a decomposition as soon as k is a continuous SPD kernel, which Matérn kernels satisfy, as proven by Proposition 3.3, and $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$. The Woodbury inversion formula is then used to rewrite the kriging estimator in a computationally efficient expression. Finally, the Karhunen-Loève decomposition is introduced as a way to represent the structure of a GP without relying on training data.

3.4.1 Mercer's Theorem

We start by generalising Definition 3.2 to functional operators on $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$.

Definition 3.5 (Self-adjoint positive semi-definite operator). *A functional operator T on $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ is self-adjoint positive semi-definite (SPSD) if:*

$$\begin{cases} \forall \phi, \phi' \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}), \langle T\phi | \phi' \rangle = \langle \phi | T\phi' \rangle & \text{self-adjoint} \\ \forall \phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}), \langle \phi | T\phi \rangle \geq 0 & \text{positive semi-definite} \end{cases} \quad (3.67)$$

where $\langle \cdot | \cdot \rangle$ is the following scalar product:

$$(\phi, \phi') \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})^2 \mapsto \langle \phi | \phi' \rangle \triangleq \int_{\mathcal{U}} \phi(x)^\top \phi'(x) dx \quad (3.68)$$

Proposition 3.11 (Mercer operator). *Assume that $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$ and k is a continuous SPD kernel. The Mercer operator:*

$$\mathcal{K} : \begin{cases} \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \rightarrow \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \\ \phi \mapsto \left(x \mapsto \int_{\mathcal{U}} k(x, y) \phi(y) dy \right) \end{cases} \quad (3.69)$$

is a compact SPSD functional operator.

Proof. See section B.8. □

Proposition 3.12 (Mercer's theorem). *Under the assumptions of Proposition 3.11, there exists a non-increasing sequence $(\lambda_i)_{i \in \mathbb{N}^*}$ of strictly positive eigenvalues of $\mathcal{K}$ converging to 0, and an associated orthonormal family $(\psi_i)_{i \in \mathbb{N}^*}$ of continuous $\mathbb{R}^{d_z}$ -valued eigenfunctions, such that:*

$$\forall x, x' \in \mathcal{U}, k(x, x') = \sum_{i=1}^{+\infty} \lambda_i \psi_i(x) \psi_i(x')^\top \quad (3.70)$$

Furthermore, this series converges absolutely and uniformly on $\mathcal{U} \times \mathcal{U}$.

Proof. See section B.9. □

Remark 3.12 (Definiteness without invertibility). *Since all the eigenvalues λ_i are strictly positive, the quadratic form:*

$$\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \mapsto \langle \phi | \mathcal{K} \phi \rangle = \sum_{i=1}^{+\infty} \lambda_i \langle \phi | \psi_i \rangle^2 \quad (3.71)$$

is strictly positive for every $\phi \neq 0$, hence positive definite. For a symmetric matrix, definiteness amounts to invertibility, but this equivalence fails in infinite dimension. In particular, since $\mathcal{K}$ is compact with $\lambda_i \rightarrow 0$, it admits no positive lower bound on the unit sphere, so 0 belongs to its spectrum. $\mathcal{K}^{-1}$ is thus unbounded, and so the Mercer operator is positive definite yet not invertible.

In finite dimension, strict positivity and invertibility coincide and jointly characterise SPD matrices, whereas this is not the case in infinite dimension. This is not restrictive, however, as only the finite Gram matrices induced by k are inverted throughout this document.

3.4.2 Efficient Kriging Estimation

For the sake of brevity, the space of the square-integrable matrix-valued kernels over $\mathcal{U} \times \mathcal{U}$ is written as:

$$\mathcal{F} \triangleq \mathcal{L}^2(\mathcal{U} \times \mathcal{U}, \mathcal{M}_{d_z, d_z}(\mathbb{R})) \quad (3.72)$$

Proposition 3.13 (Reduced-rank approximation). *For a continuous SPD kernel k , we define the rank of k as the dimension of the image of $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ by its associated Mercer operator:*

$$\text{rank}(k) \triangleq \dim(\mathcal{K}(\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}))) \quad (3.73)$$

Under the assumptions of Proposition 3.11, the truncated kernel:

$$k_{r_{\text{KL}}} : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ x, x' \mapsto \sum_{i=1}^{r_{\text{KL}}} \lambda_i \psi_i(x) \psi_i(x')^\top \end{cases} \quad (3.74)$$

is a continuous SPSPD kernel of rank at most $r_{\text{KL}} \in \mathbb{N}^*$. It converges uniformly towards k as $r_{\text{KL}} \rightarrow +\infty$, and minimises $\|k - k_{r_{\text{KL}}}\|_{\mathcal{F}}$ among all kernels of rank at most r_{KL} .

Proof. Let $r_{\text{KL}} \in \mathbb{N}^*$ and $\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. We have:

$$\mathcal{K}_{r_{\text{KL}}} \phi = \sum_{i=1}^{r_{\text{KL}}} \lambda_i \langle \phi | \psi_i \rangle \psi_i \quad (3.75)$$

where $\mathcal{K}_{r_{\text{KL}}}$ is the Mercer operator associated with $k_{r_{\text{KL}}}$. Every $\mathcal{K}_{r_{\text{KL}}} \phi$ thus belongs to the subspace of continuous functions spanned by $\psi_1, \dots, \psi_{r_{\text{KL}}}$. The orthonormality of the basis ensures that this subspace is of dimension r_{KL} , and thus the rank of $k_{r_{\text{KL}}}$ is at most r_{KL} . The uniform convergence of $(k_{r_{\text{KL}}})_{r_{\text{KL}} \in \mathbb{N}^*}$ has already been proven by Proposition 3.12.

Finally, applying the Eckart-Young-Mirsky theorem [78, 79] to $\mathcal{K}$ and using the orthonormality of $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ yields:

$$\min_{\tilde{k} \in \mathcal{F} | \text{rank}(\tilde{k}) \leq r_{\text{KL}}} \|k - \tilde{k}\|_{\mathcal{F}}^2 = \sum_{j > r_{\text{KL}}} \lambda_j^2 = \|k - k_{r_{\text{KL}}}\|_{\mathcal{F}}^2 \quad (3.76)$$

□

In compact form, the eigenfunctions are collected into:

$$\Psi(x) \triangleq \begin{bmatrix} \psi_1(x) & \cdots & \psi_{r_{\text{KL}}}(x) \end{bmatrix} \in \mathcal{M}_{d_z, r_{\text{KL}}}(\mathbb{R}) \quad (3.77)$$

and the corresponding eigenvalues are gathered as:

$$\Lambda \triangleq \begin{bmatrix} \lambda_1 & & (0) \\ & \ddots & \\ (0) & & \lambda_{r_{\text{KL}}} \end{bmatrix} \in \mathcal{M}_{r_{\text{KL}}, r_{\text{KL}}}(\mathbb{R}) \quad (3.78)$$

By Proposition 3.12, the eigenvalues λ_i are strictly positive, so that $\Lambda \succ 0$ and Λ^{-1} is well-defined. Proposition 3.13 can thus be rewritten as:

$$k(x, x') \approx \Psi(x) \Lambda \Psi(x')^\top \quad (3.79)$$

and then:

$$G(\tilde{\mathbf{x}}) \approx \Psi(\tilde{\mathbf{x}}) \Lambda \Psi(\tilde{\mathbf{x}})^\top + I_{\tilde{N}} \otimes \tilde{R}, \quad \Psi(\tilde{\mathbf{x}}) = \begin{bmatrix} \Psi(\tilde{x}_1) \\ \vdots \\ \Psi(\tilde{x}_{\tilde{N}}) \end{bmatrix} \quad (3.80)$$

Proposition 3.14 (Reduced-rank simple kriging). *Let:*

$$\tilde{G}(\tilde{\mathbf{x}}) \triangleq \Lambda^{-1} + \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{N}} \otimes \tilde{R} \right)^{-1} \Psi(\tilde{\mathbf{x}}) \in \mathcal{M}_{r_{\text{KL}}, r_{\text{KL}}}(\mathbb{R}) \quad (3.81)$$

be the reduced-rank Gram matrix. Under the reduced-rank approximation of (3.80), the centred SK estimator of Proposition 3.5 and its error covariance matrix reduce to:

$$\begin{cases} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{N}} \otimes \tilde{R} \right)^{-1} \tilde{\mathbf{z}} \\ R_{\text{SK}}(x, \tilde{\mathbf{x}}) \approx R + \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(x)^\top \end{cases} \quad (3.82)$$

which require only the inversion of $\tilde{G}(\tilde{\mathbf{x}})$.

When there exists $\sigma > 0$ such that $\tilde{R} = \sigma^2 I$, (3.82) further simplify to:

$$\begin{cases} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \Psi(\tilde{\mathbf{x}})^\top \tilde{\mathbf{z}} \\ R_{\text{SK}}(x, \tilde{\mathbf{x}}) \approx R + \sigma^2 \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \Psi(x)^\top \end{cases} \quad (3.83)$$

These expressions remain valid at $\sigma = 0$ provided $\Psi(\tilde{\mathbf{x}})$ has full column rank.

Proof. See section B.10. □

Remark 3.13 (Construction of the eigenpairs). While Mercer's theorem establishes the existence of the decomposition presented by (3.70) for any continuous SPD kernel defined on a compact subset of $\mathbb{R}^{d_x}$, it does not provide a closed-form expression for the eigenpairs $(\lambda_i, \psi_i)_{i \in \mathbb{N}^*}$. As they depend on k and $\mathcal{U}$, additional assumptions are usually made in order to derive an explicit approximation of the form of Proposition 3.13.

3.4.3 Karhunen-Loève Decomposition

While Section 3.4.1 uses the eigenpairs of the Mercer operator associated with k to decompose the kernel, they can also be used to represent the process itself. This representation is known as the Karhunen-Loève (KL) decomposition [80] and is presented by Proposition 3.15.

Proposition 3.15 (Karhunen-Loève decomposition). *Let $f \sim \mathcal{GP}(m, k)$, where $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is a continuous SPD kernel, and $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$. Let $(\lambda_i, \psi_i)_{i \in \mathbb{N}^*}$ be the Mercer eigenpairs of Proposition 3.12. Then, for all $x \in \mathcal{U}$, we have:*

$$f(x) = m(x) + \sum_{i=1}^{+\infty} \sqrt{\lambda_i} \xi_i \psi_i(x), \quad \xi_i \triangleq \frac{1}{\sqrt{\lambda_i}} \langle f - m | \psi_i \rangle \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1) \quad (3.84)$$

where the series converges in mean square uniformly over $\mathcal{U}$.

Proof. See section B.11. □

Remark 3.14 (Almost sure convergence). When f has almost surely continuous sample paths, the convergence of (3.84) strengthens from mean square to almost sure and uniform over $\mathcal{U}$: by the Itô–Nisio theorem [80], the partial sums converge to f uniformly over $\mathcal{U}$ almost surely. A single event of probability one then carries (3.84) simultaneously for all $x \in \mathcal{U}$, rather than for each x separately.

Conversely, the KL decomposition can be used to generate, from a family $(\xi_i)_{i \in \mathbb{N}^*}$ of i.i.d standard Gaussian variables, a GP of covariance kernel k . This is the object of Proposition 3.16.

Proposition 3.16 (Karhunen-Loève converse). *Assume that there exist a continuous SPD kernel $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ and a function $m : \mathcal{U} \rightarrow \mathbb{R}^{d_z}$, where $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$. Let $(\lambda_i, \psi_i)_{i \in \mathbb{N}^*}$ be the Mercer eigenpairs associated with k and $\mathcal{U}$, and let:*

$$(\xi_i)_{i \in \mathbb{N}^*} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1) \quad (3.85)$$

The stochastic process:

$$f : \begin{cases} \mathcal{U} \rightarrow \mathbb{R}^{d_z} \\ x \mapsto m(x) + \sum_{i=1}^{+\infty} \sqrt{\lambda_i} \xi_i \psi_i(x) \end{cases} \quad (3.86)$$

is a GP with mean function m and covariance kernel k .

Proof. See section B.12. □

Remark 3.15 (Finite-dimensional parametrisation). *As for section 3.4.2, Mercer's theorem can be used to represent the field by a finite approximation. Indeed, for $r_{\text{KL}} \in \mathbb{N}^*$, Ψ defined as in (3.80) and:*

$$\alpha \triangleq \begin{bmatrix} \sqrt{\lambda_1} \xi_1 & \cdots & \sqrt{\lambda_{r_{\text{KL}}}} \xi_{r_{\text{KL}}} \end{bmatrix}^\top \quad (3.87)$$

we have:

$$\forall x \in \mathcal{U}, f(x) \approx m(x) + \Psi(x) \alpha, \quad \alpha \sim \mathcal{N}(0, \Lambda) \quad (3.88)$$

Up to this approximation, inferring f thus amounts to inferring the r_{KL} coefficients of α .

3.5 Reproducing Kernel Hilbert Spaces

Beyond regression, the covariance kernel also induces a functional Hilbert space, known as reproducing kernel Hilbert space (RKHS) [81, 82]. The RKHS representation is complementary to the probabilistic viewpoint described throughout this chapter. Indeed, while the GP describes a probabilistic distribution over functions, the RKHS associated with its covariance kernel provides a deterministic functional space in which such functions can be represented, manipulated, or approximated. For the sake of simplicity, we assume that $d_z = 1$, so that the reproducing property can be written in its most natural form.

3.5.1 Reproducing Property and Definition

Let $\mathcal{U}$ be an arbitrary set, and let $(\mathcal{H}, \langle \cdot | \cdot \rangle_{\mathcal{H}})$ denote a Hilbert space of functions from $\mathcal{U}$ to $\mathbb{R}$. Definition 3.6 and proposition 3.17 describe the conditions for this space to be an RKHS.

Definition 3.6 (Reproducing kernel Hilbert space). A Hilbert space $(\mathcal{H}, \langle \cdot | \cdot \rangle_{\mathcal{H}})$ of functions from $\mathcal{U}$ to $\mathbb{R}$ is an RKHS if there exists a function $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ such that:

$$\begin{cases} \forall x \in \mathcal{U}, k(x, \cdot) \in \mathcal{H} \\ \forall x \in \mathcal{U}, \forall f \in \mathcal{H}, \langle f | k(x, \cdot) \rangle_{\mathcal{H}} = f(x) \end{cases} \quad (3.89)$$

This second equation is the reproducing property of $\mathcal{H}$.

Proposition 3.17 (Reproducing kernel). When such a k exists, it is symmetric and unique. It is called the reproducing kernel of $\mathcal{H}$.

Proof. Let k_1 satisfy Definition 3.6, and let $x, x' \in \mathcal{U}$. Then:

$$\begin{aligned} k_1(x, x') &= (k_1(x, \cdot))(x') \\ &= \langle k_1(x, \cdot) | k_1(x', \cdot) \rangle_{\mathcal{H}} \\ &= \langle k_1(x', \cdot) | k_1(x, \cdot) \rangle_{\mathcal{H}} \\ &= k_1(x', x) \end{aligned} \quad (3.90)$$

so k_1 is symmetric.

Assume that there exists another function k_2 that verifies Definition 3.6. It comes that:

$$\begin{aligned} k_2(x, x') &= (k_2(x, \cdot))(x') \\ &= \langle k_2(x, \cdot) | k_1(x', \cdot) \rangle_{\mathcal{H}} \\ &= \langle k_1(x', \cdot) | k_2(x, \cdot) \rangle_{\mathcal{H}} \\ &= k_1(x', x) \\ &= k_1(x, x') \end{aligned} \quad (3.91)$$

and so $k_2 = k_1$. □

It can be shown [83] that $\mathcal{H}$ is an RKHS if, and only if, the evaluation functional at every point of the domain:

$$e_x : f \in \mathcal{H} \mapsto f(x) \quad (3.92)$$

is continuous on $\mathcal{H}$ for every $x \in \mathcal{U}$. Its reproducing kernel is necessarily SPSP. Conversely, every SPSP kernel k admits a unique RKHS $\mathcal{H}_k$ for which k is the reproducing kernel. Such an RKHS admits an explicit construction, as detailed in Proposition 3.18.

Proposition 3.18 (Native RKHS). Let k be an SPSP kernel, let $\mathcal{H}_k$ be the completion of the set spanned by $\{k(x, \cdot), x \in \mathcal{U}\}$, and let $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ be defined on the generators by:

$$\forall x, x' \in \mathcal{U}, \langle k(x, \cdot) | k(x', \cdot) \rangle_{\mathcal{H}_k} \triangleq k(x, x') \quad (3.93)$$

Then $(\mathcal{H}_k, \langle \cdot | \cdot \rangle_{\mathcal{H}_k})$ is a Hilbert space and an RKHS. Its reproducing kernel is k , and it is the only RKHS for which k is the reproducing kernel. It is called the native RKHS of k .

Proof. See section B.13. □

The native RKHS is spanned by the functions $k(x, \cdot)$, so every $x \in \mathcal{U}$ is naturally represented in $\mathcal{H}_k$ by the function $k(x, \cdot)$ it generates. This correspondence defines the feature map, introduced by Definition 3.7, and provides the representation on which the kernel methods of the following sections operate.

Definition 3.7 (Feature map). *Let k be an SPSPD kernel on $\mathcal{U}$, and $\mathcal{H}_k$ be its native RKHS. The feature map of $\mathcal{H}_k$ is:*

$$\Phi : \begin{cases} \mathcal{U} \rightarrow \mathcal{H}_k \\ x \mapsto k(x, \cdot) \end{cases} \quad (3.94)$$

The reproducing property can thus be rewritten as:

$$\forall x \in \mathcal{U}, \forall f \in \mathcal{H}_k, f(x) = \langle f | \Phi(x) \rangle_{\mathcal{H}_k} \quad (3.95)$$

In particular, the kernel trick:

$$\forall x, x' \in \mathcal{U}, \langle \Phi(x) | \Phi(x') \rangle_{\mathcal{H}_k} = k(x, x') \quad (3.96)$$

can be used to compute inner products in $\mathcal{H}_k$ without explicitly constructing the feature map Φ or evaluating $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$.

3.5.2 Link with Gaussian Processes

In this section, we show that the GP viewpoint of kriging is equivalent to a variational reading in the native RKHS $\mathcal{H}_k$ of its covariance kernel k , as shown by Proposition 3.19. We then characterise the form of an RKHS for which the reproducing kernel admits a Mercer decomposition.

Proposition 3.19 (Regularised least-squares in RKHS). *Let $(\mathcal{H}_k, \langle \cdot | \cdot \rangle_{\mathcal{H}_k}, k)$ be an RKHS. Let $\tilde{N} \in \mathbb{N}^*$, and $\tilde{\mathbf{x}} \triangleq (\tilde{x}_1, \dots, \tilde{x}_{\tilde{N}}) \in \mathcal{U}^{\tilde{N}}$ pairwise distinct. Finally, let $\tilde{\mathbf{z}} \triangleq (\tilde{z}_1, \dots, \tilde{z}_{\tilde{N}}) \in \mathbb{R}^{\tilde{N}}$ and $\lambda \geq 0$. The regularised least-squares problem:*

$$\operatorname{argmin}_{\psi \in \mathcal{H}_k} \left(\frac{1}{\tilde{N}} \sum_{i=1}^{\tilde{N}} (\psi(\tilde{x}_i) - \tilde{z}_i)^2 + \lambda \|\psi\|_{\mathcal{H}_k}^2 \right) \quad (3.97)$$

admits a unique solution, given by:

$$\psi_{\text{RKHS}}(x) = k(x, \tilde{\mathbf{x}}) \left(k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda \tilde{N} I_{\tilde{N}} \right)^{-1} \tilde{\mathbf{z}} \quad (3.98)$$

Proof. By the representer theorem [84, 85], a minimiser lies in the span of $\{k(\cdot, \tilde{x}_i)\}_{i=1}^{\tilde{N}}$, so $\psi = \sum_{i=1}^{\tilde{N}} \alpha_i k(\cdot, \tilde{x}_i)$ for some $\alpha \in \mathbb{R}^{\tilde{N}}$. The reproducing property gives $\psi(\tilde{\mathbf{x}}) = k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \alpha$ and $\|\psi\|_{\mathcal{H}_k}^2 = \alpha^\top k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \alpha$, so the problem reduces to minimising $\frac{1}{\tilde{N}} \|k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \alpha - \tilde{\mathbf{z}}\|_2^2 + \lambda \alpha^\top k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \alpha$ over α . Cancelling its gradient and simplifying by the invertible $k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})$ gives $\alpha = \left(k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda \tilde{N} I_{\tilde{N}} \right)^{-1} \tilde{\mathbf{z}}$, hence $\psi_{\text{RKHS}}(x) = k(x, \tilde{\mathbf{x}}) \alpha$. $\square$

Remark 3.16 (Simple kriging in the RKHS). *Choosing $\lambda = \tilde{R}/\tilde{N}$ turns ψ_{RKHS} into the centred SK estimator of Proposition 3.5, so simple kriging is the regularised least-squares reconstruction of the field in $\mathcal{H}_k$ [86]. In the noise-free limit $\tilde{R} \rightarrow 0$, it becomes the minimum-norm interpolant:*

$$\operatorname{argmin}_{\psi \in \mathcal{H}_k} \left\{ \|\psi\|_{\mathcal{H}_k} \mid \forall i \in \llbracket 1, \tilde{N} \rrbracket, \psi(\tilde{x}_i) = \tilde{z}_i \right\} \quad (3.99)$$

a deterministic reading of kriging complementary to the probabilistic one of the GP.

Remark 3.17 (GP and RKHS correspondence). *The two viewpoints are complementary [87]. The GP model gives a probabilistic reading of kriging (posterior mean and variance under a Gaussian prior), while the RKHS gives a deterministic one (minimum-norm interpolant in a function space). Writing the Mercer decomposition of the kernel over its eigenpairs $(\lambda_i, \psi_i)_{i=1}^\infty$ as $k(x, x') = \sum_{i=1}^\infty \lambda_i \psi_i(x) \psi_i(x')$, a function $\psi = \sum_{i=1}^\infty c_i \psi_i$ belongs to $\mathcal{H}_k$ with squared norm $\|\psi\|_{\mathcal{H}_k}^2 = \sum_{i=1}^\infty c_i^2 / \lambda_i$; the norm thus penalises most the components carried by low-eigenvalue, rapidly oscillating eigenfunctions, so the decay of the eigenvalues λ_i sets the smoothness enforced by k . Choosing a kernel therefore amounts to choosing a regularity class for the unknown field. A smooth kernel such as the SE kernel yields infinitely differentiable interpolants, whereas a rougher Matérn kernel admits functions of limited smoothness.*

3.5.3 Kernel Mean Embedding

Just as the feature map represents a function of $\mathcal{H}_k$, it represents a probability distribution through its mean. The kernel mean embedding [88] of the law of a random variable X on $\mathcal{U}$ is:

$$\mu_X \triangleq \mathbb{E}[\Phi(X)] \in \mathcal{H}_k \quad (3.100)$$

and the reproducing property turns every expectation of an RKHS function into an inner product against it:

$$\forall f \in \mathcal{H}_k, \mathbb{E}[f(X)] = \langle f | \mu_X \rangle_{\mathcal{H}_k} \quad (3.101)$$

When k is characteristic, this embedding is injective, so μ_X retains the whole law of X , and conditioning becomes a linear operation on it. As (3.100) is no more computable than the integrals of chapter 2, it is approximated in the Monte Carlo sense of section 2.4 by a weighted particle set:

$$\hat{\mu}_X = \sum_{i=1}^N w^{(i)} \Phi(x^{(i)}) \quad (3.102)$$

3.5.4 Kernel Kalman Filter

Since conditioning acts linearly on the embedding, the Bayesian recursion of chapter 2 can be run in $\mathcal{H}_k$ as a Kalman filter. This linearity is made concrete by the conditional mean embedding [88]: from a set of training pairs, one estimates a linear operator that maps the embedding of a conditioning value to the embedding of the corresponding conditional law. Prediction and correction then become linear maps on the embedded state, so that the whole recursion amounts to propagating a finite set of weights together with their covariance.

The kernel Kalman filter (KKF) [89] applies this principle over a dictionary of reference states fixed before the recursion. It embeds the filtering distribution as a weighted combination $\sum_i w_k^{(i)} \Phi(\tilde{x}_i)$ over that dictionary, and attaches to the weights $\mathbf{w}_k$ a covariance Σ_k . A transition operator and an observation operator, both conditional mean embeddings learnt once from training data, stand for the dynamics and observation models: the prediction step pushes $\mathbf{w}_k$ and Σ_k through

the transition operator and inflates the covariance, while the correction step applies a kernel Kalman gain built from the observation operator. The dictionary and its Gram matrices are computed once, which ties the filter to a stationary model that the training set must cover.

Its adaptive variant (AKKF) [90] drops the fixed dictionary. In place of a training set it carries a particle set, regenerated at every step by propagating the particles through the dynamics and observation models, and it recomputes the Gram matrices of the state and observation kernels over that current set. The recursion can then follow a model that is time-varying, and neither Gaussian nor the image of a linear one, at the cost of the two models needed to move and score the particles.

At time step k , let $\mathbf{x}_k^+$ denote the resampled particle set, $\mathbf{w}_k^+$ and $\Sigma_{k|k}^+$ its weights and their covariance, and:

$$G_k^x \triangleq k_x(\mathbf{x}_k^+, \mathbf{x}_k^+), \quad G_k^z \triangleq k_z(\mathbf{z}_k, \mathbf{z}_k) \quad (3.103)$$

the Gram matrices of the state kernel k_x and the observation kernel k_z , regularised by $\lambda_x, \lambda_z \in \mathbb{R}_+^*$, where $\mathbf{z}_k$ gathers the observations $(z_k^{(i)})_i$ simulated at the propagated particles through the observation model. Algorithm 3.1 gives the recursion. The state estimate belongs to it rather than to its reading, since it feeds the resampling from which the next particle set is drawn.

Algorithm 3.1 Adaptive Kernel Kalman Filter

$$\begin{aligned}
& \forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot | x_{k-1}^{(i)}) &> \text{Particle propagation} \\
& D_k = (G_{k-1}^x + \lambda_x I_N)^{-1} G_{k-1}^x - I_N &> \text{Embedded prediction} \\
& \Sigma_{k|k-1} = \Sigma_{k-1|k-1} + \frac{1}{N} D_k D_k^\top \\
& \forall i \in \llbracket 1, N \rrbracket, z_k^{(i)} = h_k(x_k^{(i)}, n_k^{(i)}), n_k^{(i)} \sim p_k^{(o)} &> \text{Simulated observations} \\
& K_k = \Sigma_{k|k-1} (G_k^z \Sigma_{k|k-1} + \lambda_z I_N)^{-1} &> \text{Embedded correction} \\
& \mathbf{w}_k = \mathbf{w}_{k-1}^+ + K_k (k_z(\mathbf{z}_k, z_k) - G_k^z \mathbf{w}_{k-1}^+) \\
& \Sigma_{k|k} = \Sigma_{k|k-1} - K_k G_k^z \Sigma_{k|k-1} \\
& \hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}, \quad \hat{P}_k = \sum_{i=1}^N w_k^{(i)} (x_k^{(i)} - \hat{x}_k) (x_k^{(i)} - \hat{x}_k)^\top &> \text{State estimation} \\
& \forall i \in \llbracket 1, N \rrbracket, x_k^{+(i)} \sim \mathcal{N}(\hat{x}_k, \hat{P}_k) &> \text{Resampling} \\
& T_k = (G_k^x + \lambda_x I_N)^{-1} k_x(\mathbf{x}_k^+, \mathbf{x}_k^+) &> \text{Embedded resampling} \\
& \mathbf{w}_k^+ = T_k \mathbf{w}_k \\
& \Sigma_{k|k}^+ = T_k \Sigma_{k|k} T_k^\top
\end{aligned}$$

Remark 3.18 (Weights of either sign). *The embedded correction solves a regularised least-squares problem in $\mathcal{H}_k$, not a Bayesian update of a discrete measure, so its weights are not constrained to be positive and $\hat{\mu}_X$ is not in general the embedding of a probability measure. The estimator $\sum_i w_k^{(i)} x_k^{(i)}$ remains well defined, but the effective sample size of section 2.5 does not, and neither does the resampling it triggers. The AKKF resamples unconditionally instead, drawing a fresh particle set from the Gaussian carrying the current estimate and its dispersion.*

Part II

Navigation with Gaussian Processes

CHAPTER 4

KRIGING-BASED NAVIGATION

Contents

4.1	Introduction	72
4.2	Kriging the Observation Model	73
4.2.1	Choosing the Kriging Method	73
4.2.2	Hyperparameter Estimation	78
4.2.3	Simulation Scenario	80
4.2.4	Numerical Results	83
4.3	Adaptive Kriging	88
4.3.1	Random Subsampling	89
4.3.2	Adaptive Subsampling Window	92
4.3.3	Numerical Results	95
4.4	Non-Spherical Subsampling Windows	100
4.4.1	Nearest Neighbour Selection	101
4.4.2	Clustering	102
4.4.3	Clustered Adaptive Kriging	106
4.4.4	Numerical Results	109
4.5	Subsampling Error Bounds	109
4.5.1	Sources of Error	111
4.5.2	Subsampling Error Metrics	115
4.5.3	Importance of the Gram Matrix	116
4.5.4	Block Inversion and Residual Quantities	117
4.5.5	Relations Between the Errors	119
4.5.6	Simple Kriging and Trend Contributions	120
4.5.7	Nested Subsampling	121
4.5.8	Entrywise Bounds	122

4.1 Introduction

Assume a navigation scenario similar to that of chapter 2:

$$\begin{cases} x_k = f_k(x_{k-1}, w_k) \\ z_k = h_k(x_k, n_k) \end{cases} \quad (4.1)$$

with the exception that h_k is unknown, and the observation noise is additive for the sake of simplicity. Since the likelihood density cannot be evaluated without an observation model, Bayesian data fusion is no longer applicable. The state x_k must then be estimated from inertial measurements alone. However, inertial navigation is subject to drift and therefore cannot provide a reliable state estimate for longer missions.

In our scenarios, the observation function describes the measurement of a physical field, whose spatial distribution does not change over the course of the mission, by an embedded sensor at each time step. Furthermore, we assume that the environment has been sampled beforehand, whether during a previous reconnaissance mission, or using a known but deteriorated map. This is formally modelled by assumption 4.1.

Assumption 4.1 (Observation model). *The observation function is a time-invariant GP, and its noise is an additive Gaussian white noise of covariance matrix $R_k \succeq 0$:*

$$z_k = h(x_k) + n_k, \quad n_k \sim \mathcal{N}(0, R_k) \quad (4.2)$$

where $m : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$ is the potentially unknown mean of the GP, and $k : \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_z}$ is its chosen covariance kernel. Here, n_k plays the role of the query noise ε of section 3.3, and R_k is its associated covariance R .

We also have access to $\tilde{N} \in \mathbb{N}^*$ distinct training inputs $\tilde{\mathbf{x}}$ and their noisy observations $\tilde{\mathbf{z}}$, degraded by a noise of covariance $\tilde{R} \succeq 0$, in the sense of section 3.3.

Remark 4.1 (Time-varying observation function). *In full generality, the observed physical field could vary over time, and so the underlying GP would need to include a temporal input. While the kriging theory perfectly works with this additional parameter, it would require the training data to be sampled over time as well as space.*

This chapter is organised as follows:

- Section 4.2 combines chapters 2 and 3 to create a kriging-based framework for autonomous navigation in a poorly-known environment.
- Section 4.3 discusses non-stationary observation fields, and the necessity of training data selection for kriging.
- Section 4.4 improves upon this framework by adapting the shape of the subsampling window to that of the particle cloud to better capture the non-stationarities of the field.
- Finally, section 4.5 quantifies the estimation error committed by subsampling the training data used for kriging.

4.2 Kriging the Observation Model

Using assumption 4.1, the likelihood can be expressed as:

$$z_k | x_k, \tilde{\mathbf{z}} \sim \mathcal{N}(\hat{z}_K(x_k, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g), R_K(x_k, \tilde{\mathbf{x}}; g)) \quad (4.3)$$

where $\hat{z}_K$ denotes any of the three kriging estimators, R_K is the corresponding covariance matrix, and m and g are optional arguments depending on the chosen kriging method. In particular, R_K already carries the query noise R_k , so it must not be added a second time.

4.2.1 Choosing the Kriging Method

Section 3.3 leaves the choice between the three kriging methods open. Table 4.1 presents a $\mathbb{R} \rightarrow \mathbb{R}$ kriging scenario, from which every kriging estimator attempts to reconstruct the true function. A numerical comparison of the estimators is presented at the end of this section.

True function	$h : x \in \mathbb{R} \mapsto x \sin(x) + 10 \in \mathbb{R}$
GP mean	$m : x \mapsto 10$ or $m : x \mapsto 0$
GP covariance	$k : x, x' \in \mathbb{R} \mapsto 5^2 \exp\left(-\frac{\ x - x'\ _2^2}{2 \times 1.2^2}\right)$
Training data	$\tilde{\mathbf{x}} = \{0.5, 1.5\} \cup \{5.5, 6.5, \dots, 9.5\}$
Training noise	$\tilde{R} = 1$
Query noise	$R = 0$
Trend functions	$g : x \mapsto \begin{bmatrix} 1 & x \end{bmatrix}$ or $g : x \mapsto \begin{bmatrix} 1 & x & \dots & x^5 \end{bmatrix}$
Prediction domain	$[-3, 13]$

Table 4.1: An example kriging scenario. The SK and UK estimators are given either mean function and trend bases to illustrate their reliance on prior knowledge, as shown in their respective figures.

Five kriging estimators, two simple, one ordinary, and two universal, are evaluated on this scenario, as illustrated by Figures 4.1 to 4.3 respectively. The continuous blue line draws the true function, while the training points are represented by the black crosses. The kriged function is in dashed red, and its associated confidence interval is the pale red envelope. Since h is scalar, the confidence interval is computed as:

$$\left[\hat{z}_K(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) - 3\sqrt{R_K(x, \tilde{\mathbf{x}}; g)}, \hat{z}_K(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) + 3\sqrt{R_K(x, \tilde{\mathbf{x}}; g)} \right] \quad (4.4)$$

and contains $h(x)$ with probability $\text{erf}(3/\sqrt{2}) \approx 99.73\%$, where erf is the Gauss error function.

While it is the computationally simplest estimator, the SK estimator relies heavily on the prior knowledge of the mean function m , and poorly represents the true function in regions with a lower density of training points when that knowledge is

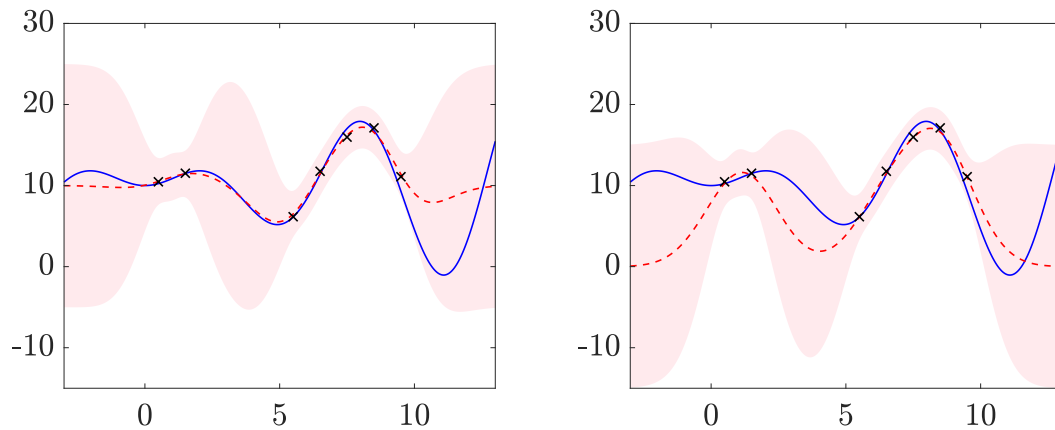


Figure 4.1: Simple kriging under a correct mean function, on the left, and under a wrong one, on the right. Since R_{SK} does not depend on m in (3.35), both shaded areas have the same width along the y -axis.

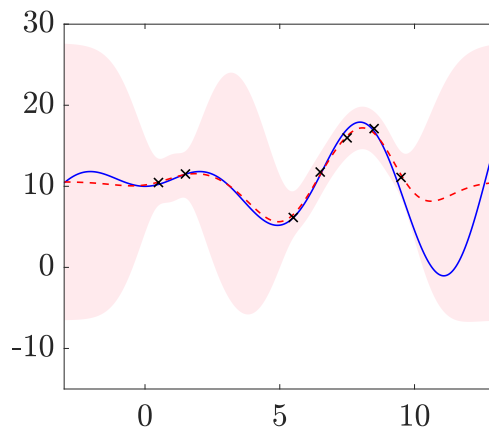


Figure 4.2: Simultaneous kriging of the training data and mean estimation by ordinary kriging. The constant mean is reconstructed here at $\hat{m}_{OK} = 10.56$.

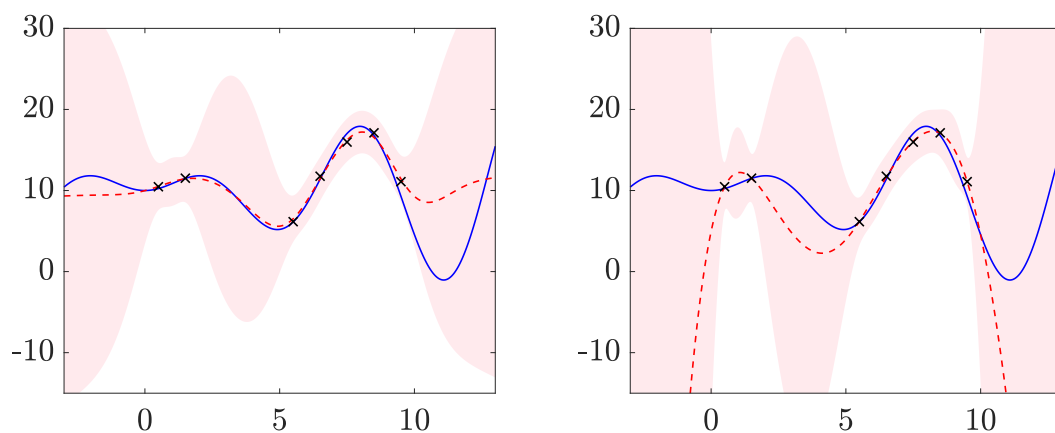


Figure 4.3: Universal kriging over an affine trend basis, on the left, and over a degree five monomial basis, on the right. The trend coefficients are estimated here at $\hat{m}_{UK} = [9.78, 0.15]^\top$ and $\hat{m}_{UK} = [4.95, 15.85, -10.82, 2.40, -0.20, 0.01]^\top$ respectively.

wrong. This is particularly apparent in the gap within the training points distribution, and outside the sampled region.

As motivated by section 3.3, ordinary kriging tackles this issue by directly estimating a constant mean function instead of relying on a given prior. Its behaviour is similar to that of the SK estimator with the correct mean function, at the cost of a slightly increased confidence interval. This is due to the additional uncertainty associated with estimating the mean, and is quantified latter in this section.

For universal kriging, we considered polynomial means of degree 1 and 5. The affine basis behaves much like ordinary kriging, which it extends by a single coefficient. The affine UK envelope is barely wider than that of OK near the samples but grows faster outside the interpolation domain.

On the other hand, the degree 5 model overfits the seven training points with six coefficients, which causes the corresponding estimator to diverge from the true function farther away from the training data. This is reflected in the UK covariance, which indicates high uncertainty in poorly sampled regions.

Since the training observations are noisy, a single draw of $\tilde{\mathbf{z}}$ cannot accurately represent the performance of the estimators above. To alleviate this issue, the training observations are redrawn over $n_{\text{MC}} = 1000$ Monte Carlo simulations, in order to compute the root mean squared error (RMSE):

$$\text{RMSE}_{\hat{z}_K} : x \mapsto \sqrt{\frac{1}{n_{\text{MC}}} \sum_{i=1}^{n_{\text{MC}}} \|\hat{z}_K(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}^{(i)}; m, g) - h(x)\|_2^2} \quad (4.5)$$

of the estimators, where $\tilde{\mathbf{z}}^{(i)}$ denotes the i -th draw of the training observations.

Their average RMSE (ARMSE):

$$\text{ARMSE}_{\hat{z}_K}(E) \triangleq \frac{\int_E \text{RMSE}_{\hat{z}_K}(x) dx}{\int_E dx} \quad (4.6)$$

is then computed over three subsets:

$$\begin{cases} E_{\text{in}} \triangleq [0.5, 1.5] \cup [5.5, 9.5] \\ E_{\text{gap}} \triangleq [1.5, 5.5] \\ E_{\text{out}} \triangleq [-3, 0.5] \cup [9.5, 13] \end{cases} \quad (4.7)$$

of the prediction domain, as presented in Table 4.2.

All five estimators perform similarly in the densely sampled region, their ARMSE values ranging from 0.85 to 0.87, with the exception of the overfitted UK estimator. In this region, the high density of training data limits the influence of the mean on the estimation, and is thus not informative for assessing the individual impact of each estimator.

In the gap region, the choice of mean becomes more influential due to the sparsity of training points. The SK estimator with the correct mean achieves an ARMSE of 1.15, while using an incorrect one increases the error to 3.69. In contrast, both the OK and affine UK estimators remain close to the correctly informed SK estimator, with an increase of less than 10% in ARMSE. The degree 5 UK estimator

Estimator	ARMSE		
	in	gap	out
Simple, correct mean	0.85	1.15	3.23
Simple, wrong mean	0.87	3.69	5.88
Ordinary	0.85	1.26	3.25
Universal, affine	0.85	1.25	3.54
Universal, degree 5	0.98	4.02	70.88

Table 4.2: ARMSE of the five estimators over the three subsets E_{in} , E_{gap} , and E_{out} of the prediction domain.

performs worst, its ARMSE representing a 250% increase over that of the correct SK estimator.

This tendency suggests that estimating the mean from the available observations generally makes the estimator more robust, provided that the mean model does not overfit the data. It is even more pronounced in the extrapolation region, where the SK estimator with an incorrect mean, and especially the high-order UK estimator, show a substantial increase in ARMSE compared with the three other estimators. However, this robustness comes at a cost, as neither the OK nor the affine UK estimator ever outperforms a correctly informed simple kriging.

Although the previous results highlight the differences in point-estimation accuracy between the five estimators, a key advantage of kriging is its ability to quantify the corresponding uncertainty. Table 4.3 compares the confidence intervals and the contribution of the mean estimation to the overall covariance for the five estimators.

Estimator	Confidence half-width			$R_{\hat{m}}/R_K$		
	in	gap	out	in	gap	out
Simple, correct mean	2.71	9.05	11.58	0.0%	0.0%	0.0%
Simple, wrong mean	2.71	9.05	11.58	0.0%	0.0%	0.0%
Ordinary	2.71	9.55	12.73	0.4%	8.9%	15.0%
Universal, affine	2.72	9.74	16.55	0.8%	11.9%	43.0%
Universal, degree 5	3.22	15.70	320.03	20.5%	59.6%	94.5%

Table 4.3: Average confidence interval half-width and average share of the covariance caused by the estimation of the mean for the five estimators over each region. The share is computed as the pointwise ratio of $R_{\hat{m}}$ over $R_K = R_{\hat{m}} + R_{\text{SK}}$.

This table confirms the trend observed in the ARMSE results, as the average confidence intervals also increase with distance from the training points. As shown by the first row, the kriging covariance grows on its own, since the lack of close samples causes the estimator to rely more on its mean. In such regions, the estimated mean becomes more influential, and thus accounts for an increasing share of the total uncertainty.

Furthermore, it must be noted that both SK confidence intervals are the same, showing that simple kriging is unable to account for an incorrectly specified prior

mean. Both present a half-width of 9.05 in the gap, well inside the sampled range, although their ARMSE differ by a factor of three. This issue is particularly relevant in autonomous navigation scenarios, as a poorly informed SK estimator will be overconfident, resulting in skewed likelihood peaks. This, in turn, may lead to relevant particles being discarded while incorrect ones are retained, and so simple kriging is set aside.

Conversely, the degree 5 UK estimator fails by trying to fit six trend coefficients to only seven training points. Outside the sampled range, the mean becomes highly uncertain, and accounts for an average of 94.5% of the total covariance. Unlike simple kriging, the UK estimator propagates this uncertainty to its confidence interval, whose half-width reaches 320.03. From a computational point of view, universal kriging requires $\tilde{N} \geq r_{\text{UK}}$, where we recall that $r_{\text{UK}} \in \mathbb{N}^*$ is the number of trend functions used for kriging. This causes the matrix $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})$ to become ill-conditioned when the training set is too small relative to the number of trend functions.

Furthermore, every estimator requires the $\mathcal{O}(\tilde{N}^3)$ inversion of $G(\tilde{\mathbf{x}})$ once, after which, solving any related system costs $\mathcal{O}(\tilde{N}^2)$. Simple kriging thus requires an additional $\mathcal{O}(\tilde{N}^2)$ at each query point. Estimating the mean amounts to solving $r_{\text{UK}} d_z$ equations, which costs $\mathcal{O}(r_{\text{UK}} \tilde{N}^2)$ once and $\mathcal{O}(r_{\text{UK}} \tilde{N})$ per query. The inversion of $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}) \in \mathcal{S}_{r_{\text{UK}} d_z}^{++}(\mathbb{R})$ does not depend on $\tilde{N}$, and is negligible as soon as $r_{\text{UK}} d_z \ll \tilde{N}$. Table 4.4 gives the empirical cost of each estimator in floating-point operations (flop), for $\tilde{N} \in \{7, 100, 1600\}$.

Estimator	Setup (flops)			Per query (flops)		
	$\tilde{N} = 7$	100	1600	$\tilde{N} = 7$	100	1600
Simple	114	333 k	1.37 G	49	10 k	2.56 M
Ordinary	+112	+20.2 k	+5.12 M	+14	+200	+3.2 k
Universal, affine	+255	+40.8 k	+10.3 M	+28	+400	+6.4 k
Universal, degree 5	+1.16 k	+127 k	+30.8 M	+84	+1.2 k	+19.2 k

Table 4.4: Computational cost of the five estimators, split into the setup and the query costs. The simple kriging row is the cost of reconstructing the field, paid by every estimator, and the following ones are what estimating the mean adds to it. Since their cost is independent of the mean they were given, the two simple estimators share the same row.

From 7 to 1600 training points, the setup cost of simple kriging rises from 114 flops to 1.37 Gflop, and its per-query cost from 49 to 2.56 Mflop. In our scenarios, every particle is a query point at each time step, so real-time constraints rule out a training set of that size. The other end of the table is no more comfortable: on seven points, the affine basis costs twice the reconstruction of the field, 255 flops against 114, and the degree 5 basis ten times.

Ordinary kriging is thus retained for the remainder of this chapter. Estimating a constant mean from the observations is robust to misinformation, its ARMSE staying within 10% of a correctly informed simple kriging everywhere, when a wrong mean triples the error in the gap. It also reports that estimation, the term $R_{\hat{m}_{\text{OK}}}$ of remark 3.8 rising from 0.4% of the covariance near the samples to 8.9% in the gap,

so its confidence interval widens exactly where its accuracy degrades. Finally, the additional $d_z \times d_z$ inversion it requires is the smallest trend a kriging estimator can carry, and remains affordable at both ends of table 4.4.

4.2.2 Hyperparameter Estimation

Recall that section 3.2.2 considered every kernel subsequently used for kriging to be fully known, with both their form and hyperparameters specified. This assumption is relaxed here by considering only the form to be known, while the hyperparameters θ must be estimated from the available data.

Chapter 3 derives every result conditionally on the kernel, and section 4.2.1 compares different estimators with that kernel held fixed. This assumption is discharged here. A kernel is specified by a vector of hyperparameters θ whose components are of two kinds. Some describe the class of fields the GP is assumed to belong to, such as a regularity or a set of active input directions, and are posited by the modeller alongside the kernel family itself. The others describe the particular field being surveyed, such as the amplitude of its variations or the distance over which they decorrelate, and are read from the training data of assumption 4.1. Only the latter are estimated below.

No closed form is available for them. The mean enters the kriging equations linearly, which is what lets proposition 3.8 recover it by a generalised least squares regression, whereas θ acts through the kernel, hence through the inverse of a matrix it parameterises. Estimating it therefore requires solving an optimisation problem, and the criterion retained is the likelihood the model assigns to the observations already collected, a procedure known as evidence maximisation [91].

That likelihood is available in closed form. Proposition 3.1 gives the joint law of any finite collection of field values, and the training observations are such a collection degraded by an independent Gaussian noise, so they are themselves jointly Gaussian. Writing $G_\theta \triangleq k_\theta(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R}$ for the noise-augmented Gram matrix, whose argument is dropped to keep the hyperparameters visible, and $\tilde{\mathbf{r}} \triangleq \tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})$ for the residual to the GP mean, their density reads:

$$p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \theta) = \frac{1}{\sqrt{(2\pi)^{\tilde{N}d_z} \det(G_\theta)}} \exp\left(-\frac{1}{2}\tilde{\mathbf{r}}^\top G_\theta^{-1}\tilde{\mathbf{r}}\right) \quad (4.8)$$

Proposition 4.1 (Maximum likelihood hyperparameters). *A hyperparameter vector θ maximising (4.8) satisfies, for each component θ_i :*

$$\text{tr}\left(\left((G_\theta^{-1}\tilde{\mathbf{r}})(G_\theta^{-1}\tilde{\mathbf{r}})^\top - G_\theta^{-1}\right) \frac{\partial G_\theta}{\partial \theta_i}\right) = 0 \quad (4.9)$$

Proof. Maximising (4.8) is equivalent to maximising its logarithm:

$$\ell(\theta) \triangleq \log p(\tilde{\mathbf{z}}|\tilde{\mathbf{x}}, \theta) = -\frac{1}{2}\left(\tilde{\mathbf{r}}^\top G_\theta^{-1}\tilde{\mathbf{r}} + \log \det(G_\theta)\right) - \frac{\tilde{N}d_z}{2} \log(2\pi) \quad (4.10)$$

A maximiser cancels every partial derivative. Using the differential of the matrix inverse, $\frac{\partial}{\partial \theta_i} G_\theta^{-1} = -G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i} G_\theta^{-1}$, and Jacobi's formula, $\frac{\partial}{\partial \theta_i} \log \det(G_\theta) = \text{tr}\left(G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i}\right)$,

it comes that:

$$\begin{aligned} \frac{\partial \ell}{\partial \theta_i} = 0 &\iff \tilde{\mathbf{r}}^\top G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i} G_\theta^{-1} \tilde{\mathbf{r}} - \text{tr} \left(G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i} \right) = 0 \\ &\iff \text{tr} \left(((G_\theta^{-1} \tilde{\mathbf{r}}) (G_\theta^{-1} \tilde{\mathbf{r}})^\top - G_\theta^{-1}) \frac{\partial G_\theta}{\partial \theta_i} \right) = 0 \end{aligned} \quad (4.11)$$

using the cyclic property of the trace to rewrite the scalar quadratic form $\tilde{\mathbf{r}}^\top G_\theta^{-1} \frac{\partial G_\theta}{\partial \theta_i} G_\theta^{-1} \tilde{\mathbf{r}}$ as $\text{tr} \left((G_\theta^{-1} \tilde{\mathbf{r}}) (G_\theta^{-1} \tilde{\mathbf{r}})^\top \frac{\partial G_\theta}{\partial \theta_i} \right)$, since G_θ^{-1} is symmetric. $\square$

Remark 4.2 (Profiled mean). *The density (4.8) is written for a known mean, whereas the estimator retained in section 4.2.1 estimates it instead. Maximising ℓ over a constant mean at fixed θ cancels $C^\top G_\theta^{-1} \tilde{\mathbf{r}}$, where C is defined as (3.49), and thus returns the very estimate $\hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ of Proposition 3.8, evaluated under the current θ . The two estimations nest rather than compete, the residual $\tilde{\mathbf{r}}$ being taken to a mean recomputed in closed form at every evaluation of (4.8).*

The two terms of ℓ pull in opposite directions, which is what makes its maximum informative rather than degenerate. The quadratic form rewards a covariance ample enough to explain the observed residual, and keeps improving as the field is allowed to vary more freely, while the log-determinant penalises exactly that freedom, measuring the volume of observation space over which the model spreads its mass. Maximising their sum thus retains the least flexible field that remains consistent with the survey.

Remark 4.3 (Non-convexity). *The log-marginal-likelihood is generally non-concave in θ , so (4.9) is solved numerically by gradient-based optimisation [92], and may admit several local maxima corresponding to different interpretations of the data, e.g. a low-noise rapidly-varying field versus a high-noise slowly-varying one.*

The hyperparameters are thus obtained by ascending ℓ numerically, one step of which is given by Algorithm 4.1. Two implementation choices are worth stating. First, G_θ is never inverted, its Cholesky factor L such that $G_\theta = LL^\top$ yielding both the quadratic form and the log-determinant. This $\mathcal{O}(\tilde{N}^3)$ factorisation is the one table 4.4 already charges to the reconstruction of the field, except that the ascent pays it once per step instead of once per mission, after which each component of the gradient costs $\mathcal{O}(\tilde{N}^2)$. Second, the ascent is carried out on $\eta \triangleq \log \theta$ rather than on θ itself, which enforces the positivity of the hyperparameters and turns each step into a multiplicative update, matching the scale invariance of quantities carrying a physical unit. The chain rule then gives $\frac{\partial G_\theta}{\partial \eta_i} = \theta_i \frac{\partial G_\theta}{\partial \theta_i}$, so that the kernel family enters the ascent through the single quantity $\frac{\partial G_\theta}{\partial \theta_i}$, and nowhere else.

The step is iterated from $\eta^{(0)} = \log \theta^{(0)}$ until the gradient falls below a chosen tolerance, the estimate being $\hat{\theta} = \exp(\eta^{(t)})$ at the last iterate. Its starting point $\theta^{(0)}$ is a free input, and the non-convexity of ℓ makes it a meaningful one, since the ascent only reaches the maximum of the basin it is started in. It is therefore read from the survey rather than fixed arbitrarily, each hyperparameter being initialised by the empirical quantity it describes, an amplitude by the sample variance of $\tilde{\mathbf{z}}$ less the known noise variance, a correlation length by the mean distance between neighbouring training inputs, which is the shortest length the survey can resolve. Should several starting points be drawn around these values, the one reaching the highest ℓ is retained.

Algorithm 4.1 Hyperparameter estimation by evidence maximisation

$G_\theta = k_{\text{exp}(\eta^{(t-1)})}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R}$	▷ Noise-augmented Gram matrix
$L \text{ s.t. } G_\theta = LL^\top$	▷ Cholesky factorisation
$\alpha = L^{-\top} (L^{-1} \tilde{\mathbf{r}})$	▷ Two triangular solves
$\ell = -\frac{1}{2} \tilde{\mathbf{r}}^\top \alpha - \sum_{j=1}^{\tilde{N}d_z} \log L_{jj} - \frac{\tilde{N}d_z}{2} \log(2\pi)$	▷ Log-evidence
$W = \alpha\alpha^\top - L^{-\top} L^{-1}$	
$\forall i, \nabla_i = \frac{1}{2} \text{tr} \left(W \frac{\partial G_\theta}{\partial \eta_i} \right)$	▷ Gradient of ℓ
$\Delta\eta = \text{Direction}(\nabla)$	▷ Gradient, conjugate gradient or quasi-Newton
$\eta^{(t)} = \eta^{(t-1)} + \rho \Delta\eta$	▷ Line search on the step length ρ

4.2.3 Simulation Scenario

Every navigation result of this chapter is produced by the same scenario, as described in this section. The observation field represented by h is a magnetic anomaly map of a 200 km×200 km region above Mexico, and extracted from [93]. It is sampled on a regular grid of 201×201 points with a resolution of 1 km×1 km, and is read by a bilinear interpolation between the points.

From this map are extracted $\tilde{N}$ training samples $(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$, where $\tilde{\mathbf{x}}$ is sampled from a regular grid, each node being independently jittered by up to 30% of the grid resolution along both axes. The grid resolution Δ_g is varied from 2.5 km to 7.5 km, with increments of 1 km. This corresponds to $\tilde{N} \in \{79^2, 56^2, 44^2, 36^2, 30^2, 26^2\}$. The corresponding observations $\tilde{\mathbf{z}}$ are corrupted by a Gaussian white noise of variance $\tilde{R} = 10^2 \text{ (nT)}^2$. The irregular grid models navigation errors during the prior survey mission, while the white noise represents the imperfect measurements of a real sensor.

The vehicle flies over the map at a constant speed of 500 m/s along a mower pattern consisting of three parallel branches, each 80 km long, and connected by half-turns with a radius of 25.5 km. The trajectory is sampled every $\Delta_t = 5$ s, and lasts $n_{\text{steps}} = 150$ steps. A single time step thus covers 2.50 km.

The vehicle state consists of its horizontal position and velocity:

$$x_k = \begin{bmatrix} p_k^{(x)} & p_k^{(y)} & v_k^{(x)} & v_k^{(y)} \end{bmatrix}^\top, \quad d_x = 4 \quad (4.12)$$

and its propagation is degraded with a Gaussian white noise of covariance matrix:

$$Q_k = \begin{bmatrix} 100^2 & 0 & 0 & 0 \\ 0 & 100^2 & 0 & 0 \\ 0 & 0 & 1^2 & 0 \\ 0 & 0 & 0 & 1^2 \end{bmatrix} \quad (4.13)$$

whose position and velocity entries are respectively expressed in m² and m²s⁻².

The observation function is the scalar ($d_z = 1$) ordinary kriging of the magnetic anomaly field, conditioned on the $\tilde{N}$ training points, and degraded by a Gaussian white noise of covariance $R_k = 5^2 \text{ (nT)}^2$. The chosen kernel is a Matérn kernel with smoothness $\nu = 3/2$, which corresponds to the smallest half-integer for which the

field and the kriging mean are differentiable, and its hyperparameters are estimated as described in section 4.2.2. A bilinear observation function on the same samples is also used as comparison.

The filter is an RPF, initialised on a cloud of $N = 5000$ particles, drawn around a point itself drawn from $\mathcal{N}(x_0, P_0)$, with:

$$P_0 = \begin{bmatrix} 8000^2 & 0 & 0 & 0 \\ 0 & 8000^2 & 0 & 0 \\ 0 & 0 & 2^2 & 0 \\ 0 & 0 & 0 & 2^2 \end{bmatrix} \quad (4.14)$$

in the same units, to prevent the particle cloud from being centred on the true initial state. The resampling threshold is fixed to $N_{\text{th}} = 0.5 N$, and the regularisation coefficient is $\alpha_{\text{reg}} = 0.3$. This scenario is illustrated by Figure 4.4 and summarised by Table 4.5.

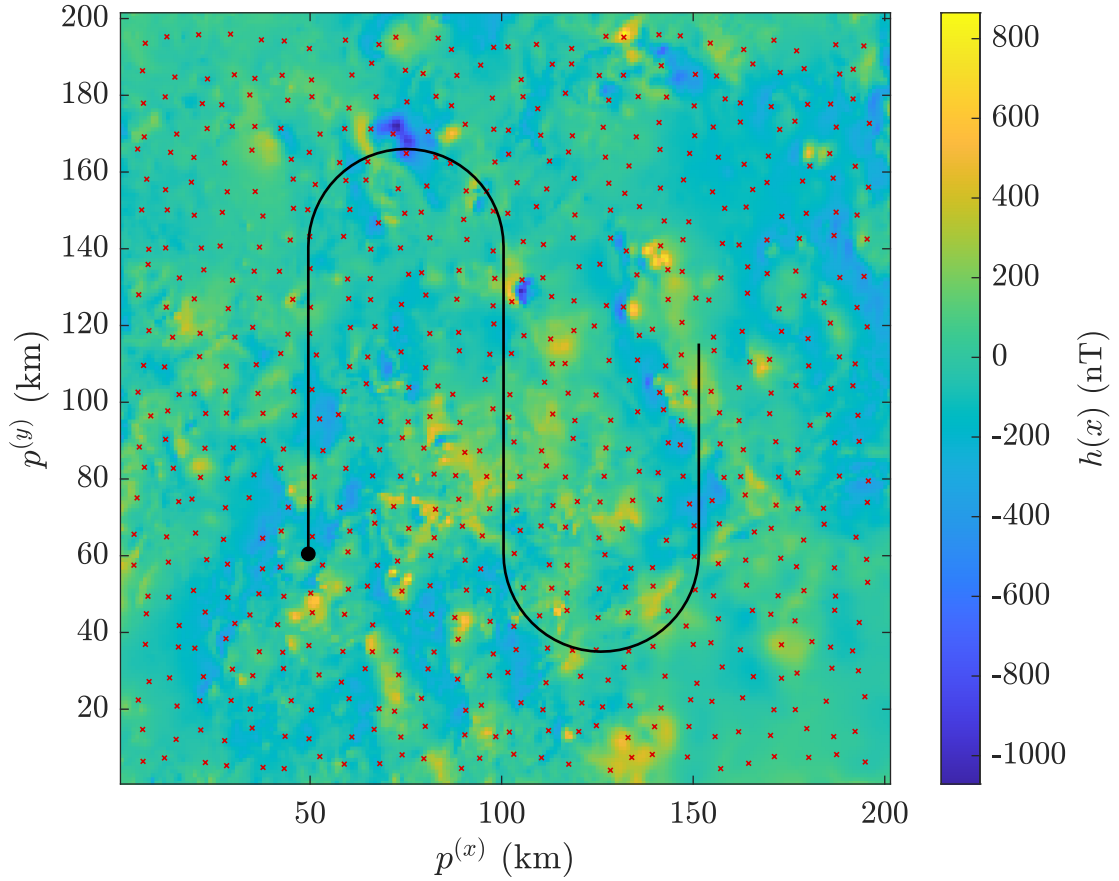


Figure 4.4: The magnetic field anomaly used throughout this chapter. The black line represents the overflow trajectory, and the red crosses correspond to the training points on a 7.5 km-resolution grid. The black dot signals the beginning of the trajectory.

Each configuration is run over $n_{\text{MC}} = 1000$ Monte Carlo realisations, redrawing the initial state and white noises. The training data are also redrawn every 10 runs so that the metrics are averaged over the reconstruction as well as the trajectory.

Map	Surface	200 km×200 km
	Resolution	1 km×1 km
Kriging	Samples	Jittered grid
	Jitter	$0.3 \Delta_g$
	Δ_g	$\in \{2.5, 3.5, \dots, 7.5\}$ km
	$\tilde{N}$	$\in \{79^2, 56^2, 44^2, 36^2, 30^2, 26^2\}$
	$\tilde{R}$	10^2 (nT)^2
Trajectory	Form	3-branch mower
	Branch length	80 km
	Turn radius	25.5 km
	Speed	500 m/s
	Δ_t	5 s
	n_{steps}	150
State	d_x	4
	x_k	$\begin{bmatrix} p_k^{(x)}, p_k^{(y)}, v_k^{(x)}, v_k^{(y)} \end{bmatrix}^\top$
	Q_k	$100^2 \text{ m}^2, 1^2 \text{ m}^2\text{s}^{-2}$
Observation	d_z	1
	Function	OK or bilinear
	Kriging kernel	Matérn, $\nu = 3/2$
	R_k	5^2 (nT)^2
Filter	Filter	RPF
	N	5000
	N_{th}	$0.5 N$
	α_{reg}	0.3
	P_0	$8000^2 \text{ m}^2, 2^2 \text{ m}^2\text{s}^{-2}$

Table 4.5: Simulation parameters for the navigation scenario. For improved readability, covariance matrices are represented by their diagonal coefficients only.

The chosen estimator is the MMSE estimator from (2.83). The performance of each compared approach is read through two quantities:

- The estimation RMSE, defined as:

$$\text{RMSE}_{\hat{x}} : k \mapsto \sqrt{\frac{1}{n_{\text{MC}}} \sum_{i=1}^{n_{\text{MC}}} \left\| \Pi_{\text{pos}} \hat{x}_k^{(i)} - \Pi_{\text{pos}} x_k \right\|_2^2} \quad (4.15)$$

where $\hat{x}_k^{(i)}$ is the state estimated by the i -th filter at time k , x_k is the corresponding true state, and Π_{pos} is the projection matrix onto the position

subspace:

$$\Pi_{\text{pos}} \triangleq \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (4.16)$$

This metric gives an absolute value for the average error made by the filter.

- The normalised estimation error squared (NEES):

$$\text{NEES}_{\hat{x}} : k \mapsto \frac{1}{n_{\text{MC}} d_x} \sum_{i=1}^{n_{\text{MC}}} \left\| \hat{x}_k^{(i)} - x_k \right\|_{\hat{P}_k^{(i)}}^2 \quad (4.17)$$

where $\| \cdot \|_{\hat{P}_k^{(i)}}$ is the Mahalanobis norm associated with the error covariance:

$$\left\| \hat{x}_k^{(i)} - x_k \right\|_{\hat{P}_k^{(i)}}^2 \triangleq \left(\hat{x}_k^{(i)} - x_k \right)^\top \left(\hat{P}_k^{(i)} \right)^{-1} \left(\hat{x}_k^{(i)} - x_k \right) \quad (4.18)$$

This metric detects overconfident filters by using the Mahalanobis norm associated with the error covariance matrix rather than the standard Euclidean distance.

Similarly, a run is declared convergent under two different criteria, as each describes a different behaviour:

- A fixed threshold of 2000 m, under which the median of the error over the last ten steps of a filter must be. This metric is simple to read and ensures that the estimation is relatively close to the true state. However, the threshold value is arbitrary.
- A Mahalanobis criterion, where the median of (4.18) over the last ten steps must remain under the 99% quantile of a χ^2 law with d_x degrees of freedom, written $\chi_{d_x}^2(99\%)$. While this criterion follows the property of a Gaussian error, it primarily assesses the consistency between the estimation error and the uncertainty reported by the filter. Consequently, a filter with a large estimation error may still satisfy this criterion if it reports a sufficiently large uncertainty.

4.2.4 Numerical Results

In this section, we compare the performance of the filter using 13 different observation functions:

- The first configuration corresponds to the known map, and serves as a reference case.
- The six next use ordinary kriging at different grid resolutions. This method is called static kriging for the purpose of distinguishing it from the adaptive methods presented in the subsequent sections.
- The final six instead use bilinear interpolation on the same grid.

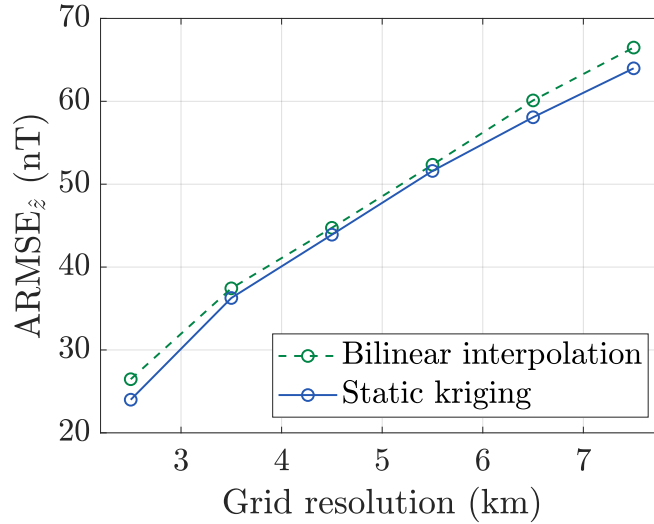


Figure 4.5: Reconstruction ARMSE of the field over the trajectory corridor for ordinary kriging and bilinear interpolation as a function of the training grid resolution. The domain E considered is a rectangular corridor around the true trajectory, with a 28 km margin. It covers 74% of the map.

Before analysing the filters themselves, it is important to assess the quality of the reconstructed maps. Figure 4.5 presents the reconstruction ARMSE along the true trajectory for ordinary kriging and bilinear interpolation across the different grid resolutions, in the sense of (4.6).

As one could have expected, denser grids lead to better reconstructions, the ARMSE evolving as the logarithm of Δ_g . However, kriging and bilinear interpolation exhibit similar errors, within 1 nT of each other at every resolution but the finest. Thus, the advantage of kriging cannot be primarily attributed to the reconstruction accuracy alone.

Focusing now on the filter performance, Figures 4.6 and 4.7 respectively compare the RMSE and NEES of the filter using the true map, kriging and bilinear interpolation on each of the six grid resolutions. For the sake of readability, both metrics are only computed on the converged runs with respect to the threshold criterion.

Three main results can be drawn from the RMSE curves. First, filters with access to more training points from a denser grid achieve better performance, with the limit case of the filter using the true map outperforming every other configuration. Second, kriging-based filters consistently achieve lower RMSEs than their bilinear counterparts despite having similar map reconstruction accuracy. This result, combined with Figure 4.5, indicates that the advantage of kriging lies in its ability to quantify reconstruction uncertainty through its covariance matrix. Finally, the time required to reach a stable estimation error increases from 13 to 25 steps as the grid resolution decreases from 2.5 to 7.5 km. In particular, the vehicle begins its first turn at $k = 33$, so any potential information gained from this turn does not affect this result.

The NEES curves further highlight the superiority of kriging methods by taking the estimated error covariance matrix into account. Kriging-based filters stay within the consistency band at every resolution, whereas bilinear interpolation lies above it even on the densest map. This shows that kriging not only leads to more accurate

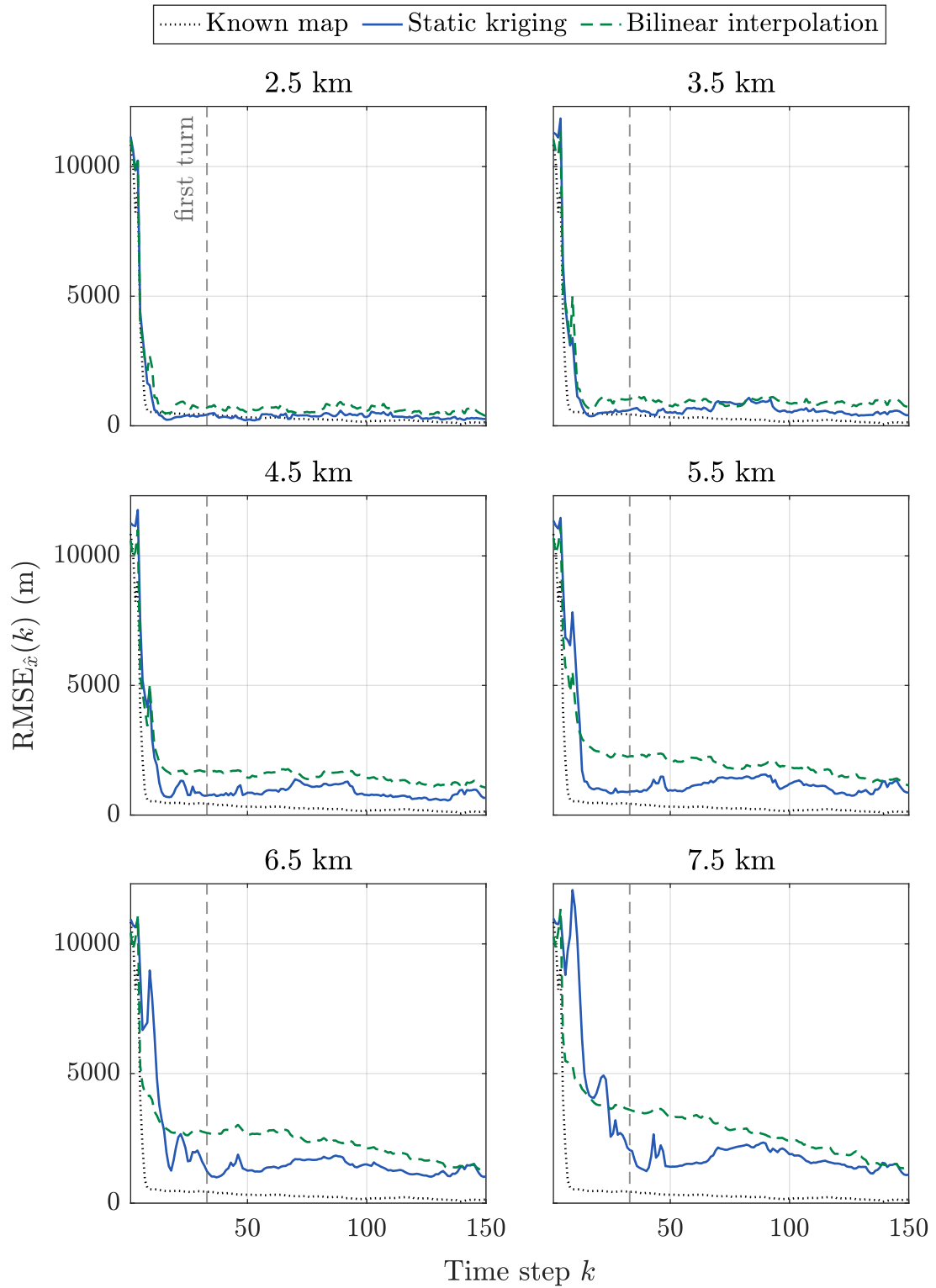


Figure 4.6: Estimation RMSE of the navigation algorithms using the known map, bilinear interpolation and static kriging. Each panel corresponds to a different grid resolution, and the true map result is repeated for comparison.

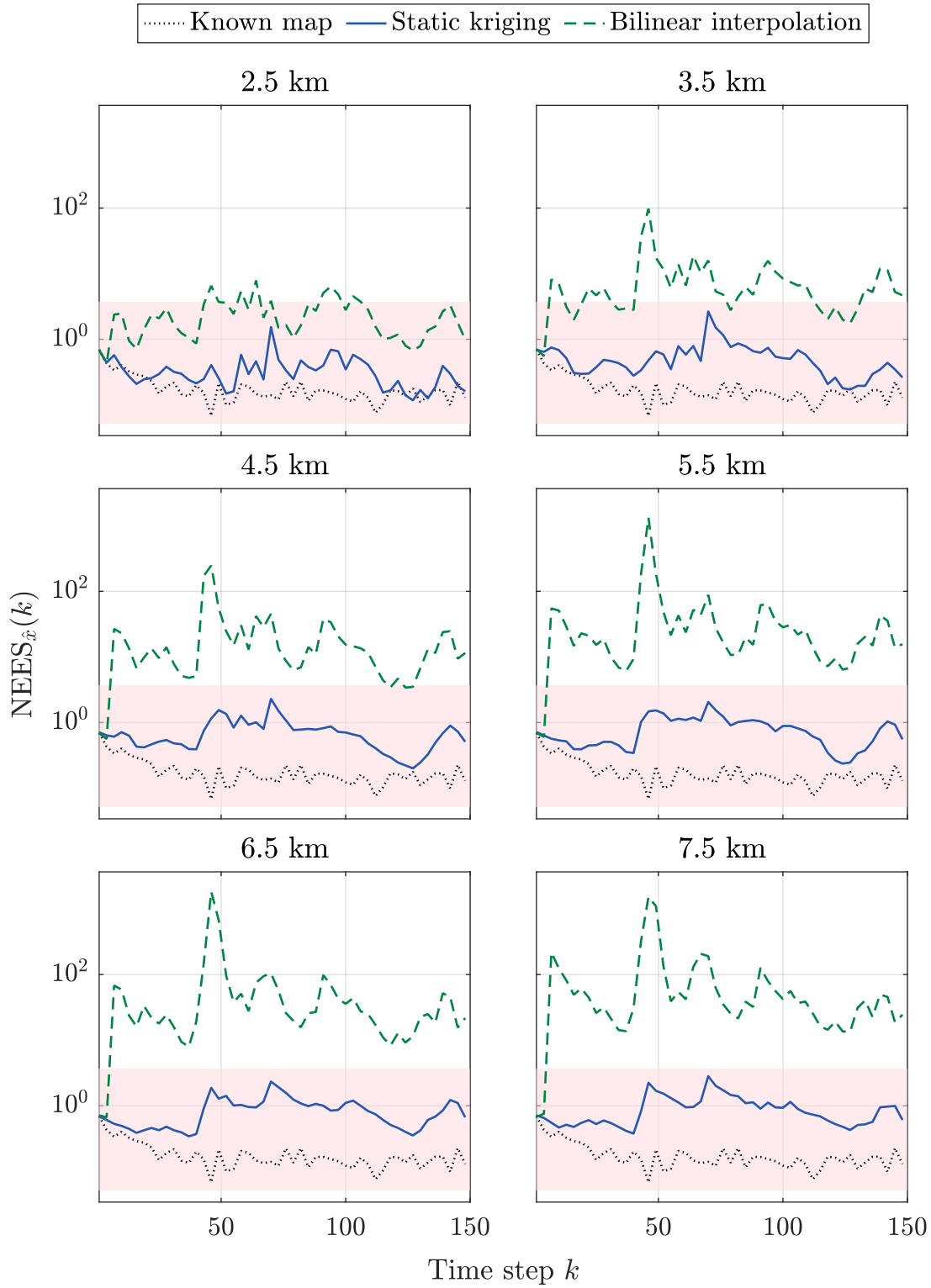


Figure 4.7: Estimation NEES of the navigation algorithms using the known map, bilinear interpolation and static kriging. The shaded area delimits the quantiles of a χ^2 law with d_x degrees of freedom, within which the NEES of a consistent filter falls with probability 0.99.

state estimates than bilinear interpolation, but also provides a better characterisation of their associated uncertainty.

In terms of convergence rates, Figure 4.8 starts by extracting the relevant the median errors from the previous curves, and Figure 4.9 computes the associated convergence rates using the two metrics introduced at the beginning of this section.

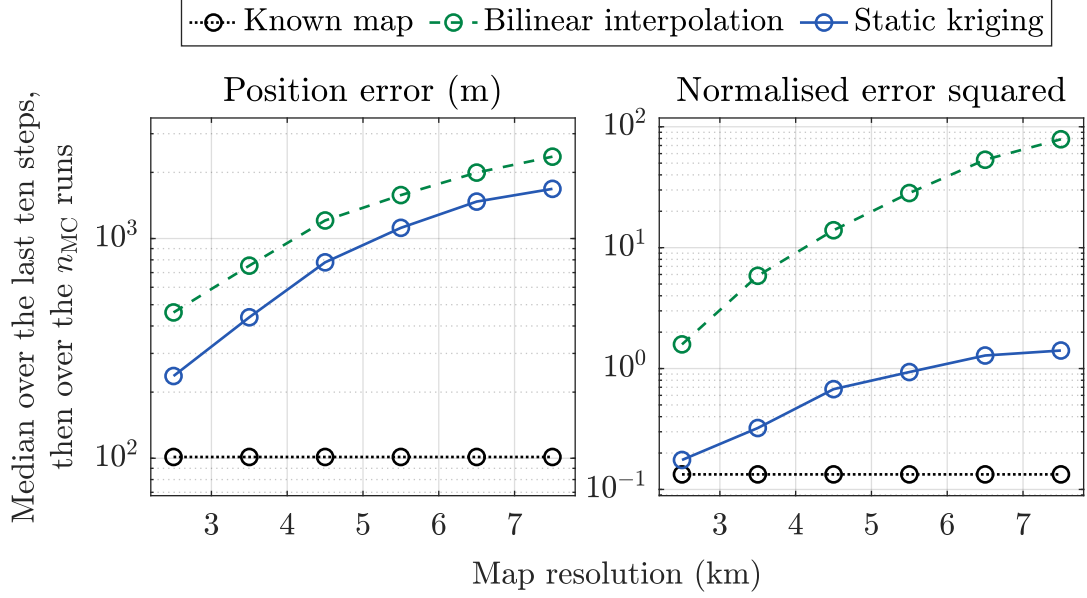


Figure 4.8: Average median RMSE and NEES at the end of the flight using the known map, bilinear interpolation and static kriging as a function of grid resolution.

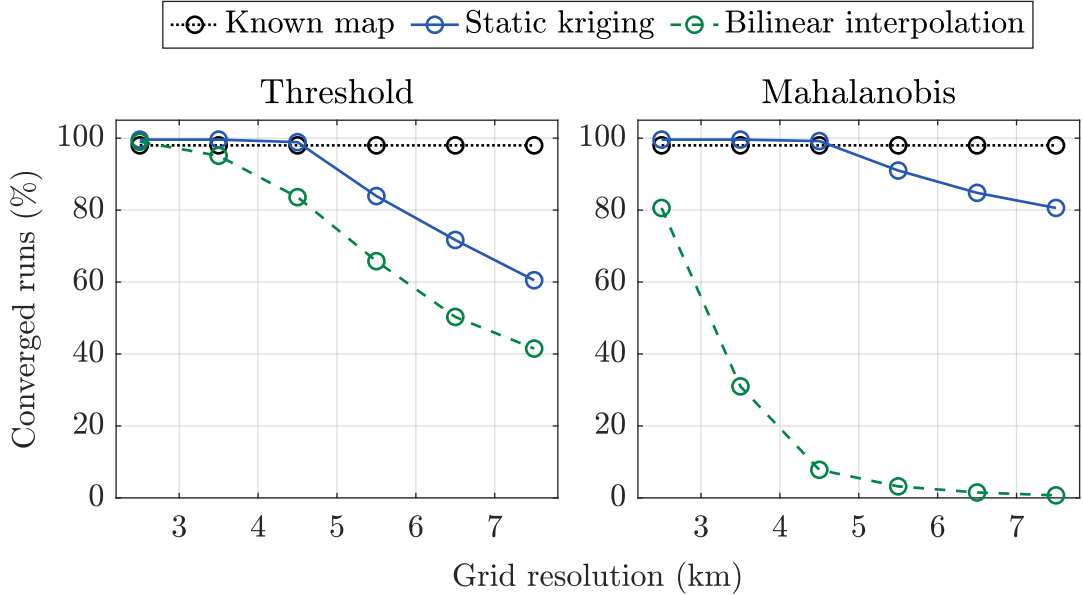


Figure 4.9: Percentage of converged runs for the known map, bilinear interpolation and static kriging against the grid resolution for the two metrics. The true map does not depend on the resolution, and has a convergence rate of 98% under both criteria.

The first two figures show that the bilinear to kriging RMSE ratio ranges from

1.4 to 1.9, and increases to 9 to 56 for the NEES. Furthermore, while the RMSE ratio tends to decrease at lower grid resolutions, the NEES ratio follows the opposite trend. Bilinear interpolation thus completely fails to measure its accuracy at lower resolutions, while kriging degrades in step with its own accuracy.

The convergence rates follow the same pattern. Under the threshold criterion, the rate for kriging decreases from 100% to 60% as the grid resolution decreases, while that of bilinear interpolation decreases from 99 to 42%. The two thus remain comparable. For the Mahalanobis criterion, however, kriging only falls to 81% while bilinear interpolation collapses to 1%, having lost every run whose reported uncertainty is consistent with its error.

However, the advantages of kriging come at an increased computational and memory costs, as described by Table 4.6.

Observation model	Cost (flops)			Memory (bytes)
	Setup	Queries	Total	
Known map	0	6 M	6 M	323 k
Bilinear interpolation	0	6 M	6 M	150 k
Static kriging, 2.5km	81 G	29.2 T	29.3 T	312 M
Static kriging, 3.5km	10.3 G	7.38 T	7.39 T	78.8 M
Static kriging, 4.5km	2.42 G	2.81 T	2.81 T	30 M
Static kriging, 5.5km	726 M	1.26 T	1.26 T	13.5 M
Static kriging, 6.5km	243 M	608 G	608 G	6.5 M
Static kriging, 7.5km	103 M	343 G	343 G	3.67 M

Table 4.6: Memory and computational costs of the navigation algorithm using the known map, bilinear interpolation, or static kriging. Since the cost of kriging depends on the number of training data, one row is assigned per grid resolution.

While the setup cost can be paid before the mission, the vehicle must still be able to store between 3.67 and 312 MB of data, and sustain an average computational rate of 0.46 to 38.9 Gflops/s over the navigation period. These requirements remain affordable at low grid resolutions, but these configurations result in poor navigation performance, as shown above. Increasing the grid resolution improves navigation accuracy, but quickly leads to prohibitive storage and computational requirements.

4.3 Adaptive Kriging

In this section, we propose to use training data subsampling, *i.e.* selecting a subset $\tilde{\mathbf{x}}^r$ of $\tilde{N}^r$ training points and using only these points for regression, as a way to alleviate the computational issues raised above. Since the computational and memory costs of kriging scale quadratically with the number of training points, discarding half the data reduces the computational load by a factor of 4 and requires 4 times less storage.

However, this method comes at a cost. In the following, we describe two subsampling strategies, each requiring to perform a subsampling step at each time step.

This means that no precomputation can be done before the mission, and in particular that inverting $G(\tilde{\mathbf{x}}^r) \in \mathcal{S}_{\tilde{N}^r d_z}^{++}(\mathbb{R})$ and $C^\top G(\tilde{\mathbf{x}}^r)^{-1} C \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ must be done online.

4.3.1 Random Subsampling

The simplest idea to reduce the number of training points is to let $p \in [0, 1]$, and draw independently $s_{i,k}$ from a Bernoulli distribution of success probability p , for each $i \in \llbracket 1, \tilde{N} \rrbracket$. Any training sample $\tilde{x}_i$ for which the corresponding $s_{i,k}$ is a success is retained to form $\tilde{\mathbf{x}}_{\text{RS},k}^r$, while the others are discarded. This operation is performed at the beginning of each time step to avoid skewing the reconstruction. Since the training points are randomly selected across the map rather than localised to a specific region of space, the hyperparameters are estimated before the mission on the entire training set, similarly to the static kriging version.

By linearity of the expectation, the number of retained data is, on average:

$$\mathbb{E} \left[\tilde{N}_{\text{RS},k}^r \right] = p \tilde{N} \quad (4.19)$$

and corresponds, on average, to the density of a regular grid of resolution $\Delta_g / \sqrt{p}$.

Similarly to the case of bilinear interpolation and static kriging presented above, the simulations described in Section 4.2.3 were run using random subsampling with:

$$p = 0.5 \quad (4.20)$$

The results are compared here with those obtained with static kriging. Figures 4.10 and 4.11 respectively compare the RMSE and NEES of the two approaches on each of the six grid resolutions, while Figure 4.12 presents the corresponding convergence rates. Finally, Table 4.7 gives the comparative computational cost of each method across the six resolutions.

For the RMSE curves, random subsampling is systematically outperformed by static kriging, especially before the first turn. From 4.5 km onward, random subsampling only reaches its error plateau during this turn, whereas static kriging stabilises before it. This indicates that, although the subsampling (RS) is randomised at each time step, and each training point is therefore available to the filter on average 50% of the time, the resulting map is too degraded to provide an accurate navigation solution. The late convergence of the filters with access to the fewest training points is likely a consequence of the additional information provided by the first turn.

Nevertheless, the similarity between random subsampling at $\Delta_g = 2.5$ and 5.5 km and static kriging at $\Delta_g = 3.5 \approx 2.5/\sqrt{0.5}$ and $7.5 \approx 5.5/\sqrt{0.5}$ km respectively, is noteworthy. This suggests that random subsampling effectively behaves as a map with a degraded resolution. Although the full training set is likely to have been sampled at least once over the course of the mission, only half of the points are available to the filter at each time step.

Regarding the NEES, the curves are consistently similar between both methods, further highlighting the ability of kriging to accurately quantify its uncertainty. As already established in Section 4.2.1, the kriging covariance increases in regions where less data are available, and thus the predicted uncertainty naturally follows the degradation of the map. Despite this, random subsampling achieves lower convergence rates at every grid resolution, showing that this method is unable to match the navigation performance of static kriging.

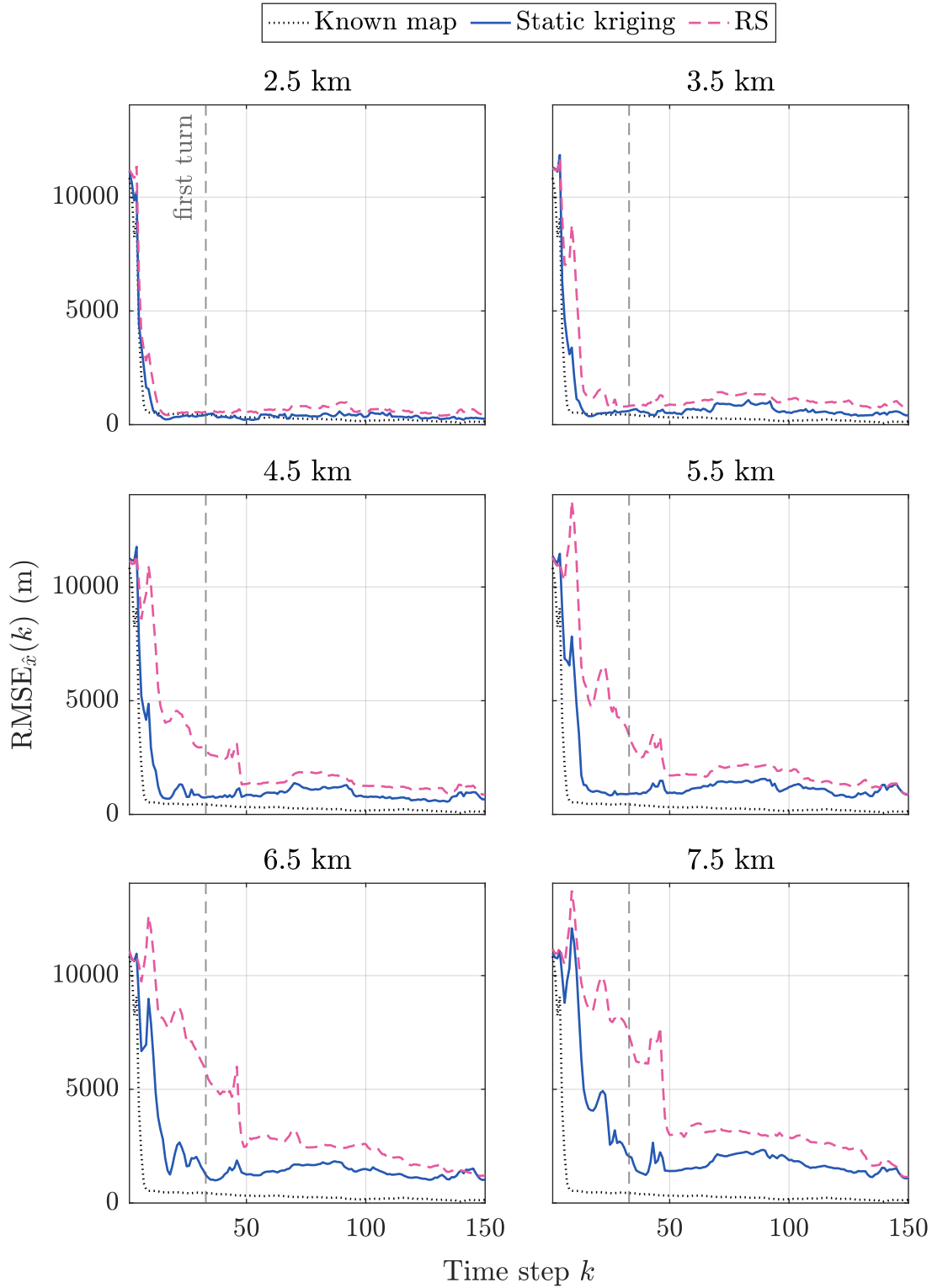


Figure 4.10: Comparative RMSE of the navigation algorithms using static kriging or random subsampling across the six grid resolutions. The performance of the filter with access to the true map is given as a reference.

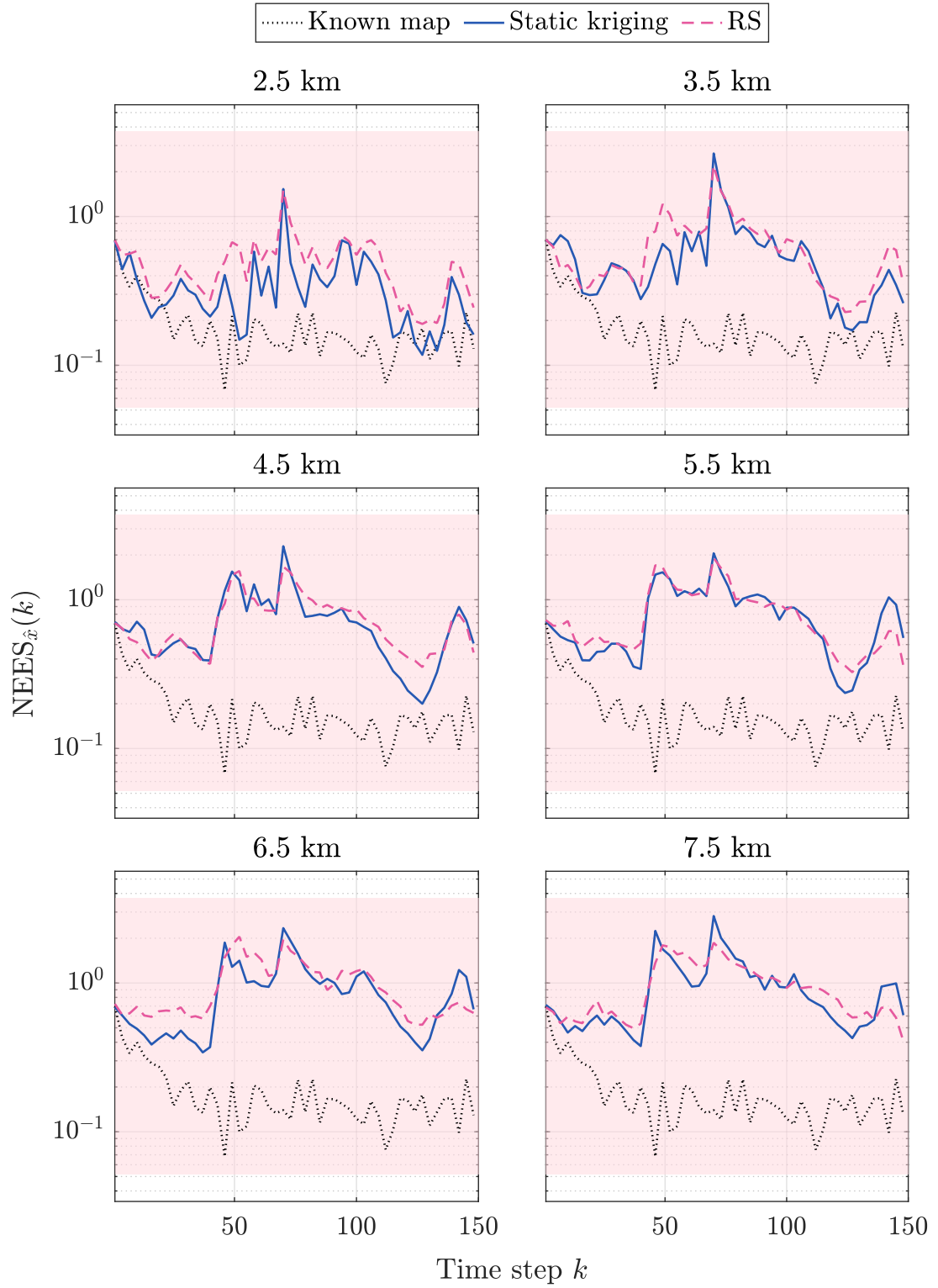


Figure 4.11: Comparative RMSE of the navigation algorithms using static kriging or random subsampling across the six grid resolutions. The shaded area corresponds to that of the previous NEES figure.

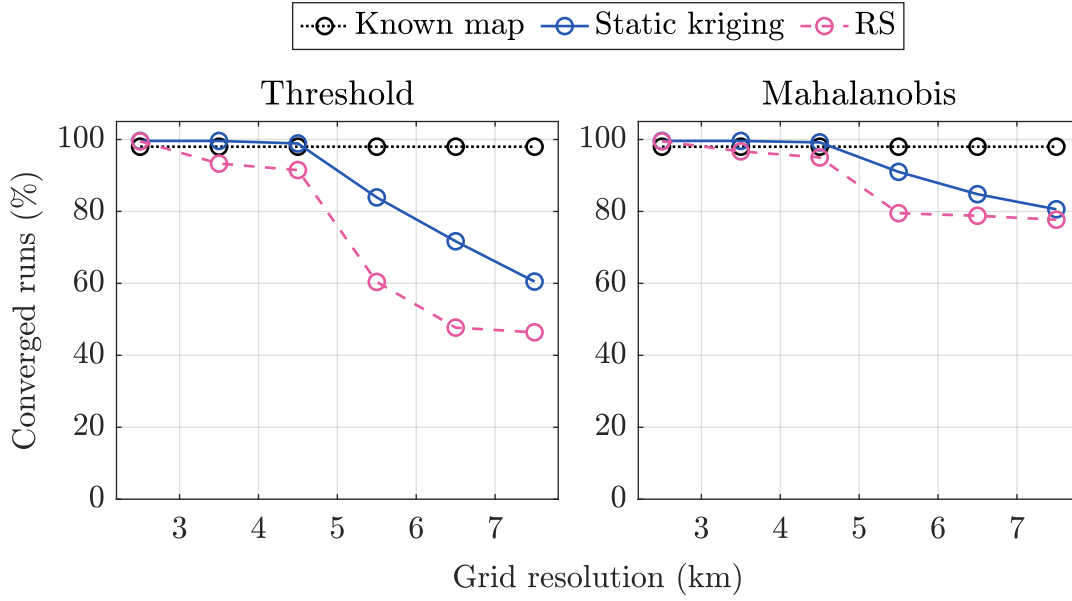


Figure 4.12: Convergence rates for the navigation algorithms using static kriging and random subsampling as a function of grid resolution.

Δ_g	Static kriging	RS
2.5km	29.3 T	7.3 T
3.5km	7.39 T	1.84 T
4.5km	2.81 T	704 G
5.5km	1.26 T	314 G
6.5km	608 G	152 G
7.5km	343 G	85.9 G

Table 4.7: Total computational cost, in flops, of using static kriging or random subsampling throughout the mission, for each grid resolution.

Finally, Table 4.7 illustrates the only advantage of random subsampling over static kriging. By reducing the number of available training points by half on average, this method reduces the computational cost by approximately 75% at every grid resolution.

4.3.2 Adaptive Subsampling Window

The above results show that, while subsampling does reduce the computational cost of the algorithm, indiscriminately discarding training points is detrimental to the navigation performance. A more suitable strategy should therefore retain only a limited number of training points while selecting those that are most relevant to the current situation.

Since, at each time step, the filter only queries the map where its particles lie, this suggests retaining the points surrounding the particle set. Beyond reducing the computational cost, a local selection also relaxes a second limitation of static kriging. Indeed, the estimators introduced in section 3.3 assume the use of a single kernel over

the entire domain, and assumption 4.1 inherits that hypothesis. In section 4.2.2, we estimated the hyperparameters of the kernel once before the mission, meaning that they are assumed to be independent of the inputs:

$$k_{3/2} : x, x' \mapsto \sigma_f^2 \left(1 + \frac{\sqrt{3} \|x - x'\|_2}{l} \right) \exp \left(-\frac{\sqrt{3} \|x - x'\|_2}{l} \right) \quad (4.21)$$

The considered kernel is thus radial, and the resulting GP is therefore stationary. This is unrealistic for a physical field, especially at large spatial scales. However, stationarity cannot be simply dropped, as the hyperparameters of such a kernel would have to be fitted at every point of the domain, where a single sample is available. We therefore assume the field to be non-stationary, but its hyperparameters to vary over a scale much larger than the grid resolution of the training data. Under this hypothesis, the hyperparameters must be estimated over a region large enough to provide sufficient data for accurate estimation, while remaining small enough to avoid averaging them over distances on which they vary significantly.

In this section, we propose an adaptive subsampling strategy to optimise the kriging performance of the filter according to four criteria:

- The retained training points must cover an area large enough to avoid extrapolation at the particle locations.
- A sufficient number of training points must be considered to ensure that the hyperparameters are accurately estimated.
- The retained training points must be confined to a sufficiently small region for the hyperparameters to be approximately constant within this region.
- Finally, their number must remain small enough to ensure that the filter can be computed within real-time constraints.

The first constraint naturally gives the shape of the retained subset. Let $\hat{x}_{k|k-1}$ be the predicted estimate of the filter at time k :

$$\hat{x}_{k|k-1} \triangleq \sum_{i=1}^N w_{k-1}^{(i)} x_k^{(i)} \approx \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k-1})}[x_k] \quad (4.22)$$

and $\hat{P}_{k|k-1}$ denote the corresponding covariance matrix:

$$\hat{P}_{k|k-1} \triangleq \sum_{i=1}^N w_{k-1}^{(i)} \left(x_k^{(i)} - \hat{x}_{k|k-1} \right) \left(x_k^{(i)} - \hat{x}_{k|k-1} \right)^\top \quad (4.23)$$

These two quantities can be computed after the prediction step at time k , but before the following weight update, and represent a Gaussian approximation of the particle set at time k before kriging the environment. It must be noted that, while the selection itself should only depend on the particle locations, both sums are weighted by the $w_{k-1}^{(i)}$ to limit the effect of resampling on the resulting moments. Although this may result in some particles being left outside the retained region, this is not detrimental when their associated weights are low.

In this approximation, each particle has a 99% chance to verify:

$$\left\| \Pi_{\text{pos}} \hat{x}_{k|k-1} - \Pi_{\text{pos}} x_k^{(i)} \right\|_{\hat{P}_{\text{pos},k|k-1}}^2 \leq \chi_{d_{\text{pos}}}^2(99\%) \quad (4.24)$$

where d_{pos} is the dimension of the position subspace, $\hat{P}_{\text{pos},k|k-1}$ is the covariance matrix associated with the projection of the predicted state onto this subspace:

$$\hat{P}_{\text{pos},k|k-1} \triangleq \Pi_{\text{pos}} \hat{P}_{k|k-1} \Pi_{\text{pos}}^\top \quad (4.25)$$

and $\chi_{d_{\text{pos}}}^2(99\%)$ is the 99% quantile of a χ^2 distribution with d_{pos} degrees of freedom. This means that the particles are likely to be contained in an ellipsoid centred on $\Pi_{\text{pos}} \hat{x}_{k|k-1}$, and whose shape is given by the eigenvectors of $\hat{P}_{\text{pos},k|k-1}$.

However, this geometry is tied to the particle set, and does not directly translate into the physical position space. Furthermore, when $\hat{P}_{\text{pos},k|k-1}$ is highly anisotropic, the resulting ellipsoid may provide insufficient coverage along directions of lower uncertainty, leading to extrapolation. To solve this issue, we use the Rayleigh quotient [94] to convert the Mahalanobis inequality into a Euclidean one:

$$\begin{aligned} \left\| \Pi_{\text{pos}} \hat{x}_{k|k-1} - \Pi_{\text{pos}} x_k^{(i)} \right\|_2^2 &\leq \lambda_{\max} \left(\hat{P}_{\text{pos},k|k-1} \right) \left\| \Pi_{\text{pos}} \hat{x}_{k|k-1} - \Pi_{\text{pos}} x_k^{(i)} \right\|_{\hat{P}_{\text{pos},k|k-1}}^2 \\ &\leq \chi_{d_{\text{pos}}}^2(99\%) \lambda_{\max} \left(\hat{P}_{\text{pos},k|k-1} \right) \end{aligned} \quad (4.26)$$

The first subsampling criterion is thus satisfied by retaining training points within a ball of radius:

$$r_{\text{cov},k} \triangleq \alpha_{\text{cov}} \sqrt{\chi_{d_{\text{pos}}}^2(99\%)} \sqrt{\lambda_{\max} \left(\hat{P}_{\text{pos},k|k-1} \right)} \quad (4.27)$$

and centred on $\Pi_{\text{pos}} \hat{x}_{k|k-1}$, where $\alpha_{\text{cov}} \geq 1$ is a dilation factor, introduced to provide a margin around the particle set and reduce the risk of extrapolation at its boundaries.

However, when the particle set collapses on a small region of space, the ball may not contain enough training points to satisfy the second criterion. We thus introduce a second radius to ensure that at least $\tilde{N}_{\min}$ training points are retained:

$$r_{\min,k} \triangleq \min_{r \in \mathbb{R}^+} \left(\text{Card} \left\{ i \in \llbracket 1, \tilde{N} \rrbracket \mid \left\| \Pi_{\text{pos}} \hat{x}_{k|k-1} - \tilde{x}_i \right\|_2 \leq r \right\} \geq \tilde{N}_{\min} \right) \quad (4.28)$$

The retained training set is thus defined at each time step k as:

$$\tilde{\mathbf{x}}_{\text{AK},k}^r \triangleq \left\{ \tilde{x}_i \in \tilde{\mathbf{x}} \mid \left\| \Pi_{\text{pos}} \hat{x}_{k|k-1} - \tilde{x}_i \right\|_2 \leq \max(r_{\text{cov},k}, r_{\min,k}) \right\} \quad (4.29)$$

and:

$$\tilde{\mathbf{z}}_{\text{AK},k}^r \triangleq \left\{ \tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}_{\text{AK},k}^r \right\} \quad (4.30)$$

At each time step, the hyperparameters $\hat{\theta}_k$ of the kernel are then re-estimated by maximum likelihood on the retained training set, as described in section 4.2.2. In particular, since the navigation estimate varies slowly between two consecutive time steps, the latter is likely to remain similar. The hyperparameters estimated at time $k-1$ can thus be used as initial values for the estimation at time k , reducing the number of iterations required for convergence of the hyperparameter estimation algorithm.

In particular, the number of retained points is, on average, the d_{pos} -volume of the window divided by that of a grid cell:

$$\mathbb{E} \left[\tilde{N}_{\text{AK},k}^r \right] = \frac{\pi^{d_{\text{pos}}/2}}{\Gamma\left(\frac{d_{\text{pos}}}{2} + 1\right)} \left(\frac{\max(r_{\text{cov},k}, r_{\text{min},k})}{\Delta_g} \right)^{d_{\text{pos}}} \quad (4.31)$$

The likelihood (4.3) is thus computed on the retained subset alone and using the updated hyperparameters:

$$z_k | x_k, \tilde{\mathbf{z}}_{\text{AK},k}^r \sim \mathcal{N} \left(\hat{z}_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{AK},k}^r, \tilde{\mathbf{z}}_{\text{AK},k}^r; \hat{\theta}_k \right), R_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{AK},k}^r; \hat{\theta}_k \right) \right) \quad (4.32)$$

This method, referred to as Adaptive Kriging (AK) and originally proposed in [95], can be integrated into any navigation algorithm presented in chapter 2. In this section, we focus on its integration into the RPF of Algorithm 2.8 using the transition density as proposal, as described in (2.97). The resulting algorithm is the Adaptive Kriging Regularised Particle Filter (AK-RPF) given in Algorithm 4.2.

Unfortunately, the third and fourth criteria may conflict with the first one in situations where the particle set spans over a large region of space. These two criteria are addressed in section 4.4.

4.3.3 Numerical Results

As for static kriging, this section evaluates the performance of the AK-RPF across the six grid resolutions, with:

$$\alpha_{\text{cov}} = 2 \quad \text{and} \quad \tilde{N}_{\text{min}} = 25 \quad (4.33)$$

The performance of the static kriging RPF is also given as a reference. Alongside the RMSE and the NEES, another metric is introduced to help explain the variations in computational cost between each configuration. This metric is the number of retained training points, *i.e.* the value of $\tilde{N}^r$, at each time step. Figures 4.13 and 4.14 present the comparative RMSE and NEES curves across the six grid resolutions, and Figure 4.15 gives the associated convergence rates. Finally, Figure 4.16 shows the evolution of the average number of retained training points throughout the mission, and Table 4.8 presents the average total computational cost of static kriging and AK during the entire mission for each of the six grid resolutions.

The first point to note is the strong similarity between the RMSE curves obtained with static kriging and AK for all grid resolutions. Apart from a slight reduction in the AK error between $k = 60$ and $k = 90$, and a slight increase during the first turn, particularly for the 7.5 km grid resolution, the curves almost completely overlap. The absence of any noticeable accuracy loss despite discarding most of the training set can be attributed to the local nature of the AK window.

Since discarded points are far from the particle cloud, they have negligible contributions to the map reconstruction at the particle locations. Their removal thus has little effect on the relative particle likelihoods. While this is only shown numerically here, its theoretical analysis is the subject of section 4.5.

On the other hand, the NEES curves of both algorithms are significantly different. Although both remain within the χ^2 confidence bounds throughout the trajectory, the NEES of static kriging rises above 2 around $k = 70$ for every resolution but

Algorithm 4.2 Adaptive Kriging Regularised Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot | x_{k-1}^{(i)})$ ▷ Particle propagation
 $\hat{x}_{k|k-1} = \sum_{i=1}^N w_{k-1}^{(i)} x_k^{(i)}$ ▷ **Predicted estimation**
 $\hat{P}_{k|k-1} = \sum_{i=1}^N w_{k-1}^{(i)} \left(x_k^{(i)} - \hat{x}_{k|k-1} \right) \left(x_k^{(i)} - \hat{x}_{k|k-1} \right)^\top$
 Compute $\tilde{\mathbf{x}}_{\text{AK},k}^r$ with (4.29) ▷ **Adaptive subsampling**
 Compute $\tilde{\mathbf{z}}_{\text{AK},k}^r$ with (4.30)
 $\hat{\theta}_k = \text{MLE} \left(\tilde{\mathbf{x}}_{\text{AK},k}^r, \tilde{\mathbf{z}}_{\text{AK},k}^r, \hat{\theta}_{k-1} \right)$ ▷ **Hyperparameter estimation**
 $\forall i \in \llbracket 1, N \rrbracket$, compute $\tilde{p} \left(z_k | x_k^{(i)} \right)$ from (4.32) ▷ **Kriging**
 $\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \tilde{p} \left(z_k | x_k^{(i)} \right)$ ▷ Weight update
 $\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$ ▷ Normalisation
 $\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N \left(w_k^{(i)} \right)^2$ ▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**
 $\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$ ▷ Whitening
 $\hat{P}_k = \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top$
 $A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top$
 $\left\{ x_k^{(j)}, w_k^{(j)}, - \right\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{ x_k^{(i)}, w_k^{(i)} \right\}_{i=1}^N \right)$ ▷ Resampling
 $\forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K$ ▷ Regularisation
 $\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)}$
end if

the finest, before decreasing again around $k = 125$. In contrast, the NEES of AK remains below 1 throughout the mission and for all grid resolutions. This may be explained by the non-stationarity of the field. The global kernel would provide a poor description of this portion of the field, whereas the local estimation of the hyperparameters would allow AK to adapt to it. It must also be noted that the RMSE of AK is lower precisely around $k = 70$, which further illustrates this explanation.

Regarding convergence, the two filters exhibit the same overall trend, with a drop in convergence for both criteria from $\Delta_g = 5.5$ km onward. Since its NEES is generally more conservative, AK achieves convergence for 87% of the runs at 7.5 km according to the Mahalanobis criterion, compared with 81% for static kriging. Their convergence rates remain similar for the threshold criterion, with the same drop at grid resolutions greater than 5.5 km.

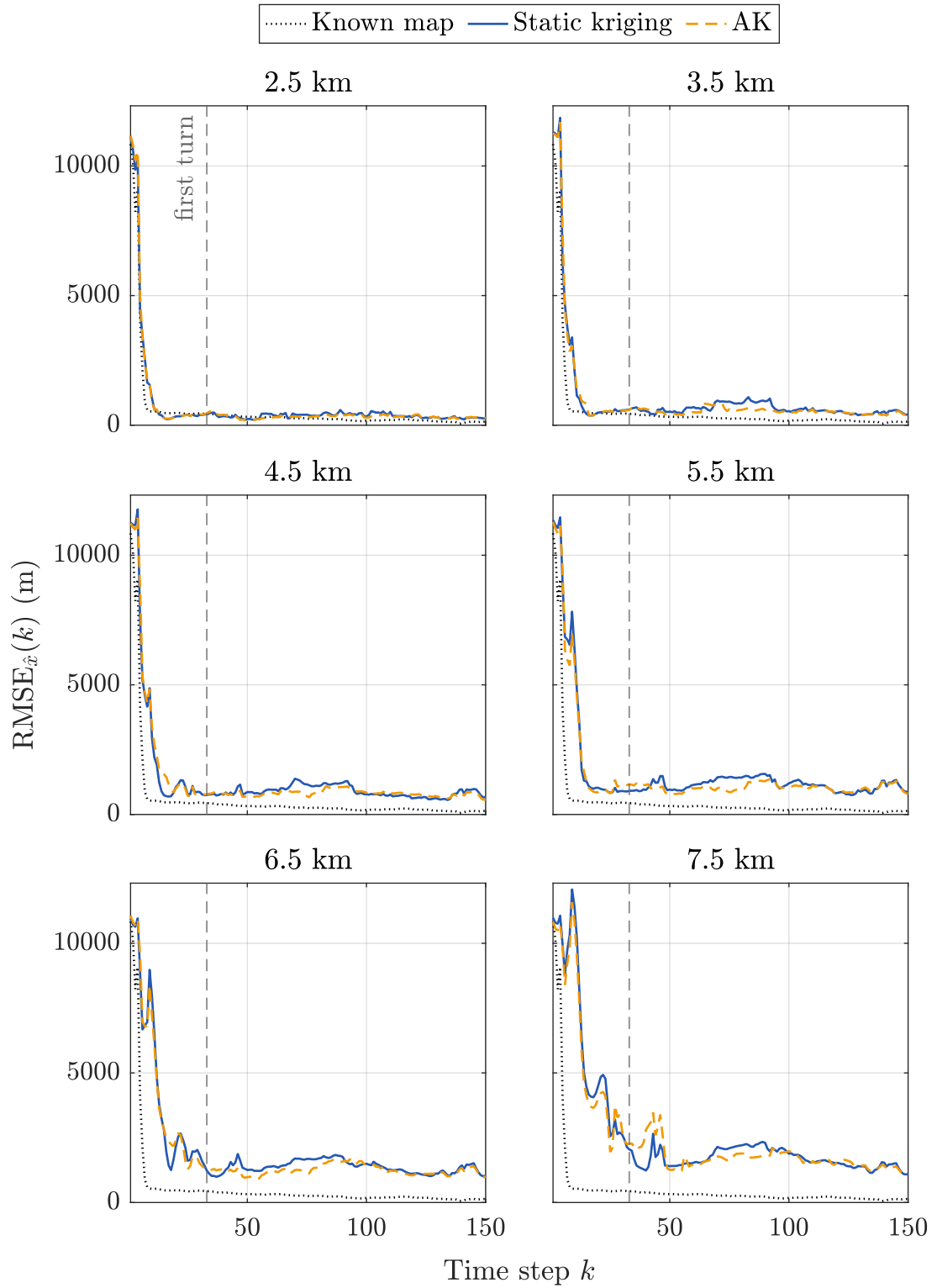


Figure 4.13: RMSE curves for the AK-RPF compared with that of an RPF using static kriging across the six grid resolutions.

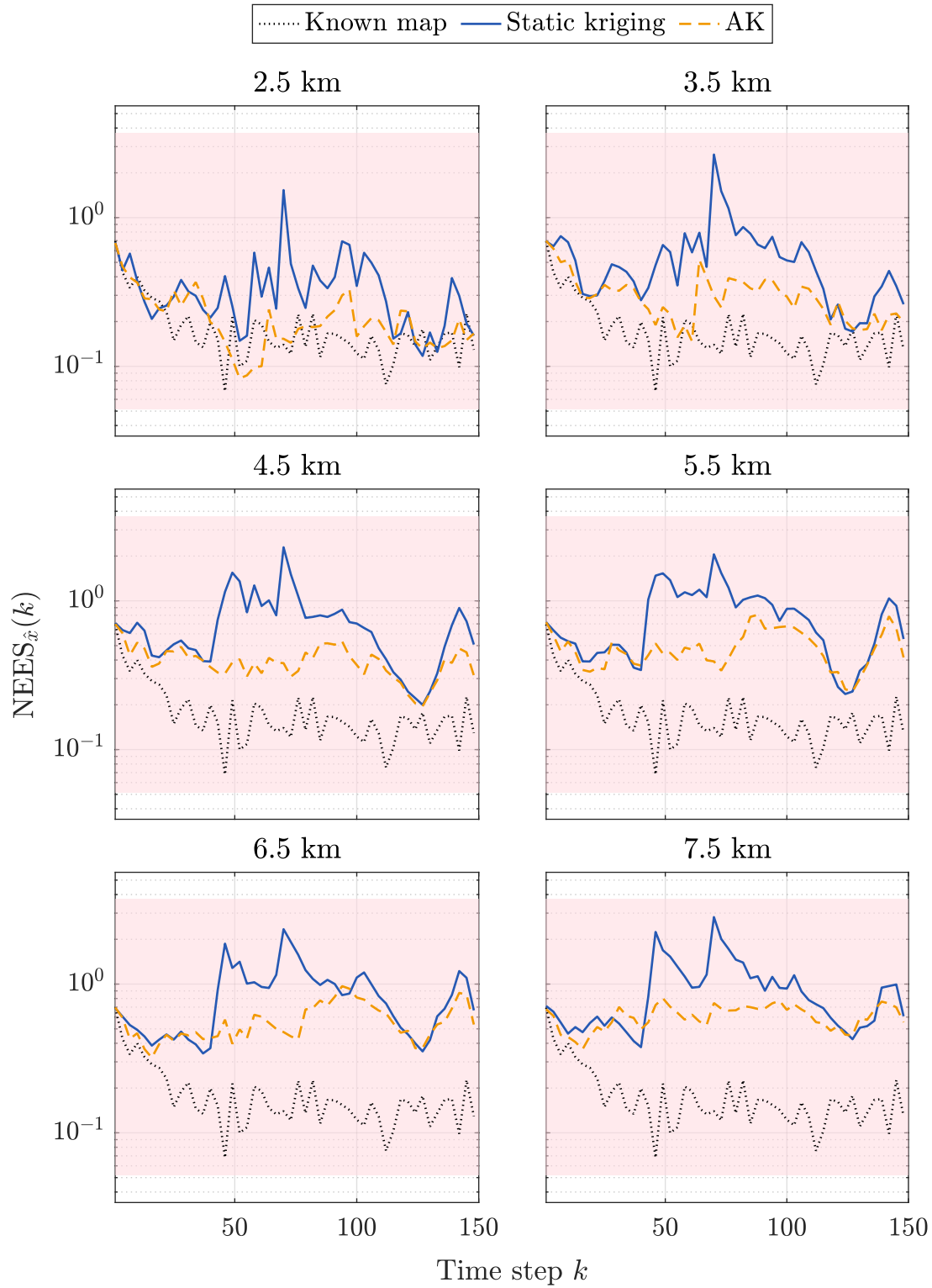


Figure 4.14: NEES curves for the AK-RPF compared with that of an RPF using static kriging across the six grid resolutions.

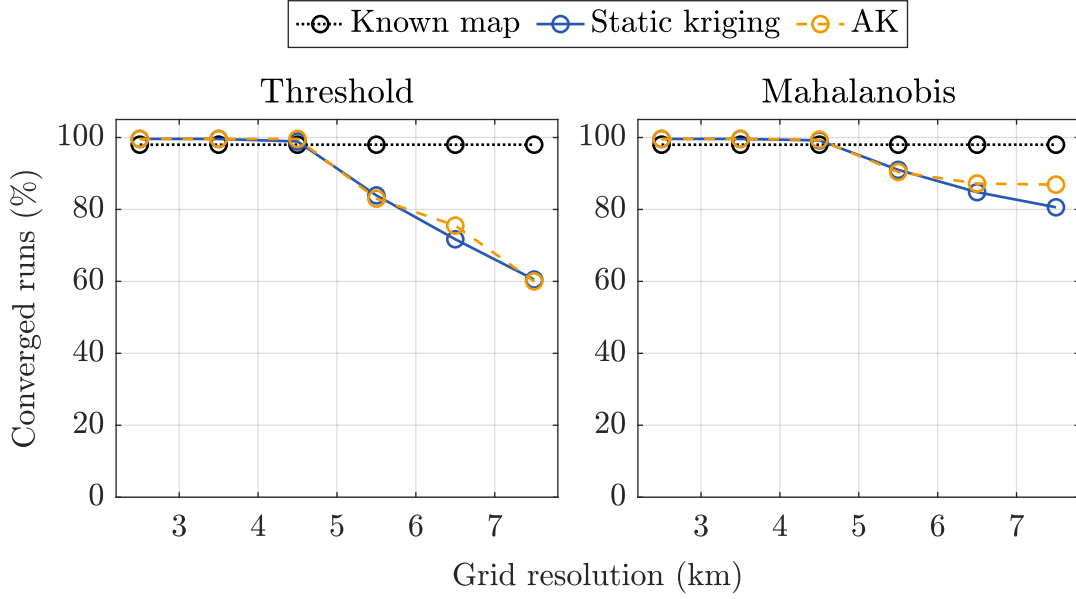


Figure 4.15: Convergence rates of the AK-RPF and the RPF using static kriging using the threshold and the Mahalanobis criteria.

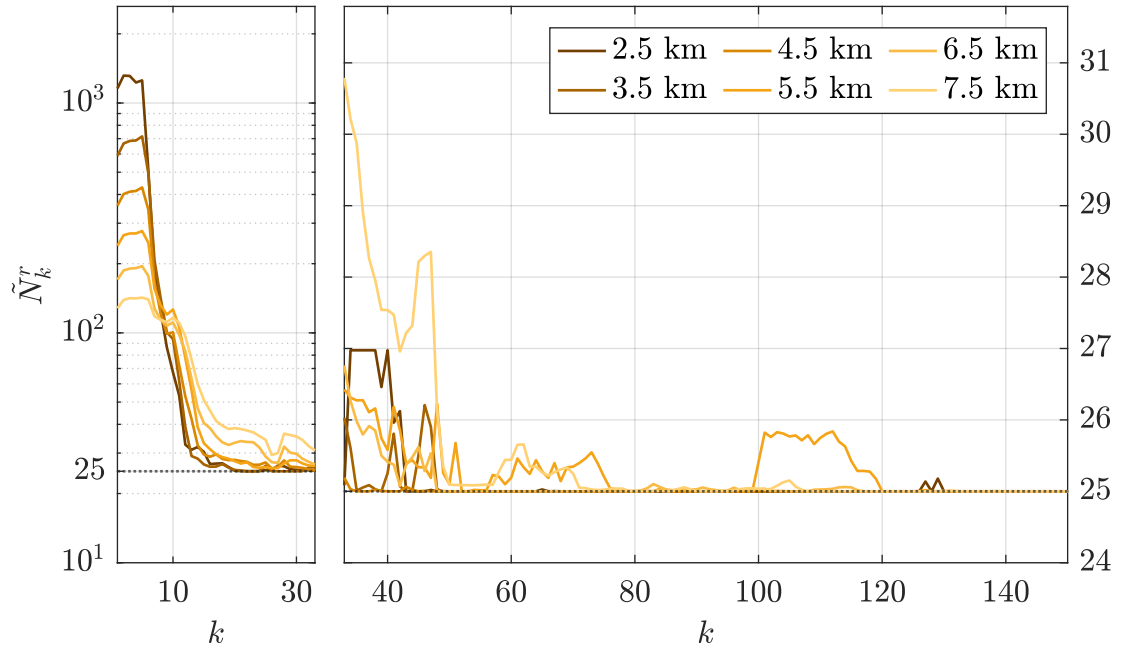


Figure 4.16: Evolution of the average number of training points retained by the AK-RPF during the mission for each grid resolution. A logarithmic scale is used before the first turn at $k = 33$ to capture the large variations in $\tilde{N}_{\text{AK},k}^r$, while a linear scale is used afterwards, where these variations are smaller.

Turning finally to the number of retained data, two regimes emerge from Figure 4.16. The first one, during which the size of the adaptive window is determined by the spatial dispersion of the particles, gives way to the second around $k = 25$, after which the number of retained points remains of the same order of magnitude as $\tilde{N}_{\text{min}} = 25$. In the first regime, using (4.31) with $d_{\text{pos}} = 2$ establishes that the

Δ_g	Static kriging	AK	
		Kriging	Hyperparameters
2.5km	29.3 T	46 G	564 M
3.5km	7.39 T	14.4 G	224 M
4.5km	2.81 T	5.75 G	121 M
5.5km	1.26 T	3.09 G	89.8 M
6.5km	608 G	1.93 G	70.1 M
7.5km	343 G	1.54 G	66.7 M

Table 4.8:

average number of retained training points is:

$$\tilde{N}_{\text{AK},k}^r \approx \frac{\pi r_{\text{cov},k}^2}{\Delta_g^2} \quad (4.34)$$

which explains the difference between the six curves at the beginning of the mission.

The rapid reduction in the number of retained training points also explains the computational cost of AK. Using the approximation that every filter uses 25 points after $k = 50$, the last hundred steps require approximately $100 \times 5000 \times 25^2 \approx 0.31$ Gflop of kriging at every grid resolution. This represents 0.7% of the AK kriging cost at $\Delta_g = 2.5$ km, up to 20% at 7.5 km. The local hyperparameter estimation only adds between 1 and 5% to these costs, so most of the computational load is incurred at the beginning of the mission, when the window is at its largest.

While AK is between 213 and 629 times less costly than static kriging for grid resolutions of 7.5 and 2.5 km respectively, this gap decreases as Δ_g increases. This happens because the cost of static kriging scales as $\tilde{N}^2$, decreasing by a factor of 85 between the finest and coarsest grids, whereas the cost of AK-RPF, bounded below by the minimum window size, decreases by only a factor of 29.

Taken together, these results show that AK removes the accuracy/cost compromise observed with static kriging, and that random subsampling failed to alleviate. Despite retaining only a small fraction of the available training set, it achieves comparable or better performance, while substantially reducing the overall cost of the filter. At $\Delta_g = 2.5$ km, its most expensive configuration, the entire AK mission only needs 46.6 Gflop, less than the 81 Gflop required by static kriging for the initial factorisation alone at the same resolution, and even less than the 343 Gflop of the least expensive static kriging configuration, at $\Delta_g = 7.5$ km.

4.4 Non-Spherical Subsampling Windows

While it outperforms static kriging, the AK window can still fail to satisfy the bottom two requirements established at the beginning of section 4.3.2. When a sufficiently large number of particles uniformly covers a wide region of the position subspace, any window that contains the particle set necessarily retains a large number of training points.

In other cases however, the limitation stems from the shape of the subsampling window. In particular, a single spherical window is poorly suited to non-Gaussian

particle distributions, especially multimodal PDFs, for which large regions within the window may contain few or no particles. This section introduces two methods to address this issue, either by adapting the shape of a single window to that of the particle distribution, or by using multiple spherical windows around clusters of particles.

4.4.1 Nearest Neighbour Selection

As established above, choosing the retained training points within a sphere centred on the predicted state provides a simple subsampling heuristic, but may include large regions without particles. A more refined selection should therefore follow the shape of the particle set more closely, avoiding the unnecessary computational cost of reconstructing the map in regions where it will not be queried.

A simple way to do this is to define, for each particle, a spherical selection window centred on it, similarly to AK but with a smaller radius, and to retain the training points contained within the union of these spheres. However, unlike the entire particle set, a single particle has no individual covariance matrix to compute (4.27). Instead, the windows are defined using only the $r_{\min,k}$ criterion, applied to a single particle:

$$\forall i \in \llbracket 1, N \rrbracket, \tilde{\mathbf{x}}_{\text{NNAK},k}^{r,(i)} \triangleq \left\{ \tilde{x}_j \in \tilde{\mathbf{x}} \mid \left\| \Pi_{\text{pos}} x_k^{(i)} - \tilde{x}_j \right\|_2 \leq r_{\min,k}^{(i)} \right\} \quad (4.35)$$

where:

$$r_{\min,k}^{(i)} \triangleq \min_{r \in \mathbb{R}^+} \left(\text{Card} \left\{ j \in \llbracket 1, \tilde{N} \rrbracket \mid \left\| \Pi_{\text{pos}} x_k^{(i)} - \tilde{x}_j \right\|_2 \leq r \right\} \geq \tilde{N}_{\min} \right) \quad (4.36)$$

Each individual selection window thus contains at least the $\tilde{N}_{\min}$ training points closest to the corresponding particle, with exactly $\tilde{N}_{\min}$ points unless several of them are equidistant from the considered particle. The global subsampling window is then defined as the union of the individual windows:

$$\tilde{\mathbf{x}}_{\text{NNAK},k}^r \triangleq \bigcup_{i=1}^N \tilde{\mathbf{x}}_{\text{NNAK},k}^{r,(i)} \quad (4.37)$$

and:

$$\tilde{\mathbf{z}}_{\text{NNAK},k}^r \triangleq \{ \tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}_{\text{NNAK},k}^r \} \quad (4.38)$$

Due to the nature of the selection, we call this method Nearest Neighbour Adaptive Kriging (NNAK). Similarly to AK, NNAK re-estimates the hyperparameters of the kernel by maximum likelihood estimation on $\tilde{\mathbf{x}}_{\text{NNAK},k}^r$ and $\tilde{\mathbf{z}}_{\text{NNAK},k}^r$, and computes the likelihood (4.3) using the entire retained data:

$$z_k | x_k, \tilde{\mathbf{z}}_{\text{NNAK},k}^r \sim \mathcal{N} \left(\hat{z}_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{NNAK},k}^r, \tilde{\mathbf{z}}_{\text{NNAK},k}^r; \hat{\theta}_k \right), R_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{NNAK},k}^r; \hat{\theta}_k \right) \right) \quad (4.39)$$

Regarding the parameters of this method, NNAK does not rely on the covariance-based criterion of AK, and has therefore no α_{cov} parameter to tune. In addition, $\tilde{N}_{\min}$ is deliberately chosen smaller than for AK, as NNAK aims to follow the shape of the particle set without unnecessarily enlarging or distorting it. It must be noted that, unlike AK, no floor is applied to the size of the entire NNAK subsampling window. The pseudo-code of a NNAK-based RPF is given by Algorithm 4.3.

Algorithm 4.3 Nearest Neighbour Adaptive Kriging RPF

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot | x_{k-1}^{(i)})$ ▷ Particle propagation
 Compute $\tilde{\mathbf{x}}_{\text{NNAK},k}^r$ with (4.37) ▷ **Adaptive subsampling**
 Compute $\tilde{\mathbf{z}}_{\text{NNAK},k}^r$ with (4.38)
 $\hat{\theta}_k = \text{MLE}(\tilde{\mathbf{x}}_{\text{NNAK},k}^r, \tilde{\mathbf{z}}_{\text{NNAK},k}^r, \hat{\theta}_{k-1})$ ▷ **Hyperparameter estimation**
 $\forall i \in \llbracket 1, N \rrbracket$, compute $\tilde{p}(z_k | x_k^{(i)})$ from (4.39) ▷ **Kriging**
 $\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \tilde{p}(z_k | x_k^{(i)})$ ▷ Weight update
 $\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$ ▷ Normalisation
 $\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N (w_k^{(i)})^2$ ▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**
 $\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$ ▷ Whitening
 $\hat{P}_k = \sum_{i=1}^N w_k^{(i)} (x_k^{(i)} - \hat{x}_k) (x_k^{(i)} - \hat{x}_k)^\top$
 $A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top$
 $\{x_k^{(j)}, w_k^{(j)}, -\}_{j=1}^N \leftarrow \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right)$ ▷ Resampling
 $\forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K$ ▷ Regularisation
 $\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)}$
end if

4.4.2 Clustering

The main difference between AK and NNAK lies in the scale at which the particle set is considered. While AK views the particle set as a whole from a macroscopic perspective, NNAK adopts a microscopic point of view by treating each particle individually. In this section, we introduce an intermediate, mesoscopic perspective by partitioning the particle set into clusters.

TODO: à reprendre

Such a partition is recomputed at every time step, and the subsampling windows are built on it, which constrains the choice of a clustering method:

- The number of clusters is unknown before the mission, and changes from one time step to the next as the ambiguity of the field is resolved. It can therefore not be an input of the method.
- The particles and their weights approximate the PDF. Therefore, clustering should account for particle weights, so that modes represented by few high-weight particles are properly identified.
- Every particle must belong to a cluster, since a particle left aside would have no window to be kriged from.

- Each cluster must come with a position and a spread, from which its window is defined by (4.27).
- The cost of the partition must remain a small fraction of the kriging cost it is meant to reduce.

The methods below are presented those exploring multiple possible partitions of the particle set to those directly producing a single partition at a computational cost compatible with the mission constraints. Throughout, the time index is dropped, and the particle set is read as the weighted point set $\{y^{(i)}, w^{(i)}\}_{i \in \llbracket 1, N \rrbracket}$, where $y^{(i)} \triangleq \Pi_{\text{pos}} x_k^{(i)}$ is the position of a particle and $w^{(i)} \triangleq w_{k-1}^{(i)}$ its weight. A partition of $\llbracket 1, N \rrbracket$ into c clusters is written $\mathcal{P} \triangleq \{\mathcal{C}^{(1)}, \dots, \mathcal{C}^{(c)}\}$, each cluster having a total weight and a weighted barycentre:

$$W^{(j)} \triangleq \sum_{i \in \mathcal{C}^{(j)}} w^{(i)} \quad \text{and} \quad \mu^{(j)} \triangleq \frac{1}{W^{(j)}} \sum_{i \in \mathcal{C}^{(j)}} w^{(i)} y^{(i)} \quad (4.40)$$

and the whole partition carrying its within-cluster inertia:

$$I(\mathcal{P}) \triangleq \sum_{j=1}^c \sum_{i \in \mathcal{C}^{(j)}} w^{(i)} \|y^{(i)} - \mu^{(j)}\|_2^2 \quad (4.41)$$

which measures how tightly each cluster is gathered around its own barycentre, and decreases as the partition is refined.

Hierarchical clustering analysis Minimising (4.41) over every partition of the point set is out of reach, their number being the Bell number B_N , which already exceeds 10^{115} for $N = 100$. Hierarchical clustering analysis (HCA) answers the problem differently, by building the whole family of nested partitions instead of selecting one of them. It starts from the finest partition, and repeatedly merges the two clusters that a dissimilarity D declares closest:

$$\mathcal{P}_0 \triangleq \{\{1\}, \dots, \{N\}\} \quad \text{and} \quad \mathcal{P}_{m+1} \triangleq (\mathcal{P}_m \setminus \{\mathcal{A}_m, \mathcal{B}_m\}) \cup \{\mathcal{A}_m \cup \mathcal{B}_m\} \quad (4.42)$$

where the merged pair and the height at which it is merged are:

$$(\mathcal{A}_m, \mathcal{B}_m) \triangleq \underset{(\mathcal{A}, \mathcal{B}) \in \mathcal{P}_m^2, \mathcal{A} \neq \mathcal{B}}{\operatorname{argmin}} D(\mathcal{A}, \mathcal{B}) \quad \text{and} \quad h_m \triangleq D(\mathcal{A}_m, \mathcal{B}_m) \quad (4.43)$$

After $N - 1$ merges a single cluster remains, and the nested partitions $(\mathcal{P}_m)_{m \in \llbracket 0, N-1 \rrbracket}$, drawn against their heights $(h_m)_{m \in \llbracket 0, N-2 \rrbracket}$, form the dendrogram of the point set. The dissimilarity need not be recomputed from the points after each merge, the usual choices obeying the recurrence [96]:

$$D(\mathcal{A} \cup \mathcal{B}, \mathcal{C}) = \alpha_{\mathcal{A}} D(\mathcal{A}, \mathcal{C}) + \alpha_{\mathcal{B}} D(\mathcal{B}, \mathcal{C}) + \beta D(\mathcal{A}, \mathcal{B}) + \gamma |D(\mathcal{A}, \mathcal{C}) - D(\mathcal{B}, \mathcal{C})| \quad (4.44)$$

whose coefficients define the linkage. Taking $\alpha_{\mathcal{A}} = \alpha_{\mathcal{B}} = \frac{1}{2}$, $\beta = 0$ and $\gamma = -\frac{1}{2}$ merges the clusters holding the two closest points, which chains elongated groups, while $\gamma = \frac{1}{2}$ merges the clusters whose farthest points are closest, which favours

compact ones. Of particular interest here is Ward’s linkage [97], which merges the pair whose merging increases the inertia (4.41) the least, that increase being:

$$\Delta I(\mathcal{A}, \mathcal{B}) = \frac{W^{\mathcal{A}} W^{\mathcal{B}}}{W^{\mathcal{A}} + W^{\mathcal{B}}} \|\mu^{\mathcal{A}} - \mu^{\mathcal{B}}\|_2^2 \quad (4.45)$$

A partition is then read from the dendrogram in one of two ways. Stopping after $N - c$ merges returns exactly c clusters, whatever their internal dissimilarity, whereas stopping at a height h returns the partition whose clusters have all been merged below h , whatever their number. Stopping after $N - c$ merges imposes a partition into exactly c clusters, whereas cutting the dendrogram at a height h determines the number of clusters according to a prescribed dissimilarity threshold.

HCA thus describes what a partition of the particle set is, at every granularity at once, and a partition can then be selected from the hierarchy, for instance by fixing either the number of clusters c or a dissimilarity threshold h . It cannot, however, be run at every time step. The merging rule needs the dissimilarities between all pairs of clusters, of which there are $N(N-1)/2$ at the first merge, that is more than twelve million for the $N = 5000$ particles of this chapter, and a hundred megabytes of memory in double precision. Maintaining them along the $N - 1$ merges costs $\mathcal{O}(N^2 \log N)$ operations, to be compared with the $\mathcal{O}\left(\left(\tilde{N}^r\right)^3\right)$ of the kriging it is meant to lighten. It therefore serves here as the reference against which the methods below are read, and not as one that can be run in flight.

k -means Once the number of clusters is fixed to c , the inertia (4.41) can be minimised directly, over the partition and the centres jointly:

$$(\mathcal{P}^*, \mu^*) \in \operatorname{argmin}_{\mathcal{P}, \mu} \left(\sum_{j=1}^c \sum_{i \in \mathcal{C}^{(j)}} w^{(i)} \|y^{(i)} - \mu^{(j)}\|_2^2 \right) \quad (4.46)$$

Neither argument is easy to optimise, but each of them becomes immediate once the other is held fixed. At fixed centres, the criterion is a sum of independent terms, minimised by assigning every point to the closest centre:

$$\mathcal{C}^{(j)} = \left\{ i \in \llbracket 1, N \rrbracket \left| j = \operatorname{argmin}_{l \in \llbracket 1, c \rrbracket} \|y^{(i)} - \mu^{(l)}\|_2 \right. \right\} \quad (4.47)$$

and at fixed partition, the criterion is quadratic in each centre, whose first-order condition returns the barycentre (4.40) of its cluster. Alternating the two is Lloyd’s algorithm [98, 99]. Each of its steps does not increase the criterion, which takes at most as many values as there are partitions, so that the alternation stops after finitely many steps, at a local minimum depending on the initial centres. One iteration costs $\mathcal{O}(N c d_{\text{pos}})$ operations, which is what makes the method affordable in real time.

The partition it returns is the Voronoi tessellation of the position subspace generated by the centres, as (4.47) makes explicit, and the criterion it minimises is the one Ward’s linkage (4.45) increases the least at every merge. Both methods thus answer the same question, one at a fixed number of clusters, the other along the whole hierarchy. That number is precisely what is unknown here, and a wide unimodal particle set is cut by (4.47) as readily as two distant modes are separated.

Mean shift The particle set being a weighted sample of the PDF, its clusters can be sought as the modes of that PDF rather than as the cells of a tessellation. Smoothing the sample with a kernel of profile κ and bandwidth r_{ms} gives the estimated density:

$$\hat{p}(y) \triangleq \frac{1}{r_{\text{ms}}^{d_{\text{pos}}}} \sum_{i=1}^N w^{(i)} \kappa \left(\left\| \frac{y - y^{(i)}}{r_{\text{ms}}} \right\|_2^2 \right) \quad (4.48)$$

whose gradient factorises into a non-negative scalar and a displacement [100]:

$$\nabla \hat{p}(y) = \frac{2}{r_{\text{ms}}^{d_{\text{pos}}+2}} \left(\sum_{i=1}^N w^{(i)} g^{(i)}(y) \right) \left(\frac{\sum_{i=1}^N w^{(i)} g^{(i)}(y) y^{(i)}}{\sum_{i=1}^N w^{(i)} g^{(i)}(y)} - y \right) \quad (4.49)$$

where $g^{(i)}(y) \triangleq -\kappa' \left(\left\| (y - y^{(i)}) / r_{\text{ms}} \right\|_2^2 \right)$ weighs each particle according to its distance to y .

The second factor of (4.49) points from y towards the weighted barycentre of its neighbourhood, and the first one is a non-negative scalar. The mean-shift vector therefore points in the direction of the density gradient, leading to the following fixed-point iteration:

$$m^{(n+1)} \triangleq \sum_{i=1}^N w^{(i)} g^{(i)}(m^{(n)}) y^{(i)} / \sum_{i=1}^N w^{(i)} g^{(i)}(m^{(n)}) \quad (4.50)$$

Using a uniform kernel, the mean-shift update reduces to the weighted barycentre of the particles lying within a ball of radius r_{ms} around the current iterate:

$$\mathcal{B}(m) \triangleq \{i \in \llbracket 1, N \rrbracket \mid \|y^{(i)} - m\|_2 \leq r_{\text{ms}}\} \quad (4.51)$$

so that (4.50) is the weighted barycentre of the particles lying inside it:

$$m^{(n+1)} = \sum_{i \in \mathcal{B}(m^{(n)})} w^{(i)} y^{(i)} / \sum_{i \in \mathcal{B}(m^{(n)})} w^{(i)} \quad (4.52)$$

The iteration seeks a mode of the estimated density, and particles converging to the same mode can be associated with the same cluster [101]. This is the mean shift. Its parameter is a length rather than a count, the weights enter it through (4.50), every point is assigned, and one iteration of one seed costs $\mathcal{O}(N)$ distance evaluations.

DBSCAN A density can be thresholded rather than climbed. Writing $\mathcal{V}_\varepsilon(y)$ for the particles lying within a distance ε of y , a particle is declared a core particle when the weight gathered in its own neighbourhood reaches a threshold W_ε :

$$\mathcal{V}_\varepsilon(y) \triangleq \{i \in \llbracket 1, N \rrbracket \mid \|y^{(i)} - y\|_2 \leq \varepsilon\} \quad \text{and} \quad \sum_{l \in \mathcal{V}_\varepsilon(y^{(i)})} w^{(l)} \geq W_\varepsilon \quad (4.53)$$

Two core particles are said to be connected when a chain of core particles, each lying within ε of the next, joins them. Connectedness is an equivalence relation over the core particles, whose classes are the clusters, and every remaining particle lying within ε of a core particle joins the cluster of the latter. What no chain reaches is labelled as noise. This is DBSCAN [102], whose parameters are again a length and a weight rather than a count, and which costs $\mathcal{O}(N \log N)$ when the neighbourhoods of (4.53) are searched through a spatial index.

Mean shift and DBSCAN thus answer the same question, both reading the clusters in the density of the point set instead of imposing their number, which is what the particle set requires. They differ in what they return. The first assigns every point to a mode and hands back that mode as the position of the cluster, whereas the second recovers clusters of arbitrary shape but returns neither a position nor a spread, leaves the points of the sparse regions out of the partition, and rests on a radius that would have to suit a particle density varying by orders of magnitude between the acquisition and the tracking phases.

Returning to the notation of the chapter, the c_k modes found at time k are written $\mu_k^{(j)}$, and the clusters of the particle set $\mathcal{C}_k^{(j)}$. Attaching every particle to the closest mode, rather than to the mode its own ascent reaches, replaces the basins of attraction by their Voronoi approximation, and extends the partition to any point of the position subspace:

$$\mathcal{C}_k^{(j)} \triangleq \left\{ i \in \llbracket 1, N \rrbracket \left| j = \underset{l \in \llbracket 1, c_k \rrbracket}{\operatorname{argmin}} \left\| \Pi_{\text{pos}} x_k^{(i)} - \mu_k^{(l)} \right\|_2 \right. \right\} \quad (4.54)$$

4.4.3 Clustered Adaptive Kriging

AK and NNAK can thus be seen as limiting cases of clustering, corresponding respectively to a single cluster containing the entire particle set and to one cluster per particle. The framework introduced in Section 4.4.2 can be used to construct the intermediate, mesoscopic view. However, the question of which clustering algorithm is best suited for this view must be addressed.

This choice is constrained by both computational and structural considerations. First, the heavy computational cost of HCA makes it unsuitable for real-time applications. Furthermore, the number of clusters at each time step is unknown prior to the mission, which prevents the use of k -means. Finally, neither Mean-shift nor DBSCAN require the number of clusters to be specified in advance, and both are computationally affordable at each time step.

Mean-shift introduces a characteristic scale through its bandwidth, which may artificially limit the size of each cluster and split a single, spatially extended cluster into several ones, whose windows may then overlap. However, this bound helps satisfy the last two requirements of section 4.3.2. On the other hand, DBSCAN identifies clusters based on the connected components of the neighbourhood graph of the particle set. A chain of dense regions is thus returned as a single cluster, which is precisely the situation that the clustering step was introduced to prevent. The clustering step is thus performed using Mean-shift on the weighted particle set, with the weights from the previous time step carrying the particle density between two resamplings.

While substantial work on clustered particle filters exists [103], this work focuses on the use of clustering for adaptive kriging rather than for particle management

and mode preservation. Nonetheless, the two approaches are not mutually exclusive and can be applied jointly.

Similarly to AK and NNAK, the clustering step must be performed after the particle propagation, but before the correction step. Let C_k be the number of clusters at time step k , and $\mathcal{C}_k^{(c)}$ denote the indices of the particles belonging to the c -th. For $k \geq 0$ and $c \in \llbracket 1, C_k \rrbracket$, the barycentre and empirical covariance matrix of the c -th cluster are respectively defined as:

$$\hat{x}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} x_k^{(i)} \quad (4.55)$$

and:

$$\hat{P}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)} \right) \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)} \right)^\top \quad (4.56)$$

where:

$$\forall i \in \mathcal{C}_k^{(c)}, w_{k-1}^{(i,c)} \triangleq w_{k-1}^{(i)} / \sum_{j \in \mathcal{C}_k^{(c)}} w_{k-1}^{(j)} \quad (4.57)$$

is the weight of the i -th particle, renormalised within its associated cluster.

Applying (4.27) and (4.28) to these moments respectively gives the two radii of the cluster:

$$r_{\text{cov},k}^{(c)} \triangleq \alpha_{\text{cov}} \sqrt{\chi_{d_{\text{pos}}}^2(99\%)} \sqrt{\lambda_{\max} \left(\Pi_{\text{pos}} \hat{P}_{k|k-1}^{(c)} \Pi_{\text{pos}}^\top \right)} \quad (4.58)$$

and:

$$r_{\text{min},k}^{(c)} \triangleq \min_{r \in \mathbb{R}^+} \left(\text{Card} \left\{ i \in \llbracket 1, \tilde{N} \rrbracket \mid \left\| \Pi_{\text{pos}} \hat{x}_{k|k-1}^{(c)} - \tilde{x}_i \right\|_2 \leq r \right\} \geq \tilde{N}_{\text{min}} \right) \quad (4.59)$$

Each cluster thus defines its own subsampling window:

$$\tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)} \triangleq \left\{ \tilde{x}_i \in \tilde{\mathbf{x}} \mid \left\| \Pi_{\text{pos}} \hat{x}_{k|k-1}^{(c)} - \tilde{x}_i \right\|_2 \leq \max \left(r_{\text{cov},k}^{(c)}, r_{\text{min},k}^{(c)} \right) \right\} \quad (4.60)$$

and:

$$\tilde{\mathbf{z}}_{\text{ICAK},k}^{r,(c)} \triangleq \left\{ \tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)} \right\} \quad (4.61)$$

The C_k windows can be handled in two ways. The first one, called Independent Clustering Adaptive Kriging (ICAK), keeps them separate in order to preserve the locality induced by the clustering step. Every cluster estimates its hyperparameters $\hat{\theta}_k^{(c)}$ on its own window, and the likelihood of a particle of $\mathcal{C}_k^{(c)}$ is computed using that window alone:

$$z_k | x_k, \tilde{\mathbf{z}}_{\text{ICAK},k}^{r,(c)} \sim \mathcal{N} \left(\hat{z}_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}, \tilde{\mathbf{z}}_{\text{ICAK},k}^{r,(c)}; \hat{\theta}_k^{(c)} \right), R_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}; \hat{\theta}_k^{(c)} \right) \right) \quad (4.62)$$

TODO: du coup la vraisemblance dépend du nuage de particules ??? en fait c'était déjà un problème depuis AK, et je pense même entre "hey on a un GP" et "hey en fait on n'a que qqs points"

Unlike AK, the hyperparameters of a cluster cannot be directly inherited from a unique cluster at the previous time step. Instead, for each current cluster, they are initialised from those estimated for the parent cluster associated with the largest

number of particles in the current cluster. A weighted average of the hyperparameters could also be considered, but may yield values that are not representative of any individual region. Additionally, each new particle inherits the cluster label of its parent during resampling. The pseudo-code for the ICAK-RPF is given by Algorithm 4.4.

Algorithm 4.4 Independent Clustering Adaptive Kriging RPF

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot | x_{k-1}^{(i)})$ ▷ Particle propagation
 $\left\{ \mathcal{C}_k^{(c)} \right\}_{c=1}^{C_k} = \text{Clustering} \left(\left\{ \Pi_{\text{pos}} x_k^{(i)}, w_{k-1}^{(i)} \right\}_{i=1}^N \right)$ ▷ **Clustering**
for $c \in \llbracket 1, C_k \rrbracket$ **do**
 $\hat{x}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} x_k^{(i)}$ ▷ **Predicted estimation**
 $\hat{P}_{k|k-1}^{(c)} = \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)} \right) \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)} \right)^\top$
Compute $\tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}$ with (4.60) ▷ **Adaptive subsampling**
Compute $\tilde{\mathbf{z}}_{\text{ICAK},k}^{r,(c)}$ with (4.61)
 $\hat{\theta}_k^{(c)} = \text{MLE} \left(\tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}, \tilde{\mathbf{z}}_{\text{ICAK},k}^{r,(c)}, \hat{\theta}_{k-1}^{(c)} \right)$ ▷ **Hyperparameter estimation**
 $\forall i \in \mathcal{C}_k^{(c)}, \text{ compute } \tilde{p} \left(z_k | x_k^{(i)} \right) \text{ from (4.62)}$ ▷ **Kriging**
end for
 $\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \tilde{p} \left(z_k | x_k^{(i)} \right)$ ▷ Weight update
 $\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$ ▷ Normalisation
 $\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N \left(w_k^{(i)} \right)^2$ ▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**
 $\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$ ▷ Whitening
 $\hat{P}_k = \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top$
 $A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top$
 $\left\{ x_k^{(j)}, w_k^{(j)}, - \right\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{ x_k^{(i)}, w_k^{(i)} \right\}_{i=1}^N \right)$ ▷ Resampling
 $\forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K$ ▷ Regularisation
 $\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)}$
end if

On the other hand, the clustering step was originally introduced to bridge the gap between AK and NNAK. While AK is indeed the limit case for a single cluster, ICAK does not reduce to NNAK when each particle is given its own cluster. Indeed, ICAK treats each cluster independently, while NNAK considers the union

of all the windows jointly. **à l'issue** This motivates the introduction of the Merged Clustering Adaptive Kriging (MCAK), which performs the regression on the union of the clustered windows.

Formally, MCAK still computes (4.60) as its independent counterpart, but also defines:

$$\tilde{\mathbf{x}}_{\text{MCAK},k}^r \triangleq \bigcup_{c=1}^{C_k} \tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)} \quad (4.63)$$

and:

$$\tilde{\mathbf{z}}_{\text{MCAK},k}^r \triangleq \{ \tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}_{\text{MCAK},k}^r \} \quad (4.64)$$

Since a single window is considered in this method, the hyperparameters can be estimated as for AK, and the likelihood (4.3) becomes:

$$z_k | x_k, \tilde{\mathbf{z}}_{\text{MCAK},k}^r \sim \mathcal{N} \left(\hat{z}_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{MCAK},k}^r, \tilde{\mathbf{z}}_{\text{MCAK},k}^r; \hat{\theta}_k \right), R_{\text{OK}} \left(x_k, \tilde{\mathbf{x}}_{\text{MCAK},k}^r; \hat{\theta}_k \right) \right) \quad (4.65)$$

Algorithm 4.5 presents the pseudo-code for the MCAK-RPF.

The two variants differ in the law under which the particles are compared. ICAK gives each cluster its own training set and its own hyperparameters, so that the likelihood becomes a piecewise function of the position, discontinuous across the boundaries between clusters, and a particle is only ever compared with the model of the cluster it fell in. MCAK applies a single law to the whole particle set, at the price of a kriging system as large as the union of the windows. Since the weights of a particle filter are a comparison between particles, this distinction bears on the filter itself, and not only on its computational cost. **TODO: à reprendre mais très bonne remarque, je crois qu'on en a parlé plus haut**

4.4.4 Numerical Results

Figure 4.17 compares the shapes of the subsampling windows produced by AK, NNAK and MCAK using the data generated by the corresponding algorithms on the simulation scenario presented in section 4.2.3.

4.5 Subsampling Error Bounds

The setting and the notations are those of proposition 3.9, the covariance of the estimated trend coefficients $\hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)$ being written $M(\tilde{\mathbf{x}}) \triangleq (g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}))^{-1}$.

This section rests on the bordered form of the kriging equations, which gathers the unbiasedness constraints into the Gram matrix. Let $\mathcal{G}(\tilde{\mathbf{x}})$ be the bordered counterpart of $G(\tilde{\mathbf{x}})$, and $\bar{k}$ and $\bar{\mathbf{z}}$ the covariance and observation matrices bordered accordingly:

$$\mathcal{G}(\tilde{\mathbf{x}}) \triangleq \begin{bmatrix} G(\tilde{\mathbf{x}}) & g(\tilde{\mathbf{x}}) \\ g(\tilde{\mathbf{x}})^\top & 0 \end{bmatrix}, \quad \bar{k}(\tilde{\mathbf{x}}, x) \triangleq \begin{bmatrix} k(\tilde{\mathbf{x}}, x) \\ g(x)^\top \end{bmatrix}, \quad \bar{\mathbf{z}} \triangleq \begin{bmatrix} \tilde{\mathbf{z}} \\ 0 \end{bmatrix} \quad (4.66)$$

with $\bar{k}(x, \tilde{\mathbf{x}}) \triangleq \bar{k}(\tilde{\mathbf{x}}, x)^\top$. Handling the unbiasedness constraints with Lagrange multipliers rewrites (3.58) as:

$$\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{\mathbf{z}} \quad (4.67)$$

Algorithm 4.5 Merged Clustering Adaptive Kriging RPF

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot | x_{k-1}^{(i)})$ ▷ Particle propagation
 $\left\{ \mathcal{C}_k^{(c)} \right\}_{c=1}^{C_k} = \text{Clustering} \left(\left\{ \Pi_{\text{pos}} x_k^{(i)}, w_{k-1}^{(i)} \right\}_{i=1}^N \right)$ ▷ **Clustering**
for $c \in \llbracket 1, C_k \rrbracket$ **do**
 $\hat{x}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} x_k^{(i)}$ ▷ **Predicted estimation**
 $\hat{P}_{k|k-1}^{(c)} = \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)} \right) \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)} \right)^\top$
 Compute $\tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}$ with (4.60) ▷ **Adaptive subsampling**
end for
 $\tilde{\mathbf{x}}_{\text{MCAK},k}^r = \bigcup_{c=1}^{C_k} \tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}$ ▷ **Cluster merging**
 $\tilde{\mathbf{z}}_{\text{MCAK},k}^r \triangleq \left\{ \tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}_{\text{MCAK},k}^r \right\}$
 $\hat{\theta}_k = \text{MLE} \left(\tilde{\mathbf{x}}_{\text{MCAK},k}^r, \tilde{\mathbf{z}}_{\text{MCAK},k}^r, \hat{\theta}_{k-1} \right)$ ▷ **Hyperparameter estimation**
 $\forall i \in \llbracket 1, N \rrbracket$, compute $\tilde{p} \left(z_k | x_k^{(i)} \right)$ from (4.65) ▷ **Kriging**
 $\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \tilde{p} \left(z_k | x_k^{(i)} \right)$ ▷ Weight update
 $\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$ ▷ Normalisation
 $\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N \left(w_k^{(i)} \right)^2$ ▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**
 $\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$ ▷ Whitening
 $\hat{P}_k = \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top$
 $A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top$
 $\left\{ x_k^{(j)}, w_k^{(j)}, - \right\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{ x_k^{(i)}, w_k^{(i)} \right\}_{i=1}^N \right)$ ▷ Resampling
 $\forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K$ ▷ Regularisation
 $\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)}$
end if

The associated error **covariance** is:

$$R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) \triangleq \mathbb{V} [\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - z] = k(x, x) + R - \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{k}(\tilde{\mathbf{x}}, x) \quad (4.68)$$

A major limitation of this method is the $\mathcal{O}(\tilde{N}^3 d_z^3)$ computational cost, stemming from the inversion of the Gram matrix $G(\tilde{\mathbf{x}})$, **or equivalently of $\mathcal{G}(\tilde{\mathbf{x}})$** , in both (4.67) and (4.68). One common approach to mitigate this is to omit certain elements of $\tilde{\mathbf{x}}$ to reduce the size of $G(\tilde{\mathbf{x}})$. However, this approximation introduces a loss of

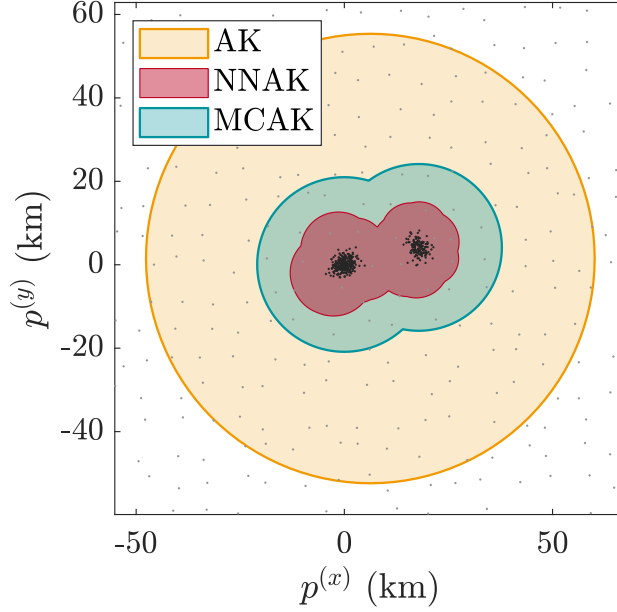


Figure 4.17:

accuracy in (4.67). This section quantifies the error induced by such subsampling approximations. It first situates that error among the other approximations the estimator rests on, then defines two metrics for it and rewrites them in terms of the information carried by the discarded data alone, and finally derives bounds that avoid inverting any matrix of the size of that data.

4.5.1 Sources of Error

Three distinct approximations separate $\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$ from the field it estimates. The process is assumed to be a GP of known kernel, whereas the field is a fixed function of the position; the field is only known through $\tilde{N}$ noisy observations; and only $\tilde{N}^r$ of them are used. Only the last one is the object of this section, and the two others are recalled here to situate it.

The first approximation is a modelling choice, and the only one the estimator does not report. Under the probabilistic reading, it does not exist: if the field is a trajectory of the assumed GP, then (4.68) is exact. That assumption cannot be verified on a single field, and the deterministic reading of the same estimator is the one that quantifies its failure. Whenever the field belongs to the native RKHS $\mathcal{H}_k$ of section 3.5, the interpolation error at any query point is bounded by:

$$\|f(x) - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)\|_2 \leq \|f\|_{\mathcal{H}_k} \cdot \sqrt{\|R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)\|_{\text{op}}} \quad (4.69)$$

so that the model error has the very shape the estimator already reports, up to the single unknown factor $\|f\|_{\mathcal{H}_k}$. Two caveats attach to this reading. The trajectories of a GP almost surely do not belong to $\mathcal{H}_k$, so the deterministic and the probabilistic statements bracket the truth rather

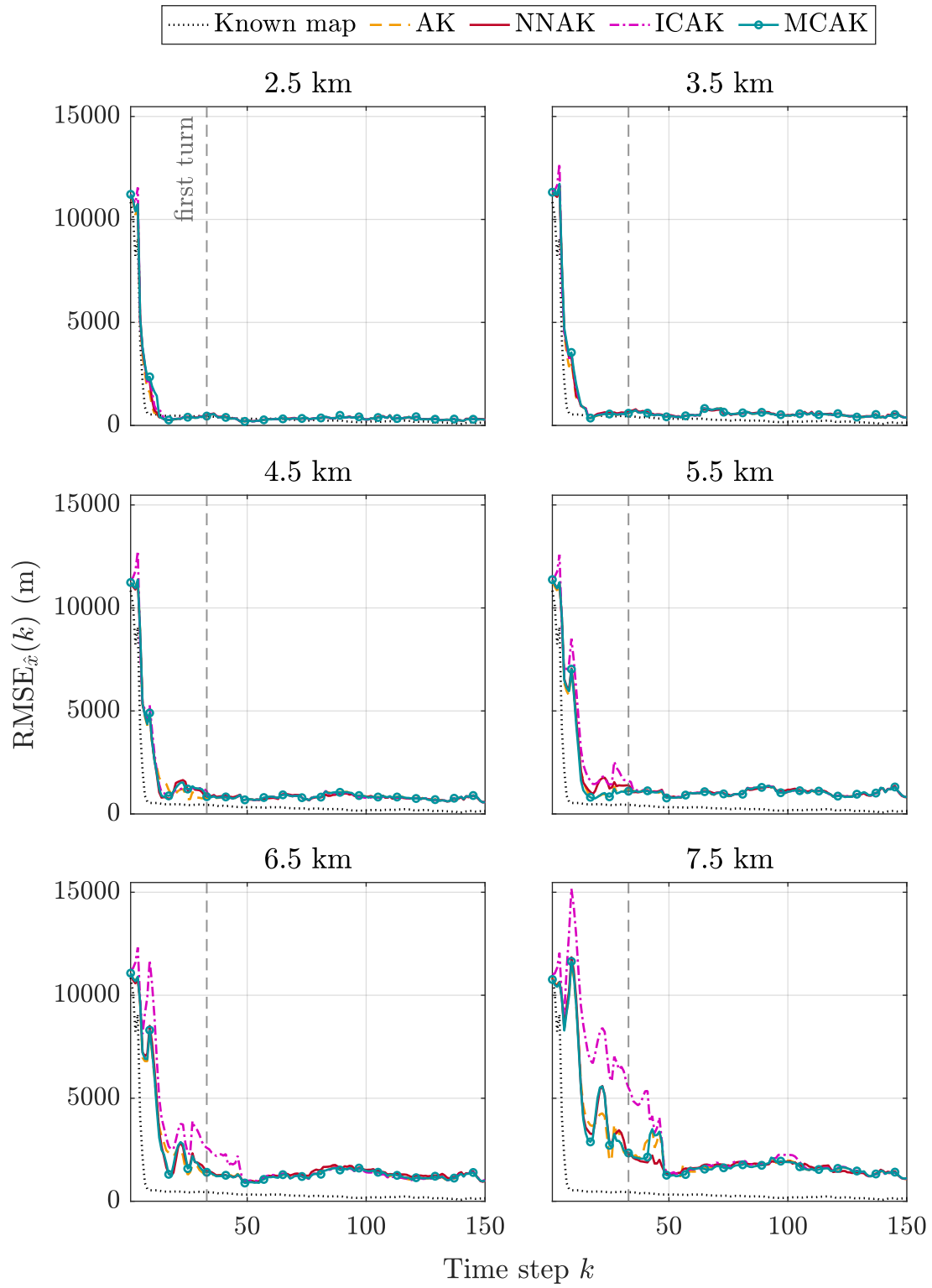


Figure 4.18:

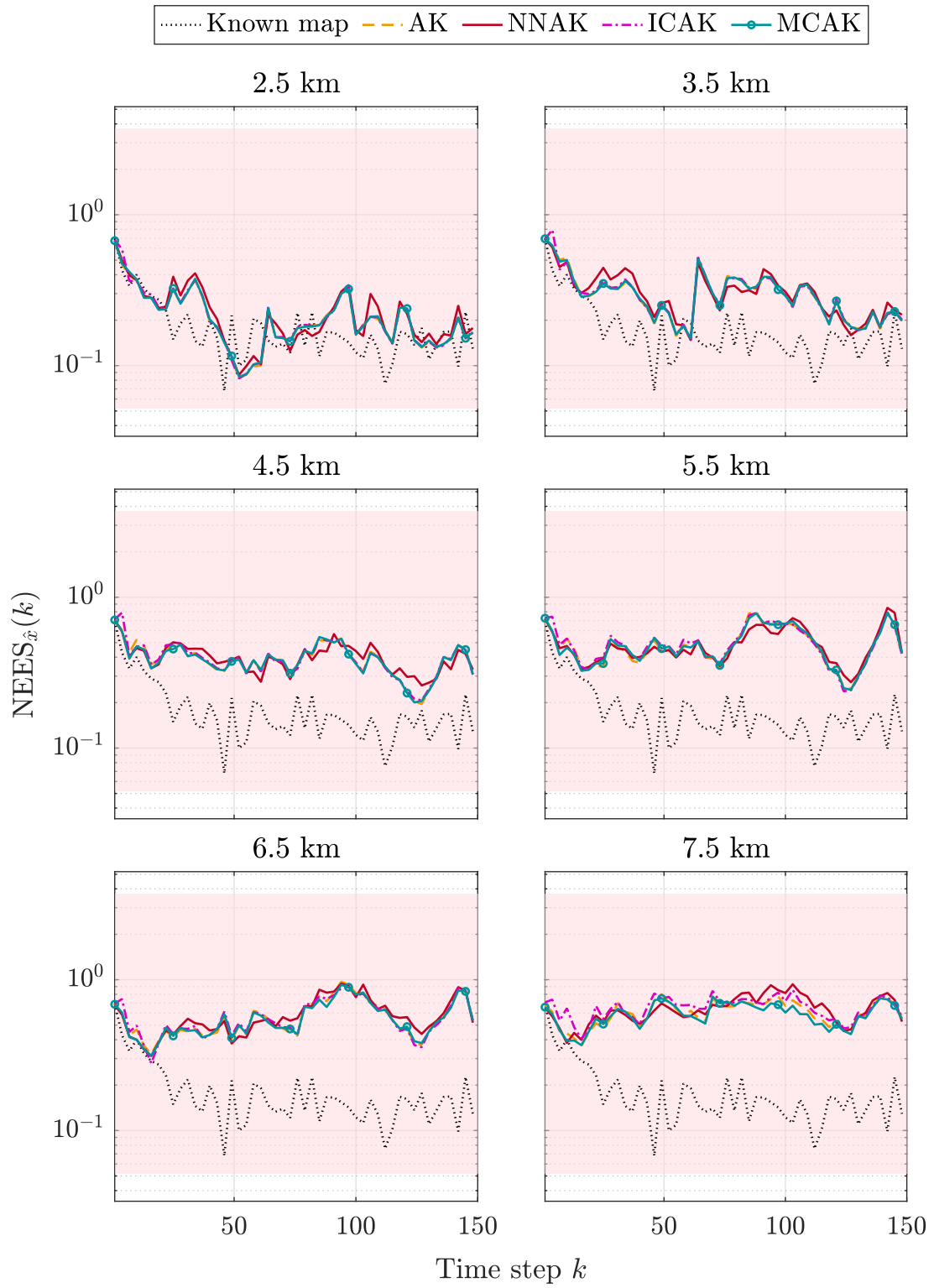


Figure 4.19:

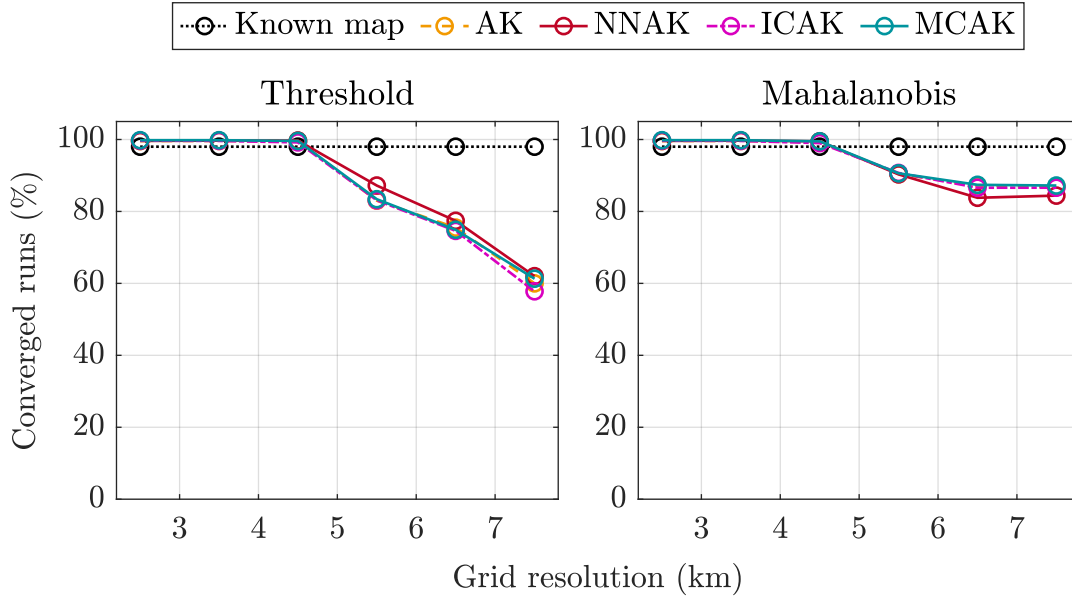


Figure 4.20:

Δ_g	AK		NNAK		
	Kriging	HP	Kriging	HP	NN
2.5km	46 G	564 M	4.3 G	57 M	47.3 M
3.5km	14.4 G	224 M	2.05 G	36.1 M	43.6 M
4.5km	5.75 G	121 M	1.23 G	25.9 M	40.9 M
5.5km	3.09 G	89.8 M	885 M	21.7 M	38.8 M
6.5km	1.93 G	70.1 M	705 M	19.4 M	36.8 M
7.5km	1.54 G	66.7 M	603 M	18.1 M	35.3 M

Δ_g	ICAK			MCAK		
	Kriging	HP	Clustering	Kriging	HP	Clustering
2.5km	2.26 G	87.3 M	114 M	2.47 G	25.3 M	114 M
3.5km	901 M	83.2 M	158 M	811 M	21.5 M	158 M
4.5km	616 M	83.3 M	194 M	496 M	17.2 M	195 M
5.5km	547 M	91.4 M	223 M	352 M	11.9 M	224 M
6.5km	513 M	107 M	251 M	241 M	7.12 M	253 M
7.5km	499 M	131 M	270 M	150 M	3.4 M	274 M

Table 4.9:

than coincide; and $\|f\|_{\mathcal{H}_k}$ is not accessible for a physical field, which makes (4.69) a statement about the shape of the error and not about its magnitude. In practice, the binding part of this approximation is not the Gaussian assumption but the stationarity of the kernel, which section 4.3.2 addresses by refitting the hyperparameters locally.

The second approximation is reported in full. The covariance (4.68) is exactly the ignorance left by $\tilde{N}$ noisy observations: it grows where the data are sparse, and $\tilde{R}$ keeps it away from zero however dense they become. Deterministically, it is governed by the fill distance of $\tilde{\mathbf{x}}$, the radius of the largest hole left by the training set, to a power set by the smoothness of the kernel. This is what the resolution of a survey buys, and section 4.2.4 measures it directly on the field of this chapter.

The third approximation is reported as well, and this is the observation on which the rest of this section rests: remark 4.4 shows that discarding data inflates the reported covariance by exactly the error it commits. A subsampled estimator is therefore never overconfident, and the question is not whether its covariance covers its error, but by how much it has grown — which is what the following metrics measure.

4.5.2 Subsampling Error Metrics

Formally, subsampling is achieved by splitting $\tilde{\mathbf{x}}$ into $\begin{bmatrix} \tilde{\mathbf{x}}^r & \tilde{\mathbf{x}}^d \end{bmatrix}^\top$, with $\tilde{\mathbf{x}}^r$ being the $\tilde{N}^r$ retained data, usually close to the considered kriging location $x \in \mathcal{U}$, and $\tilde{\mathbf{x}}^d$ being the $\tilde{N}^d$ discarded data, usually far from x . A similar operation is performed on $\tilde{\mathbf{z}}$. This decomposition provides the foundation for all subsequent error bounds, as it explicitly isolates the predictive contribution of the discarded points from that of the retained training set.

Two error metrics between $\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)$ and $\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$ are considered here, each capturing a distinct aspect of the kriging approximation. The first metric corresponds to **the covariance of their difference, of which Definition 4.1 also retains the largest standard deviation over the directions of $\mathbb{R}^{d_z}$** . In contrast, since the kriging estimator is a deterministic function conditionally to $\tilde{\mathbf{z}}$, a natural alternative is to measure the **distance** between the two estimators, as proposed by Definition 4.2.

Definition 4.1. *Let E_r denote the covariance of the difference between the full and subsampled estimators, and e_r the largest error it induces in $\mathcal{L}^2(\mathbb{R})$ over the unit directions a of $\mathbb{R}^{d_z}$:*

$$\begin{cases} E_r : x \in \mathcal{U} \mapsto \mathbb{V}[\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)] \\ e_r : x \in \mathcal{U} \mapsto \max_{\|a\|_2=1} \|a^\top (\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g))\|_{\mathcal{L}^2(\mathbb{R})} = \sqrt{\lambda_{\max}(E_r(x))} \end{cases}$$

Definition 4.2. *The interpolation error e_f between the full and subsampled estimators is defined as:*

$$e_f : x \in \mathcal{U} \mapsto \|\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)\|_2$$

Both errors are well defined whatever the value of β : the two estimators being unbiased, their difference is a linear combination $V^\top \tilde{\mathbf{z}}$ with $V^\top g(\tilde{\mathbf{x}}) = 0$, whose distribution therefore does not depend on β . The subsampled estimator exists as long as $g(\tilde{\mathbf{x}}^r)$ has full column rank, which requires $\tilde{N}^r \geq r_{\text{UK}}$.

Since $\tilde{R}$ may be singular, k is assumed SPD in the sense of definition 3.2, which makes $G(\tilde{\mathbf{x}})$ invertible, and $\mathcal{G}(\tilde{\mathbf{x}})$ with it, its Schur complement with respect to $G(\tilde{\mathbf{x}})$ being the negative definite matrix $-M(\tilde{\mathbf{x}})^{-1}$.

4.5.3 Importance of the Gram Matrix

As stated above, most of the computational cost comes from inverting the Gram matrix. **Ordering the rows and columns of $\mathcal{G}(\tilde{\mathbf{x}})$ as $\tilde{\mathbf{x}}^r$, the $r_{\text{UK}}d_z$ unbiasedness constraints, then $\tilde{\mathbf{x}}^d$, the decomposition introduced in section 4.5.2 leads to the following block partitioning:**

$$\mathcal{G}(\tilde{\mathbf{x}}) = \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} = \begin{bmatrix} \mathcal{G}(\tilde{\mathbf{x}}^r) & \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)^\top & G(\tilde{\mathbf{x}}^d) \end{bmatrix} \quad \text{with} \quad \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \triangleq \begin{bmatrix} k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ g(\tilde{\mathbf{x}}^d)^\top \end{bmatrix} \quad (4.70)$$

the bordered vectors being reordered accordingly:

$$\bar{k}(\tilde{\mathbf{x}}, x) = \begin{bmatrix} \bar{k}(\tilde{\mathbf{x}}^r, x)^\top & k(\tilde{\mathbf{x}}^d, x)^\top \end{bmatrix}^\top, \quad \bar{\mathbf{z}} = \begin{bmatrix} (\bar{\mathbf{z}}^r)^\top & (\tilde{\mathbf{z}}^d)^\top \end{bmatrix}^\top, \quad \bar{\mathbf{z}}^r \triangleq \begin{bmatrix} (\tilde{\mathbf{z}}^r)^\top & 0 \end{bmatrix}^\top \quad (4.71)$$

Let $\mathcal{G}_h(\tilde{\mathbf{x}}^r)$ be the hollowed-out Gram matrix, **of the same size $(\tilde{N} + r_{\text{UK}})d_z$ as $\mathcal{G}(\tilde{\mathbf{x}})$:**

$$\mathcal{G}_h(\tilde{\mathbf{x}}^r) \triangleq \begin{bmatrix} \mathcal{G}(\tilde{\mathbf{x}}^r) & 0_{(\tilde{N}+r_{\text{UK}})d_z, \tilde{N}d_z} \\ 0_{\tilde{N}d_z, (\tilde{N}+r_{\text{UK}})d_z} & 0_{\tilde{N}d_z, \tilde{N}d_z} \end{bmatrix} \quad (4.72)$$

Proposition 4.2 states that $\mathcal{G}_h(\tilde{\mathbf{x}}^r)$ acts as a substitute for $\mathcal{G}(\tilde{\mathbf{x}})$ in the subsampled case with respect to (4.67) and (4.68).

Proposition 4.2. *The subsampled kriging estimator verifies:*

$$\begin{cases} \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) = \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger \bar{\mathbf{z}} \\ R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) = k(x, x) + R - \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger \bar{k}(\tilde{\mathbf{x}}, x) \end{cases}$$

where $\dagger$ denotes the Moore-Penrose pseudo-inverse symbol [7].

Proof. The result comes from applying (4.67) and (4.68) to $\tilde{\mathbf{x}}^r$, and noting that:

$$\mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger = \begin{bmatrix} \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} & 0 \\ 0 & 0 \end{bmatrix}$$

□

Although it is not directly used in computations, Proposition 4.2 further highlights the importance of the Gram matrix in the subsampling problem. In particular, Definition 4.3 serves as the foundation for the key reformulations of the studied errors established by Proposition 4.3.

Definition 4.3. *The Gram inverse difference matrix $\mathfrak{D}_G$ is the quantity defined as:*

$$\mathfrak{D}_G \triangleq \mathcal{G}(\tilde{\mathbf{x}})^{-1} - \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger$$

Proposition 4.3. *The two subsampling errors can be respectively expressed as:*

$$\begin{cases} E_r(x) = \bar{k}(x, \tilde{\mathbf{x}}) \mathfrak{D}_G \bar{k}(\tilde{\mathbf{x}}, x) \\ e_f(x) = \|\bar{k}(x, \tilde{\mathbf{x}}) \mathfrak{D}_G \bar{\mathbf{z}}\|_2 \end{cases}$$

Proof. See proof in section B.14.

□

Remark 4.4. *An alternative reformulation of $E_r(x)$ is obtained in the demonstration of Proposition 4.3, and is worth comparing with the original definition of $e_f(x)$ given in Definition 4.2.*

$$E_r(x) = R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)$$

In particular, $R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) \succeq R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)$: subsampling can only inflate the covariance reported by the estimator, and by exactly the error it commits.

4.5.4 Block Inversion and Residual Quantities

To isolate the impact of discarding $\tilde{\mathbf{x}}^d$ on the kriging estimator, $\mathfrak{D}_G$ is decomposed using the Woodbury block matrix inversion formula [94]:

$$\begin{aligned} \mathfrak{D}_G &= \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix}^{-1} - \begin{bmatrix} A^{-1} & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} A^{-1}BS_{d|r}^{-1}B^\top A^{-1} & -A^{-1}BS_{d|r}^{-1} \\ -S_{d|r}^{-1}B^\top A^{-1} & S_{d|r}^{-1} \end{bmatrix} \\ &= \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^d d_z} \end{bmatrix} S_{d|r}^{-1} \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^d d_z} \end{bmatrix}^\top \end{aligned} \quad (4.73)$$

where $S_{d|r}$ is the Schur complement of $\mathcal{G}(\tilde{\mathbf{x}})$ with respect to $A = \mathcal{G}(\tilde{\mathbf{x}}^r)$, **invertible since both $\mathcal{G}(\tilde{\mathbf{x}})$ and A are**:

$$S_{d|r} \triangleq D - B^\top A^{-1}B \quad (4.74)$$

This decomposition, combined with Proposition 4.3, reveals two residual quantities that contribute to the error introduced by subsampling. The first one, called residual covariance function, quantifies the portion of the covariance between x and the discarded training set $\tilde{\mathbf{x}}^d$ not accounted for by the retained set $\tilde{\mathbf{x}}^r$. It is the object of Definition 4.4. Similarly, the residual observation vector captures the information contained within the discarded observations $\tilde{\mathbf{z}}^d$ and unexplained by the retained observations $\tilde{\mathbf{z}}^r$. Its expression is given by Definition 4.5.

Definition 4.4. *The residual covariance function $k_{d|r}$ is given by:*

$$k_{d|r} : x \in \mathcal{U} \mapsto k(\tilde{\mathbf{x}}^d, x) - \bar{k}(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r) \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} \bar{k}(\tilde{\mathbf{x}}^r, x)$$

with $\bar{k}(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r) \triangleq \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)^\top$.

Definition 4.5. *The residual observation vector $\tilde{\mathbf{z}}_{d|r}$ is defined as:*

$$\tilde{\mathbf{z}}_{d|r} \triangleq \tilde{\mathbf{z}}^d - \bar{k}(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r) \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} \tilde{\mathbf{z}}^r$$

Proposition 4.4 gives relevant identities that link (4.73) with $k_{d|r}$ and $\tilde{\mathbf{z}}_{d|r}$. Proposition 4.5 and Proposition 4.7 push this further by showing that the studied metrics can be fully expressed as functions of those residual quantities, **which Proposition 4.6 relates to their simple kriging counterparts.**

Proposition 4.4. *The following residual identities hold:*

$$\begin{cases} k_{d|r}(x) = \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^{d_{d_z}}} \end{bmatrix}^\top \bar{k}(\tilde{\mathbf{x}}, x) \\ k_{d|r}(\tilde{\mathbf{x}}^d) = S_{d|r} - I_{\tilde{N}^d} \otimes \tilde{R} \\ \tilde{\mathbf{z}}_{d|r} = \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^{d_{d_z}}} \end{bmatrix}^\top \tilde{\mathbf{z}} \end{cases}$$

Proof. Recall that $A = \mathcal{G}(\tilde{\mathbf{x}}^r)$ and $B = \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)$. Direct calculations give the desired results. $\square$

Proposition 4.5. *The residual forms of the two errors are given below:*

$$\begin{cases} E_r(x) = k_{d|r}(x)^\top \left(k_{d|r}(\tilde{\mathbf{x}}^d) + I_{\tilde{N}^d} \otimes \tilde{R} \right)^{-1} k_{d|r}(x) \\ e_f(x) = \left\| k_{d|r}(x)^\top \left(k_{d|r}(\tilde{\mathbf{x}}^d) + I_{\tilde{N}^d} \otimes \tilde{R} \right)^{-1} \tilde{\mathbf{z}}_{d|r} \right\|_2 \end{cases}$$

Proof. These expressions are obtained by applying (4.73) and Proposition 4.4 to Proposition 4.3. $\square$

Proposition 4.6. *Let $k_{d|r}^0$, $S_{d|r}^0$ and $\tilde{\mathbf{z}}_{d|r}^0$ be the residual quantities of simple kriging, obtained by replacing the bordered Gram matrix with the Gram matrix:*

$$\begin{cases} k_{d|r}^0 : x \in \mathcal{U} \mapsto k(\tilde{\mathbf{x}}^d, x) - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)G(\tilde{\mathbf{x}}^r)^{-1}k(\tilde{\mathbf{x}}^r, x) \\ S_{d|r}^0 \triangleq G(\tilde{\mathbf{x}}^d) - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)G(\tilde{\mathbf{x}}^r)^{-1}k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ \tilde{\mathbf{z}}_{d|r}^0 \triangleq \tilde{\mathbf{z}}^d - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)G(\tilde{\mathbf{x}}^r)^{-1}\tilde{\mathbf{z}}^r \end{cases}$$

and let $U : x \in \mathcal{U} \mapsto g(x) - k(x, \tilde{\mathbf{x}}^r)G(\tilde{\mathbf{x}}^r)^{-1}g(\tilde{\mathbf{x}}^r)$ be the weights given to the estimated trend coefficients by $\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$, so that $U(\tilde{\mathbf{x}}^d) = g(\tilde{\mathbf{x}}^d) - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)G(\tilde{\mathbf{x}}^r)^{-1}g(\tilde{\mathbf{x}}^r)$ stacks its values on $\tilde{\mathbf{x}}^d$. The residual quantities verify:

$$\begin{cases} k_{d|r}(x) = k_{d|r}^0(x) + U(\tilde{\mathbf{x}}^d)M(\tilde{\mathbf{x}}^r)U(x)^\top \\ S_{d|r} = S_{d|r}^0 + U(\tilde{\mathbf{x}}^d)M(\tilde{\mathbf{x}}^r)U(\tilde{\mathbf{x}}^d)^\top \\ \tilde{\mathbf{z}}_{d|r} = \tilde{\mathbf{z}}_{d|r}^0 - U(\tilde{\mathbf{x}}^d)\hat{m}_{\text{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) \end{cases}$$

In particular, $S_{d|r}$ is SPD, and $\tilde{\mathbf{z}}_{d|r}$ is unchanged when a trend $g(\tilde{\mathbf{x}})\gamma$, $\gamma \in \mathbb{R}^{r_{\text{UK}^{d_z}}}$, is added to the observations.

Proof. See proof in section B.15. $\square$

Proposition 4.7. *Let $\|\cdot\|_{\text{op}}$ denote the operator norm associated with the euclidean norm $\|\cdot\|_2$, and let $\langle \cdot | \cdot \rangle_{S_{d|r}}$ be the Mahalanobis inner product weighted by the inverse of $S_{d|r} = k_{d|r}(\tilde{\mathbf{x}}^d) + I_{\tilde{N}^d} \otimes \tilde{R}$, which is SPD by Proposition 4.6:*

$$\forall n \in \mathbb{N}^*, \forall A, B \in \mathcal{M}_{\tilde{N}^{d_{d_z}}, n}(\mathbb{R}), \langle A | B \rangle_{S_{d|r}} \triangleq A^\top S_{d|r}^{-1} B, \quad \|A\|_{S_{d|r}} \triangleq \left\| S_{d|r}^{-\frac{1}{2}} A \right\|_{\text{op}}$$

The kriging errors can be interpreted as metrics in the euclidean space $(\mathbb{R}^{\tilde{N}^{d_{d_z}}}, \langle \cdot | \cdot \rangle_{S_{d|r}})$, to which the columns of $k_{d|r}(x)$ and $\tilde{\mathbf{z}}_{d|r}$ belong:

$$\begin{cases} E_r(x) = \langle k_{d|r}(x) | k_{d|r}(x) \rangle_{S_{d|r}}, & e_r(x) = \|k_{d|r}(x)\|_{S_{d|r}} \\ e_f(x) = \left\| \langle k_{d|r}(x) | \tilde{\mathbf{z}}_{d|r} \rangle_{S_{d|r}} \right\|_2 \end{cases}$$

Remark 4.5. *Contrary to simple kriging, discarding data uncorrelated with both $\tilde{\mathbf{x}}^r$ and x is not free. If $k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r) = 0$ and $k(\tilde{\mathbf{x}}^d, x) = 0$, then $k_{d|r}^0(x) = 0$, $U(\tilde{\mathbf{x}}^d) = g(\tilde{\mathbf{x}}^d)$ and $M(\tilde{\mathbf{x}})^{-1} = M(\tilde{\mathbf{x}}^r)^{-1} + g(\tilde{\mathbf{x}}^d)^\top G(\tilde{\mathbf{x}}^d)^{-1} g(\tilde{\mathbf{x}}^d)$. Applying the Woodbury formula [94] to $S_{d|r} = G(\tilde{\mathbf{x}}^d) + g(\tilde{\mathbf{x}}^d)M(\tilde{\mathbf{x}}^r)g(\tilde{\mathbf{x}}^d)^\top$ in Proposition 4.7 yields:*

$$\begin{cases} E_r(x) = U(x) (M(\tilde{\mathbf{x}}^r) - M(\tilde{\mathbf{x}})) U(x)^\top \\ e_f(x) = \|U(x) (\hat{m}_{\text{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g))\|_2 \end{cases}$$

The discarded data still reduce the covariance $M(\tilde{\mathbf{x}})$ of the estimated trend coefficients, and both errors measure what is lost from that estimation, seen through the reliance $U(x)$ of the subsampled estimator on the trend. For ordinary kriging, $E_r(x)$ is the increase in covariance of the estimated mean, seen through $U(x)$.

4.5.5 Relations Between the Errors

This section introduces two relations between $e_r(x)$ and $e_f(x)$. The first relation is the concentration inequality described by Proposition 4.8, while the second comes from algebraic considerations, and is given by Proposition 4.9.

Proposition 4.8. *Let $\chi_{d_z}^2(1 - \delta)$ denote the quantile of order $1 - \delta$ of the chi-squared distribution with d_z degrees of freedom. For $\delta \in (0, 1]$:*

$$e_f(x) \leq \sqrt{\chi_{d_z}^2(1 - \delta)} \cdot e_r(x)$$

with a probability of at least $1 - \delta$.

Proof. Whatever the value of β , $\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$ follows a centred normal distribution of covariance $E_r(x)$, and can therefore be written $E_r(x)^{\frac{1}{2}}\xi$ with $\xi \sim \mathcal{N}(0, I_{d_z})$. Thus:

$$e_f(x)^2 = \xi^\top E_r(x) \xi \leq \lambda_{\max}(E_r(x)) \|\xi\|_2^2 = e_r(x)^2 \|\xi\|_2^2$$

and $\|\xi\|_2^2$ follows a chi-squared distribution with d_z degrees of freedom, so that $\|\xi\|_2^2 \leq \chi_{d_z}^2(1 - \delta)$ with a probability of $1 - \delta$. $\square$

Remark 4.6. *For a scalar field, $\sqrt{\chi_1^2(1 - \delta)} = \Phi^{-1}(1 - \frac{\delta}{2})$, where Φ is the cumulative distribution function of the standard normal distribution, and a simpler, albeit more pessimistic, version of Proposition 4.8 can be derived from the inverse form of Mill's inequality [104]:*

$$\forall \delta \in (0, 1], \mathbb{P} \left(e_f(x) \leq \sqrt{2 \ln \left(\frac{2}{\delta} \right)} \cdot e_r(x) \right) \geq 1 - \delta$$

Proposition 4.9. *$e_r(x)$ and $e_f(x)$ are linked by the following inequality:*

$$e_f(x) \leq e_r(x) \cdot \|\tilde{\mathbf{z}}_{d|r}\|_{S_{d|r}}$$

Proof. Applying the Cauchy-Schwarz inequality [7] to Proposition 4.7 proves the relation. $\square$

The residual forms of Proposition 4.7 still involve $S_{d|r}^{-1}$, whose size $\tilde{N}^d d_z$ is that of the discarded observations, so that computing $E_r(x)$ exactly is as costly as the full kriging estimator that subsampling aims to avoid. This section derives upper bounds that do not require this inversion, first by separating the contributions of simple kriging and of the trend estimation, then by nesting the retained data into a larger set whose own subsampling error is small, and finally from entrywise bounds on the covariances between the data.

4.5.6 Simple Kriging and Trend Contributions

Proposition 4.6 splits the residual covariance function into the residual $k_{d|r}^0(x)$ of simple kriging and a trend contribution. Proposition 4.10 bounds these two parts separately, the latter being handled exactly as in Remark 4.5.

Proposition 4.10. *Let $e_{r,0}(x) \triangleq \|k_{d|r}^0(x)\|_{S_{d|r}^0}$ be the error that simple kriging would commit if the trend were known, and $P \triangleq U(\tilde{\mathbf{x}}^d)^\top \left(S_{d|r}^0\right)^{-1} U(\tilde{\mathbf{x}}^d)$. Then:*

$$e_r(x) \leq e_{r,0}(x) + \sqrt{\|U(x) (M(\tilde{\mathbf{x}}^r) - (M(\tilde{\mathbf{x}}^r)^{-1} + P)^{-1}) U(x)^\top\|_{\text{op}}}$$

with equality when $k_{d|r}^0(x) = 0$.

Proof. See proof in section B.16. $\square$

The simple kriging term $e_{r,0}(x)$ still requires $\left(S_{d|r}^0\right)^{-1}$. When the observations are noisy, Proposition 4.11 removes this last inversion.

Proposition 4.11. *Assume that $\tilde{R} \succ 0$, and let $\tilde{\lambda} \triangleq \lambda_{\min}(\tilde{R})$. Then $S_{d|r}^0 \succeq I_{\tilde{N}^d} \otimes \tilde{R} \succeq \tilde{\lambda} I_{\tilde{N}^d d_z}$, and:*

$$e_r(x) \leq e_r^U(x) \triangleq \frac{\|k_{d|r}^0(x)\|_{\text{op}}}{\sqrt{\tilde{\lambda}}} + \sqrt{\left\|U(x) \left(M(\tilde{\mathbf{x}}^r) - \left(M(\tilde{\mathbf{x}}^r)^{-1} + \frac{U(\tilde{\mathbf{x}}^d)^\top U(\tilde{\mathbf{x}}^d)}{\tilde{\lambda}}\right)^{-1}\right) U(x)^\top\right\|_{\text{op}}}$$

Proof. See proof in section B.16. $\square$

Unlike the residual forms, $e_r^U(x)$ only involves $G(\tilde{\mathbf{x}}^r)^{-1}$, the cross covariances $k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)$ and $k(\tilde{\mathbf{x}}^d, x)$, and the trend matrices, at a cost of $\mathcal{O}\left(d_z^3 \left(\left(\tilde{N}^r\right)^3 + \tilde{N}^r \tilde{N}^d\right)\right)$ that grows linearly with the number of discarded data. Through Proposition 4.8, it also bounds the interpolation error, $e_f(x) \leq \sqrt{\chi_{d_z}^2 (1 - \delta)} \cdot e_r^U(x)$ holding with a probability of at least $1 - \delta$.

Remark 4.7. *Along a fixed direction $a \in \mathbb{R}^{d_z}$, $a^\top E_r(x)a$ is the scalar quadratic form $(k_{d|r}(x)a)^\top S_{d|r}^{-1}(k_{d|r}(x)a)$. The cruder bound $a^\top E_r(x)a \leq \|k_{d|r}(x)a\|_2^2/\tilde{\lambda}$, which follows from $S_{d|r} \succeq S_{d|r}^0 \succeq I_{\tilde{N}^d} \otimes \tilde{R}$, is its Gauss-Radau estimate with a single node prescribed at $\tilde{\lambda} \leq \lambda_{\min}(S_{d|r})$ [105]. Running m iterations of the Lanczos algorithm on $S_{d|r}$ from $k_{d|r}(x)a$ yields Gauss-Radau upper bounds that converge to $a^\top E_r(x)a$ as m grows, at the cost of m products by $S_{d|r}$ instead of its inversion.*

Remark 4.8. *The two terms of Proposition 4.10 behave differently as the observation noise grows. When the noise is scaled up, $\tilde{R} = \rho \tilde{R}_0$ with $\rho \rightarrow +\infty$ and the other quantities fixed, $e_{r,0}(x)$ vanishes, discarded and retained data becoming equally uninformative about $f(x)$, whereas the trend term does not. Since $U(x) \rightarrow g(x)$, $M(\tilde{\mathbf{x}}^r)^{-1} \sim g(\tilde{\mathbf{x}}^r)^\top (I_{\tilde{N}^r} \otimes \tilde{R})^{-1} g(\tilde{\mathbf{x}}^r)$ and $P \sim g(\tilde{\mathbf{x}}^d)^\top (I_{\tilde{N}^d} \otimes \tilde{R})^{-1} g(\tilde{\mathbf{x}}^d)$, it comes that, as soon as the right-hand side does not vanish:*

$$E_r(x) \sim g(x) \left(\left(g(\tilde{\mathbf{x}}^r)^\top (I_{\tilde{N}^r} \otimes \tilde{R})^{-1} g(\tilde{\mathbf{x}}^r) \right)^{-1} - \left(g(\tilde{\mathbf{x}})^\top (I_{\tilde{N}} \otimes \tilde{R})^{-1} g(\tilde{\mathbf{x}}) \right)^{-1} \right) g(x)^\top$$

For ordinary kriging, this equivalent is $(1/\tilde{N}^r - 1/\tilde{N})\tilde{R}$: both estimators become averages of the observations, and discarding data increases the covariance of the estimated mean.

4.5.7 Nested Subsampling

The bound of Proposition 4.11 replaces $(S_{d|r}^0)^{-1}$ by $(I_{\tilde{N}^d} \otimes \tilde{R})^{-1}$, which is pessimistic when $k_{d|r}^0(x)$ lies along the dominant eigenvectors of $S_{d|r}^0$. This happens when the discarded data are strongly correlated with the retained ones, typically just outside a spatial window, where the bound of Proposition 4.11 is at its loosest. Proposition 4.12 handles these data exactly, and only applies the bound to the remaining ones.

Proposition 4.12. *Let $\tilde{\mathbf{x}}^e$ be a set of training data such that $\tilde{\mathbf{x}}^r \subset \tilde{\mathbf{x}}^e \subset \tilde{\mathbf{x}}$, and $E_r^{(e)}(x)$ be the error covariance committed by subsampling $\tilde{\mathbf{x}}$ into $\tilde{\mathbf{x}}^e$. Then, writing $\Delta_e(x) \triangleq R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}^e, g)$:*

$$E_r(x) = \Delta_e(x) + E_r^{(e)}(x)$$

Consequently, if $\tilde{R} \succ 0$ and $e_r^{U,(e)}(x)$ denotes the bound of Proposition 4.11 for the retained set $\tilde{\mathbf{x}}^e$:

$$\|\Delta_e(x)\|_{\text{op}} \leq e_r(x)^2 \leq \|\Delta_e(x)\|_{\text{op}} + e_r^{U,(e)}(x)^2$$

Proof. Applying Remark 4.4 to the subsamplings of $\tilde{\mathbf{x}}$ into $\tilde{\mathbf{x}}^r$ and into $\tilde{\mathbf{x}}^e$ gives $E_r(x) = R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)$ and $E_r^{(e)}(x) = R_{\text{UK}}(x, \tilde{\mathbf{x}}^e, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)$, whose difference is the announced identity. Both terms being SPSPD, $e_r(x)^2 = \lambda_{\max}(E_r(x))$ lies between $\lambda_{\max}(\Delta_e(x))$ and $\lambda_{\max}(\Delta_e(x)) + \lambda_{\max}(E_r^{(e)}(x))$, and Proposition 4.11 bounds the latter eigenvalue by $e_r^{U,(e)}(x)^2$. $\square$

The identity states that the subsampling error accumulates along a nesting, each step adding the covariance it loses to the ones before it. Three properties follow. First, $\Delta_e(x)$ is computed exactly on the $\tilde{N}^e$ data of $\tilde{x}^e$, at the $\mathcal{O}\left(\left(\tilde{N}^e\right)^3 d_z^3\right)$ cost of two kriging covariances, and is a lower bound on the error: it says how much of the loss is already certain, which no upper bound alone provides. Second, only the data outside $\tilde{x}^e$ are left to Proposition 4.11, so the bracket tightens as $\tilde{x}^e$ grows, and its width is chosen by the user rather than dictated by the configuration. Third, when $\tilde{x}^e$ gathers $\tilde{x}^r$ and the discarded data lying within a few correlation lengths of x , the remaining data are, for most kernels used in practice, screened by the neighbouring ones [106], so that $e_r^{U,(e)}(x)$ becomes small at a cost still far below that of the full estimator, $\tilde{N}^e$ remaining far below $\tilde{N}$.

4.5.8 Entrywise Bounds

The previous bounds are evaluated on the data at hand. Bounds on the residual quantities can also be derived from bounds on the covariance blocks and on the observations, which make explicit how the errors depend on the correlations between the data. Since $\tilde{z}_{d|r}$ is unchanged when a trend is added to the observations, the latter are measured from an arbitrary reference trend $g(\tilde{x})\beta_0$, $\beta_0 \in \mathbb{R}^{r_{\text{UK}}d_z}$, and the trend functions are assumed to verify $\|g(x')\|_{\text{op}} \leq g_{\text{max}}$ for $x' = x$ and for every entry x' of $\tilde{x}$. Writing $\tilde{x}_i^j$ and $\tilde{z}_i^j$ for the i -th entries of $\tilde{x}^j$ and $\tilde{z}^j$, and $\|K\|_{\text{max}}$ for the largest operator norm of the $d_z \times d_z$ blocks of a matrix K , which is its largest entry in absolute value when $d_z = 1$, we assume the following blockwise bounds, for $j \in \{r, d\}$ and for all the indices involved:

$$\left\{ \begin{array}{l} \|k(\tilde{x}^d, \tilde{x}^r)\|_{\text{max}} \leq \alpha, \quad \|k(\tilde{x}^r, x)\|_{\text{max}} \leq \kappa, \quad \|k(\tilde{x}^d, x)\|_{\text{max}} \leq \eta \\ \|k(\tilde{x}_i^j, \tilde{x}_l^j)\|_{\text{op}} \leq \kappa \text{ for } i \neq l, \quad \|k(\tilde{x}_i^j, \tilde{x}_i^j)\|_{\text{op}} \leq k_{\text{max}} \\ \|\tilde{z}^j - g(\tilde{x}^j)\beta_0\|_{\text{max}} \leq \tilde{z}_{\text{max}} \end{array} \right. \quad (4.75)$$

Here, κ bounds the covariances within $\tilde{x}^r$, within $\tilde{x}^d$, and between $\tilde{x}^r$ and x , while α and η respectively bound the covariances between $\tilde{x}^r$ and $\tilde{x}^d$, and between $\tilde{x}^d$ and x . The constants k_{max} and $\tilde{z}_{\text{max}}$ respectively bound the prior covariances of the training data and the deviation of the observations from the reference trend, which can be minimised over β_0 , the mid-range of $\tilde{z}$ being optimal for ordinary kriging on a scalar field. The subsampling strategy described in section 4.5.2, which retains the data close to x and discards the farther ones, aims at the regime:

$$0 < \eta, \alpha \ll \kappa \leq k_{\text{max}} \quad (4.76)$$

We further define:

$$\lambda_j^- \triangleq \lambda_{\min}(G(\tilde{x}^j)), \quad \lambda_j^+ \triangleq k_{\text{max}} + (\tilde{N}^j - 1)\kappa + \lambda_{\max}(\tilde{R}), \quad j \in \{r, d\} \quad (4.77)$$

where λ_j^+ bounds $\lambda_{\max}(G(\tilde{x}^j))$ by the blockwise Gershgorin circle theorem [7], recalled in section B.17. The conditioning of the trend on the retained

data is measured by:

$$\tau_r \triangleq \frac{\tilde{N}^r g_{\max}^2}{\lambda_{\min}(g(\tilde{\mathbf{x}}^r)^\top g(\tilde{\mathbf{x}}^r))} \geq r_{\text{UK}} \quad (4.78)$$

which is equal to 1 for ordinary kriging. Proposition 4.13 bounds the residual quantities, from which Proposition 4.14 deduces bounds on the two errors.

Proposition 4.13. *Assume that (4.75) holds and that $\lambda_r^- \lambda_d^- > \alpha^2 \tilde{N}^r \tilde{N}^d$. Then:*

$$\left\{ \begin{array}{l} \|k_{d|r}(x)\|_{\text{op}} \leq k_{d|r}^{\max} \triangleq \sqrt{\tilde{N}^d} \left(\frac{\lambda_r^- \eta + \alpha \tilde{N}^r \kappa}{\lambda_r^-} + \frac{\tau_r \lambda_r^+ \rho_r (\lambda_r^- + \tilde{N}^r \kappa)}{\tilde{N}^r \lambda_r^-} \right) \\ \|S_{d|r}^{-1}\|_{\text{op}} \leq s_{\max} \triangleq \frac{\lambda_r^-}{\lambda_r^- \lambda_d^- - \alpha^2 \tilde{N}^r \tilde{N}^d} \\ \|\tilde{\mathbf{z}}_{d|r}\|_2 \leq \tilde{\mathbf{z}}_{d|r}^{\max} \triangleq \sqrt{\tilde{N}^d} \rho_r \frac{\lambda_r^- + \tau_r \lambda_r^+}{\lambda_r^-} \tilde{z}_{\max} \end{array} \right.$$

where $\rho_r \triangleq (\lambda_r^- + \alpha \tilde{N}^r) / \lambda_r^-$.

Proof. See proof in section B.17. □

Proposition 4.14. *Under the assumptions of Proposition 4.13, the two errors verify:*

$$\left\{ \begin{array}{l} e_r(x) \leq k_{d|r}^{\max} \sqrt{s_{\max}} \\ e_f(x) \leq k_{d|r}^{\max} \tilde{\mathbf{z}}_{d|r}^{\max} s_{\max} \end{array} \right.$$

and $\|\tilde{\mathbf{z}}_{d|r}\|_{S_{d|r}} \leq \tilde{\mathbf{z}}_{d|r}^{\max} \sqrt{s_{\max}}$.

Proof. For A and B with $\tilde{N}^d d_z$ rows, $\left\| \langle A|B \rangle_{S_{d|r}} \right\|_{\text{op}} \leq \|A\|_{\text{op}} \cdot \|B\|_{\text{op}} \cdot \|S_{d|r}^{-1}\|_{\text{op}}$. Applying this inequality and Proposition 4.13 to Proposition 4.7 concludes the proof. □

As in Proposition 4.9, the bound on $e_f(x)$ is the product of the bounds on $e_r(x)$ and $\|\tilde{\mathbf{z}}_{d|r}\|_{S_{d|r}}$. Unlike their simple kriging counterparts, these bounds do not vanish when α and η tend to 0, their second term accounting for the estimation of the trend described by Remark 4.5. They also decrease with λ_r^- and λ_d^- , and thus remain valid when these eigenvalues are replaced by lower bounds such as $\tilde{R}$. Moreover, [107] shows that the smallest eigenvalue of a covariance matrix remains bounded away from 0 as the number of data grows, provided the training inputs are separated by a minimal distance.

Remark 4.9. *The condition $\lambda_r^- \lambda_d^- > \alpha^2 \tilde{N}^r \tilde{N}^d$ requires a gap between the retained and the discarded data, wide enough for their cross covariances to be small compared with the eigenvalues of the Gram matrices. When $\tilde{\mathbf{x}}^r$ is a window cut out of a continuous set of training data, α is reached by the closest data on either side of its boundary, and is close to κ , so that Proposition 4.14 does not apply and the bounds of sections 4.5.6 and 4.5.7 are to be used instead. Even when it applies, its successive triangular and Cauchy-Schwarz inequalities make it loose, and its interest lies in the dependence it exhibits on α , η and κ rather than in its numerical value.*

CHAPTER 5

SIMULTANEOUS LOCALISATION AND MAPPING

Contents

5.1	Introduction	126
5.2	The RBPF Approach	126
5.2.1	A Finite-Dimensional Map	127
5.2.2	A Conditionally Linear Model	128
5.3	The RKHS Approach	130
5.3.1	Embedding the Filter	130
5.3.2	Random Fourier Features	131
5.3.3	Recursive Least Squares	132
5.3.4	The AKKF-SLAM Algorithm	132
5.3.5	Numerical Results	133
5.4	The Learning Bias	135

5.1 Introduction

TODO: il faut dire qu'on peut utiliser une base de fonctions quelconques aussi, cf Camille et son polynôme + partir d'une zon krigée ce serait chouette et ça ferait le lien avec ce qui précède

Throughout chapter 4, we assumed to have access to at least some prior knowledge of the environment in the form of a sparse and noisy set of training data. Such training points were the base on which we were able to reconstruct the observation function using kriging without which no Bayesian correction could be performed.

In this chapter, we attempt to solve the navigation problem described by (2.1) without any prior data. This situation is called Simultaneous Localisation and Mapping (SLAM), and is a well-known problem in autonomous navigation. A standard solution is to use artificial vision, whether by using the data from an embedded stereo camera, a lidar, a sonar, or some other sensor able to probe its environment beyond its current location.

In our case however, we focus on local sensors, such as a magnetometer or a gravitometer, that only provide a point measurement of the surrounding field, meaning that no information on the neighbourhood of the field is gained after a sensor measurement, and so the likelihood of any reading can only be computed at the true location, which a Bayesian filter is precisely trying to recover.

Similarly to chapter 4, additional assumptions on the environment must be made in order to be able to do something. In this chapter, we assume the field to be stationary, such that it can be fully represented by the same GP on the whole of $\mathbb{R}^{d_x}$ and across time.

This chapter is organised as follows:

- Section 5.2 presents an existing SLAM method based on an RBPF formulation to estimate the coefficients of the Karhunen-Loève decomposition associated with the environment.
- Section 5.3 uses the AKKF to embed the field into an RKHS, and solve for its coefficients in this space.
- Section 5.4 examines the bias both methods incur by training their map along the very trajectory they are estimating.

5.2 The RBPF Approach

Without a prior survey, everything the vehicle knows about the field comes from the observations it has already collected. The map is therefore no longer an input of the filter, as assumption 4.1 assumed, but an unknown to be estimated jointly with the state. Carrying it in the state raises two difficulties. A GP is an infinite-dimensional object, whereas a filter propagates a finite state, and the estimators of section 3.3 represent the field through their whole training set, which would here grow by one observation at every time step.

Section 3.4 answers both at once. Truncating the Mercer decomposition of the kernel at rank r_{KL} replaces the field by a linear combination of basis functions chosen before the mission, so that the map reduces to the r_{KL} coefficients of that combination whatever the number of observations gathered. This section follows the

construction of [108], which builds such a basis explicitly and exploits the linearity it creates.

5.2.1 A Finite-Dimensional Map

Proposition 3.12 requires the input set to be compact, so let the mission be confined to a bounded domain:

$$\mathcal{U} \triangleq \prod_{d=1}^{d_x} [a_d, b_d], \quad L_d \triangleq b_d - a_d \quad (5.1)$$

As noted by Remark 3.13, the eigenfunctions of the Mercer operator are generally not available in closed form. For a radial kernel over such a domain, they are however well approximated by the eigenfunctions of the Dirichlet Laplacian on $\mathcal{U}$ [109], which are known explicitly. Indexing them by a multi-index $j \in (\mathbb{N}^*)^{d_x}$ and gathering the associated frequencies as:

$$\omega_j \triangleq \pi \left[\frac{j_1}{L_1} \quad \cdots \quad \frac{j_{d_x}}{L_{d_x}} \right]^\top \in \mathbb{R}^{d_x} \quad (5.2)$$

the corresponding basis functions read:

$$\psi_j : \begin{cases} \mathcal{U} \rightarrow \mathbb{R} \\ x \mapsto \prod_{d=1}^{d_x} \sqrt{\frac{2}{L_d}} \sin(\omega_{j,d}(x_d - a_d)) \end{cases} \quad (5.3)$$

and satisfy $-\Delta \psi_j = \|\omega_j\|_2^2 \psi_j$, where Δ is the Laplace operator.

This family is orthonormal on $\mathcal{U}$ and does not depend on the kernel, which only enters through the weight given to each of its members. Writing S_ν for the spectral density of k_ν , that is, the Fourier transform computed in section B.4, the kernel is approximated by:

$$k_\nu(x, x') \approx \sum_j S_\nu(\omega_j) \psi_j(x) \psi_j(x') \quad (5.4)$$

which is (3.70) with $S_\nu(\omega_j)$ in the role of the eigenvalue λ_j . Retaining the r_{KL} largest weights and renumbering the multi-indices accordingly, this is exactly the reduced-rank approximation of Proposition 3.13, with:

$$\Psi(x) = \begin{bmatrix} \psi_1(x) & \cdots & \psi_{r_{\text{KL}}}(x) \end{bmatrix}, \quad \Lambda = \begin{bmatrix} S_\nu(\omega_1) & & (0) \\ & \ddots & \\ (0) & & S_\nu(\omega_{r_{\text{KL}}}) \end{bmatrix} \quad (5.5)$$

Two properties make this basis usable for navigation. It is built from the domain and the kernel alone, so it is entirely known before the first observation and never has to be recomputed. And it carries the field itself rather than only its covariance, since:

$$h(x) = \Psi(x) \alpha, \quad \alpha \sim \mathcal{N}(0, \Lambda) \quad (5.6)$$

defines a centred GP of covariance $\Psi(x) \Lambda \Psi(x')^\top$, which is the truncated kernel of Proposition 3.13. The map is thus fully described by the r_{KL} coefficients α , and estimating the field amounts to estimating a vector of fixed size.

Remark 5.1 (Boundary conditions). *Every function of the basis (5.3) vanishes on the boundary of $\mathcal{U}$, and so does every field it represents. The domain must therefore be taken larger than the region actually flown over, with a margin of the order of the correlation length, so that the reconstruction is not pulled towards zero where the vehicle navigates.*

This margin is not free. The frequencies (5.2) are spaced by π/L_d along each axis, so enlarging the domain refines the spectrum and more basis functions are needed to reach the same maximal frequency.

Remark 5.2 (Rank and resolution). *The smallest wavelength the basis can represent along an axis is $2L_d/j_d$, so resolving a field of correlation length l requires indices up to roughly L_d/l along that axis. The rank therefore scales as the product of these bounds:*

$$r_{\text{KL}} \sim \prod_{d=1}^{d_x} \frac{L_d}{l} \quad (5.7)$$

and grows exponentially with the dimension of the domain. A two-dimensional map covering ten correlation lengths per axis is within reach, whereas the same resolution in three dimensions costs a thousand basis functions. This is the trade-off three-dimensional magnetic SLAM faces, and it is why the domain is there kept small and the map local by construction.

5.2.2 A Conditionally Linear Model

The interest of (5.6) is that the observation is linear in α once the state is known, which is precisely the structure (2.139) requires. Let the augmented state gather the vehicle state and the coefficients of its map:

$$x_k^n \triangleq \text{vehicle state}, \quad x_k^l \triangleq \alpha \quad (5.8)$$

Under Assumption 4.1 and (5.6), the navigation problem (2.1) is then the system (2.139) with:

$$\begin{cases} F_k^n = 0 & f_k^n = f_k & Q_k^n = Q_k & Q_k^{nl} = 0 \\ F_k^l = I_{r_{\text{KL}}} & f_k^l = 0 & Q_k^l = 0 \\ H_k(x_k^n) = \Psi(x_k^n) & h_k = 0 \end{cases} \quad (5.9)$$

provided the evolution noise is additive and Gaussian, as (2.139) requires. The map is static, since the field does not change over the course of the mission, and it does not drive the vehicle, since F_k^n is null. A known mean function would enter as $h_k = m$, and is null here because (5.6) defines a centred field. Both the evolution and the observation equations are thus linear and Gaussian in the coefficients conditionally on the trajectory, and the RBPF of section 2.5 applies.

The identification (5.9) cancels two of the five steps of Algorithm 2.10. The pre-correction (2.149) through (2.154) has a null gain, as the propagation of a vehicle that does not sense the field teaches nothing about it, and the Kalman prediction (2.155) through (2.160) leaves the coefficients and their covariance unchanged. What remains at each time step is a weight update and a correction of the map carried by each particle.

Writing $\Psi_k^{(i)}$ for $\Psi(x_k^{n(i)})$, the innovation covariance and the Kalman gain of the i -th particle read:

$$\begin{cases} S_k^{(i)} = \Psi_k^{(i)} P_{k-1|k-1}^{l(i)} \left(\Psi_k^{(i)}\right)^\top + R_k \\ K_k^{(i)} = P_{k-1|k-1}^{l(i)} \left(\Psi_k^{(i)}\right)^\top \left(S_k^{(i)}\right)^{-1} \end{cases} \quad (5.10)$$

and its map is corrected as:

$$\begin{cases} \hat{x}_{k|k}^{l(i)} = \hat{x}_{k-1|k-1}^{l(i)} + K_k^{(i)} \left(z_k - \Psi_k^{(i)} \hat{x}_{k-1|k-1}^{l(i)}\right) \\ P_{k|k}^{l(i)} = \left(I_{r_{\text{KL}}} - K_k^{(i)} \Psi_k^{(i)}\right) P_{k-1|k-1}^{l(i)} \end{cases} \quad (5.11)$$

The likelihood entering the weight update (2.161) is the marginal of the observation under the map of that particle:

$$\tilde{p}\left(z_k | x_k^{n(i)}\right) = \mathcal{N}\left(z_k; \Psi_k^{(i)} \hat{x}_{k-1|k-1}^{l(i)}, S_k^{(i)}\right) \quad (5.12)$$

and Algorithm 5.1 summarises the resulting filter. As noted for Algorithm 2.10, the map correction (5.11) follows the weight update, so that a particle discarded by the resampling never pays for the correction of a map that is about to be deleted. The resampling itself carries the maps along with the particles, each duplicated particle inheriting a copy of the coefficients and of their covariance.

Algorithm 5.1 Rao-Blackwellised SLAM

$\forall i \in \llbracket 1, N \rrbracket, x_k^{n(i)} \sim p\left(\cdot x_{k-1}^{n(i)}\right)$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \text{compute } S_k^{(i)} \text{ from (5.10)}$	▷ Innovation covariance
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \tilde{p}\left(z_k x_k^{n(i)}\right) \text{ from (5.12)}$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation
$\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N \left(w_k^{(i)}\right)^2$	▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ then	
$\left\{x_k^{n(j)}, w_k^{(j)}, i_j\right\}_{j=1}^N \leftarrow \text{Resampling}\left(\left\{x_k^{n(i)}, w_k^{(i)}\right\}_{i=1}^N\right)$	▷ Resampling
$\forall j \in \llbracket 1, N \rrbracket, \left(\hat{x}_{k-1 k-1}^{l(j)}, P_{k-1 k-1}^{l(j)}\right) \leftarrow \left(\hat{x}_{k-1 k-1}^{l(i_j)}, P_{k-1 k-1}^{l(i_j)}\right)$	▷ Map duplication
end if	
$\forall i \in \llbracket 1, N \rrbracket, \text{compute } K_k^{(i)} \text{ from (5.10)}$	▷ Map correction
$\forall i \in \llbracket 1, N \rrbracket, \text{compute } \hat{x}_{k k}^{l(i)} \text{ and } P_{k k}^{l(i)} \text{ from (5.11)}$	

Remark 5.3 (Kriging in disguise). Equation (5.12) has the form of the kriged likelihood (4.3) of section 4.2, a Gaussian whose mean reconstructs the field at the queried point and whose covariance carries both the reconstruction error and the observation noise. Starting from the prior $\mathcal{N}(0, \Lambda)$ of (5.6) and conditioning successively

on $z_1, \dots, z_{k-1}$, the pair $\left(\hat{x}_{k-1|k-1}^{l(i)}, P_{k-1|k-1}^{l(i)}\right)$ is the reduced-rank estimator of Proposition 3.14 applied to the training set $\left(\mathbf{x}_{0:k-1}^{n(i)}, \mathbf{z}_{1:k-1}\right)$, the Kalman correction (5.11) playing the part of the sequential update of that estimator.

What changes with respect to chapter 4 is the training set. There, it is the survey, shared by every particle and fixed for the whole mission. Here, it is what the vehicle has measured along the trajectory of that particle, so each particle krige its own history, and the training inputs are exact under the conditioning that defines them.

Remark 5.4 (Cost of one map per particle). Each particle stores its coefficients and their covariance, that is $r_{\text{KL}}(r_{\text{KL}} + 3)/2$ scalars once symmetry is exploited, and (5.10) and (5.11) cost $\mathcal{O}(r_{\text{KL}}^2 d_z)$ flops. A time step of Algorithm 5.1 is thus $\mathcal{O}(N r_{\text{KL}}^2 d_z)$ in time and $\mathcal{O}(N r_{\text{KL}}^2)$ in memory.

Neither grows with the number of observations already assimilated, which is what the truncation bought, but both are paid at every step and for every particle. The memory is the binding constraint in practice, a cloud of 5000 particles carrying a rank 64 map already storing more than ten million scalars.

Remark 5.5 (Vector fields and the magnetic potential). A magnetometer measures three components rather than one, and those components are not independent. In a current-free region, the magnetic field derives from a scalar potential u through $h = \nabla u$, and u is the natural quantity to model as a GP [110]. The basis of the field is then $(\nabla \psi_j)_j$ instead of $(\psi_j)_j$, and (5.6) becomes $h(x) = \nabla \Psi(x) \alpha$ with $\nabla \Psi(x) \in \mathcal{M}_{3, r_{\text{KL}}}(\mathbb{R})$, everything else being unchanged. Differentiating the basis rather than the reconstruction is what enforces the physical constraint exactly, and it also divides by three the number of coefficients a vector measurement has to estimate [111].

5.3 The RKHS Approach

Algorithm 5.1 owes its tractability to two choices, a basis fixed before the mission and a map entering the observation linearly. It pays for them with a map per particle, and with a basis tied to a rectangular domain on whose boundary the field is forced to vanish. This section keeps the linear map and abandons the rest, replacing the particle filter by a Kalman recursion cast into an RKHS [112].

5.3.1 Embedding the Filter

The kernel mean embedding (3.100) introduced in Section 3.5.4 carries a probability distribution into $\mathcal{H}_k$, where conditioning acts linearly. The adaptive kernel Kalman filter (AKKF) of Algorithm 3.1 runs the Bayesian recursion in that space: it represents the state distribution by the embedding of its particle set, propagates the attached weights $\mathbf{w}_k$ and their covariance Σ_k rather than the particles, and recomputes the state and observation Gram matrices at every step, so as to follow a model that is time-varying and neither Gaussian nor the image of a linear one. Its embedded correction is a regularised least-squares problem rather than a Bayesian update, so, as Remark 3.18 recalls, the weights it returns are of either sign and the filter resamples unconditionally, drawing a fresh particle set from the Gaussian that carries the current estimate.

5.3.2 Random Fourier Features

The filter above requires sampling from the likelihood, hence evaluating the observation function, which is the very quantity assumed unknown here. Since the state space is already embedded into $\mathcal{H}_k$, it is natural to look for the map inside the same space. By Proposition 3.18, $\mathcal{H}_k$ is the closure of the span of $\{\Phi(x), x \in \mathcal{U}\}$, so the map admits an arbitrarily good approximation as a finite combination of feature maps, the particles being the natural points at which to take them. This representation carries the same defect as kriging, however, its size being that of the particle set, and it also requires Φ to be evaluated explicitly, which Definition 3.7 precisely avoids.

An explicit finite basis is thus needed once more, and Bochner's theorem provides one for a radial kernel. Writing S_ν for its spectral density as in section 5.2.1, and normalising it into the probability density:

$$p_\nu \triangleq S_\nu / \int_{\mathbb{R}^{d_x}} S_\nu(\omega) d\omega \quad (5.13)$$

which is legitimate since $S_\nu > 0$, the kernel is the characteristic function of p_ν :

$$k_\nu(x, x') = \sigma_f^2 \int_{\mathbb{R}^{d_x}} \exp(i \omega^\top (x - x')) p_\nu(\omega) d\omega = \sigma_f^2 \mathbb{E}_{p_\nu} [\cos(\omega^\top (x - x'))] \quad (5.14)$$

the imaginary part vanishing by symmetry of p_ν , as k_ν is real-valued.

Equation (5.14) is an expectation, and Proposition 2.4 applies. Drawing r_{RFF} frequencies ω_j from p_ν and as many phases b_j from $\mathcal{U}([0, 2\pi])$, and letting:

$$\psi_j : \begin{cases} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ x \mapsto \sqrt{\frac{2}{r_{\text{RFF}}}} \cos(\omega_j^\top x + b_j) \end{cases} \quad (5.15)$$

the product-to-sum identity gives $\mathbb{E}_{b_j} [\psi_j(x) \psi_j(x')] = \cos(\omega_j^\top (x - x')) / r_{\text{RFF}}$, the term carried by $x + x'$ averaging out over the phase, so that the sum of these r_{RFF} products estimates (5.14). Gathering them in $\Psi(x)$ as in section 5.2.1 yields the random Fourier features approximation [113, 114]:

$$k_\nu(x, x') \approx \sigma_f^2 \Psi(x) \Psi(x')^\top \quad (5.16)$$

and the map is again of the form (5.6), this time with $\Lambda = \sigma_f^2 I_{r_{\text{RFF}}}$.

Remark 5.6 (Two ways of choosing the frequencies). *Equations (5.3) and (5.15) provide two finite-dimensional approximations of the same stationary kernel, but differ in how their frequencies are selected and weighted. The Laplace basis places them on the deterministic grid (5.2) imposed by the domain, and weights each of them by $S_\nu(\omega_j)$. The random features draw them from S_ν itself and weight them uniformly, which is the Monte Carlo approximation of section 2.4 applied to the spectrum.*

Sampling the frequencies removes the two constraints of Remark 5.1, since nothing forces the reconstruction to vanish anywhere and no domain has to be declared in advance. It also replaces the deterministic truncation error of Proposition 3.13 by a Monte Carlo approximation error, whose rate is $\mathcal{O}(r_{\text{RFF}}^{-1/2})$ instead of following the decay of the spectrum. The comparison is therefore not uniformly in favour of either, a grid being the better choice when the domain is known and small enough for the spectrum to be covered.

5.3.3 Recursive Least Squares

The basis (5.15) is fixed once the frequencies are drawn, so learning the map amounts to estimating the coefficients $\alpha \in \mathbb{R}^{r_{\text{RFF}}}$ of (5.6) from the pairs $(\hat{x}_l, z_l)$ the filter produces, where $\hat{x}_l$ is its state estimate at time l and z_l the observation it received. The criterion minimised at time k is the regularised and exponentially discounted least-squares cost:

$$J_k : \alpha \mapsto \sum_{l=1}^k \beta^{k-l} \|z_l - \Psi(\hat{x}_l) \alpha\|_2^2 + \beta^k \lambda_{\text{RLS}} \|\alpha\|_2^2 \quad (5.17)$$

where $\lambda_{\text{RLS}} \in \mathbb{R}_+^*$ is a regularisation coefficient and $\beta \in (0, 1]$ a forgetting factor.

Recursive least squares [115] minimise (5.17) without storing the past data, by carrying the estimate $\hat{\alpha}_k$ together with the inverse P_k^α of the information matrix of (5.17), and updating both from the current pair alone:

$$\begin{cases} K_k^\alpha = P_{k-1}^\alpha \Psi(\hat{x}_k)^\top \left(\beta + \Psi(\hat{x}_k) P_{k-1}^\alpha \Psi(\hat{x}_k)^\top \right)^{-1} \\ \hat{\alpha}_k = \hat{\alpha}_{k-1} + K_k^\alpha (z_k - \Psi(\hat{x}_k) \hat{\alpha}_{k-1}) \\ P_k^\alpha = \frac{1}{\beta} (I_{r_{\text{RFF}}} - K_k^\alpha \Psi(\hat{x}_k)) P_{k-1}^\alpha \end{cases} \quad (5.18)$$

initialised at $\hat{\alpha}_0 = 0$ and $P_0^\alpha = I_{r_{\text{RFF}}} / \lambda_{\text{RLS}}$.

Remark 5.7 (One regression, two training sets). Equation (5.18) is the correction (5.11) written for a single map, and the two recursions coincide exactly when $\beta = 1$ and $\lambda_{\text{RLS}} I_{r_{\text{RFF}}} = R_k \Lambda^{-1}$, which the random features allow since their Λ is a multiple of the identity. Their gains are then equal, P_k^α is the coefficient covariance $P_{k|k}^l$ divided by R_k , and the minimiser of (5.17) is the reduced-rank estimator of Proposition 3.14. Both sections therefore perform the same regression of the field onto a finite basis, and what distinguishes them is the training set each feeds it, the trajectory of a particle in one case and the sequence of state estimates in the other.

A forgetting factor $\beta < 1$ breaks this correspondence by discounting the past, which no Bayesian update of a static field would do. It buys robustness to the drift of the estimated inputs at the price of discarding information the field has not invalidated, and it makes the reconstruction local in time rather than in space.

5.3.4 The AKKF-SLAM Algorithm

Since (5.18) consumes one pair per time step, the map can be learnt at the rate of the filtering recursion, and both can be interleaved into a single loop. Algorithm 5.2 presents the resulting algorithm, in which $\mathbf{x}_k^+$ denotes the particle set after resampling, $\mathbf{w}_k^+$ and $\Sigma_{k|k}^+$ its weights and their covariance, and:

$$G_k^x \triangleq k_x (\mathbf{x}_k^+, \mathbf{x}_k^+), \quad G_k^z \triangleq k_z (\mathbf{z}_k, \mathbf{z}_k) \quad (5.19)$$

the Gram matrices of the state and observation kernels, whose regularisation coefficients are λ_x and λ_z .

Two points deserve attention. The correction step draws its simulated observations from the current map, since the true one is unavailable, whereas the map update uses the actual observation z_k and the estimate produced by the correction. And the state estimate is required by (5.18), so it belongs to the recursion rather than to its reading, as the predicted estimate does in Algorithm 4.2.

Algorithm 5.2 AKKF-SLAM

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p\left(\cdot x_{k-1}^{+(i)}\right)$	▷ Particle propagation
$D_k = \left(G_{k-1}^x + \lambda_x I_N\right)^{-1} G_{k-1}^x - I_N$	▷ Embedded prediction
$\Sigma_{k k-1} = \Sigma_{k-1 k-1}^+ + \frac{1}{N} D_k D_k^\top$	
$\forall i \in \llbracket 1, N \rrbracket, n_k^{(i)} \sim p_k^{(o)}$	▷ Simulated observations
$\forall i \in \llbracket 1, N \rrbracket, z_k^{(i)} = \Psi\left(x_k^{(i)}\right) \hat{\alpha}_{k-1} + n_k^{(i)}$	
$K_k = \Sigma_{k k-1} \left(G_k^z \Sigma_{k k-1} + \lambda_z I_N\right)^{-1}$	▷ Embedded correction
$\mathbf{w}_k = \mathbf{w}_{k-1}^+ + K_k \left(k_z(\mathbf{z}_k, z_k) - G_k^z \mathbf{w}_{k-1}^+\right)$	
$\Sigma_{k k} = \Sigma_{k k-1} - K_k G_k^z \Sigma_{k k-1}$	
$\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$	▷ State estimation
$\hat{P}_k = \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right) \left(x_k^{(i)} - \hat{x}_k\right)^\top$	
Compute K_k^α , $\hat{\alpha}_k$ and P_k^α from (5.18)	▷ Map update
$\forall i \in \llbracket 1, N \rrbracket, x_k^{+(i)} \sim \mathcal{N}\left(\hat{x}_k, \hat{P}_k\right)$	▷ Resampling
$T_k = \left(G_k^x + \lambda_x I_N\right)^{-1} k_x\left(\mathbf{x}_k^+, \mathbf{x}_k\right)$	▷ Embedded resampling
$\mathbf{w}_k^+ = T_k \mathbf{w}_k$	
$\Sigma_{k k}^+ = T_k \Sigma_{k k} T_k^\top$	

5.3.5 Numerical Results

Algorithm 5.2 is evaluated on the two-dimensional scenario of Table 5.1, in which a vehicle follows a contracting spiral over an analytic field and observes it through a scalar magnetometer-like sensor. The field is deliberately non-injective, so that a single reading is compatible with several positions, and the initial estimate is offset from the true state.

Three particle filters serve as a baseline, each given a known but degraded observation function:

$$h_\gamma : x \mapsto (1 + \varepsilon(x)) h(x), \quad \varepsilon(x) \sim \mathcal{N}(0, \gamma) \quad (5.20)$$

with $\gamma \in \{0, 0.1, 0.5\}$, the first of them navigating on the exact map. All four filters are run over $n_{\text{MC}} = 100$ Monte Carlo simulations, and compared through the trajectory error:

$$\text{RMSE}_{\hat{x}} : k \mapsto \sqrt{\frac{1}{n_{\text{MC}}} \sum_{i=1}^{n_{\text{MC}}} \left\| \hat{x}_k^{(i)} - x_k \right\|_2^2} \quad (5.21)$$

where $\hat{x}_k^{(i)}$ is the estimate of the i -th run at time k , and its average over a set E of time steps:

$$\text{ARMSE}_{\hat{x}}(E) \triangleq \frac{1}{|E|} \sum_{k \in E} \text{RMSE}_{\hat{x}}(k) \quad (5.22)$$

which is (4.6) over a discrete index set. Table 5.2 reports the maximum of (5.21) over the first ten steps and its average over the last twenty.

True function	$h : x \mapsto \sin(0.7p^{(x)}) + 0.5 \cos(1.2p^{(y)}) + 0.3 \sin(0.4(p^{(x)} + p^{(y)}))$
State	$x_k = \begin{bmatrix} p_k^{(x)} & p_k^{(y)} \end{bmatrix}^\top$, $d_x = 2$, $d_z = 1$
Trajectory	contracting spiral of $n_{\text{steps}} = 100$ steps over $[-5, 5]^2$
Evolution noise	$Q_k = 0.01^2 I_2$
Observation noise	$R_k = 0.01^2$
Initial covariance	$P_0 = 0.1^2 I_2$
Kernels	$k_x = k_z = k_{\text{SE}}$ with $l = 1.5$ and $\sigma_f^2 = 1$
Regularisation	$\lambda_x = \lambda_z = 0.1$, $\lambda_{\text{RLS}} = 0.01$, $\beta = 0.995$
Features	$r_{\text{RFF}} = 500$, frequencies drawn from $\mathcal{N}(0, 1.5)$
Particles	$N = 1000$

Table 5.1: The SLAM scenario. The observation function is scalar and non-injective, so that the likelihood is multimodal at every time step.

	AKKF-SLAM	$\gamma = 0.5$	$\gamma = 0.1$	$\gamma = 0$
$\max_{k \leq 10} \text{RMSE}_{\hat{x}}(k)$	0.56	0.18	0.15	0.13
$\text{ARMSE}_{\hat{x}}(\llbracket 80, 100 \rrbracket)$	0.19	0.18	0.11	0.04
Ratio	2.88	0.96	1.38	3.15

Table 5.2: Transient and asymptotic trajectory errors of the four filters. The three rightmost columns are particle filters navigating on a known map degraded at level γ .

The baselines behave as expected, their error decreasing with γ , and the most degraded of them merely maintaining its initial error instead of converging. AKKF-SLAM starts markedly worse, at 0.56 against 0.13 for the best-informed filter, which is the price of having no map at all during the first steps. It then divides its error by almost three, ending at the level of the filter navigating on the most degraded map, whereas the ratio of the same two quantities is close to one for that filter. The shape of the decrease, not its level, is what the comparison establishes, and it matches that of the filter given the exact map.

The reconstruction is evaluated through the spatial error defined as in (4.5), computed between h and the final map $x \mapsto \Psi(x) \hat{\alpha}_{n_{\text{steps}}}$ of each run. Its average is 0.21 over $[-5, 5]^2$ and 0.07 along the true trajectory, the gap between the two being extrapolation, as most of the domain is never approached closer than a correlation length. Three of the four peaks of the spatial error indeed lie outside the spiral. The fourth lies inside it, where the trajectory does pass, and corresponds to a reconstruction shifted with respect to the true field rather than to a loss of amplitude, which section 5.4 attributes to the way the map is trained.

5.4 The Learning Bias

Both algorithms of this chapter estimate the map from observations attached to positions they do not know. Algorithm 5.1 handles this by carrying the joint law of the trajectory and the map, factorised as in section 2.5:

$$p(\alpha, \mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) = p(\alpha | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k}) p(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) \quad (5.23)$$

so that each map is conditioned on a trajectory which is exact under that conditioning, as Remark 5.3 observes. The coupling between the two is represented by the particle set itself, a particle whose trajectory is wrong carrying a map that is wrong in a consistent way, and the weights arbitrating between them.

Algorithm 5.2 carries a single map instead, and trains it on the pairs $(\hat{x}_l, z_l)$. This is the construction (5.17) assumes, and it breaks two of the hypotheses under which the regression of section 3.3 is unbiased.

The training inputs are no longer exact. Kriging assumes noiseless inputs, and an error on the input is not interchangeable with an error on the output. Writing $\hat{x}_l = x_l + \delta_l$ with δ_l centred of covariance $\hat{P}_l$, a second-order expansion of a twice-differentiable field gives:

$$\mathbb{E}[h(\hat{x}_l)] \approx h(x_l) + \frac{1}{2} \text{tr} \left(\hat{P}_l \nabla^2 h(x_l) \right) \quad (5.24)$$

so the regression sees a field smoothed by the estimation error rather than the field itself. Extrema are attenuated and gradients flattened, which an inflation of R_k would not reproduce, since it widens the confidence band without moving its centre.

More importantly, the two members of the pair are correlated. The estimate $\hat{x}_l$ is produced by a correction step that has already used z_l , so the map is taught to agree with the very estimate that observation produced, and confirms it at the following step. The covariance P_k^α then shrinks with no evidence behind it, and the filter grows confident in a map that its own estimates wrote. The interior peak of section 5.3.5 is its signature, a reconstruction displaced along the trajectory rather than degraded in amplitude.

What Algorithm 5.2 performs is therefore mapping with known poses, applied to poses that are merely estimated. Genuine simultaneous localisation and mapping carries the joint law of the trajectory and the map, with the cross-covariances between them, which is what the particle set of Algorithm 5.1 represents and what an extended Kalman filter holding pose-landmark blocks [116] keeps explicitly. Dropping those terms is what produces the bias, and the cost of keeping them is the one Remark 5.4 measures.

Three corrections suggest themselves, none of which is implemented here. The correlation is the easiest of the two defects to remove, since training the map on the predicted estimate $\hat{x}_{k|k-1}$ rather than on the corrected one uses an input that does not depend on z_k , at the cost of a less accurate position. The input error calls instead for a regression that accepts a distribution over its inputs, integrating the likelihood against $\mathcal{N}(\hat{x}_k, \hat{P}_k)$ instead of evaluating it at $\hat{x}_k$. Carrying the cross-covariances explicitly is the third, and is what separates the two algorithms of this chapter.

Remark 5.8 (Trajectory degeneracy). *Algorithm 5.1 is not exempt from a related defect. Every resampling discards trajectories, and after enough steps all surviving*

particles descend from a single ancestor, so the distant past of the trajectory is represented by one sample whatever the value of N . The maps of all particles are then conditioned on a common early history whose uncertainty has disappeared from the particle set, and the confidence they express on the corresponding region of the field is overstated.

This is the counterpart, for a filter carrying the joint law, of the bias described above, and it bounds the duration over which Algorithm 5.1 can be trusted to quantify its own map uncertainty.

CHAPTER 6

CONCLUSION

This thesis set out to navigate autonomously on physical fields when neither satellite positioning nor a reliable embedded map can be taken for granted. The guiding idea has been to reconstruct the observation function of the field with Gaussian processes and to embed this reconstruction within a Bayesian, particle-based navigation filter, so that a single probabilistic language carries both the uncertainty of the state and the uncertainty of the map.

The foundations laid in the first part recalled the Bayesian filters that cope with the non-linear and multimodal densities of terrain-aided navigation – from the Kalman filter to the regularised and Rao-Blackwellised particle filters – and the Gaussian-process machinery – kriging, its reduced-rank approximations and the reproducing-kernel-Hilbert-space viewpoint – that turns field data into an observation function endowed with its own uncertainty. The second part brought the two together. Kriging was integrated into the particle filter, yielding a navigation scheme that no longer requires a dense and exact embedded map but reconstructs the field from whatever data are available, and that feeds the uncertainty of this reconstruction back into the state estimate. To keep the reconstruction affordable and to respect the non-stationarity of real fields, an adaptive kriging scheme restricts it to the data relevant to the current region, the resulting subsampling error being bounded theoretically. Finally, for the case of an altogether unknown field, localisation and mapping were carried out jointly – through a Rao-Blackwellised filter and a reproducing-kernel-Hilbert-space reconstruction from a dictionary of basis functions – and the learning bias inherent to feeding the map with the vehicle’s own estimated positions was identified and characterised.

TODO: bilan quantitatif – résumer ici les résultats des campagnes (précision, robustesse, coût AK/statique), une fois les chiffres figés.

6.1 Limitations

Several limitations temper these results. Adaptive kriging trades exactness for tractability: restricting the reconstruction to a local window rests on a local-stationarity assumption that has the status of a heuristic rather than of a guarantee, and the cost of the reconstruction, although reduced, remains higher than the mere reading of a stored map. In the map-free case, the learning bias was identified

and characterised but not corrected; in the absence of a joint law over the state and the map, the joint estimate is moreover left without a reference against which its consistency could be certified. The observation function, finally, is treated as time-invariant, and the results rest on simulation rather than on data from a field campaign.

6.2 Perspectives

Several directions extend this work. On the side of the models, a spatially varying set of hyperparameters, fitted offline, would relax the local-stationarity heuristic on which adaptive kriging rests, and richer covariance families would widen the range of fields the method can handle. On the side of the methodology, the learning bias of the map-free case calls for a correction rather than a mere characterisation, and for a way of certifying the consistency of a joint estimate that no joint law underwrites.

The most natural opening, however, is the fusion of several complementary physical fields – magnetic, gravimetric, bathymetric. Their joint informativeness could be quantified through the Fisher information, and their combination could resolve the multimodal likelihoods that a single field leaves ambiguous, extending the reconstruction pursued throughout this thesis from one field to several.

Part III

Appendices

APPENDIX A

USEFUL MATRIX IDENTITIES

Contents

A.1 Gaussian Conditioning Formula	142
A.2 Schur Inversion Formula	143
A.3 Woodbury Inversion Formula	144

A.1 Gaussian Conditioning Formula

Proposition A.1 (Gaussian conditioning [117]). *For $n, p \in \mathbb{N}$, let $U \in \mathbb{R}^n$ and $V \in \mathbb{R}^p$ be jointly Gaussian random variables, of respective means μ_U, μ_V and covariance matrices $A \succ 0$ and $D \succ 0$. There exists $B \in \mathcal{M}_{n,p}(\mathbb{R})$ such that:*

$$\begin{bmatrix} U \\ V \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_U \\ \mu_V \end{bmatrix}, \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \right) \quad (\text{A.1})$$

We assume B to be such that this joint covariance is SPD in the sense of Definition 2.1. The conditional distribution of U given $V = v$ is then:

$$U|V = v \sim \mathcal{N}(\mu_U + BD^{-1}(v - \mu_V), P_{U|V}) \quad (\text{A.2})$$

with:

$$P_{U|V} \triangleq A - BD^{-1}B^\top \succ 0 \quad (\text{A.3})$$

Proof. The existence of B is direct by definition of jointly Gaussian random variables. As $D \succ 0$ is invertible, $S \triangleq A - BD^{-1}B^\top$ the Schur complement with respect to D is well-defined.

Let $u \neq 0$:

$$\begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix}^\top \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix} = u^\top (A - BD^{-1}B^\top) u > 0 \quad (\text{A.4})$$

by SPDness of the joint covariance, the left-hand vector being non-zero. S is thus also invertible, and we can apply the inferior right Schur inversion formula (see Proposition A.2) to obtain:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \triangleq \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix}^{-1} \quad (\text{A.5})$$

For some realisation u of U and v of V , we define $\tilde{u} \triangleq u - \mu_U$ and $\tilde{v} \triangleq v - \mu_V$, and we compute:

$$\begin{aligned} \alpha &\triangleq \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix}^\top \begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix} \\ &= \tilde{u}^\top \bar{A} \tilde{u} + \tilde{u}^\top \bar{B} \tilde{v} + \tilde{v}^\top \bar{C} \tilde{u} + \tilde{v}^\top \bar{D} \tilde{v} \\ &= \tilde{u}^\top S^{-1} \tilde{u} + \tilde{u}^\top S^{-1} (-BD^{-1} \tilde{v}) + (-BD^{-1} \tilde{v})^\top S^{-1} \tilde{u} + \\ &\quad (-BD^{-1} \tilde{v})^\top S^{-1} (-BD^{-1} \tilde{v}) + \tilde{v}^\top D^{-1} \tilde{v} \\ &= (\tilde{u} - BD^{-1} \tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1} \tilde{v}) + \tilde{v}^\top D^{-1} \tilde{v} \end{aligned} \quad (\text{A.6})$$

The conditional density is thus:

$$\begin{aligned} p(\tilde{u}|\tilde{v}) &\propto \frac{p(\tilde{u}, \tilde{v})}{p(\tilde{v})} \\ &\propto \frac{\exp\left(-\frac{1}{2}\alpha\right)}{\exp\left(-\frac{1}{2}\tilde{v}^\top D^{-1} \tilde{v}\right)} \\ &\propto \exp\left(-\frac{1}{2}(\tilde{u} - BD^{-1} \tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1} \tilde{v})\right) \end{aligned} \quad (\text{A.7})$$

With this formulation, it comes that $U|V = v$ is Gaussian with mean:

$$\mu_{U|V} \triangleq \mu_U + BD^{-1}(v - \mu_V) \quad (\text{A.8})$$

and covariance:

$$P_{U|V} \triangleq S = A - BD^{-1}B^\top \succ 0 \quad (\text{A.9})$$

□

A.2 Schur Inversion Formula

Proposition A.2 (Schur inversion formula [118]). *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$, with A invertible. We define the matrix M and its Schur complement with respect to A as:*

$$M \triangleq \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \quad M/A \triangleq D - CA^{-1}B \quad (\text{A.10})$$

If M/A is invertible, then M is invertible and:

$$M^{-1} = \begin{bmatrix} A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} & -A^{-1}B(M/A)^{-1} \\ -(M/A)^{-1}CA^{-1} & (M/A)^{-1} \end{bmatrix} \quad (\text{A.11})$$

Proof. Let $\bar{M}$ be the matrix defined by the right-hand side of the formula, and write its blocks as:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} = \bar{M} \quad (\text{A.12})$$

The identity $M\bar{M} = I$ yields:

$$\begin{cases} A\bar{A} + B\bar{C} = I \\ A\bar{B} + B\bar{D} = 0 \\ C\bar{A} + D\bar{C} = 0 \\ C\bar{B} + D\bar{D} = I \end{cases} \quad (\text{A.13})$$

In particular, the second equation can be rewritten as:

$$\bar{B} = -A^{-1}B\bar{D} \quad (\text{A.14})$$

which can be plugged into the fourth equation to give:

$$\begin{aligned} C\bar{B} + D\bar{D} &= I \\ \iff -CA^{-1}B\bar{D} + D\bar{D} &= I \\ \iff (D - CA^{-1}B)\bar{D} &= I \\ \iff (M/A)\bar{D} &= I \\ \iff \bar{D} &= (M/A)^{-1} \end{aligned} \quad (\text{A.15})$$

(A.14) can then be solved:

$$\bar{B} = -A^{-1}B(M/A)^{-1} \quad (\text{A.16})$$

A similar operation can be performed on the two remaining equations to obtain:

$$\bar{A} = A^{-1}(I - B\bar{C}) \quad (\text{A.17})$$

and:

$$\begin{aligned} & C\bar{A} + D\bar{C} = 0 \\ \iff & CA^{-1}(I - B\bar{C}) + D\bar{C} = 0 \\ \iff & CA^{-1} + (D - CA^{-1}B)\bar{C} = 0 \\ \iff & CA^{-1} + (M/A)\bar{C} = 0 \\ \iff & \bar{C} = -(M/A)^{-1}CA^{-1} \end{aligned} \quad (\text{A.18})$$

Finally:

$$\bar{A} = A^{-1}(I - B\bar{C}) = A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} \quad (\text{A.19})$$

It can be verified by direct calculations that:

$$M\bar{M} = \bar{M}M = I \quad (\text{A.20})$$

which proves that M is indeed invertible with inverse $\bar{M}$. $\square$

Remark A.1 (Inversion with respect to D). *If D is invertible, one can instead introduce the Schur complement with respect to D :*

$$M/D \triangleq A - BD^{-1}C \quad (\text{A.21})$$

When M/D is also invertible, an analogous block inversion formula holds:

$$M^{-1} = \begin{bmatrix} (M/D)^{-1} & -(M/D)^{-1}BD^{-1} \\ -D^{-1}C(M/D)^{-1} & D^{-1} + D^{-1}C(M/D)^{-1}BD^{-1} \end{bmatrix} \quad (\text{A.22})$$

Remark A.2 (Combining both formulations). *When both A and D are invertible, the Woodbury lemma proves that the two formulations give the same matrices for each block. In particular, they can be combined block by block at will.*

A.3 Woodbury Inversion Formula

Proposition A.3 (Inversion identity). *For $n \in \mathbb{N}$, let $P \in \mathcal{M}_{n,n}(\mathbb{R})$ such that $I + P$ is invertible. Then:*

$$(I + P)^{-1} = I - (I + P)^{-1}P \quad (\text{A.23})$$

Proof. This result is direct from:

$$I = (I + P)^{-1}(I + P) = (I + P)^{-1} + (I + P)^{-1}P \quad (\text{A.24})$$

$\square$

Proposition A.4 (Push-through identity [94]). *For $n, p \in \mathbb{N}$, let $P \in \mathcal{M}_{n,p}(\mathbb{R})$ and $Q \in \mathcal{M}_{p,n}(\mathbb{R})$ such that $I_n + PQ$ is invertible. Then $I_p + QP$ is also invertible and:*

$$(I_n + PQ)^{-1}P = P(I_p + QP)^{-1} \quad (\text{A.25})$$

Proof. We start by using the identity:

$$P + PQP = P(I_p + QP) = (I_n + PQ)P \quad (\text{A.26})$$

Thus:

$$\begin{aligned} P(I_p + QP) &= (I_n + PQ)P \\ \iff P &= (I_n + PQ)P(I_p + QP)^{-1} \\ \iff (I_n + PQ)^{-1}P &= P(I_p + QP)^{-1} \end{aligned} \quad (\text{A.27})$$

with $I_p + QP$ admitting a right inverse, hence being invertible. $\square$

Proposition A.5 (Generalised Woodbury formula [119]). *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$. We assume that A , D , and $D + CA^{-1}B$ are invertible. Then $A + BD^{-1}C$ is also invertible, and:*

$$(A + BD^{-1}C)^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \quad (\text{A.28})$$

Proof. We first factorise:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= (A(I + A^{-1}BD^{-1}C))^{-1} \\ &= (I + A^{-1}BD^{-1}C)^{-1}A^{-1} \end{aligned} \quad (\text{A.29})$$

Applying Proposition A.3 to $A^{-1}BD^{-1}C$ yields:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= \left(I - (I + A^{-1}BD^{-1}C)^{-1}A^{-1}BD^{-1}C \right) A^{-1} \\ &= A^{-1} - (I + A^{-1}BD^{-1}C)^{-1}A^{-1}BD^{-1}CA^{-1} \end{aligned} \quad (\text{A.30})$$

Using Proposition A.4 with $P = A^{-1}B$ and $Q = D^{-1}C$ further gives:

$$(I + A^{-1}BD^{-1}C)^{-1}A^{-1}B = A^{-1}B(I + D^{-1}CA^{-1}B)^{-1} \quad (\text{A.31})$$

Therefore:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= A^{-1} - A^{-1}B(I + D^{-1}CA^{-1}B)^{-1}D^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B(D(I + D^{-1}CA^{-1}B))^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \end{aligned} \quad (\text{A.32})$$

$\square$

APPENDIX B

DEFERRED PROOFS

Contents

B.1 Proof of Proposition 2.3	148
B.2 Proof of Proposition 2.10	149
B.3 Proof of Proposition 3.2	150
B.4 Proof of Proposition 3.3	151
B.5 Proof of Proposition 3.8	152
B.6 Proof of Proposition 3.9	153
B.7 Proof of Proposition 3.10	154
B.8 Proof of Proposition 3.11	156
B.9 Proof of Proposition 3.12	157
B.10 Proof of Proposition 3.14	159
B.11 Proof of Proposition 3.15	160
B.12 Proof of Proposition 3.16	161
B.13 Proof of Proposition 3.18	163
B.14 Proof of Proposition 4.3	163
B.15 Proof of Lemma 4.2	164
B.16 Proof of Proposition 4.7 and Corollary 4.2	165
B.17 Proof of Lemma 4.3	165

B.1 Proof of Proposition 2.3

Prediction. Let $\mathbb{V}[X]$ denote the variance of the random variable X :

$$\mathbb{V}[X] \triangleq \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top] \quad (\text{B.1})$$

Since $x_k = F_k x_{k-1} + b_k^{(e)} + w_k$ is an affine transformation of two independent Gaussian random variables $x_{k-1} | \mathbf{z}_{1:k-1} \sim \mathcal{N}(\hat{x}_{k-1|k-1}, P_{k-1|k-1})$ and $w_k \sim \mathcal{N}(0, Q_k)$, the predicted state $x_k | \mathbf{z}_{1:k-1}$ is also Gaussian, with:

$$\begin{aligned} \hat{x}_{k|k-1} &\triangleq \mathbb{E}[x_k | \mathbf{z}_{1:k-1}] \\ &= F_k \mathbb{E}[x_{k-1} | \mathbf{z}_{1:k-1}] + b_k^{(e)} + \mathbb{E}[w_k | \mathbf{z}_{1:k-1}] \\ &= F_k \hat{x}_{k-1|k-1} + b_k^{(e)} \end{aligned} \quad (\text{B.2})$$

and:

$$\begin{aligned} P_{k|k-1} &\triangleq \mathbb{V}[x_k | \mathbf{z}_{1:k-1}] \\ &= F_k \mathbb{V}[x_{k-1} | \mathbf{z}_{1:k-1}] F_k^\top + \mathbb{V}[w_k | \mathbf{z}_{1:k-1}] \\ &= F_k P_{k-1|k-1} F_k^\top + Q_k \end{aligned} \quad (\text{B.3})$$

The predicted covariance is thus SPD as the sum of an SPSD matrix and the SPD matrix Q_k .

Correction. The joint distribution of (x_k, z_k) given $\mathbf{z}_{1:k-1}$ is also Gaussian, since $z_k = H_k x_k + b_k^{(o)} + n_k$ is an affine transformation of the independent Gaussian variables $x_k | \mathbf{z}_{1:k-1} \sim \mathcal{N}(\hat{x}_{k|k-1}, P_{k|k-1})$ and $n_k \sim \mathcal{N}(0, R_k)$. It comes that:

$$\begin{bmatrix} x_k \\ z_k \end{bmatrix} \Big| \mathbf{z}_{1:k-1} \sim \mathcal{N} \left(\begin{bmatrix} \hat{x}_{k|k-1} \\ H_k \hat{x}_{k|k-1} + b_k^{(o)} \end{bmatrix}, \begin{bmatrix} P_{k|k-1} & P_{k|k-1} H_k^\top \\ H_k P_{k|k-1} & S_k \end{bmatrix} \right) \quad (\text{B.4})$$

where:

$$S_k \triangleq H_k P_{k|k-1} H_k^\top + R_k \quad (\text{B.5})$$

is called the innovation covariance, and is the covariance associated with the innovation vector:

$$y_k \triangleq z_k - \left(H_k \hat{x}_{k|k-1} + b_k^{(o)} \right) \quad (\text{B.6})$$

Since $R_k \succ 0$, the independent pair (x_k, n_k) has the covariance:

$$\begin{bmatrix} P_{k|k-1} & 0 \\ 0 & R_k \end{bmatrix} \succ 0 \quad (\text{B.7})$$

Its image under the invertible linear map $(x, n) \mapsto (x, H_k x + n)$ therefore has an SPD covariance as well. The Gaussian conditioning formula (see Proposition A.1) can thus be applied to give the posterior distribution $x_k | \mathbf{z}_{1:k}$ is Gaussian, with:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k y_k \quad (\text{B.8})$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (\text{B.9})$$

where:

$$K_k \triangleq P_{k|k-1} H_k^\top S_k^{-1} \quad (\text{B.10})$$

is the Kalman gain.

Finally, Proposition A.1 proves that the Schur complement of the SPD joint covariance:

$$P_{k|k-1} - (P_{k|k-1} H_k^\top) S_k^{-1} (H_k P_{k|k-1}) = P_{k|k-1} - K_k H_k P_{k|k-1} = P_{k|k} \quad (\text{B.11})$$

is itself SPD, so $P_{k|k} \succ 0$.

B.2 Proof of Proposition 2.10

As a linear combination of the centred noises w_k^l and w_k^n , $\bar{w}_k^l$ is centred. Its covariance develops as:

$$\begin{aligned} \bar{Q}_k^l &= \mathbb{E} [\bar{w}_k^l \bar{w}_k^{l\top}] \\ &= \mathbb{E} [(w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] \\ &= \mathbb{E} [w_k^l (w_k^l)^\top] - \mathbb{E} [w_k^l (Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] - \mathbb{E} [(Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^l)^\top] \\ &\quad + \mathbb{E} [(Q_k^{nl} (Q_k^n)^{-1} w_k^n) (Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] \\ &= \mathbb{E} [w_k^l (w_k^l)^\top] - \mathbb{E} [w_k^l (w_k^n)^\top] (Q_k^n)^{-\top} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^l)^\top] \\ &\quad + Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^n)^\top] (Q_k^n)^{-\top} (Q_k^{nl})^\top \\ &= Q_k^l - Q_k^{nl} (Q_k^n)^{-\top} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top + Q_k^{nl} (Q_k^n)^{-1} Q_k^n (Q_k^n)^{-\top} (Q_k^{nl})^\top \\ &= Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top + Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \\ &= Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \end{aligned} \quad (\text{B.12})$$

Finally, computing:

$$\begin{aligned} \mathbb{E} [\bar{w}_k^l (w_k^n)^\top] &= \mathbb{E} [(w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^n)^\top] \\ &= \mathbb{E} [w_k^l (w_k^n)^\top] - Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^n)^\top] \\ &= Q_k^{nl} - Q_k^{nl} (Q_k^n)^{-1} Q_k^n \\ &= 0 \end{aligned} \quad (\text{B.13})$$

shows that $\bar{w}_k^l$ and w_k^n are uncorrelated.

B.3 Proof of Proposition 3.2

Forward implication. Let $n \in \mathbb{N}^*$ and $\mathbf{x} \in \mathcal{U}^n$. It comes from the symmetry of k that:

$$\begin{aligned}
 k(\mathbf{x}, \mathbf{x}) &= \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \\
 &= \begin{bmatrix} k(x_1, x_1)^\top & \cdots & k(x_n, x_1)^\top \\ \vdots & \ddots & \vdots \\ k(x_1, x_n)^\top & \cdots & k(x_n, x_n)^\top \end{bmatrix} \\
 &= \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix}^\top \\
 &= k(\mathbf{x}, \mathbf{x})^\top
 \end{aligned} \tag{B.14}$$

Let $a \in \mathbb{R}^{nd_z}$. We decompose $a = [a_1^\top, \dots, a_n^\top]^\top$ with $a_i \in \mathbb{R}^{d_z}$, $i \in \llbracket 1, n \rrbracket$ to obtain:

$$\begin{aligned}
 a^\top k(\mathbf{x}, \mathbf{x}) a &= \begin{bmatrix} a_1^\top & \cdots & a_n^\top \end{bmatrix} \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\
 &= \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j \\
 &\geq 0
 \end{aligned} \tag{B.15}$$

Reverse implication. Let $x, x' \in \mathcal{U}$. By symmetry of $k((x, x'), (x, x'))$, it comes that:

$$\begin{aligned}
 \begin{bmatrix} k(x, x) & k(x, x') \\ k(x', x) & k(x', x') \end{bmatrix} &= \begin{bmatrix} k(x, x) & k(x, x') \\ k(x', x) & k(x', x') \end{bmatrix}^\top \\
 &= \begin{bmatrix} k(x, x)^\top & k(x', x)^\top \\ k(x, x')^\top & k(x', x')^\top \end{bmatrix}
 \end{aligned} \tag{B.16}$$

In particular, the upper-right block gives:

$$k(x, x') = k(x', x)^\top \tag{B.17}$$

Let $n \in \mathbb{N}^*$, $\mathbf{x} \in \mathcal{U}^n$, and $a_1, \dots, a_n \in \mathbb{R}^{d_z}$, and set $a \triangleq [a_1^\top, \dots, a_n^\top]^\top \in \mathbb{R}^{nd_z}$. Then:

$$\begin{aligned}
 \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j &= \begin{bmatrix} a_1^\top & \cdots & a_n^\top \end{bmatrix} \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\
 &= a^\top k(\mathbf{x}, \mathbf{x}) a \\
 &\geq 0
 \end{aligned} \tag{B.18}$$

Strict version. The same computations hold with strict inequalities, provided the inputs are pairwise distinct and the coefficients not all zero.

B.4 Proof of Proposition 3.3

Assume that $\nu \in \mathbb{R}_+^*$ and let k_ν denote such a function. We also let:

$$g : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} r^\nu K_\nu(r) \end{cases} \quad \text{and} \quad \tilde{r} : \begin{cases} \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}_+ \\ x, x' \mapsto \frac{\sqrt{2\nu}}{l} \|x - x'\| \end{cases} \quad (\text{B.19})$$

$\tilde{r}$ and K_ν are continuous on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$ and $\mathbb{R}_+^*$ respectively, and the latter admits the following asymptotic expansion as $r \rightarrow 0^+$:

$$K_\nu(r) = 2^{\nu-1} \Gamma(\nu) r^{-\nu} + o(r^{-\nu}) \quad (\text{B.20})$$

Therefore, f can be continuously extended to 0 by setting $f(0) = \sigma_f^2$, and k_ν is thus continuous on the entirety of $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$.

Let $x, x' \in \mathbb{R}^{d_x}$. By symmetry of the norm, $\tilde{r}(x, x') = \tilde{r}(x', x)$, and thus:

$$k_\nu = g \circ \tilde{r} \quad (\text{B.21})$$

is symmetric, with $\circ$ being the composition operator.

Furthermore, Bochner's theorem [120] states that a continuous integrable function with a strictly positive Fourier transform is SPD. It can be shown [121] that the Fourier transform of k_ν on $\mathbb{R}^{d_x}$ is:

$$S_\nu : \begin{cases} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ \omega \mapsto C \left(\frac{2\nu}{l^2} + \|\omega\|^2 \right)^{-(\nu+d_x/2)} \end{cases} \quad (\text{B.22})$$

where $C > 0$ is a normalising constant which only depends on d_x and the parameters of k_ν . Since $\nu > 0$, $l > 0$, it comes that:

$$\forall \omega \in \mathbb{R}^{d_x}, S_\nu(\omega) > 0 \quad (\text{B.23})$$

and thus k_ν is a SPD kernel. By definition, it is also a radial kernel.

The case $\nu = +\infty$ is more direct. The squared exponential kernel is continuous on the whole of $\mathbb{R}_+$ without any extension, and symmetric by the same composition argument. Its Fourier transform on $\mathbb{R}^{d_x}$ is:

$$S_\nu : \begin{cases} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ \omega \mapsto C \exp \left(-\frac{l^2 \|\omega\|^2}{2} \right) \end{cases} \quad (\text{B.24})$$

which is again strictly positive, so Bochner's theorem applies as above.

B.5 Proof of Proposition 3.8

Let:

$$\mathcal{L}(W, \xi) \triangleq \text{tr}(\mathbb{V}[W^\top \tilde{\mathbf{z}} - z]) + 2 \text{tr}(\xi^\top (C^\top W - I_{d_z})) \quad (\text{B.25})$$

be the Lagrangian associated with the optimisation problem (3.32) under the constraint (3.48), and $\xi \in \mathcal{M}_{d_z, d_z}(\mathbb{R})$ the corresponding Lagrange multiplier.

This problem is solved by finding the minimum of $\mathcal{L}$, *i.e.* finding a zero of its derivative. Writing G for $G(\tilde{\mathbf{x}})$, the same properties used in the proof of Proposition 3.5 give:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial W}(W, \xi) &= 0 \\ \iff 2G W - 2k(\tilde{\mathbf{x}}, x) + 2 \frac{\partial}{\partial W} (\text{tr}(\xi^\top (C^\top W - I_{d_z}))) &= 0 \\ \iff 2G W - 2k(\tilde{\mathbf{x}}, x) + 2C\xi &= 0 \\ \iff G W + C\xi &= k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (\text{B.26})$$

Together with the constraint $C^\top W = I_{d_z}$, this yields the block system:

$$\begin{bmatrix} G & C \\ C^\top & 0 \end{bmatrix} \begin{bmatrix} W \\ \xi \end{bmatrix} = \begin{bmatrix} k(\tilde{\mathbf{x}}, x) \\ I_{d_z} \end{bmatrix} \quad (\text{B.27})$$

The left matrix block can be inverted using the upper left Schur inversion formula from Proposition A.2. Its Schur complement,

$$S \triangleq 0 - C^\top G^{-1} C = -C^\top G^{-1} C, \quad (\text{B.28})$$

is invertible since $G \succ 0$ and C has full column rank. The convention of Proposition A.2 gives the four blocks of the inverse:

$$\begin{cases} \bar{A} = G^{-1} + G^{-1} C S^{-1} C^\top G^{-1} \\ \bar{B} = -G^{-1} C S^{-1} \\ \bar{C} = -S^{-1} C^\top G^{-1} \\ \bar{D} = S^{-1} \end{cases} \quad (\text{B.29})$$

Let $M \triangleq -S^{-1} = (C^\top G^{-1} C)^{-1}$ and $U \triangleq I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)$. The weight matrix can be derived from the upper blocks of the inverse:

$$\begin{aligned} W &= \bar{A} k(\tilde{\mathbf{x}}, x) + \bar{B} I_{d_z} \\ &= (G^{-1} + G^{-1} C S^{-1} C^\top G^{-1}) k(\tilde{\mathbf{x}}, x) - G^{-1} C S^{-1} \\ &= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C S^{-1} (C^\top G^{-1} k(\tilde{\mathbf{x}}, x) - I_{d_z}) \\ &= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C M (I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)) \\ &= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C M U \end{aligned} \quad (\text{B.30})$$

and ξ from the lower blocks:

$$\begin{aligned} \xi &= \bar{C} k(\tilde{\mathbf{x}}, x) + \bar{D} I_{d_z} \\ &= -S^{-1} C^\top G^{-1} k(\tilde{\mathbf{x}}, x) + S^{-1} \\ &= S^{-1} (I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)) \\ &= -M (I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)) \\ &= -M U \end{aligned} \quad (\text{B.31})$$

Since M is symmetric, the OK estimator is:

$$\begin{aligned}
 \hat{z}_{\text{OK}} &= W^\top \tilde{\mathbf{z}} \\
 &= k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + U^\top M C^\top G^{-1} \tilde{\mathbf{z}} \\
 &= k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + (I_{d_z} - k(x, \tilde{\mathbf{x}}) G^{-1} C) M C^\top G^{-1} \tilde{\mathbf{z}} \\
 &= M C^\top G^{-1} \tilde{\mathbf{z}} + k(x, \tilde{\mathbf{x}}) G^{-1} (\tilde{\mathbf{z}} - C M C^\top G^{-1} \tilde{\mathbf{z}})
 \end{aligned} \tag{B.32}$$

which is exactly (3.50), the estimated mean being $\hat{m}_{\text{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = M C^\top G^{-1} \tilde{\mathbf{z}}$.

Since R_{OK} is a centred moment of the estimation error, it is insensitive to the unknown mean and (3.36) can be applied. Using (B.26) and the constraint $W^\top C = I_{d_z}$ gives the identity:

$$\begin{aligned}
 W^\top G W &= W^\top (k(\tilde{\mathbf{x}}, x) - C \xi) \\
 &= W^\top k(\tilde{\mathbf{x}}, x) - W^\top C \xi \\
 &= W^\top k(\tilde{\mathbf{x}}, x) - \xi
 \end{aligned} \tag{B.33}$$

Combining these results yields:

$$\begin{aligned}
 R_{\text{OK}} &= W^\top G W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\
 &= W^\top k(\tilde{\mathbf{x}}, x) - \xi - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\
 &= -\xi - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\
 &= M U - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) G^{-1} C M U + k(x, x) + R \\
 &= (I_{d_z} - k(x, \tilde{\mathbf{x}}) G^{-1} C) M U + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) \\
 &= U^\top M U + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x)
 \end{aligned} \tag{B.34}$$

Finally, recognising $R_{\hat{m}_{\text{OK}}}(x, \tilde{\mathbf{x}}) = U^\top M U \succeq 0$, since $M \succ 0$, gives the desired expression.

B.6 Proof of Proposition 3.9

The unbiased assumption now reads:

$$g(\tilde{\mathbf{x}})^\top W - g(x)^\top = 0 \tag{B.35}$$

and the Lagrange multiplier method of Proposition 3.8 can be transposed to universal kriging under the substitutions:

$$\begin{cases} C \longrightarrow g(\tilde{\mathbf{x}}) \\ I_{d_z} \longrightarrow g(x)^\top \\ \xi \in \mathcal{M}_{r_{\text{UK}} d_z, d_z}(\mathbb{R}) \end{cases} \tag{B.36}$$

The full column rank of $g(\tilde{\mathbf{x}})$ ensures that S is invertible, and that M is SPD. Writing G for $G(\tilde{\mathbf{x}})$, the shorthands of that proof thus become:

$$\begin{cases} M \triangleq -S^{-1} = (g(\tilde{\mathbf{x}})^\top G^{-1} g(\tilde{\mathbf{x}}))^{-1} \\ U \triangleq g(x)^\top - g(\tilde{\mathbf{x}})^\top G^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \tag{B.37}$$

so that the weight matrix and Lagrange multiplier are:

$$\begin{cases} W = G^{-1}k(\tilde{\mathbf{x}}, x) + G^{-1}g(\tilde{\mathbf{x}}) M U \\ \xi = -M U \end{cases} \quad (\text{B.38})$$

Finally, the bilinear identity (B.33) becomes:

$$W^\top G W = W^\top k(\tilde{\mathbf{x}}, x) - g(x) \xi \quad (\text{B.39})$$

The UK estimator and covariance matrices are derived exactly like for Proposition 3.8, using the corrected results:

$$\begin{aligned} \hat{z}_{\text{UK}} &= k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + U^\top M g(\tilde{\mathbf{x}})^\top G^{-1} \tilde{\mathbf{z}} \\ &= g(x) M g(\tilde{\mathbf{x}})^\top G^{-1} \tilde{\mathbf{z}} + k(x, \tilde{\mathbf{x}}) G^{-1} (\tilde{\mathbf{z}} - g(\tilde{\mathbf{x}}) M g(\tilde{\mathbf{x}})^\top G^{-1} \tilde{\mathbf{z}}) \end{aligned} \quad (\text{B.40})$$

and:

$$R_{\text{UK}} = U^\top M U + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) \quad (\text{B.41})$$

B.7 Proof of Proposition 3.10

Using (3.54), it comes that:

$$f \mid \beta \sim \mathcal{GP}(g \beta, k) \quad (\text{B.42})$$

and thus:

$$f - g \beta \mid \beta \sim \mathcal{GP}(0, k) \quad (\text{B.43})$$

Since this law is free of β , $f - g \beta$ is independent of β , which gives:

$$f - g \beta \sim \mathcal{GP}(0, k) \quad (\text{B.44})$$

Let $x, x' \in \mathcal{U}$. The mean of f can thus be computed by decomposing:

$$\mathbb{E}[f(x)] = g(x) \mathbb{E}[\beta] + \mathbb{E}[f(x) - g(x) \beta] = 0 \quad (\text{B.45})$$

By independence of $f - g \beta$ and β , it follows that:

$$\begin{aligned} \mathbb{E}[f(x) f(x')^\top] &= \mathbb{E}[(f(x) - g(x) \beta)(f(x') - g(x') \beta)^\top] + g(x)^\top \mathbb{E}[\beta \beta^\top] g(x') + \\ &\quad \mathbb{E}[(f(x) - g(x) \beta) \beta^\top] g(x')^\top + g(x') \mathbb{E}[\beta (f(x) - g(x) \beta)^\top] \\ &= k(x, x') + g(x) B g(x')^\top + 0 + 0 \\ &= k_B(x, x') \end{aligned} \quad (\text{B.46})$$

As a linear combination of the jointly Gaussian distributions $f - g \beta$ and β , we finally have:

$$f \sim \mathcal{GP}(0, k_B) \quad (\text{B.47})$$

In this form, the mean function of f is known, and Proposition 3.7 states that simple kriging using k_B gives the conditional distribution of $z \mid \tilde{\mathbf{z}}$:

$$z \mid \tilde{\mathbf{z}} \sim \mathcal{N}(\hat{z}_{\text{SK},B}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0), R_{\text{SK},B}(x, \tilde{\mathbf{x}})) \quad (\text{B.48})$$

The SK estimator and error covariance matrix can be expressed as follows:

$$\begin{cases} \hat{z}_{\text{SK},B}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) = k_B(x, \tilde{\mathbf{x}}) G_B(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ R_{\text{SK},B}(x, \tilde{\mathbf{x}}) = k_B(x, x) + R - k_B(x, \tilde{\mathbf{x}}) G_B(\tilde{\mathbf{x}})^{-1} k_B(\tilde{\mathbf{x}}, x) \end{cases} \quad (\text{B.49})$$

where:

$$G_B \triangleq k_B(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R} = G + g(\tilde{\mathbf{x}}) B g(\tilde{\mathbf{x}})^\top \quad (\text{B.50})$$

For the sake of clarity, we let G and g respectively denote $G(\tilde{\mathbf{x}})$ and $g(\tilde{\mathbf{x}})$, and we reuse:

$$\begin{cases} M \triangleq (g^\top G^{-1} g)^{-1} \\ U \triangleq g(x)^\top - g^\top G^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (\text{B.51})$$

from section B.6.

Using Proposition A.5, it comes that:

$$G_B^{-1} = G^{-1} - G^{-1} g M_B g^\top G^{-1} \quad (\text{B.52})$$

with:

$$M_B \triangleq (B^{-1} + g^\top G^{-1} g)^{-1} = (B^{-1} + M^{-1})^{-1} \quad (\text{B.53})$$

By definition of M_B , we have:

$$\begin{aligned} (B^{-1} + M^{-1}) M_B &= I \\ \iff M^{-1} M_B &= I - B^{-1} M_B \\ \iff B M^{-1} M_B &= B - M_B \end{aligned} \quad (\text{B.54})$$

We also precompute:

$$\begin{aligned} g(x) B g^\top G_B^{-1} &= g(x) B g^\top (G^{-1} - G^{-1} g M_B g^\top G^{-1}) \\ &= g(x) (B g^\top G^{-1} - B g^\top G^{-1} g M_B g^\top G^{-1}) \\ &= g(x) (B g^\top G^{-1} - B M^{-1} M_B g^\top G^{-1}) \\ &= g(x) (B g^\top G^{-1} - (B - M_B) g^\top G^{-1}) \\ &= g(x) M_B g^\top G^{-1} \end{aligned} \quad (\text{B.55})$$

It comes that:

$$\begin{aligned} k_B(x, \tilde{\mathbf{x}}) G_B^{-1} &= (k(x, \tilde{\mathbf{x}}) + g(x) B g^\top) G_B^{-1} \\ &= k(x, \tilde{\mathbf{x}}) G_B^{-1} + g(x) B g^\top G_B^{-1} \\ &= k(x, \tilde{\mathbf{x}}) (G^{-1} - G^{-1} g M_B g^\top G^{-1}) + g(x) M_B g^\top G^{-1} \\ &= k(x, \tilde{\mathbf{x}}) G^{-1} + (-k(x, \tilde{\mathbf{x}}) G^{-1} g + g(x)) M_B g^\top G^{-1} \\ &= k(x, \tilde{\mathbf{x}}) G^{-1} + U^\top M_B g^\top G^{-1} \end{aligned} \quad (\text{B.56})$$

and thus:

$$\hat{z}_{\text{SK},B} = k_B(x, \tilde{\mathbf{x}}) G_B^{-1} \tilde{\mathbf{z}} = k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + U^\top M_B g^\top G^{-1} \tilde{\mathbf{z}} \quad (\text{B.57})$$

For the covariance, we compute:

$$\begin{aligned}
k_B(x, \tilde{\mathbf{x}}) G_B^{-1} k_B(\tilde{\mathbf{x}}, x) &= (k(x, \tilde{\mathbf{x}}) G^{-1} + U^\top M_B g^\top G^{-1}) (k(\tilde{\mathbf{x}}, x) + g B g(x)^\top) \\
&= k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M_B g^\top G^{-1} k(\tilde{\mathbf{x}}, x) + \\
&\quad k(x, \tilde{\mathbf{x}}) G^{-1} g B g(x)^\top + U^\top M_B g^\top G^{-1} g B g(x)^\top \\
&= k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M_B g^\top G^{-1} k(\tilde{\mathbf{x}}, x) + \\
&\quad k(x, \tilde{\mathbf{x}}) G^{-1} g B g(x)^\top + U^\top M_B M^{-1} B g(x)^\top \\
&= k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M_B g^\top G^{-1} k(\tilde{\mathbf{x}}, x) + \\
&\quad k(x, \tilde{\mathbf{x}}) G^{-1} g B g(x)^\top + U^\top (B - M_B) g(x)^\top
\end{aligned} \tag{B.58}$$

and:

$$g(x) B g(x)^\top - k(x, \tilde{\mathbf{x}}) G^{-1} g B g(x)^\top = U^\top g B g(x)^\top \tag{B.59}$$

to obtain:

$$\begin{aligned}
R_{\text{SK},B} &= k_B(x, x) + R - k_B(x, \tilde{\mathbf{x}}) G_B(\tilde{\mathbf{x}})^{-1} k_B(\tilde{\mathbf{x}}, x) \\
&= k(x, x) + g(x) B g(x)^\top + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) - U^\top M_B g^\top G^{-1} k(\tilde{\mathbf{x}}, x) - \\
&\quad k(x, \tilde{\mathbf{x}}) G^{-1} g B g(x)^\top - U^\top (B - M_B) g(x)^\top \\
&= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) - U^\top M_B g^\top G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M_B g(x)^\top \\
&= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M_B U
\end{aligned} \tag{B.60}$$

Since $M_B \rightarrow M$ as $B^{-1} \rightarrow 0$, we finally have:

$$\begin{cases} \hat{z}_{\text{SK},B}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \xrightarrow{B^{-1} \rightarrow 0} k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + U^\top M g^\top G^{-1} \tilde{\mathbf{z}} \\ R_{\text{SK},B}(x, \tilde{\mathbf{x}}) \xrightarrow{B^{-1} \rightarrow 0} k(x, x) + R - k(x, \tilde{\mathbf{x}}) G^{-1} k(\tilde{\mathbf{x}}, x) + U^\top M U \end{cases} \tag{B.61}$$

the limit of which are exactly (B.40) and (B.41).

B.8 Proof of Proposition 3.11

As $\mathcal{U}$ is compact and k continuous, k is bounded on $\mathcal{U} \times \mathcal{U}$, and $\mathcal{K}$ is compact.

Let $\phi, \phi' \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. Cauchy-Schwarz inequality ensures that there exists a constant $c \in \mathbb{R}_+$ such that:

$$\iint_{\mathcal{U} \times \mathcal{U}} |\phi(y)^\top k(x, y)^\top \phi'(x)| \, dx \, dy \leq c \sqrt{\langle \phi | \phi \rangle} \sqrt{\langle \phi' | \phi' \rangle} < +\infty \tag{B.62}$$

Fubini's integration theorem [122] can thus be used to give:

$$\begin{aligned}
\langle \mathcal{K}\phi | \phi' \rangle &= \int_{\mathcal{U}} (\mathcal{K}\phi)(x)^\top \phi'(x) dx \\
&= \int_{\mathcal{U}} \left(\int_{\mathcal{U}} k(x, y) \phi(y) dy \right)^\top \phi'(x) dx \\
&= \int_{\mathcal{U}} \int_{\mathcal{U}} \phi(y)^\top k(x, y)^\top \phi'(x) dy dx \\
&= \int_{\mathcal{U}} \int_{\mathcal{U}} \phi(y)^\top k(y, x) \phi'(x) dx dy \\
&= \int_{\mathcal{U}} \phi(y)^\top \left(\int_{\mathcal{U}} k(y, x) \phi'(x) dx \right) dy \\
&= \int_{\mathcal{U}} \phi(y)^\top (\mathcal{K}\phi')(y) dy \\
&= \langle \phi | \mathcal{K}\phi' \rangle
\end{aligned} \tag{B.63}$$

Let $\phi : \mathcal{U} \rightarrow \mathbb{R}^{d_z}$ be a continuous function, and let $n \in \mathbb{N}^*$. We define $\left(\mathcal{U}_i^{(n)}\right)_{i \in \llbracket 1, n \rrbracket}$ a partition of $\mathcal{U}$ such that the diameter of each $\mathcal{U}_i^{(n)}$, $i \in \llbracket 1, n \rrbracket$ tends to 0 when $n \rightarrow +\infty$. Here, the diameter of a subset of the Euclidean space $\mathbb{R}^{d_x}$ is defined as the maximum distance between two points of this subset. For $i \in \llbracket 1, n \rrbracket$, we define $x_i^{(n)} \in \mathcal{U}_i^{(n)}$, and we let:

$$a_i^{(n)} \triangleq \left(\int_{\mathcal{U}_i^{(n)}} dx \right) \phi(x_i^{(n)}) \tag{B.64}$$

By continuity of ϕ , the function:

$$(x, y) \mapsto \phi(x)^\top k(x, y) \phi(y) \tag{B.65}$$

is continuous on the compact $\mathcal{U} \times \mathcal{U}$, so its Riemann sums converge:

$$\sum_{i=1}^n \sum_{j=1}^n \left(a_i^{(n)} \right)^\top k(x_i^{(n)}, x_j^{(n)}) a_j^{(n)} \xrightarrow{n \rightarrow +\infty} \iint_{\mathcal{U} \times \mathcal{U}} \phi(x)^\top k(x, y) \phi(y) dx dy = \langle \phi | \mathcal{K}\phi \rangle \tag{B.66}$$

The SPSPDness of k ensures that each Riemann sum is non-negative. By passing to the limit, it follows that:

$$\langle \phi | \mathcal{K}\phi \rangle \geq 0 \tag{B.67}$$

for every continuous function ϕ .

Finally, recall that the set of continuous functions of $\mathcal{U}$ is dense in $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. By continuity of:

$$\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \mapsto \langle \phi | \mathcal{K}\phi \rangle \tag{B.68}$$

the inequality (B.67) extends to all $\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$.

B.9 Proof of Proposition 3.12

By Proposition 3.11, $\mathcal{K}$ is a compact and self-adjoint operator on $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. The spectral theorem [123] thus states that there exists an orthonormal basis $(\psi_i)_{i \in \mathbb{N}^*}$

of $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ consisting of eigenfunctions of $\mathcal{K}$, with each corresponding eigenvalue $(\lambda_i)_{i \in \mathbb{N}^*}$ being real. Up to reordering, the sequence of eigenvalues can be considered to be non-increasing.

Since k is uniformly continuous on the compact $\mathcal{U} \times \mathcal{U}$, $\mathcal{K}\phi$ is continuous for every $\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. In particular, for each $i \in \mathbb{N}^*$ such that $\lambda_i \neq 0$, the eigenfunction:

$$\psi_i = \lambda_i^{-1} \mathcal{K}\psi_i \quad (\text{B.69})$$

is continuous. Without loss of generality, it must be noted that the ψ_i associated with a null eigenvalue do not contribute to (3.70) and may thus be dismissed.

Since $(\psi_i)_{i \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$, the family $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{F}$. Let $\langle \cdot | \cdot \rangle_{\mathcal{U} \times \mathcal{U}}$ denote:

$$(\phi, \phi') \in \mathcal{F}^2 \mapsto \langle \phi | \phi' \rangle_{\mathcal{U} \times \mathcal{U}} \triangleq \iint_{\mathcal{U} \times \mathcal{U}} \text{tr}(\phi(x, y)^\top \phi'(x, y)) \, dx \, dy \quad (\text{B.70})$$

Let $i, j \in \mathbb{N}^*$. We have:

$$\begin{aligned} \langle k | \psi_i \psi_j^\top \rangle_{\mathcal{U} \times \mathcal{U}} &= \iint_{\mathcal{U} \times \mathcal{U}} \text{tr}(k(x, y)^\top \psi_i(x) \psi_j(y)^\top) \, dx \, dy \\ &= \iint_{\mathcal{U} \times \mathcal{U}} \psi_j(y)^\top k(x, y)^\top \psi_i(x) \, dx \, dy \\ &= \int_{\mathcal{U}} \psi_j(y)^\top \left(\int_{\mathcal{U}} k(y, x) \psi_i(x) \, dx \right) dy \\ &= \int_{\mathcal{U}} \psi_j(y)^\top \mathcal{K} \psi_i(y) \, dy \\ &= \lambda_i \int_{\mathcal{U}} \psi_j(y)^\top \psi_i(y) \, dy \\ &= \lambda_i \delta_{i,j} \end{aligned} \quad (\text{B.71})$$

where:

$$\delta_{i,j} \triangleq \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (\text{B.72})$$

is the Kronecker symbol.

Because $k \in \mathcal{F}$, it follows from the completeness of $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ that:

$$k_{r_{\text{KL}}}(x, x') \triangleq \sum_{i=1}^{r_{\text{KL}}} \lambda_i \psi_i(x) \psi_i(x')^\top \quad (\text{B.73})$$

converges to k in the $\mathcal{F}$ sense.

It must be noted that $\mathcal{L}^2$ convergence only identifies k up to a Lebesgue-null subset of $\mathcal{U} \times \mathcal{U}$, and gives no control at any specific point of the domain. In particular, it is not enough to justify substituting $k_{r_{\text{KL}}}$ for k in kriging scenarios. However, since k is continuous on $\mathcal{U} \times \mathcal{U}$, the convergence can be extended to absolute and uniform convergence on the entire domain [80]. The argument applies Dini's theorem [124] to the non-decreasing sequence $(x \mapsto \text{tr} k_{r_{\text{KL}}}(x, x))_{r_{\text{KL}} \in \mathbb{N}^*}$, whose limit is continuous on the compact $\mathcal{U}$, and extends the result off the diagonal through the Cauchy-Schwarz inequality.

Since $\mathcal{K}$ is SPSP, every eigenvalue is non-negative. Assume that there exists a rank $r_0 \in \mathbb{N}^*$ such that $\lambda_i = 0$ as soon as $i > r_0$. Then the decomposition (3.70) would reduce to the finite sum:

$$k = \sum_{i=1}^{r_0} \lambda_i \psi_i \psi_i^\top \quad (\text{B.74})$$

whose Gram matrices have rank at most r_0 and are thus singular at any n distinct points with $nd_z > r_0$, contradicting the SPDness of k by Proposition 3.2. Consequently:

$$\lambda_1 \geq \lambda_2 \geq \dots > 0 \quad (\text{B.75})$$

B.10 Proof of Proposition 3.14

As (3.80) is a rank r_{KL} perturbation of $A \triangleq I_{\tilde{N}} \otimes \tilde{R}$, the Woodbury inversion formula (see Proposition A.5) can be used to reduce the $\tilde{N}d_z \times \tilde{N}d_z$ inversion of $G(\tilde{\mathbf{x}})$ to an $r_{\text{KL}} \times r_{\text{KL}}$ one:

$$G(\tilde{\mathbf{x}})^{-1} \approx A^{-1} - A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \quad (\text{B.76})$$

Computing the centred SK estimator from Proposition 3.5 using this approximation and the push-through identity (see Proposition A.4) yields:

$$\begin{aligned} \hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) &= k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ &\approx \Psi(x) \Lambda \Psi(\tilde{\mathbf{x}})^\top \left(I - A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top \right) A^{-1} \tilde{\mathbf{z}} \\ &= \Psi(x) \Lambda \left(I - \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \right) \Psi(\tilde{\mathbf{x}})^\top A^{-1} \tilde{\mathbf{z}} \end{aligned} \quad (\text{B.77})$$

Recall that, by definition of $\tilde{G}(\tilde{\mathbf{x}})$:

$$\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} = \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \quad (\text{B.78})$$

which gives:

$$\begin{aligned} I - \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} &= I - \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} \right) \tilde{G}(\tilde{\mathbf{x}})^{-1} \\ &= I - I + \Lambda^{-1} \tilde{G}(\tilde{\mathbf{x}})^{-1} \\ &= \Lambda^{-1} \tilde{G}(\tilde{\mathbf{x}})^{-1} \end{aligned} \quad (\text{B.79})$$

Plugging this result into (B.77) and using $A^{-1} = I_{\tilde{N}} \otimes \tilde{R}^{-1}$ finally gives:

$$\hat{z}_{\text{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \tilde{\mathbf{z}} \quad (\text{B.80})$$

For R_{SK} , reusing:

$$k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \quad (\text{B.81})$$

together with (B.79) yields:

$$\begin{aligned} R_{\text{SK}} &= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &\approx \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \times \Psi(\tilde{\mathbf{x}}) \Lambda \Psi(x)^\top \\ &= \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} \right) \Lambda \Psi(x)^\top \\ &= \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \left(\Lambda - \tilde{G}(\tilde{\mathbf{x}})^{-1} \right) \Psi(x)^\top \\ &= R + \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(x)^\top \end{aligned} \quad (\text{B.82})$$

Finally, when $\tilde{R} = \sigma^2 I_{d_z}$ with $\sigma > 0$, substituting $A^{-1} = \sigma^{-2} I_{\tilde{N}d_z}$ yields the announced simplifications. For such $\sigma > 0$, the matrix:

$$\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \succ 0 \quad (\text{B.83})$$

is SPD, as the sum of the SPSD matrix $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}})$ and the SPD matrix $\sigma^2 \Lambda^{-1}$. The map:

$$\sigma \in \mathbb{R}_+^* \mapsto (\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1})^{-1} \quad (\text{B.84})$$

is thus well-defined and continuous, and extends to $\sigma = 0$ when $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) \succ 0$, *i.e.* when $\Psi(\tilde{\mathbf{x}})$ has full column rank. As $\hat{z}_{\text{SK}}$ and R_{SK} depend on σ only through this inverse and an explicit factor σ^2 , both expressions remain valid at $\sigma = 0$ under this condition.

B.11 Proof of Proposition 3.15

Let $i \in \mathbb{N}^*$. Since ψ_i is bounded on the compact $\mathcal{U}$, Tonelli's theorem [122] and the continuity of k give:

$$\mathbb{E} \left[\int_{\mathcal{U}} \|f(x) - m(x)\|^2 dx \right] = \int_{\mathcal{U}} \text{tr } k(x, x) dx < \infty \quad (\text{B.85})$$

so that $f - m \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ a.s.. Each $\langle f - m | \psi_i \rangle$ is thus well-defined and finite. This integrability also justifies the exchanges of expectation and integration below.

Since any finite linear combination of $(\xi_i)_{i \in \mathbb{N}^*}$ is itself a linear combination of the values of f , they are jointly Gaussian. Let $j \in \mathbb{N}^*$. It comes that:

$$\mathbb{E}[\xi_i] = \frac{1}{\sqrt{\lambda_i}} \int_{\mathcal{U}} \mathbb{E}[f(x) - m(x)]^\top \psi_i(x) dx = 0 \quad (\text{B.86})$$

and:

$$\begin{aligned} \mathbb{E}[\xi_i \xi_j] &= \frac{1}{\sqrt{\lambda_i \lambda_j}} \iint_{\mathcal{U} \times \mathcal{U}} \psi_i(x)^\top \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))^\top] \psi_j(x') dx dx' \\ &= \frac{1}{\sqrt{\lambda_i \lambda_j}} \iint_{\mathcal{U} \times \mathcal{U}} \psi_i(x)^\top k(x, x') \psi_j(x') dx dx' \\ &= \frac{1}{\sqrt{\lambda_i \lambda_j}} \int_{\mathcal{U}} \psi_i(x)^\top \left(\int_{\mathcal{U}} k(x, x') \psi_j(x') dx' \right) dx \\ &= \frac{\lambda_j}{\sqrt{\lambda_i \lambda_j}} \int_{\mathcal{U}} \psi_i(x)^\top \psi_j(x) dx \\ &= \frac{\lambda_j}{\sqrt{\lambda_i \lambda_j}} \langle \psi_i | \psi_j \rangle \\ &= \delta_{i,j} \end{aligned} \quad (\text{B.87})$$

using the fact that ψ_i is an eigenfunction of the Mercer operator $\mathcal{K}$. Together with the jointly Gaussian nature of $(\xi_i)_{i \in \mathbb{N}^*}$, this proves that they are i.i.d standard Gaussian variables.

Finally, for $r \in \mathbb{N}^*$, let:

$$f_r : \begin{cases} \mathcal{U} \rightarrow \mathbb{R}^{d_z} \\ x \mapsto m(x) + \sum_{i=1}^r \sqrt{\lambda_i} \xi_i \psi_i(x) \end{cases} \quad (\text{B.88})$$

Fubini's theorem can be used to compute the following cross-covariance:

$$\begin{aligned} \mathbb{E}[(f(x) - m(x)) \xi_i] &= \frac{1}{\sqrt{\lambda_i}} \mathbb{E}[(f(x) - m(x)) \langle f - m | \psi_i \rangle] \\ &= \frac{1}{\sqrt{\lambda_i}} \mathbb{E} \left[(f(x) - m(x)) \int_{\mathcal{U}} (f(y) - m(y))^{\top} \psi_i(y) dy \right] \\ &= \frac{1}{\sqrt{\lambda_i}} \int_{\mathcal{U}} \mathbb{E}[(f(x) - m(x))(f(y) - m(y))^{\top}] \psi_i(y) dy \quad (\text{B.89}) \\ &= \frac{1}{\sqrt{\lambda_i}} \int_{\mathcal{U}} k(x, y) \psi_i(y) dy \\ &= \sqrt{\lambda_i} \psi_i(x) \end{aligned}$$

which can be used to obtain:

$$\begin{aligned} \mathbb{E}[\|f(x) - f_r(x)\|^2] &= \mathbb{E}[\|f(x) - m(x)\|^2] - 2 \sum_{i=1}^r \sqrt{\lambda_i} \psi_i(x)^{\top} \mathbb{E}[(f(x) - m(x)) \xi_i] \\ &\quad + \sum_{i=1}^r \sum_{j=1}^r \sqrt{\lambda_i \lambda_j} \mathbb{E}[\xi_i \xi_j] \psi_i(x)^{\top} \psi_j(x) \\ &= \text{tr } k(x, x) - 2 \sum_{i=1}^r \lambda_i \text{tr}(\psi_i(x) \psi_i(x)^{\top}) + \sum_{i=1}^r \lambda_i \text{tr}(\psi_i(x) \psi_i(x)^{\top}) \\ &= \text{tr } k(x, x) - \sum_{i=1}^r \lambda_i \text{tr}(\psi_i(x) \psi_i(x)^{\top}) \\ &= \sum_{i=r+1}^{+\infty} \lambda_i \text{tr}(\psi_i(x) \psi_i(x)^{\top}) \quad (\text{B.90}) \end{aligned}$$

By Mercer's theorem, this last sum is the tail of:

$$\text{tr } k(x, x) = \sum_{i=1}^{+\infty} \lambda_i \text{tr}(\psi_i(x) \psi_i(x)^{\top}) \quad (\text{B.91})$$

and therefore converges to 0. The partial sums thus converge to f in mean square, uniformly over $\mathcal{U}$.

B.12 Proof of Proposition 3.16

For $r \in \mathbb{N}^*$, let:

$$f_r : \begin{cases} \mathcal{U} \rightarrow \mathbb{R}^{d_z} \\ x \mapsto m(x) + \sum_{i=1}^r \sqrt{\lambda_i} \xi_i \psi_i(x) \end{cases} \quad (\text{B.92})$$

be the partial sum of the series.

Let $r' \in \mathbb{N}^*$ such that $r' > r$ and $x \in \mathcal{U}$. Since the $(\xi_i)_{i \in \mathbb{N}^*}$ are i.i.d standard Gaussian variables, it follows that:

$$\begin{aligned} \mathbb{E} [\|f_r(x) - f_{r'}(x)\|^2] &= \sum_{i=r+1}^{r'} \lambda_i \|\psi_i(x)\|^2 \\ &= \sum_{i=r+1}^{r'} \lambda_i \operatorname{tr} (\psi_i(x) \psi_i(x)^\top) \end{aligned} \quad (\text{B.93})$$

As for Proposition 3.15, this sum is the tail of the convergent series:

$$\operatorname{tr} k(x, x) = \sum_{i=1}^{+\infty} \lambda_i \operatorname{tr} (\psi_i(x) \psi_i(x)^\top) \quad (\text{B.94})$$

and tends to 0. The sequence $(f_r(x))_{r \in \mathbb{N}^*}$ is thus a Cauchy sequence in the space of square-integrable $\mathbb{R}^{d_z}$ -valued random variables, which is complete. The sequence thus converges, and $f(x)$ is well-defined as its limit.

Let $n \in \mathbb{N}^*$, $x_1, \dots, x_n \in \mathcal{U}$ and $a_1, \dots, a_n \in \mathbb{R}^{d_z}$. For every $r \in \mathbb{N}^*$, the sum:

$$\sum_{i=1}^n a_i^\top f_r(x_i) \quad (\text{B.95})$$

is affine in $\xi_1, \dots, \xi_r$, and is thus Gaussian. Passing to the limit, this proves that:

$$\sum_{i=1}^n a_i^\top f(x_i) \quad (\text{B.96})$$

is also Gaussian, and so f is a GP by Definition 3.1.

Since mean-square convergence carries the first two moments to the limit, the mean of f is:

$$\begin{aligned} \mathbb{E}[f(x)] &= \lim_{r \rightarrow +\infty} \mathbb{E}[f_r(x)] \\ &= m(x) + \lim_{r \rightarrow +\infty} \sum_{i=1}^r \sqrt{\lambda_i} \mathbb{E}[\xi_i] \psi_i(x) \\ &= m(x) \end{aligned} \quad (\text{B.97})$$

and its covariance is:

$$\begin{aligned} \mathbb{E} [(f(x) - m(x))(f(x') - m(x'))^\top] &= \lim_{r \rightarrow +\infty} \sum_{i=1}^r \sum_{j=1}^r \sqrt{\lambda_i \lambda_j} \mathbb{E}[\xi_i \xi_j] \psi_i(x) \psi_j(x')^\top \\ &= \lim_{r \rightarrow +\infty} \sum_{i=1}^r \sum_{j=1}^r \sqrt{\lambda_i \lambda_j} \delta_{i,j} \psi_i(x) \psi_j(x')^\top \\ &= \sum_{i=1}^{+\infty} \lambda_i \psi_i(x) \psi_i(x')^\top \\ &= k(x, x') \end{aligned} \quad (\text{B.98})$$

B.13 Proof of Proposition 3.18

TODO: à relire

The form is well defined. Let $\mathcal{H}_0$ denote the span of $\{k(x, \cdot), x \in \mathcal{U}\}$, and extend $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ to it by bilinearity. For $\psi = \sum_{i=1}^n a_i k(x_i, \cdot)$ and $\varphi = \sum_{j=1}^p b_j k(x'_j, \cdot)$ in $\mathcal{H}_0$, it comes that:

$$\langle \psi | \varphi \rangle_{\mathcal{H}_k} = \sum_{i=1}^n \sum_{j=1}^p a_i b_j k(x_i, x'_j) = \sum_{j=1}^p b_j \psi(x'_j) = \sum_{i=1}^n a_i \varphi(x_i) \quad (\text{B.99})$$

The last two expressions involve one of the two functions through its values alone, so neither representation matters and the form is well defined. It is symmetric and bilinear by construction, and positive since k is SPSPD. Taking $\varphi = k(x, \cdot)$ gives the reproducing property on $\mathcal{H}_0$:

$$\forall \psi \in \mathcal{H}_0, \forall x \in \mathcal{U}, \langle \psi | k(x, \cdot) \rangle_{\mathcal{H}_k} = \psi(x) \quad (\text{B.100})$$

It is definite. The Cauchy-Schwarz inequality holds for any positive semi-definite bilinear form, so (B.100) bounds every evaluation:

$$\forall \psi \in \mathcal{H}_0, \forall x \in \mathcal{U}, |\psi(x)|^2 \leq \langle \psi | \psi \rangle_{\mathcal{H}_k} k(x, x) \quad (\text{B.101})$$

A function of null norm thus vanishes everywhere, and $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ is an inner product on $\mathcal{H}_0$.

The completion is a space of functions. Equation (B.101) makes every Cauchy sequence of $\mathcal{H}_0$ pointwise Cauchy, hence pointwise convergent, which sends each class of the abstract completion to a function on $\mathcal{U}$. That map is injective. Indeed, let $(\psi_n)_n$ be a Cauchy sequence converging pointwise to 0, let $\varepsilon > 0$, and let N be such that $\|\psi_n - \psi_m\|_{\mathcal{H}_k} \leq \varepsilon$ for $n, m \geq N$. Fixing $n \geq N$ and writing $\psi_n = \sum_{i=1}^{n'} a_i k(x_i, \cdot)$, (B.99) gives:

$$\|\psi_n\|_{\mathcal{H}_k}^2 = \langle \psi_n | \psi_n - \psi_m \rangle_{\mathcal{H}_k} + \sum_{i=1}^{n'} a_i \psi_m(x_i) \quad (\text{B.102})$$

whose first term is at most $\varepsilon \sup_n \|\psi_n\|_{\mathcal{H}_k}$, finite since a Cauchy sequence is bounded, and whose second is a finite sum tending to 0 with m . Hence $\|\psi_n\|_{\mathcal{H}_k} \rightarrow 0$, and only the zero class is sent to the zero function. Passing (B.100) to the limit, $\mathcal{H}_k$ is an RKHS of reproducing kernel k .

Uniqueness. Let $\mathcal{H}$ be an RKHS of reproducing kernel k . It contains every $k(x, \cdot)$, hence $\mathcal{H}_0$, and its inner product agrees with $\langle \cdot | \cdot \rangle_{\mathcal{H}_k}$ on the generators, hence on $\mathcal{H}_0$ by bilinearity. A function of $\mathcal{H}$ orthogonal to $\mathcal{H}_0$ verifies $\psi(x) = \langle \psi | k(x, \cdot) \rangle_{\mathcal{H}} = 0$ for every $x \in \mathcal{U}$, so $\mathcal{H}_0$ is dense in $\mathcal{H}$, which is therefore its completion $\mathcal{H}_k$.

B.14 Proof of Proposition 4.3

The weights of (4.67) and the Lagrange multipliers ξ of the unbiasedness constraints solve $\mathcal{G}(\tilde{\mathbf{x}}) \begin{bmatrix} W \\ \xi \end{bmatrix} = \bar{k}(\tilde{\mathbf{x}}, x)$, whose first block row reads

$G(\tilde{\mathbf{x}})W + g(\tilde{\mathbf{x}})\xi = k(\tilde{\mathbf{x}}, x)$. The error $e_{\tilde{\mathbf{x}}} \triangleq z - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)$ being centred whatever the value of β , and ε being independent of $\tilde{\mathbf{z}}$, it comes that:

$$\mathbb{E} [\tilde{\mathbf{z}} e_{\tilde{\mathbf{x}}}^\top] = k(\tilde{\mathbf{x}}, x) - G(\tilde{\mathbf{x}})W = g(\tilde{\mathbf{x}})\xi$$

so that $e_{\tilde{\mathbf{x}}}$ is uncorrelated with every linear combination $V^\top \tilde{\mathbf{z}}$ such that $V^\top g(\tilde{\mathbf{x}}) = 0$. The difference between the two interpolators can be decomposed as follows:

$$\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) = \underbrace{(z - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g))}_{e_{\tilde{\mathbf{x}}^r} \triangleq} - \underbrace{(z - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g))}_{e_{\tilde{\mathbf{x}}} \triangleq}$$

Thus:

$$\begin{aligned} E_r(x) &= \mathbb{E} \left[(e_{\tilde{\mathbf{x}}^r} - e_{\tilde{\mathbf{x}}}) (e_{\tilde{\mathbf{x}}^r} - e_{\tilde{\mathbf{x}}})^\top \right] \\ &= \mathbb{E} [e_{\tilde{\mathbf{x}}^r} e_{\tilde{\mathbf{x}}^r}^\top] + \mathbb{E} [e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}}^\top] - \mathbb{E} [e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}^r}^\top] - \mathbb{E} [e_{\tilde{\mathbf{x}}^r} e_{\tilde{\mathbf{x}}}^\top] \end{aligned}$$

The difference of the two interpolators being such a combination, $\mathbb{E} [e_{\tilde{\mathbf{x}}} (e_{\tilde{\mathbf{x}}^r} - e_{\tilde{\mathbf{x}}})^\top] = 0$, hence $\mathbb{E} [e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}^r}^\top] = \mathbb{E} [e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}}^\top]$, and so does its transpose. Moreover, both interpolators being unbiased, the second moments of their errors are their covariances. Therefore:

$$E_r(x) = \mathbb{E} [e_{\tilde{\mathbf{x}}^r} e_{\tilde{\mathbf{x}}^r}^\top] - \mathbb{E} [e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}}^\top] = R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)$$

Using Proposition 4.2 yields:

$$\begin{aligned} E_r(x) &= k(x, x) + R - \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger \bar{k}(\tilde{\mathbf{x}}, x) - k(x, x) - R + \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{k}(\tilde{\mathbf{x}}, x) \\ &= \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{D}_G \bar{k}(\tilde{\mathbf{x}}, x) \end{aligned}$$

For $e_f(x)$, direct calculations give:

$$\begin{aligned} e_f(x) &= \|\hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\text{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)\|_2 \\ &= \|\bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} - \bar{k}(x, \tilde{\mathbf{x}}) \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger \tilde{\mathbf{z}}\|_2 \\ &= \|\bar{k}(x, \tilde{\mathbf{x}}) \mathcal{D}_G \tilde{\mathbf{z}}\|_2 \end{aligned}$$

B.15 Proof of Proposition 4.6

Writing $G_r \triangleq G(\tilde{\mathbf{x}}^r)$, $g_r \triangleq g(\tilde{\mathbf{x}}^r)$ and $M_r \triangleq M(\tilde{\mathbf{x}}^r)$ for brevity, the block inversion of $A = \mathcal{G}(\tilde{\mathbf{x}}^r)$ gives:

$$\mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} = \begin{bmatrix} G_r^{-1} - G_r^{-1} g_r M_r g_r^\top G_r^{-1} & G_r^{-1} g_r M_r \\ M_r g_r^\top G_r^{-1} & -M_r \end{bmatrix}$$

Consequently, for $x' \in \mathcal{U}$ and for any matrices y and t with respectively $\tilde{N}^r d_z$ and $r_{\text{UK}} d_z$ rows:

$$\bar{k}(x', \tilde{\mathbf{x}}^r) \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} \begin{bmatrix} y \\ t \end{bmatrix} = k(x', \tilde{\mathbf{x}}^r) G_r^{-1} y - U(x') M_r (t - g_r^\top G_r^{-1} y)$$

Taking $y = k(\tilde{\mathbf{x}}^r, x)$ and $t = g(x)^\top$ for every $x' \in \tilde{\mathbf{x}}^d$ gives the expression of $k_{d|r}(x)$, since $t - g_r^\top G_r^{-1} y$ is then equal to $U(x)^\top$. Taking the columns of

$k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)$ and of $g(\tilde{\mathbf{x}}^d)^\top$ for y and t gives the expression of $S_{d|r} = D - B^\top A^{-1} B$ in the same way, and $y = \tilde{\mathbf{z}}^r$ with $t = 0$ gives that of $\tilde{\mathbf{z}}_{d|r}$, since $g_r^\top G_r^{-1} \tilde{\mathbf{z}}^r = M_r^{-1} \hat{m}_{\text{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$.

$S_{d|r}^0$ is the Schur complement of the SPD matrix $G(\tilde{\mathbf{x}})$ with respect to $G(\tilde{\mathbf{x}}^r)$, and is therefore SPD [7]. Adding the SPSP matrix $U(\tilde{\mathbf{x}}^d) M_r U(\tilde{\mathbf{x}}^d)^\top$ keeps $S_{d|r}$ SPD. Finally, adding $g(\tilde{\mathbf{x}}) \gamma$ to the observations adds $U(\tilde{\mathbf{x}}^d) \gamma$ to $\tilde{\mathbf{z}}_{d|r}^0$ and γ to $\hat{m}_{\text{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$, which leaves $\tilde{\mathbf{z}}_{d|r}$ unchanged.

B.16 Proof of Proposition 4.10 and Proposition 4.11

By Proposition 4.6, $k_{d|r}(x) = k_{d|r}^0(x) + U(\tilde{\mathbf{x}}^d) M(\tilde{\mathbf{x}}^r) U(x)^\top$ and $S_{d|r} = S_{d|r}^0 + U(\tilde{\mathbf{x}}^d) M(\tilde{\mathbf{x}}^r) U(\tilde{\mathbf{x}}^d)^\top \succeq S_{d|r}^0$, so that $\|\cdot\|_{S_{d|r}} \leq \|\cdot\|_{S_{d|r}^0}$. Applying the triangular inequality to the norm $\|\cdot\|_{S_{d|r}}$ in Proposition 4.7 thus gives:

$$e_r(x) \leq \|k_{d|r}^0(x)\|_{S_{d|r}^0} + \|U(\tilde{\mathbf{x}}^d) M(\tilde{\mathbf{x}}^r) U(x)^\top\|_{S_{d|r}}$$

with equality when $k_{d|r}^0(x) = 0$. Writing $U_d \triangleq U(\tilde{\mathbf{x}}^d)$ and $M_r \triangleq M(\tilde{\mathbf{x}}^r)$ for brevity, the Woodbury formula [94] applied to $S_{d|r} = S_{d|r}^0 + U_d M_r U_d^\top$ yields $U_d^\top S_{d|r}^{-1} U_d = P - P(M_r^{-1} + P)^{-1} P = P(M_r^{-1} + P)^{-1} M_r^{-1}$, hence:

$$\begin{aligned} \|U_d M_r U(x)^\top\|_{S_{d|r}}^2 &= \|U(x) M_r P (M_r^{-1} + P)^{-1} U(x)^\top\|_{\text{op}} \\ &= \|U(x) (M_r - (M_r^{-1} + P)^{-1}) U(x)^\top\|_{\text{op}} \end{aligned}$$

which proves Proposition 4.10.

For Proposition 4.11, note that $S_{d|r}^0 - I_{\tilde{N}^d} \otimes \tilde{R}$ is the Schur complement of $k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \begin{bmatrix} I_{\tilde{N}^r} & 0 \\ 0 & 0 \end{bmatrix} \otimes \tilde{R}$ with respect to $G(\tilde{\mathbf{x}}^r)$. This matrix being SPSP, so is its Schur complement [7], and $S_{d|r}^0 \succeq I_{\tilde{N}^d} \otimes \tilde{R} \succeq \tilde{\lambda} I_{\tilde{N}^d d_z}$. Consequently, $\|k_{d|r}^0(x)\|_{S_{d|r}^0} \leq \|k_{d|r}^0(x)\|_{\text{op}} / \sqrt{\tilde{\lambda}}$ and $P \preceq U_d^\top U_d / \tilde{\lambda}$. Since $P \mapsto M_r - (M_r^{-1} + P)^{-1}$ is non-decreasing for the Loewner order, applying these inequalities to Proposition 4.10 concludes the proof.

B.17 Proof of Proposition 4.13

Let $N(K)$ gather the operator norms of the $d_z \times d_z$ blocks of a matrix K . Since $\|Kv\|_2 \leq \|N(K)u\|_2$, where u gathers the euclidean norms of the blocks of v , it comes that:

$$\|K\|_{\text{op}} \leq \|N(K)\|_{\text{op}} \leq \sqrt{\|N(K)\|_1 \|N(K)\|_\infty}$$

the last two norms being the largest column and row sums of $N(K)$ [7]. Applied to the symmetric $G(\tilde{\mathbf{x}}^j)$, this gives the blockwise Gershgorin bound $\lambda_{\max}(G(\tilde{\mathbf{x}}^j)) \leq \lambda_j^+$, and applied to (4.75), it gives $\|k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\|_{\text{op}} \leq$

$\sqrt{\tilde{N}^r \tilde{N}^d} \alpha$, $\|k(\tilde{\mathbf{x}}^r, x)\|_{\text{op}} \leq \sqrt{\tilde{N}^r} \kappa$, $\|k(\tilde{\mathbf{x}}^d, x)\|_{\text{op}} \leq \sqrt{\tilde{N}^d} \eta$, $\|g(\tilde{\mathbf{x}}^j)\|_{\text{op}} \leq \sqrt{\tilde{N}^j} g_{\max}$ and $\|\tilde{\mathbf{z}}^j - g(\tilde{\mathbf{x}}^j) \beta_0\|_2 \leq \sqrt{\tilde{N}^j} \tilde{z}_{\max}$. Since $G(\tilde{\mathbf{x}}^r)$ is SPD, it follows that:

$$\begin{aligned} \|k_{d|r}^0(x)\|_{\text{op}} &\leq \|k(\tilde{\mathbf{x}}^d, x)\|_{\text{op}} + \frac{\|k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\|_{\text{op}} \|k(\tilde{\mathbf{x}}^r, x)\|_{\text{op}}}{\lambda_r^-} \\ &\leq \sqrt{\tilde{N}^d} \eta + \frac{\sqrt{\tilde{N}^r \tilde{N}^d} \alpha \cdot \sqrt{\tilde{N}^r} \kappa}{\lambda_r^-} = \sqrt{\tilde{N}^d} \frac{\lambda_r^- \eta + \alpha \tilde{N}^r \kappa}{\lambda_r^-} \end{aligned}$$

The same steps, applied to $U(\tilde{\mathbf{x}}^d)$ and $U(x)$, give:

$$\|U(\tilde{\mathbf{x}}^d)\|_{\text{op}} \leq \sqrt{\tilde{N}^d} g_{\max} \rho_r, \quad \|U(x)\|_{\text{op}} \leq g_{\max} \frac{\lambda_r^- + \tilde{N}^r \kappa}{\lambda_r^-}$$

Besides, for every unit vector $v \in \mathbb{R}^{r_{\text{UK}} d_z}$, the Rayleigh quotient [7] and the definitions of λ_r^+ and τ_r give:

$$v^\top M(\tilde{\mathbf{x}}^r)^{-1} v = (g(\tilde{\mathbf{x}}^r) v)^\top G(\tilde{\mathbf{x}}^r)^{-1} (g(\tilde{\mathbf{x}}^r) v) \geq \frac{\|g(\tilde{\mathbf{x}}^r) v\|_2^2}{\lambda_r^+} \geq \frac{\tilde{N}^r g_{\max}^2}{\tau_r \lambda_r^+}$$

hence $\|M(\tilde{\mathbf{x}}^r)\|_{\text{op}} \leq \tau_r \lambda_r^+ / (\tilde{N}^r g_{\max}^2)$. Combining these inequalities with $k_{d|r}(x) = k_{d|r}^0(x) + U(\tilde{\mathbf{x}}^d) M(\tilde{\mathbf{x}}^r) U(x)^\top$ from Proposition 4.6 through the triangular inequality gives the bound on $\|k_{d|r}(x)\|_{\text{op}}$.

Since $S_{d|r} - S_{d|r}^0$ is SPSPD by Proposition 4.6, $\lambda_{\min}(S_{d|r}) \geq \lambda_{\min}(S_{d|r}^0)$. For every unit vector $v \in \mathbb{R}^{\tilde{N}^d d_z}$:

$$\begin{aligned} v^\top S_{d|r}^0 v &= v^\top G(\tilde{\mathbf{x}}^d) v - (k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) v)^\top G(\tilde{\mathbf{x}}^r)^{-1} (k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) v) \\ &\geq \lambda_d^- - \frac{\|k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)\|_{\text{op}}^2}{\lambda_r^-} \geq \frac{\lambda_r^- \lambda_d^- - \alpha^2 \tilde{N}^r \tilde{N}^d}{\lambda_r^-} \end{aligned}$$

This lower bound does not depend on v , and thus also bounds $\lambda_{\min}(S_{d|r}^0)$, the minimum of $v^\top S_{d|r}^0 v$ over the unit vectors. It is positive by assumption, so that $\|S_{d|r}^{-1}\|_{\text{op}} = \lambda_{\min}(S_{d|r})^{-1} \leq s_{\max}$.

Finally, since $\tilde{\mathbf{z}}_{d|r}$ is unchanged when a trend is added to the observations, Proposition 4.6 can be applied to the observations $\tilde{\mathbf{z}} - g(\tilde{\mathbf{x}}) \beta_0$:

$$\begin{aligned} \tilde{\mathbf{z}}_{d|r} &= (\tilde{\mathbf{z}}^d - g(\tilde{\mathbf{x}}^d) \beta_0) - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r) G(\tilde{\mathbf{x}}^r)^{-1} (\tilde{\mathbf{z}}^r - g(\tilde{\mathbf{x}}^r) \beta_0) \\ &\quad - U(\tilde{\mathbf{x}}^d) (\hat{m}_{\text{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) - \beta_0) \end{aligned}$$

The steps used for $k_{d|r}^0(x)$ bound the norm of the first two terms by $\sqrt{\tilde{N}^d} \rho_r \tilde{z}_{\max}$, while:

$$\begin{aligned} \|\hat{m}_{\text{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) - \beta_0\|_2 &= \|M(\tilde{\mathbf{x}}^r) g(\tilde{\mathbf{x}}^r)^\top G(\tilde{\mathbf{x}}^r)^{-1} (\tilde{\mathbf{z}}^r - g(\tilde{\mathbf{x}}^r) \beta_0)\|_2 \\ &\leq \frac{\tau_r \lambda_r^+}{\tilde{N}^r g_{\max}^2} \cdot \frac{\sqrt{\tilde{N}^r} g_{\max} \cdot \sqrt{\tilde{N}^r} \tilde{z}_{\max}}{\lambda_r^-} = \frac{\tau_r \lambda_r^+}{g_{\max} \lambda_r^-} \tilde{z}_{\max} \end{aligned}$$

which concludes with the triangular inequality.

BIBLIOGRAPHY

- [1] O. Cappé, E. Moulines, and T. Rydén. *Inference in Hidden Markov Models*. Springer Series in Statistics. New York, NY: Springer, 2005.
- [2] G. A. Einicke. “Nonlinear Prediction, Filtering and Smoothing”. In: *Smoothing, Filtering and Prediction: Estimating the Past, Present and Future*. 2nd. Prime Publishing, 2019, pp. 293–322.
- [3] H. L. Van Trees. “Classical Detection and Estimation Theory”. In: *Detection, Estimation, and Modulation Theory, Part I*. New York, NY: John Wiley & Sons, 1968.
- [4] S. Särkkä. “Bayesian Filtering Equations and Exact Solutions”. In: *Bayesian Filtering and Smoothing*. Cambridge, UK: Cambridge University Press, 2013.
- [5] M. S. Arulampalam, S. Maskell, N. Gordon, and T. Clapp. “A Tutorial on Particle Filters for Online Nonlinear/Non-Gaussian Bayesian Tracking”. In: *IEEE Transactions on Signal Processing* 50.2 (2002), pp. 174–188.
- [6] R. E. Kalman. “A New Approach to Linear Filtering and Prediction Problems”. In: *Journal of Basic Engineering* 82.1 (1960), pp. 35–45.
- [7] R. A. Horn and C. R. Johnson. “Positive Definite and Semidefinite Matrices”. In: *Matrix Analysis*. 2nd. Cambridge, UK: Cambridge University Press, 2013.
- [8] B. D. O. Anderson and J. B. Moore. “The Discrete-Time Kalman Filter”. In: *Optimal Filtering*. Englewood Cliffs, NJ: Prentice-Hall, 1979.
- [9] R. S. Bucy and P. D. Joseph. *Filtering for Stochastic Processes with Applications to Guidance*. New York, NY: Interscience Publishers, 1968.
- [10] B. D. O. Anderson and J. B. Moore. “Computational Aspects”. In: *Optimal Filtering*. Englewood Cliffs, NJ: Prentice-Hall, 1979.
- [11] S. Särkkä. “Extended and Unscented Kalman Filtering”. In: *Bayesian Filtering and Smoothing*. Cambridge, UK: Cambridge University Press, 2013.
- [12] J. Havlík and O. Straka. “Performance Evaluation of Iterated Extended Kalman Filter with Variable Step-Length”. In: *Journal of Physics: Conference Series* 659 (2015), p. 012022.
- [13] B. M. Bell and F. W. Cathey. “The Iterated Kalman Filter Update as a Gauss-Newton Method”. In: *IEEE Transactions on Automatic Control* 38.2 (1993), pp. 294–297.

- [14] A. H. Jazwinski. “Nonlinear Filtering”. In: *Stochastic Processes and Filtering Theory*. New York, NY: Academic Press, 1970.
- [15] S. J. Julier and J. K. Uhlmann. “A New Extension of the Kalman Filter to Nonlinear Systems”. In: *Proceedings of AeroSense: The 11th International Symposium on Aerospace/Defense Sensing, Simulation and Controls*. 1997, pp. 182–193.
- [16] G. H. Golub and C. F. Van Loan. *Matrix Computations*. 4th. Baltimore, MD: Johns Hopkins University Press, 2013.
- [17] S. J. Julier and J. K. Uhlmann. “Unscented Filtering and Nonlinear Estimation”. In: *Proceedings of the IEEE* 92.3 (2004), pp. 401–422.
- [18] R. van der Merwe and E. A. Wan. “The Square-Root Unscented Kalman Filter for State and Parameter-Estimation”. In: *2001 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. Vol. 6. 2001, pp. 3461–3464.
- [19] G. Evensen. “The Ensemble Kalman Filter: Theoretical Formulation and Practical Implementation”. In: *Ocean Dynamics* 53.4 (2003), pp. 343–367.
- [20] P. L. Houtekamer and F. Zhang. “Review of the Ensemble Kalman Filter for Atmospheric Data Assimilation”. In: *Monthly Weather Review* 144.12 (2016), pp. 4498–4532.
- [21] N. Metropolis and S. Ulam. “The Monte Carlo Method”. In: *Journal of the American Statistical Association* 44.247 (1949), pp. 335–341.
- [22] F. Le Gland. “Méthodes de Monte Carlo”. In: *Filtrage bayésien et approximation particulière*. Cours SOD333. ENSTA Paris, 2021.
- [23] C. P. Robert and G. Casella. “Monte Carlo Integration”. In: *Monte Carlo Statistical Methods*. 2nd. New York, NY: Springer, 2004.
- [24] H. Kahn and A. W. Marshall. “Methods of Reducing Sample Size in Monte Carlo Computations”. In: *Journal of the Operations Research Society of America* 1.5 (1953), pp. 263–278.
- [25] J. Geweke. “Bayesian Inference in Econometric Models Using Monte Carlo Integration”. In: *Econometrica* 57.6 (1989), pp. 1317–1339.
- [26] A. Doucet, S. Godsill, and C. Andrieu. “On Sequential Monte Carlo Sampling Methods for Bayesian Filtering”. In: *Statistics and Computing* 10.3 (2000), pp. 197–208.
- [27] A. Doucet, N. de Freitas, and N. Gordon. “An Introduction to Sequential Monte Carlo Methods”. In: *Sequential Monte Carlo Methods in Practice*. New York, NY: Springer New York, 2001, pp. 3–14.
- [28] P. Del Moral. *Feynman-Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Probability and its Applications. New York, NY: Springer, 2004.
- [29] F. Le Gland. “Généralisation : distributions de Feynman-Kac”. In: *Filtrage bayésien et approximation particulière*. Cours SOD333. ENSTA Paris, 2021.
- [30] N. J. Higham. “Computing a Nearest Symmetric Positive Semidefinite Matrix”. In: *Linear Algebra and its Applications* 103 (1988), pp. 103–118.

- [31] A. Kong, J. S. Liu, and W. H. Wong. “Sequential Imputations and Bayesian Missing Data Problems”. In: *Journal of the American Statistical Association* 89.425 (1994), pp. 278–288.
- [32] D. B. Rubin. “Using the SIR Algorithm to Simulate Posterior Distributions”. In: *Bayesian Statistics 3*. Oxford, UK: Oxford University Press, 1988, pp. 395–402.
- [33] N. J. Gordon, D. J. Salmond, and A. F. M. Smith. “Novel Approach to Nonlinear/Non-Gaussian Bayesian State Estimation”. In: *IEE Proceedings F – Radar and Signal Processing* 140.2 (1993), pp. 107–113.
- [34] T. Li, M. Bolić, and P. M. Djurić. “Resampling Methods for Particle Filtering: Classification, Implementation, and Strategies”. In: *IEEE Signal Processing Magazine* 32.3 (2015), pp. 70–86.
- [35] P. Kozierski, M. Lis, and J. Ziętkiewicz. “Resampling in Particle Filtering – Comparison”. In: *Studia z Automatyki i Informatyki* 38 (2013), pp. 35–64.
- [36] G. Kitagawa. “Monte Carlo Filter and Smoother for Non-Gaussian Nonlinear State Space Models”. In: *Journal of Computational and Graphical Statistics* 5.1 (1996), pp. 1–25.
- [37] D. Crisan and A. Doucet. “A Survey of Convergence Results on Particle Filtering Methods for Practitioners”. In: *IEEE Transactions on Signal Processing* 50.3 (2002), pp. 736–746.
- [38] N. Oudjane. “Stabilité et approximations particulières en filtrage non linéaire. Application au pistage”. PhD thesis. Université de Rennes 1, 2000.
- [39] C. Snyder, T. Bengtsson, P. Bickel, and J. Anderson. “Obstacles to High-Dimensional Particle Filtering”. In: *Monthly Weather Review* 136.12 (2008), pp. 4629–4640.
- [40] B. Ristic, S. Arulampalam, and N. Gordon. “A Tutorial on Particle Filters”. In: *Beyond the Kalman Filter: Particle Filters for Tracking Applications*. Boston, MA: Artech House, 2004.
- [41] T. Li, S. Sun, T. P. Sattar, and J. M. Corchado. “Fight Sample Degeneracy and Impoverishment in Particle Filters: A Review of Intelligent Approaches”. In: *Expert Systems with Applications* 41.8 (2014), pp. 3944–3954.
- [42] S. Särkkä. “Particle Filtering”. In: *Bayesian Filtering and Smoothing*. Cambridge, UK: Cambridge University Press, 2013.
- [43] J. S. Liu and R. Chen. “Sequential Monte Carlo Methods for Dynamic Systems”. In: *Journal of the American Statistical Association* 93.443 (1998), pp. 1032–1044.
- [44] C. Musso, N. Oudjane, and F. Le Gland. “Improving Regularised Particle Filters”. In: *Sequential Monte Carlo Methods in Practice*. New York, NY: Springer New York, 2001, pp. 247–271.
- [45] M. Rosenblatt. “Remarks on Some Nonparametric Estimates of a Density Function”. In: *The Annals of Mathematical Statistics* 27.3 (1956), pp. 832–837.
- [46] E. Parzen. “On Estimation of a Probability Density Function and Mode”. In: *The Annals of Mathematical Statistics* 33.3 (1962), pp. 1065–1076.

- [47] B. W. Silverman. “The Kernel Method for Multivariate Data”. In: *Density Estimation for Statistics and Data Analysis*. London: Chapman and Hall, 1986.
- [48] V. A. Epanechnikov. “Non-Parametric Estimation of a Multivariate Probability Density”. In: *Theory of Probability and Its Applications* 14.1 (1969), pp. 153–158.
- [49] L. Devroye. “Nonuniform Random Variate Generation”. In: *Handbooks in Operations Research and Management Science, Vol. 13: Simulation*. Amsterdam: Elsevier, 2006, pp. 83–121.
- [50] M. K. Pitt and N. Shephard. “Filtering via Simulation: Auxiliary Particle Filters”. In: *Journal of the American Statistical Association* 94.446 (1999), pp. 590–599.
- [51] R. Douc, E. Moulines, and J. Olsson. “Optimality of the Auxiliary Particle Filter”. In: *Probability and Mathematical Statistics* 29.1 (2009), pp. 1–28.
- [52] T. Schön, F. Gustafsson, and P.-J. Nordlund. “Marginalized Particle Filters for Mixed Linear/Nonlinear State-Space Models”. In: *IEEE Transactions on Signal Processing* 53.7 (2005), pp. 2279–2289.
- [53] G. Casella and C. P. Robert. “Rao-Blackwellisation of Sampling Schemes”. In: *Biometrika* 83.1 (1996), pp. 81–94.
- [54] G. Hendeby, R. Karlsson, and F. Gustafsson. “A New Formulation of the Rao-Blackwellized Particle Filter”. In: *2007 IEEE/SP 14th Workshop on Statistical Signal Processing (SSP)*. 2007, pp. 84–88.
- [55] R. Chen and J. S. Liu. “Mixture Kalman Filters”. In: *Journal of the Royal Statistical Society: Series B* 62.3 (2000), pp. 493–508.
- [56] J. Ko and D. Fox. “GP-BayesFilters: Bayesian Filtering Using Gaussian Process Prediction and Observation Models”. In: *Autonomous Robots* 27.1 (2009), pp. 75–90.
- [57] C. Palmier, A. Giremus, P. Minvielle, C. P. Vacar, and G. Bourmaud. “Terrain-Aided SLAM with Limited-Size Reference Maps Using Gaussian Processes”. In: *2023 31st European Signal Processing Conference (EUSIPCO)*. 2023, pp. 1–5.
- [58] R. J. Adler and J. E. Taylor. “Gaussian Fields”. In: *Random Fields and Geometry*. Springer Monographs in Mathematics. New York, NY: Springer, 2007.
- [59] H. Cramér and H. Wold. “Some Theorems on Distribution Functions”. In: *Journal of the London Mathematical Society* s1-11.4 (1936), pp. 290–294.
- [60] I. J. Schoenberg. “Metric Spaces and Positive Definite Functions”. In: *Transactions of the American Mathematical Society* 44.3 (1938), pp. 522–536.
- [61] M. L. Stein. “Linear Prediction”. In: *Interpolation of Spatial Data: Some Theory for Kriging*. Springer Series in Statistics. New York, NY: Springer, 1999.
- [62] B. Matérn. *Spatial Variation*. Tech. rep. 5. Stockholm: Statens Skogsforskningsinstitut, 1960.

- [63] M. Abramowitz and I. A. Stegun. *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Vol. 55. Applied Mathematics Series. Washington, D.C.: National Bureau of Standards, 1964.
- [64] C. E. Rasmussen and C. K. I. Williams. “Covariance Functions”. In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.
- [65] G. Matheron. “Principles of Geostatistics”. In: *Economic Geology* 58.8 (1963), pp. 1246–1266.
- [66] C. E. Rasmussen and C. K. I. Williams. “Regression”. In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.
- [67] M. Ebden. *Gaussian Processes for Regression: A Quick Introduction*. arXiv: 1505.02965. 2015.
- [68] G. S. Kimeldorf and G. Wahba. “A Correspondence Between Bayesian Estimation on Stochastic Processes and Smoothing by Splines”. In: *The Annals of Mathematical Statistics* 41.2 (1970), pp. 495–502.
- [69] D. G. Luenberger. “Local Theory of Constrained Optimization”. In: *Optimization by Vector Space Methods*. New York, NY: John Wiley & Sons, 1969.
- [70] G. Matheron. *Le krigeage universel*. Tech. rep. 1. Fontainebleau: École des Mines de Paris, 1969.
- [71] P. S. Palar and K. Shimoyama. “On Efficient Global Optimization via Universal Kriging Surrogate Models”. In: *Structural and Multidisciplinary Optimization* (2018).
- [72] M. Syirojudin, E. Haryono, and S. Ahadi. “The Accuracy of Geostatistics for Regional Geomagnetic Modeling in an Archipelago Setting”. In: *Scientific Reports* 12 (2022), p. 7441.
- [73] H. Omre and K. B. Halvorsen. “The Bayesian Bridge Between Simple and Universal Kriging”. In: *Mathematical Geology* 21.7 (1989), pp. 767–786.
- [74] A. O’Hagan. “Curve Fitting and Optimal Design for Prediction”. In: *Journal of the Royal Statistical Society, Series B* 40 (1978). With discussion, pp. 1–42.
- [75] C. K. I. Williams and M. Seeger. “Using the Nyström Method to Speed Up Kernel Machines”. In: *Advances in Neural Information Processing Systems* 13. 2001, pp. 682–688.
- [76] M. K. Titsias. “Variational Learning of Inducing Variables in Sparse Gaussian Processes”. In: *Proceedings of the 12th International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2009, pp. 567–574.
- [77] J. Mercer. “Functions of Positive and Negative Type, and their Connection with the Theory of Integral Equations”. In: *Philosophical Transactions of the Royal Society A* 209 (1909), pp. 415–446.
- [78] C. Eckart and G. Young. “The Approximation of One Matrix by Another of Lower Rank”. In: *Psychometrika* 1.3 (1936), pp. 211–218.

- [79] L. Mirsky. “Symmetric Gauge Functions and Unitarily Invariant Norms”. In: *The Quarterly Journal of Mathematics* 11.1 (1960), pp. 50–59.
- [80] R. J. Adler and J. E. Taylor. “Orthogonal Expansions”. In: *Random Fields and Geometry*. Springer Monographs in Mathematics. New York, NY: Springer, 2007.
- [81] N. Aronszajn. “Theory of Reproducing Kernels”. In: *Transactions of the American Mathematical Society* 68.3 (1950), pp. 337–404.
- [82] L. Dümbgen. *Reproducing Kernel Hilbert Spaces*. Reading notes, Universität Bern. 2021.
- [83] H. Daumé III. *From Zero to Reproducing Kernel Hilbert Spaces in Twelve Pages or Less*. 2004.
- [84] G. Kimeldorf and G. Wahba. “Some Results on Tchebycheffian Spline Functions”. In: *Journal of Mathematical Analysis and Applications* 33.1 (1971), pp. 82–95.
- [85] B. Schölkopf, R. Herbrich, and A. J. Smola. “A Generalized Representer Theorem”. In: *Proceedings of the 14th Annual Conference on Computational Learning Theory (COLT)*. 2001, pp. 416–426.
- [86] M. Kanagawa, P. Hennig, D. Sejdinovic, and B. K. Sriperumbudur. *Gaussian Processes and Kernel Methods: A Review on Connections and Equivalences*. arXiv: 1807.02582. 2018.
- [87] C. E. Rasmussen and C. K. I. Williams. “Relationships between GPs and Other Models”. In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.
- [88] L. Song, K. Fukumizu, and A. Gretton. “Kernel Embeddings of Conditional Distributions”. In: *IEEE Signal Processing Magazine* 30.4 (2013), pp. 98–111.
- [89] G. H. W. Gebhardt, A. Kupcsik, and G. Neumann. “The Kernel Kalman Rule”. In: *Machine Learning* 108 (2019), pp. 2113–2157.
- [90] M. Sun, M. E. Davies, I. K. Proudler, and J. R. Hopgood. “Adaptive Kernel Kalman Filter”. In: *IEEE Transactions on Signal Processing* 71 (2023), pp. 713–726.
- [91] C. E. Rasmussen and C. K. I. Williams. “Model Selection and Adaptation of Hyperparameters”. In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.
- [92] D. J. C. MacKay. “Comparison of Approximate Methods for Handling Hyperparameters”. In: *Neural Computation* 11.5 (1999), pp. 1035–1068.
- [93] V. Bankey et al. *Digital Data Grids for the Magnetic Anomaly Map of North America*. Tech. rep. Open-File Report 02-414. <https://pubs.usgs.gov/publication/ofr02414>. U.S. Geological Survey, 2002.
- [94] K. B. Petersen and M. S. Pedersen. *The Matrix Cookbook*. Technical University of Denmark. 2012.
- [95] B. Hubert, K. Dahia, N. Merlinge, and A. Giremus. “Adaptive Kriging Particle Filter and its Application to Terrain-Aided Navigation”. In: *2024 27th International Conference on Information Fusion (FUSION)*. 2024, pp. 1–7.

- [96] G. N. Lance and W. T. Williams. “A General Theory of Classificatory Sorting Strategies: 1. Hierarchical Systems”. In: *The Computer Journal* 9.4 (1967), pp. 373–380.
- [97] J. H. Ward. “Hierarchical Grouping to Optimize an Objective Function”. In: *Journal of the American Statistical Association* 58.301 (1963), pp. 236–244.
- [98] J. MacQueen. “Some Methods for Classification and Analysis of Multivariate Observations”. In: *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*. Vol. 1. 1967, pp. 281–297.
- [99] S. P. Lloyd. “Least Squares Quantization in PCM”. In: *IEEE Transactions on Information Theory* 28.2 (1982), pp. 129–137.
- [100] K. Fukunaga and L. Hostetler. “The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition”. In: *IEEE Transactions on Information Theory* 21.1 (1975), pp. 32–40.
- [101] D. Comaniciu and P. Meer. “Mean Shift: A Robust Approach Toward Feature Space Analysis”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 24.5 (2002), pp. 603–619.
- [102] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu. “A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise”. In: *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*. 1996, pp. 226–231.
- [103] A. Murangira. “Nouvelles approches en filtrage particulière. Application au recalage de la navigation inertielle”. PhD thesis. Université de Technologie de Troyes, 2014.
- [104] A. Gut. *Probability: A Graduate Course*. 2nd. Springer, 2013.
- [105] G. H. Golub and G. Meurant. *Matrices, Moments and Quadrature with Applications*. Princeton University Press, 2010.
- [106] M. L. Stein. “The Screening Effect in Kriging”. In: *The Annals of Statistics* 30.1 (2002), pp. 298–323.
- [107] F. Bachoc and R. Furrer. “On the Smallest Eigenvalues of Covariance Matrices of Multivariate Spatial Processes”. In: *Stat* (2016).
- [108] M. Kok and A. Solin. “Scalable Magnetic Field SLAM in 3D Using Gaussian Process Maps”. In: *2018 21st International Conference on Information Fusion (FUSION)*. 2018, pp. 1353–1360.
- [109] A. Solin and S. Särkkä. “Hilbert Space Methods for Reduced-Rank Gaussian Process Regression”. In: *Statistics and Computing* 30 (2020), pp. 419–446.
- [110] N. Wahlström, M. Kok, T. B. Schön, and F. Gustafsson. “Modeling Magnetic Fields Using Gaussian Processes”. In: *2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2013, pp. 3522–3526.
- [111] A. Solin, M. Kok, N. Wahlström, T. B. Schön, and S. Särkkä. “Modeling and Interpolation of the Ambient Magnetic Field by Gaussian Processes”. In: *IEEE Transactions on Robotics* 34.4 (2018), pp. 1112–1127.

- [112] B. Hubert, N. Merlinge, K. Dahia, and A. Giremus. “Learning Observation Model for SLAM via Random Fourier Features in a Kernel Kalman Filter”. In: *2026 29th International Conference on Information Fusion (FUSION)*. 2026, pp. 1–8.
- [113] A. Rahimi and B. Recht. “Random Features for Large-Scale Kernel Machines”. In: *Advances in Neural Information Processing Systems 20*. 2007, pp. 1177–1184.
- [114] F. Liu, X. Huang, Y. Chen, and J. A. K. Suykens. “Random Features for Kernel Approximation: A Survey on Algorithms, Theory, and Beyond”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44.10 (2022), pp. 7128–7148.
- [115] Z. Qin, B. Chen, and N. Zheng. “Random Fourier Feature Kernel Recursive Least Squares”. In: *2017 International Joint Conference on Neural Networks (IJCNN)*. 2017, pp. 2881–2886.
- [116] F. Viset, R. Helmons, and M. Kok. “An Extended Kalman Filter for Magnetic Field SLAM Using Gaussian Process Regression”. In: *Sensors* 22.8 (2022), p. 2833.
- [117] T. W. Anderson. “The Multivariate Normal Distribution”. In: *An Introduction to Multivariate Statistical Analysis*. 3rd. Hoboken, NJ: John Wiley & Sons, 2003.
- [118] F. Zhang, ed. *The Schur Complement and Its Applications*. Numerical Methods and Algorithms. New York, NY: Springer, 2005.
- [119] M. A. Woodbury. *Inverting Modified Matrices*. Tech. rep. 42. Princeton, NJ: Statistical Research Group, Princeton University, 1950.
- [120] N. I. Akhiezer and I. M. Glazman. *Theory of Linear Operators in Hilbert Space*. New York, NY: Dover Publications, 1993.
- [121] P. Guttorp and T. Gneiting. “On the Matérn Correlation Family”. In: *Biometrika* 93.4 (2006), pp. 989–995.
- [122] W. Rudin. “Integration on Product Spaces”. In: *Real and Complex Analysis*. 3rd. International Series in Pure and Applied Mathematics. New York, NY: McGraw-Hill, 1987, pp. 160–174.
- [123] H. Brezis. “Compact Operators. Spectral Decomposition of Self-Adjoint Compact Operators”. In: *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Universitext. New York, NY: Springer, 2011.
- [124] W. Rudin. “Sequences and Series of Functions”. In: *Principles of Mathematical Analysis*. 3rd. International Series in Pure and Applied Mathematics. New York, NY: McGraw-Hill, 1976, pp. 143–171.